Existential Subtype Entailment Is Decidable for Non-Structural Subtyping

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Existential subtype entailment asks whether every solution of a finite system φ of subtype constraints extends to a solution of a second system ψ over additional hidden variables, that is, whether $\varphi \models \exists \bar{z}\, \psi$. Subsumption between constrained type schemes and the correctness of constraint simplification are instances of this question. For non-structural subtyping the problem has been open since 1996, and it has been suggested that it might be undecidable. We prove that it is decidable for one constructor f of every arity and every variance, with bottom and top, over finite, regular, and arbitrary trees, and also with bottom only when f is covariant. In particular, subsumption of constrained types over arrows, bottom and top is decidable, with no restriction on the polarity of the variables. When both systems are satisfiable in the tree class considered and the entailment fails, a counterexample word of length triply exponential in the number of variables after flattening certifies the failure, uniformly in the arity.

We represent each tree by two cones of words, those that meet a leaf of either sign, which turns every system into a Horn system with unary rules, and we characterize the entailment by three conditions on single seeds. Two of them become hole problems for finite readers: one for a single word, and one for two words that share a suffix, in which a word may also obstruct itself through a comparison with its own suffixes. For two words, outside one hard case, finite automata and a pumping argument give singly exponential holes. In the hard case, the mortal holes, we write the two words as one word around a separator, bound its hierarchical depth, and extend the bounded-depth compression from the decision procedure for plain entailment so that it preserves a comparison anchored at the separator. The decidability proof is mechanized in Lean 4. In a separate part that is not mechanized, we show that a search over compression summaries decides the problem in 2-EXPSPACE, and in PSPACE for a unary constructor, that shortest counterexamples can be exponentially long, and that for a covariant constructor the problem is PSPACE-hard for every fixed arity at least two.

1 Introduction

Type inference with subtyping expresses the requirements of a program as subtype constraints between type expressions with unknowns [Aiken and Wimmers 1993; Eifrig et al. 1995; Fuh and Mishra 1990; Mitchell 1991; Pottier 1998a]. An inferred *constrained type* $\forall \bar{\alpha}.\, C \Rightarrow \tau$ pairs a type τ with a conjunction C of constraints; its instances are the types $\rho(\tau)$ for the solutions ρ of C [Trifonov and Smith 1996]. Inferred constraint sets grow quickly, and practical systems simplify them [Aiken et al. 1999; Fähndrich and Aiken 1996; Flanagan and Felleisen 1999; Pottier 1996, 2001; Trifonov and Smith 1996]. Whether a simplification is correct, and whether one constrained type may be used where another is expected, are questions about the solution sets of constraint systems.

The simplest such question is *entailment*: does every solution of a conjunction φ satisfy an inequality $x \le y$? Constrained types, however, also involve variables that occur on one side only. Let $\varphi(\bar{x})$ and $\psi(\bar{x}, \bar{z})$ be finite conjunctions of subtype constraints, where the *hidden* variables $\bar{z}$ do not occur in φ. *Existential entailment* $\varphi \models \exists \bar{z}\, \psi$ holds if every solution of φ extends to a solution of ψ by some choice of trees for $\bar{z}$ (Definition 2.4); logically, it is the validity of $\forall \bar{x}\, (\varphi \Rightarrow \exists \bar{z}\, \psi)$. Plain entailment is the case without hidden variables, and satisfiability of ψ is the case in which φ is empty and every variable is hidden.

Subsumption and Simplification. Trifonov and Smith [1996] define subtyping between constrained types semantically: $\forall \bar{\alpha}.\, C_1 \Rightarrow \tau_1$ is below $\forall \bar{\beta}.\, C_2 \Rightarrow \tau_2$ if every instance of the right-hand type is a supertype of some instance of the left-hand type. With $\bar{\alpha}$ and $\bar{\beta}$ renamed apart from each other and from the free variables of both types (Proposition 12.1), this is the existential entailment

$$\forall \bar{\alpha}.\, C_1 \Rightarrow \tau_1 \;\le\; \forall \bar{\beta}.\, C_2 \Rightarrow \tau_2 \qquad \Longleftrightarrow \qquad C_2 \models \exists \bar{\alpha}\, (C_1 \wedge \tau_1 \le \tau_2), \tag{1}$$

in which the variables of the right-hand type are universally quantified and the witnesses belong to the smaller, more general type. For example, $\forall\alpha.\,\alpha \to \alpha \le \beta \to \beta$ holds with the witness $\alpha = \beta$, while the converse fails; Proposition 12.1 adds the assumptions on program variables of Trifonov and Smith [1996]. Likewise, replacing a system $C(\bar{x}, \bar{z})$ by $D(\bar{x}, \bar{k})$, where $\bar{x}$ remain observable, is correct when $C \models \exists\bar{k}\, D$ and $D \models \exists\bar{z}\, C$. Removing a redundant constraint needs only plain entailment; eliminating or merging internal variables needs the existential form (Section 12). The arrow is contravariant in its domain, so a variable of a constrained type of a higher-order program typically occurs both co- and contravariantly, as α does in $\alpha \to \alpha$.

Why Plain Entailment Does Not Suffice. Consider trees built from a bottom element $\bot$ and a binary covariant constructor f, and the consequent

$$\psi \;=\; x \le z \;\wedge\; z \le y \;\wedge\; z \le f(z, z).$$

The literal $z \le f(z, z)$ says that all subtrees of z lie below z. Write $x \cdot \eta$ for the subtree of x at the path $\eta \in \{0, 1\}^*$, with $x \cdot \eta = \bot$ when x has no node at η. Then $\exists z\, \psi$ holds exactly when $x \cdot \eta \le y$ for every path η: necessity follows from $x \cdot \eta \le z \cdot \eta \le z \le y$, and the least upper bound of all the subtrees $x \cdot \eta$ is a witness. Thus $\varphi \models \exists z\, \psi$ is an infinite family of plain entailments, and the witness for z depends on the solution of φ. Sound tests replace hidden variables by canonical bounds or search for syntactic derivations [Pottier 2001; Trifonov and Smith 1996]; they are incomplete, and Section 12 recalls a correct simplification of Pottier [1998b] that such a test misses.

Example 1.1 (Shift closure: the running example). Take ψ as above, with x, y visible and z hidden. Let $K(x) \subseteq \{0, 1\}^*$ be the set of positions of the f-nodes of x, so that $x \le y$ iff $K(x) \subseteq K(y)$ (Lemma 2.3), and let $\mathrm{Suf}(L)$ be the set of suffixes of words of L. Since $K(x \cdot \eta) = \eta^{-1}K(x)$, the condition "$x \cdot \eta \le y$ for every η" says that $\exists z\, \psi$ holds iff $\mathrm{Suf}(K(x)) \subseteq K(y)$. For $\varphi_1 = x \le y$ the entailment fails: the solution $x = y = f(f(\bot, f(\bot, \bot)), \bot)$ of φ_1, with $K(x) = \{\varepsilon, 0, 01\}$, has the position 01, whose suffix 1 is not in $K(y)$. For $\varphi_2 = x \le y \wedge x \le f(x, b) \wedge b = \bot$, the entailment holds: $z = x$ is a witness, because $x \le f(x, b) \le f(x, x)$. For $\varphi_3 = x \le y \wedge x \le f(f(\bot, f(\bot, \bot)), \bot)$ the entailment fails with the same countermodel as for φ_1, in which x takes the largest value that φ_3 allows.

Figure 1 draws two solutions of φ_1. For the first, the join of all subtrees of x fits below y and is the least witness for z; for the second it does not, and no witness exists. The choice of z is made after x and y are known, from their values.

An Open Problem. In non-structural subtyping, $\bot \le t \le \top$ relates trees of different shapes. Plain non-structural subtype entailment was posed as an open problem in 1996 [Rehof 1996], and Zhao [2026] decides it for one constructor of any arity and variance. The existential problem has been open since the same year. Trifonov and Smith [1996, Section 1] "leave open the question whether it is decidable" and decide a sound approximation. Fähndrich and Aiken [1996, Section 3.2] write that "no complete algorithm is known". Su [2002, Section 7.2.2] writes that "for existential entailment and subtyping constrained types, virtually nothing is known at this point, except the lower bounds carried over from entailment", and that for non-structural subtyping the problem "may in fact be undecidable", because an alternative undecidability proof for the first-order theory appears to use a single quantifier alternation. Priesnitz [2004, Section 1.4.2] calls subtyping of constrained types a "prominent open problem". The first-order theory is indeed undecidable, already over $\{\bot, f\}$ with f binary and covariant, in all three tree classes [Su et al. 2002], but with six quantifier blocks and a matrix with disjunctions and negations, whereas existential entailment has one alternation, and its antecedent and consequent are conjunctions. Over feature trees, existential entailment is decided

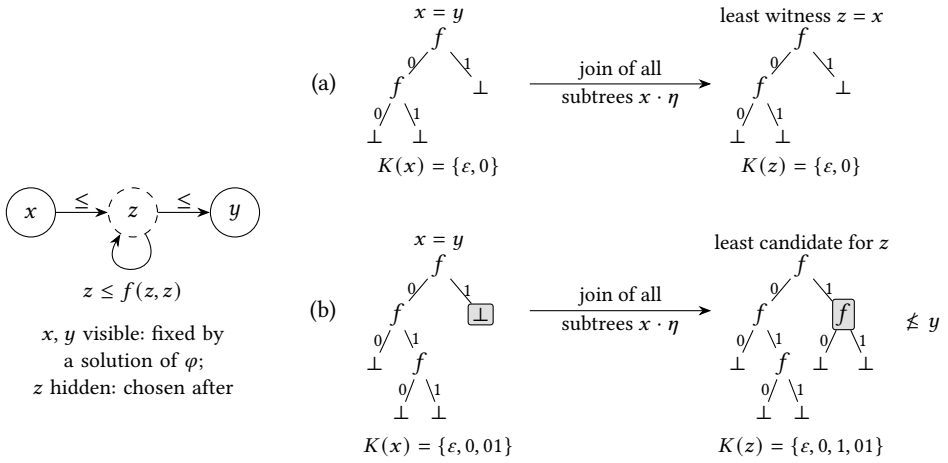


Fig. 1. The shift closure of Example 1.1 over $\{\bot, f\}$. A solution of φ fixes the visible variables x and y; the hidden z is chosen afterwards and depends on it. The literal $z \le f(z,z)$ makes the constructor positions of z closed under suffixes, so z must lie above every subtree of x, and their join is the least candidate. (a) For $x = y = f(f(\bot,\bot),\bot)$ the candidate is x itself, which fits below y. (b) For $x = y = f(f(\bot, f(\bot,\bot)),\bot)$, the countermodel of φ_1, the position 01 of x forces a constructor of z at its suffix 1 (shaded), where y has the leaf $\bot$; no z exists.

by an inclusion of finite automata [Müller et al. 1998, 2001], in a setting where a fresh feature is always available, which a constructor of fixed arity does not offer (Section 15).

Our Result. We fix one constructor f of arity $n \ge 0$ and a *variance* $c : \{0,\dots,n-1\} \to \{0,1\}$, which makes each argument covariant or contravariant; the arrow is $n = 2$ with $c(0) = 1$ and $c(1) = 0$ (Section 2.2). Trees may be infinite unless stated otherwise.

Theorem 1.2 (Main theorem). *For every arity $n \ge 0$ and every variance c, existential entailment $\varphi \models \exists \bar{z}\, \psi$ over $\{\bot, f, \top\}$ is decidable over arbitrary, over regular, and over finite trees. Over $\{\bot, f\}$, the same holds when f is covariant. If φ and ψ are satisfiable in the tree class considered and the entailment fails, then the failure is certified by a* counterexample word *of length at most $E_3(O(s))$: a word over the argument positions of f, or two such words joined by a separator, which determines a countermodel (Definition 11.1). Here $E_3(x) = 2^{2^{2^x}}$, s is the number of variables after flattening, every introduced name and every hidden variable included, and the constant does not depend on n.*

Section 11 defines counterexample words (Definition 11.1), states the theorem precisely (Theorem 11.2), shows that every counterexample word yields a countermodel (Lemma 11.3), and assembles the proof. The theorem needs no restriction on the polarity of the variables: a variable may occur both co- and contravariantly, in the antecedent and in the consequent. In particular, it decides the subsumption of Trifonov and Smith [1996] between constrained types over $\bot$, $\to$ and $\top$, and the correctness of simplifications of such types (Section 12). We do not claim the signature $\{\bot, f\}$ with a contravariant argument; Example 5.19 explains why the reduction to $\{\bot, f, \top\}$ needs covariance.

Bounds and Complexity. Proposition 11.5 refines the theorem. Only two kinds of failure, both described in Section 1.1, have triply exponential bounds: failures of condition (U) and *mortal holes*. The larger bound belongs to the mortal holes, the only case that needs a new compression technique; all other failures have singly exponential witnesses. The complexity of the problem

is treated in a separate part of the paper, which is not mechanized and which the main theorem does not use (Section 13). There we show that the decision procedure need not enumerate words: a nondeterministic search builds a witness level by level, keeping only a summary of each level, and runs in 2-EXPSPACE, hence in 3-EXPTIME (Proposition 13.8 and Theorem 13.9); for plain entailment this saves one exponential in space over the procedure of Zhao [2026] (Remark 13.10). For a unary constructor, $n = 1$, witnesses are singly exponential and the problem is in PSPACE, for both variances and all three tree classes (Theorem 13.11), and for $n = 0$ it is a finite enumeration (Remark 13.12). On the other side, shortest counterexample words can have length $2^{\Omega(s)}$ (Proposition 13.13), and the problem is PSPACE-hard for every fixed arity $n \geq 2$, already with f covariant, by explicit reductions that use the additional arguments of f; bare padding of a binary construction does not suffice (Theorem 13.16 and Example 13.14). The gap between PSPACE and 2-EXPSPACE, the gap between singly and triply exponential witnesses, and hardness for $n = 1$ remain open.

Mechanization. The decidability results are checked in the Lean 4 proof assistant [de Moura and Ullrich 2021] with Mathlib [The mathlib Community 2020]: over $\{\bot, f, \top\}$ for every arity, arity zero included, every variance and all three tree classes, and over $\{\bot, f\}$ with f covariant (Section 14). Each of the six procedures is a computable definition with a correctness theorem without premises, and plain entailment is decided as the special case without hidden variables. The paper and Lean follow the same proof routes, including anchored compression, the explicit search bounds, flattening, and the applications. Every numbered result before Section 13 has a corresponding declaration in the paper–Lean contract (Appendix I); the contract checks labels and numbered items. The complexity results and the bounds that are sharper than those of the main theorem are collected in Section 13, a separate part that is not mechanized and that no earlier result uses. The worked examples illustrate the definitions and the role of the hypotheses, and the remarks marked as informal give intuition; the proofs do not depend on them.

1.1 Overview of the Proof

The proof follows a single route for all arities and variances (Figure 2). Right after the characterization, it reduces the other tree classes and the signature $\{\bot, f\}$ to $\{\bot, f, \top\}$ and arbitrary trees, and it works in this one setting from then on. We end in the hole problem of the plain-entailment proof [Zhao 2026], whose readers, holes and bounds Section 2 recalls: a *reader* is a deterministic automaton with rejected states and an admission relation on pairs of states; a *comparison* of a word w is a pair of cuts (s, e), $s < e$, such that the suffix after e is a prefix of the suffix after s; and a *hole* is a rejected word without an admitted comparison. That proof bounds a shortest hole in three steps: a short rejected tail, a bound on the *depth*, a measure of nested repetition, and a compression of words of bounded depth. What is new here is that some of our holes must also *contain* a comparison of a particular kind, and that a single word may be dead because of a comparison with itself.

Leaf cones. Over $\{\bot, f\}$ with f covariant, a tree is its set of constructor positions, solution sets are closed under unions and intersections, and every literal becomes a set of implications with one premise (Remark 4.6). With $\top$ and variance, a literal can relate a leaf $\top$ of one tree to a leaf $\bot$ of another, and one set of positions no longer describes a tree. We restore a Horn structure for every arity and variance by describing a tree t through its two *leaf cones*: the words whose path meets a leaf of t of *sign* 0, and those whose path meets a leaf of sign 1, where the sign of a leaf combines its label, $\top$ or $\bot$, with the polarity of its path. The two cones are disjoint and closed under extension, every such pair of sets is the pair of exactly one tree, and the order of variance c adds leaves of sign 0 and removes leaves of sign 1 (Theorem 3.2). An *atom* $(v^{\pm}, w)$ says that w lies in a cone of v. Every literal becomes *pure moves*, implications between atoms that copy, push or pop the first letter of the word as on a stack, together with *facts*, atoms that every solution contains, and

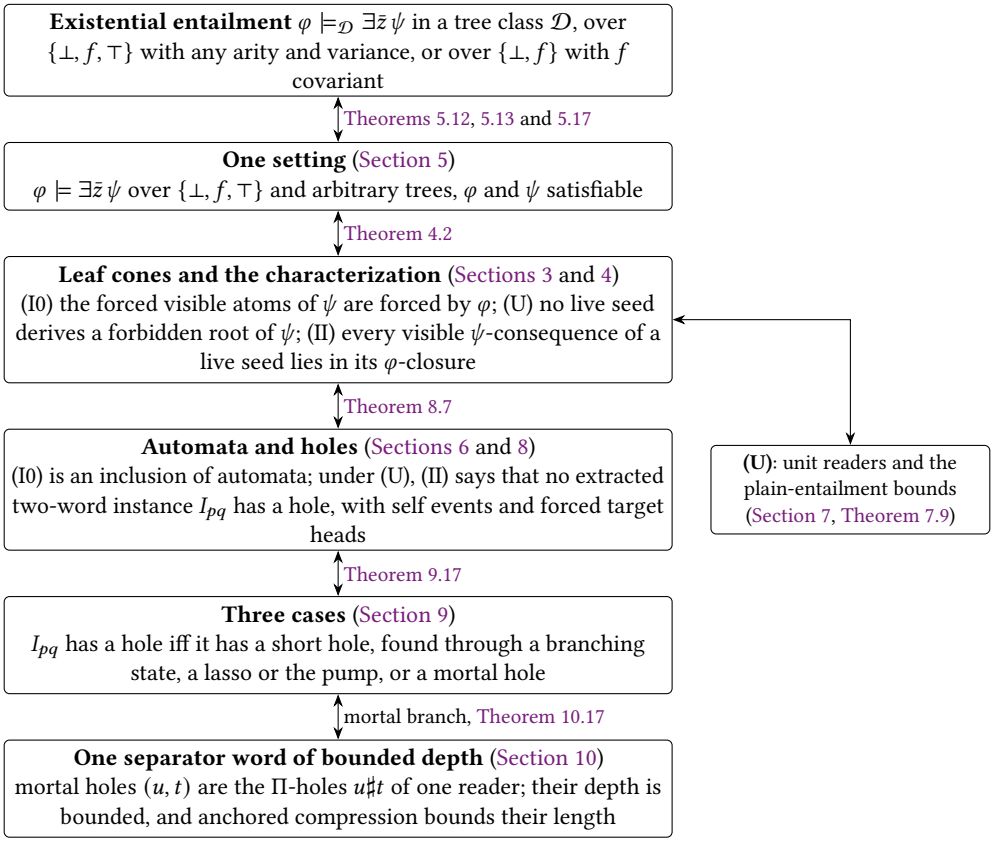


Fig. 2. The route of the proof. The first double arrow reduces every tree class, and the signature $\{\bot, f\}$ with f covariant, to satisfiable systems over $\{\bot, f, \top\}$ and arbitrary trees: regular trees agree with arbitrary trees, finite trees add two finite-satisfiability tests, and a gadget on every variable reduces $\{\bot, f\}$ to $\{\bot, f, \top\}$. Each further double arrow is an equivalence, labeled with the result that proves it, or for condition (U) with its box; the last vertical one concerns only the mortal branch of the case split, whose short holes are found directly. Condition (U) leaves the main route: it is a hole problem of the kind solved for plain entailment, decided by the bounds of Zhao [2026]. Here p, q range over the visible signed variables. Section 11 assembles the steps.

forbidden roots, atoms that none contains (Lemma 3.5). The only all-negative binary clauses are the exclusions of opposite atoms, which are the same for every system. So least models are closures of seeds, every atom of a closure has a single seed (Lemma 3.8), every pure move has a dual of opposite sign (Lemma 3.10), and two derivations that collide in an exclusion can be merged into one (Lemma 3.12).

Three conditions. Collision return turns every obstruction to extending a solution of φ into an obstruction at a single seed. If both systems are satisfiable, $\varphi \models \exists \bar{z}\, \psi$ holds iff three conditions hold (Theorem 4.2): (I0), every visible atom that ψ forces is forced by φ; (U), ψ derives no forbidden root from a visible atom that some solution of φ contains; and (II), every visible atom that ψ derives from such an atom lies in the least solution of φ that contains it.

Tree classes and the signature without top. The characterization concerns arbitrary trees over $\{\bot, f, \top\}$, and this one setting suffices (Section 5). A failure to extend has at most two seeds, and

the at most two observations behind it are expressed by finite ground bounds. Adding finitely many regular ground bounds preserves the existence of a regular solution, by the graph equations of their presentations, so regular and arbitrary trees give the same answer (Theorem 5.12). Adding finite ground bounds preserves finite satisfiability, for every variance (Theorem 5.7), which gives the dispatch of Theorem 5.13; least models do not suffice here, since they are infinite in general. For $\{\bot, f\}$ with f covariant, the literals $v \le y_v$ and $y_v \le f(y_v, \dots, y_v)$ force v to be free of $\top$, and placing them on every visible and every hidden variable reduces $\{\bot, f\}$ to $\{\bot, f, \top\}$ in each tree class (Theorem 5.17). From here on, the proof works over $\{\bot, f, \top\}$ and arbitrary trees.

Pushdown summaries. The pure moves form a pushdown system whose control states are the signed variables. Two deterministic automata summarize it: after reading a word γ, the *upper automaton* of p holds the signed variables reached from p by popping γ, and after reading β the *lower automaton* of q holds those from which q is reached by pushing β. Every pure derivation splits at its lowest point into a *pop head*, read by the first, and a *push head*, read by the second, above a common suffix (Lemma 6.3). Variance adds a new reason for an atom to be dead. Besides a reachable forbidden root, or an opposite extension forced by the facts, which make an atom *dead by support*, the closure of a seed (p, w) may contain the opposite of an extension of the seed. For a seed that is not already dead by support, this happens exactly when two cuts of the single word w form a comparison whose automaton states are related: a *self event* (Theorem 6.11). Intuitively, self events need contravariant directions (Remark 6.13), and the set of live words need not be regular (Example 6.12). Condition (I0) is an inclusion of automata, decided by a bounded search (Proposition 6.16).

Condition (U) as a hole problem. A failure of (U) is a word w such that (p, w) is live in φ and leads to a forbidden root in ψ. Liveness is the absence of deadness by support, a regular property, and of self events, which are comparisons of w. So the failures are exactly the holes of a single reader, the *unit reader*, whose admitted comparisons are the self events (Lemma 7.5). It has the rejected-path property of Zhao [2026], and the hole search of that paper, applied to the unit reader, decides (U) (Theorem 7.9).

Two-word holes with self events. A failure of (II) consists of a live seed (p, t), a pure derivation of ψ from it to a visible atom (q, u), and the fact that (q, u) lies outside the closure of the seed in φ. Each part is read by automata on the two words: the derivation is a *head event*, a pair of cuts at which t and u have equal suffixes; membership in the closure is an *event*, a pair of cuts (s, e) at which the suffix of t after e is a prefix of the suffix of u after s; and liveness of the seed forbids self events of t. So, once (U) holds, (II) fails iff an *extracted instance* I_{pq} of a hole problem for two words has a *hole*: a safe spine u, a rejected target t, a head event, and neither an event nor a self event (Definition 8.1 and Theorem 8.7). The target automaton marks a cut of the seed as rejected only while the seed is still a candidate, neither dead by support in φ nor root-dead in ψ. Under (U) this enriched rejection loses nothing, and it forces every pop head of ψ to end before the first rejected cut, condition (FH) (Lemma 8.6).

Holes that are not mortal. Cut a hole (u, t) at the first rejected cut of its target. By (FH) the rest θ of the target is a common suffix of both words, and what remains is a *pre-hole*. Appending θ to a pre-hole creates no new event and no new self event, and it keeps an old one exactly when θ continues its *overhang*, the part of one suffix that the other does not cover, periodically (Lemma 9.1). So the question is whether one safe continuation breaks every such periodicity obligation. If the spine ends in a *branching* state, which reaches a cycle with a choice of letters, a breaking continuation exists by the theorem of Fine and Wilf [1965] (Lemma 9.6). If it ends in a *thin* state, long safe continuations follow finitely many lassos; a lasso that is not purely periodic breaks everything, and

for a purely periodic lasso the *pump* shortens the pre-hole itself, keeping the obligations broken (Theorem 9.14). Short pre-holes are found by a finite graph (Proposition 9.12). In the remaining case the hole is *mortal*: its spine ends in a state without an infinite safe continuation, and its target ends shortly after its first rejected cut (Theorem 9.17). Outside the mortal case, holes are polynomial in the sizes of the automata, hence singly exponential in s.

Mortal holes. In a mortal hole the continuation is short, but the common suffix of the two words can be long, so the pair itself must be shortened, with both words together. We pack them into one word $u\sharp t$ around a separator. Events and self events become comparisons of this word, and a head event becomes an *anchored* comparison, one whose matched factor is followed by the separator (Lemma 10.4). Mortal holes are exactly the Π-*holes* of one reader, its holes with an anchored comparison whose state pair lies in a relation Π (Lemma 10.5), and the short rejected tail bounds their depth by the depth theorem of Zhao [2026] (Lemma 10.6). The bounded-depth compression of that paper keeps every negative condition but can delete the letters on which the anchored comparison depends (Example 10.13). A *positive channel* records the anchored witness at each level of the hierarchy (Lemma 10.10). The channel is added to the information that this compression compares before it merges letters or deletes columns, so every shortening keeps some anchored witness. This gives the anchored compression theorem (Theorem 10.14) and a triply exponential bound on mortal holes (Corollary 10.19), the largest of our closed forms, which decides the two-word problem under (FH) (Theorem 10.17).

The running examples. Two examples recur through the paper (Table 1). The first is the shift closure of Example 1.1, for a covariant constructor; the second follows the antecedent $x \le x \to x$ for the arrow, in which x occurs at both polarities. Informally, for a covariant constructor, and more generally on instances with a joint parity assignment, self events do not arise (Remarks 4.5, 6.13 and 8.10); over $\{\bot, f\}$ the three conditions compare unions of solutions, least solutions, and closures of single positions (Remark 4.6).

Contributions.

- *Decidability* (Theorem 1.2). Existential entailment is decidable for one constructor of every arity and every variance over $\{\bot, f, \top\}$, and over $\{\bot, f\}$ with f covariant, in all three tree classes, with triply exponential counterexample words. This settles the decidability question of Trifonov and Smith [1996] for semantic subsumption of constrained types.
- *Signed leaf cones* (Sections 3 and 4). A representation of trees of any variance in which every system is a Horn system with unary rules, and a characterization of existential entailment by three conditions on single seeds.
- *Tree classes and* $\{\bot, f\}$ (Section 5). Regular and finite transfers for every variance, and a reduction of covariant $\{\bot, f\}$ to $\{\bot, f, \top\}$, so that the rest of the proof works in one setting.
- *Self events* (Sections 6 to 8). Deadness is deadness by support or a self event, a comparison of one word with itself. Both (U) and (II) become hole problems whose admissions include self events, the second with an enriched rejection that makes target heads forced.
- *Non-mortal holes* (Section 9). Breakers, lassos and a pump with polynomial capacity give singly exponential witnesses outside the mortal case.
- *Anchored compression* (Section 10). Positive channels extend the bounded-depth compression of Zhao [2026] to holes that must contain an anchored comparison.
- *Applications* (Section 12). Subsumption of constrained type schemes and correctness of simplification are decidable, including arrow types without any restriction on polarity.
- *Mechanization* (Section 14). Computable decision procedures for all six cases of Theorem 1.2, checked in Lean 4, with a checked correspondence for every numbered result before Section 13.

Table 1. The running examples.

Example	Sections	What it shows
Shift closure: $\psi = x \le z \wedge z \le y \wedge z \le f(z,z)$, with f binary and covariant		
Example 1.1	Section 1	an existential entailment that is an infinite family of plain entailments
Example 6.5	Section 6	the signed system over $\{\bot, f, \top\}$ and its pure derivations
Example 7.7	Section 7	(U) fails at $x = y = \top$, and holds once y is restricted to trees without $\top$
Example 8.11	Section 8	(II) fails by the hole $(01, 1)$ of an extracted instance, whose spine ends in a branching state
Example 9.4	Section 9	the pre-hole $(0, \varepsilon)$, its single obligation, and, after Lemma 9.6, a breaker
Example 11.4	Section 11	the procedure end to end: the hole word $01\sharp 1$, its two pins, and the finite countermodel of Example 1.1
The arrow: the antecedent $x \le x \to x$		
Examples 3.7 and 3.9	Section 3	a single-seed inconsistency
Example 4.4	Section 4	the entailment $x \le x \to x \models \exists z\,(x \le z \to z)$, whose failure without the antecedent needs collision return
Example 6.12	Section 6	a nonregular set of dead words
Examples 7.8 and 8.12	Sections 7 and 8	why admitted self events and the enrichment of the target automaton are needed

- *Complexity*, not mechanized (Section 13). A 2-EXPSPACE decision by summary search; PSPACE for $n \le 1$; exponential shortest witnesses; and PSPACE-hardness for every fixed arity $n \ge 2$, already with f covariant.

Organization. Section 2 fixes notation and recalls the results of Zhao [2026] that we use. The sections that follow form one argument. Sections 3 and 4 turn trees into leaf cones and characterize existential entailment by three conditions on single seeds. Section 5 shows that the tree class and the signature $\{\bot, f\}$ do not change the problem, so that the remaining sections work over $\{\bot, f, \top\}$ and arbitrary trees. Sections 6 to 8 turn the three conditions into automata and hole problems, deciding (U) on the way, and Sections 9 and 10, the technical core, bound the holes of the two-word problem; Section 10 assumes Section 2.6. Section 11 assembles the decision procedure and proves the main theorem, and Section 12 applies it to constrained types. All of this is mechanized. Section 13, a separate part that is not mechanized and that nothing before it uses, gives sharper bounds and the complexity of the problem. Sections 14 to 16 describe the mechanization and discuss related work and open problems. Omitted proofs are in the appendices, in the same order. A reader interested in the result and its use may go from Section 4 to Sections 5, 11 and 12.

2 Background

Throughout the paper we fix an arity $n \ge 0$ and a variance c of the constructor f. This section fixes the notation for words, trees, and constraints, defines existential entailment, and states the results that we use from the decidability proof for plain entailment of Zhao [2026]. That paper treats one constructor of every arity and every variance; we cite its results in that generality. Short proofs are repeated; all other cited results are stated exactly and proved there. Appendix A collects the further results of that paper used later: the satisfiability criteria, the cores lemma, the ordinary and symbolic states of derived monitors, the structure of the bounded-depth search, and its constants.

2.1 Words

Put $\Sigma = \{0, \dots, n-1\}$, the alphabet of *directions*; the letter i names the ith argument of f. For $n = 0$ the alphabet is empty and ε is the only word over it. Readers (Section 2.5) also work over other finite alphabets, written Γ, for instance $\Sigma \sqcup \{\sharp\}$.

Words are finite unless they are explicitly called infinite. For a word w, a *cut* is an integer j with $0 \le j \le |w|$; it separates the first j letters from the rest. We write $w[i]$ for the letter at position i, where $0 \le i < |w|$, and $w[:j]$, $w[j{:}h]$, and $w[j:]$ for the prefix, factor, and suffix at the indicated cuts. The empty word is ε, and $u \preceq w$ means that u is a prefix of w. For a set L of words and a word β, put $\beta L = \{\beta\sigma : \sigma \in L\}$ and $\beta^{-1}L = \{\sigma : \beta\sigma \in L\}$. A positive integer ϖ is a *period* of w if $w[i] = w[i - \varpi]$ for $\varpi \le i < |w|$. For a nonempty word o, we write $\theta \preceq o^\omega$ if $\theta \preceq o^k$ for some $k \ge 0$.

Lemma 2.1 (Periodic prefixes [Zhao 2026, Lemma "Periodic prefixes"]). *For every nonempty word o and every word θ, $\theta \preceq o\theta$ iff $\theta \preceq o^\omega$.*

Proof. If $\theta \preceq o^k$, then $o\theta \preceq o^{k+1}$ and $\theta \preceq o^{k+1}$; of two prefixes of one word the shorter is a prefix of the longer, so $\theta \preceq o\theta$. Conversely, let $\theta \preceq o\theta$. If $|\theta| \le |o|$, then $\theta \preceq o$. Otherwise $\theta = o\theta'$ for a word θ', and $o\theta' \preceq oo\theta'$ gives $\theta' \preceq o\theta'$. By induction on the length, $\theta' \preceq o^k$ for some k, so $\theta \preceq o^{k+1}$. □

2.2 Trees, Variance, and Normalization

Trees. Order the labels by $\bot < f < \top$. A *tree* is a partial map $t : \Sigma^* \rightharpoonup \{\bot, f, \top\}$ whose domain $\mathrm{dom}(t)$ contains ε, is prefix closed, and satisfies, for every $w \in \mathrm{dom}(t)$ and every $i \in \Sigma$,

$$wi \in \mathrm{dom}(t) \quad \Longleftrightarrow \quad t(w) = f.$$

The elements of $\mathrm{dom}(t)$ are the *positions* of t. We write $f(t_0, \dots, t_{n-1})$, or $f(t_i)_i$, for the tree with root label f and subtrees t_i at the positions i. The *subtree* of t at a position β is the tree $\beta^{-1}t : w \mapsto t(\beta w)$. The full constructor tree $\Omega = f(\Omega, \dots, \Omega)$ has label f at every word; it is infinite for $n \ge 1$, and $\Omega = f()$ for $n = 0$. At arity zero the only trees are $\bot$, $f()$, and $\top$.

Variance and the order. A *variance* is a map $c : \Sigma \to \{0, 1\}$; argument i is *covariant* if $c(i) = 0$ and *contravariant* if $c(i) = 1$, and f is covariant if $c = 0$. For the arrow, $n = 2$, $f(a_0, a_1) = a_0 \to a_1$, $c(0) = 1$, and $c(1) = 0$. The *polarity* of a word counts its contravariant letters modulo two:

$$\mathrm{par}(w) = \bigoplus_{j < |w|} c(w[j]), \qquad \mathrm{par}(\varepsilon) = 0, \qquad \mathrm{par}(uv) = \mathrm{par}(u) \oplus \mathrm{par}(v). \tag{2}$$

The *non-structural order* of variance c is

$$t \le_c u \quad \Longleftrightarrow \quad \forall w \in \mathrm{dom}(t) \cap \mathrm{dom}(u) \quad \begin{cases} t(w) \le u(w), & \mathrm{par}(w) = 0, \\ u(w) \le t(w), & \mathrm{par}(w) = 1. \end{cases} \tag{3}$$

For the covariant variance $c = 0$ we write $\le$ and call the order *covariant*. The relation $\le_c$ is a partial order, and it is the greatest relation R such that $R(t, u)$ implies $t = \bot$, or $u = \top$, or $t = f(a_i)_i$ and $u = f(b_i)_i$ with $R(a_i, b_i)$ for every i with $c(i) = 0$ and $R(b_i, a_i)$ for every i with $c(i) = 1$ [Zhao 2026, Sect. "Non-Structural Subtyping"]. In particular, $\bot \le_c t \le_c \top$ for every tree t, and $f(a_i)_i \le_c f(b_i)_i$ iff each pair of arguments is related in the direction given by its variance. For the arrow this is the usual rule: $a_0 \to a_1 \le_c b_0 \to b_1$ iff $b_0 \le_c a_0$ and $a_1 \le_c b_1$.

Duality and normalization. The *dual* t^∂ of a tree t has the domain of t and the labels of t with $\bot$ and $\top$ exchanged; labels f are unchanged. For a *sign* $s \in \{0, 1\}$, the *normalization* $\nu_s t$ has the domain of t; its label at a position w is that of t with $\bot$ and $\top$ exchanged if $s \oplus \mathrm{par}(w) = 1$, and unchanged otherwise. Thus ν_0 exchanges the extreme labels at the positions of odd polarity and ν_1

at those of even polarity. The polarity is taken at actual positions of t only; a leaf is never followed by further letters. For covariant f, $\nu_0 t = t$ and $\nu_1 t = t^\partial$. We also write + for the sign 0 and − for the sign 1.

Lemma 2.2 (Normalization [Zhao 2026, Lemma "Normalization"]). *For every variance c, all trees $t, u, t_0, \ldots, t_{n-1}$, and every sign s:*

(1) $\nu_s f(t_i)_i = f(\nu_{s \oplus c(i)} t_i)_i$, $\quad \nu_1 t = (\nu_0 t)^\partial = \nu_0(t^\partial)$, $\quad$ and $\quad \nu_s \nu_s t = t$;
(2) $t \leq_c u$ iff $\nu_0 t \leq \nu_0 u$ iff $\nu_1 u \leq \nu_1 t$;
(3) for the covariant order, $t \leq u$ iff $u^\partial \leq t^\partial$;
(4) the subtree of $\nu_s t$ at a position w is $\nu_{s \oplus \mathrm{par}(w)}(w^{-1} t)$, and ν_s and duality preserve domains, finiteness, and regularity.

Proof. (1) The root of $f(t_i)_i$ is labeled f and is never changed. At a position iw we have $\mathrm{par}(iw) = c(i) \oplus \mathrm{par}(w)$, so the condition $s \oplus \mathrm{par}(iw) = 1$ for ν_s at iw is the condition for $\nu_{s \oplus c(i)}$ at the position w of t_i. The conditions of ν_0 and ν_1 at a position are complementary, so $\nu_1 t$ is $\nu_0 t$ with every extreme label exchanged, and exchanging all extreme labels commutes with exchanging some of them. Applying ν_s twice exchanges the same labels twice.

(3) Exchanging $\bot$ and $\top$ reverses the chain $\bot < f < \top$, and t and t^∂ have the same domain, so every comparison in Eq. (3), read covariantly, is reversed.

(2) The trees t and $\nu_0 t$ have the same domain. At a common position w with $\mathrm{par}(w) = 0$, ν_0 changes neither label; at one with $\mathrm{par}(w) = 1$ it applies the order-reversing exchange to both labels, so $u(w) \leq t(w)$ iff $(\nu_0 t)(w) \leq (\nu_0 u)(w)$. Hence Eq. (3) for $\leq_c$ is the covariant comparison of $\nu_0 t$ and $\nu_0 u$ at every common position. The second equivalence follows from (1) and (3) for the covariant order.

(4) The first assertion follows from (1) by induction on w. So the subtrees of $\nu_s t$ are among the trees $\nu_0 t'$ and $\nu_1 t'$ for subtrees t' of t, which are finitely many if t is regular. Domains are preserved by definition, and duality is handled in the same way. □

Thus ν_0 is an order isomorphism from the trees ordered by $\leq_c$ onto the trees ordered covariantly, and ν_1 is an order anti-isomorphism. This is how variance is removed: a constraint of variance c about t becomes a pair of covariant constraints about $\nu_0 t$ and $\nu_1 t$ (Lemma 2.5).

Tree classes. We consider three *tree classes*: the class $\mathcal{T}$ of all trees, possibly infinite; the class $\mathcal{T}_{\mathrm{reg}}$ of *regular* trees, which have finitely many distinct subtrees; and the class $\mathcal{T}_{\mathrm{fin}}$ of *finite* trees, which have finite domain. Then $\mathcal{T}_{\mathrm{fin}} \subseteq \mathcal{T}_{\mathrm{reg}} \subseteq \mathcal{T}$, and Ω is regular but not finite for $n \geq 1$. A tree is regular iff it is the unfolding of a finite graph whose constructor nodes have n ordered successors [Zhao 2026, Sect. "Words and Trees"]. Constructor application and the constants preserve each class. At arity zero the three classes coincide.

Trees over $\{\bot, f\}$. A *tree over* $\{\bot, f\}$ is a tree without the label $\top$. For a tree class $\mathcal{D}$, its trees over $\{\bot, f\}$ form the corresponding class over $\{\bot, f\}$; constructor application and $\bot$ preserve it. We use the signature $\{\bot, f\}$ only with f covariant. Then a tree t over $\{\bot, f\}$ is determined by the set $K(t) = \{w : t(w) = f\}$ of its constructor positions, which is prefix closed, and every prefix-closed set of words is $K(t)$ for exactly one tree t over $\{\bot, f\}$.

Lemma 2.3 (Order over $\{\bot, f\}$). *For covariant trees t, u over $\{\bot, f\}$, $t \leq u$ iff $K(t) \subseteq K(u)$.*

Proof. Suppose $K(t) \subseteq K(u)$ and let w be a common position. If $t(w) = f$, then $u(w) = f$; otherwise $t(w) = \bot$. In both cases $t(w) \leq u(w)$. Conversely, suppose $t \leq u$ and $w \in K(t)$. Every prefix of w lies in $K(t)$. By induction on the prefixes w' of w, each of them is a position of u, and $f = t(w') \leq u(w')$ forces $u(w') = f$, since u has no label $\top$. Hence $w \in K(u)$. □

2.3 Constraints and Existential Entailment

Flat systems. All variable sets in this paper are finite. Fix such a set. A *flat literal* has one of the forms

$$u \le v, \qquad f(a_i)_i \le u, \qquad u \le f(a_i)_i, \qquad u = \bot, \qquad \top \le u,$$

where $u, v, a_0, \dots, a_{n-1}$ are variables, not necessarily distinct. We call $u \le v$ a *copy literal*, $f(a_i)_i \le u$ a *lower constructor literal* and $u \le f(a_i)_i$ an *upper constructor literal*, both with *root* u and *children* a_i, and $u = \bot$ and $\top \le u$ *extreme literals*. Over $\{\bot, f\}$ the literal $\top \le u$ is not used. A *system* is a finite conjunction of flat literals over a finite set of variables, also read as a finite set of literals; a system *over* V has its variables in V.

Literals are always written with $\le$ and are read with the order of the variance at hand: an assignment ρ of trees to the variables of a system χ is a *solution* of χ for the variance c if $\rho(u) \le_c \rho(v)$ for every $u \le v \in \chi$, $f(\rho(a_i))_i \le_c \rho(u)$ for every $f(a_i)_i \le u \in \chi$, $\rho(u) \le_c f(\rho(a_i))_i$ for every $u \le f(a_i)_i \in \chi$, $\rho(u) = \bot$ for every $u = \bot \in \chi$, and $\rho(u) = \top$ for every $\top \le u \in \chi$. A *covariant system* is read with the covariant order. For a tree class $\mathcal{D}$, $\mathrm{Sol}_{\mathcal{D}}(\chi)$ is the set of solutions with values in $\mathcal{D}$, and χ is *$\mathcal{D}$-satisfiable*, written $\mathrm{Sat}_{\mathcal{D}}(\chi)$, if this set is nonempty. We write Sat, $\mathrm{Sat}_{\mathrm{reg}}$, and $\mathrm{Sat}_{\mathrm{fin}}$ for the three tree classes, and add the subscript c, as in Sat_c, when the variance must be named. Over $\{\bot, f\}$, all values are trees over $\{\bot, f\}$.

Definition 2.4 (Existential entailment). Let X and Z be disjoint finite sets of variables, let φ be a system over X, and let ψ be a system over $X \cup Z$. For a tree class $\mathcal{D}$, the *existential entailment*

$$\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$$

holds if every solution $\rho : X \to \mathcal{D}$ of φ extends to a solution $\rho' : X \cup Z \to \mathcal{D}$ of ψ. The variables in X are *visible*, those in Z *hidden*. Over $\{\bot, f\}$, both ρ and ρ' take values in the trees of $\mathcal{D}$ over $\{\bot, f\}$.

We omit the subscript for $\mathcal{D} = \mathcal{T}$, and write $\models_{\mathrm{reg}}$ and $\models_{\mathrm{fin}}$ for the other two classes. As predicates of the pair of systems, $E_{\mathrm{all}}(\varphi, \psi)$, $E_{\mathrm{reg}}(\varphi, \psi)$ and $E_{\mathrm{fin}}(\varphi, \psi)$ denote the existential entailment $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ for $\mathcal{D} = \mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$ and $\mathcal{T}_{\mathrm{fin}}$, respectively. The subscript of $\models$ always names a tree class; the signature, $\{\bot, f\}$ or $\{\bot, f, \top\}$, and the variance are stated in words. Without loss of generality X contains every variable of φ and every variable of ψ outside Z; a visible variable that does not occur in φ is unconstrained there. If φ is not $\mathcal{D}$-satisfiable, the entailment holds vacuously; if φ is $\mathcal{D}$-satisfiable and ψ is not, it fails. *Plain entailment* $\varphi \models_{\mathcal{D}} x \le y$ is the special case $\psi = \{x \le y\}$ with $Z = \varnothing$; it is the non-structural subtype entailment problem, which Zhao [2026] decides.

The size parameter. All size bounds of this paper are stated in terms of

$$s = |X| + |Z|,$$

the total number of variables, counted after flattening (see below) with every introduced name included. The bounds on the lengths of witness words do not depend on the arity n.

Signed systems. The *signed variables* of a set V are the pairs $v^s = (v, s)$ with $v \in V$ and a sign s; we write $v^+ = v^0$, $v^- = v^1$, $V^{\pm}$ for their set, and $\overline{v^s} = v^{1-s}$ for the *flip* of a signed variable. The *signed system* $\chi^{\pm}$ of a system χ over V, for the variance c, is the covariant system over $V^{\pm}$ that contains, for every literal of χ, the two literals listed in Table 2. It extends the signed translation of the plain-entailment proof [Zhao 2026, Sect. "The Signed System"] by copy literals.

Lemma 2.5 (Signed correspondence [Zhao 2026, Lemma "Signed correspondence"]). *Let χ be a system over V and c a variance. For an assignment ρ of trees to V, define $\rho^{\pm}$ on $V^{\pm}$ by $\rho^{\pm}(v^s) = v_s \rho(v)$. Then ρ is a solution of χ for the variance c iff $\rho^{\pm}$ is a solution of the covariant system $\chi^{\pm}$. The map*

Table 2. The signed literals of a literal of variance c; here $\bar{c}(i) = 1 \oplus c(i)$.

Literal of χ	Literals of $\chi^{\pm}$	
$u \le v$	$u^+ \le v^+$	$v^- \le u^-$
$f(a_i)_i \le u$	$f(a_i^{c(i)})_i \le u^+$	$u^- \le f(a_i^{\bar{c}(i)})_i$
$u \le f(a_i)_i$	$u^+ \le f(a_i^{c(i)})_i$	$f(a_i^{\bar{c}(i)})_i \le u^-$
$u = \bot$	$u^+ = \bot$	$\top \le u^-$
$\top \le u$	$\top \le u^+$	$u^- = \bot$

$\rho \mapsto \rho^{\pm}$ is a bijection from the assignments to V onto the assignments σ to $V^{\pm}$ with $\sigma(v^-) = \sigma(v^+)^{\partial}$ for every v, and both $\rho \mapsto \rho^{\pm}$ and its inverse preserve each tree class.

Proof. Each literal of χ is equivalent to each of its two signed literals under $\rho^{\pm}$, by Lemma 2.2. For $u \le v$, part (2) gives $\rho u \le_c \rho v$ iff $\nu_0 \rho u \le \nu_0 \rho v$ iff $\nu_1 \rho v \le \nu_1 \rho u$. For $f(a_i)_i \le u$, part (1) gives $\nu_0 f(\rho a_i)_i = f(\rho^{\pm}(a_i^{c(i)}))_i$ and $\nu_1 f(\rho a_i)_i = f(\rho^{\pm}(a_i^{\bar{c}(i)}))_i$, and part (2) applies again; the upper constructor literals are symmetric. For the extreme literals, ν_0 maps the one-node trees $\bot$ and $\top$ to themselves and ν_1 exchanges them. By part (1), $\rho^{\pm}(v^-) = \rho^{\pm}(v^+)^{\partial}$; conversely, if σ has this property, then $\rho(v) = \nu_0 \sigma(v^+)$ defines the unique assignment with $\rho^{\pm} = \sigma$, because ν_0 is an involution. Part (4) gives the preservation of the tree classes. □

A solution σ of $\chi^{\pm}$ need not satisfy $\sigma(v^-) = \sigma(v^+)^{\partial}$, and only the solutions with this property correspond to solutions of χ.

Flattening. Inequalities and equalities between arbitrary finite type expressions reduce to flat literals by naming subexpressions. Let φ and ψ be finite conjunctions of inequalities $e \le e'$ and equalities $e = e'$ between finite expressions built from variables, $\bot$, f of arity n, and, over $\{\bot, f, \top\}$, $\top$; the variables of ψ are visible or hidden as in Definition 2.4. Treat φ and ψ separately. In each, give every distinct nonvariable subexpression e a fresh name v_e, visible for φ and hidden for ψ; a subexpression of both systems receives one name of each kind. Let $v_e = e$ when e is a variable. For $e = f(e_i)_i$ add the *defining literals* $v_e \le f(v_{e_i})_i$ and $f(v_{e_i})_i \le v_e$, and for a constant add $v_e = \bot$ or $\top \le v_e$. Replace $e \le e'$ by $v_e \le v_{e'}$, and $e = e'$ by $v_e \le v_{e'}$ and $v_{e'} \le v_e$. Write φ' and ψ' for the resulting systems, and $\bar{z}'$ for the hidden variables of ψ together with the names introduced in ψ.

Lemma 2.6 (Flattening). *For every variance c and both signatures, $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ iff $\varphi' \models_{\mathcal{D}} \exists \bar{z}'\, \psi'$ in every class of trees $\mathcal{D}$ that contains the values of the subexpressions under assignments in $\mathcal{D}$, in particular in each of the three tree classes. For each input system, flattening introduces at most one name per symbol and produces at most six variable occurrences per symbol, independently of the arity. Thus its size is linear in the input size.*

Proof. The order $\le_c$ is antisymmetric, so the two defining literals of v_e say $v_e = f(v_{e_i})_i$, and the defining literal of a constant fixes its value. By induction on subexpressions, every assignment to the variables of a system has exactly one extension to its names that satisfies the defining literals, namely the one that assigns to v_e the value of e, and under it the replaced literals say exactly the original ones. These values lie in $\mathcal{D}$ by hypothesis. In particular, constructor application and the constants preserve each of the three tree classes, and preserve trees over $\{\bot, f\}$.

Suppose $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$, and let ρ' be a $\mathcal{D}$-solution of φ'. Its restriction to the original visible variables solves φ, since ρ' assigns to each name of φ the value of its subexpression. Extend this restriction to the hidden variables of ψ so that ψ holds, and then to the names of ψ by the values of their subexpressions; the result solves ψ' and agrees with ρ' on the visible variables of φ'.

Conversely, suppose $\varphi' \models_{\mathcal{D}} \exists \bar{z}'\ \psi'$, and let ρ be a $\mathcal{D}$-solution of φ. Its extension to the names of φ solves φ', so it extends further to a solution of ψ', which assigns to each name of ψ the value of its subexpression; hence its restriction to the variables of ψ solves ψ. For the size, count the variable, constant, and constructor occurrences in the input expressions. There is at most one new name per occurrence. The defining literals contribute at most two root occurrences per nonvariable symbol and two child occurrences per edge of an expression tree, hence at most four variable occurrences per input symbol. The replaced inequalities and equalities contribute at most two occurrences per expression root, hence at most two more per input symbol. Sharing equal subexpressions can only decrease these counts. □

From now on all systems are flat, and s counts the variables *after flattening*. The bounds of this paper would be false in terms of the variables of unflattened expressions. For example, let f be covariant with $n \geq 1$, and put $T_0 = \bot$ and $T_{k+1} = f(T_k, \ldots, T_k)$, so that $K(T_k)$ is the set of words of length less than k. Over $\{\bot, f\}$, the query true $\models x \leq T_k$ has one variable, and by Lemma 2.3 a solution violates $x \leq T_k$ exactly when x has a constructor position of length k; so every countermodel has a constructor at depth k. Flattening names the $k+1$ distinct subexpressions $T_0, \ldots, T_k$, so that $s = k + 2$, and the bounds in s then apply.

2.4 Results on Constraints from the Plain-Entailment Proof

The flat syntax of Zhao [2026] has no copy literals: its literals are $f(a_i)_i \leq u$, $u \leq f(a_i)_i$, $u = \bot$, and $u = \top$, the last being our $\top \leq u$. At positive arity copy literals are eliminated by a sandwich, a variant of the one of [Zhao 2026, Proposition "Flattening"].

Lemma 2.7 (Sandwich elimination). *Let $n \geq 1$. Replace a literal $u \leq v$ by the two literals $f(a_i)_i \leq y_{uv}$ and $y_{uv} \leq f(b_i)_i$, where y_{uv} is a fresh variable for the ordered pair (u, v), $a_i = b_i = u$ for $i \geq 1$, and $(a_0, b_0) = (u, v)$ if $c(0) = 0$ and $(a_0, b_0) = (v, u)$ if $c(0) = 1$. An assignment satisfies $u \leq v$ iff some value of y_{uv} satisfies the two new literals.*

Proof. If the two literals hold, transitivity gives $f(a_i)_i \leq_c f(b_i)_i$, whose argument 0 gives $u \leq_c v$ in both cases of $c(0)$. Conversely, put $y_{uv} = f(a_i)_i$; then $f(a_i)_i \leq_c f(b_i)_i$, because the arguments 0 compare in the direction of their variance and the other arguments are equal. □

The variable u itself fills the arguments $1, \ldots, n-1$, so no further fresh variable is needed. Applying Lemma 2.7 to every copy literal of a system χ gives a system $\chi^\flat$ in the syntax of Zhao [2026], with one fresh variable for each ordered pair of variables. Every solution ρ of χ extends to a solution of $\chi^\flat$, by $y_{uv} = f(\rho(a_i))_i$, and every solution of $\chi^\flat$ restricts to a solution of χ. The extension of a finite solution is finite, so χ and $\chi^\flat$ are finitely satisfiable together.

Remark 2.8 (Arity zero). For $n = 0$ the trees are $\bot < f() < \top$, the only word is ε, the order $\leq_c$ is this chain for the empty variance, and the three tree classes coincide. Existential entailment $\varphi \models \exists \bar{z}\ \psi$ is then decided by enumerating the 3^s assignments.

Theorem 2.9 (Regular and finite solutions [Zhao 2026, Theorem "Least shape and satisfiability", Lemma "Finite signed correspondence"]). *For every arity $n \geq 0$ and every variance c, the following hold for systems χ over $\{\bot, f, \top\}$ with finite variable sets, copy literals included.*

(1) $\mathrm{Sat}_{\mathrm{reg}}(\chi) \iff \mathrm{Sat}(\chi)$: a satisfiable system has a regular solution.
(2) Finite satisfiability is decidable.

Proof. For $n = 0$ every tree is finite and regular, and there are finitely many assignments. For $n \geq 1$, replace χ by $\chi^\flat$. If χ has a solution, so has $\chi^\flat$, which by Zhao [2026] has a regular solution (Theorem A.1(3)); its restriction is a regular solution of χ. Finite satisfiability of χ is finite satisfiability of $\chi^\flat$, which Zhao [2026] decides (Theorem A.1(1,2)). □

Satisfiability itself is decided in Section 6.7 (Proposition 6.16), with the leaf cones of Section 3. Plain entailment is the special case of existential entailment with no hidden variables and the consequent $x \le y$, so its decidability is a corollary of our main theorem (Corollary 11.6).

These results concern solutions over $\{\bot, f, \top\}$. A system without literals $\top \le u$ can still have solutions that use $\top$, so they do not decide finite satisfiability over $\{\bot, f\}$: for $n \ge 1$ and f covariant, $\{f(x, \dots, x) \le x\}$ has the finite solution $x = \top$, but its only solution over $\{\bot, f\}$ is $x = \Omega$. The signature $\{\bot, f\}$ is reduced to $\{\bot, f, \top\}$ in Section 5, before the hole problems.

2.5 Readers and Holes

Sections 3 to 5 do not use this subsection or the next, and Section 6 uses only comparisons (Definition 2.10). Readers and the tail, depth and search theorems are used from Section 7 on; lettered monitors, derivatives and the outline of the search at the end of Section 2.6 are used only in Section 10, and the first two also in the summary search of the part that is not mechanized (Section 13).

Zhao [2026] reduces plain entailment to a word problem for finite automata, the hole problem below, and proves bounds that hold for every such automaton. We use the same notions and bounds. Throughout, Γ is a finite alphabet.

Definition 2.10 (Comparison). A *comparison* of a word w is a pair of cuts (s, e) with $0 \le s < e \le |w|$ and $w[e{:}] \preceq w[s{:}]$. Its *period* is $e - s$, and it is *terminal* if $e = |w|$.

Equivalently, (s, e) is a comparison iff the suffix $w[s{:}]$ has period $e - s$: reading w from e repeats what it reads from s, until the word ends. Every pair $(s, |w|)$ with $s < |w|$ is a terminal comparison. For example, the comparisons of 0100 are $(0, 3)$, $(2, 3)$, and the four terminal pairs $(s, 4)$; the pair $(0, 2)$ is not one, since 00 is not a prefix of 0100. If (s, e) is a comparison of w and $e \le h \le |w|$, then (s, e) is a comparison of $w[{:}h]$.

Definition 2.11 (Reader). A *reader* $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ consists of a deterministic finite automaton $\mathcal{M} = (Q, q_0, \delta)$ over Γ, with $\delta : Q \times \Gamma \to Q$ extended to words, an *admission* relation $\mathrm{Adm} \subseteq Q \times Q$, and a set $\mathrm{Rej} \subseteq Q$ of *rejected* states. Its *size* is $N = |Q| \ge 1$. For a word w, we write $q_i = \delta(q_0, w[{:}i])$ for the state at cut i.

Definition 2.12 (Hole). A word w is *rejected* if $q_{|w|} \in \mathrm{Rej}$; then $J(w)$ is the least cut J with $q_J \in \mathrm{Rej}$, the *first rejected cut*, and $r(w) = |w| - J(w)$ is the *rejected tail* of w. A rejected word w is a *hole* of the reader if no comparison (s, e) of w satisfies $\mathrm{Adm}(q_s, q_e)$; a comparison that satisfies it is *admitted*.

A reader thus reads a word once and tests every comparison by the pair of states at its two cuts; a hole is a rejected word that passes all these tests. The names follow Zhao [2026]. Rejection is the regular part of a failure, and an admitted comparison is a non-regular exception that rescues the word; a hole is a word that the regular test condemns and no comparison rescues. In the readers of later sections, a rejected word is a candidate failure, and an admitted comparison shows that the candidate is not one. The empty word has no comparison, so it is a hole iff $q_0 \in \mathrm{Rej}$. The tail bound below needs two properties that the readers for plain entailment have and that our readers will be shown to have.

Definition 2.13 (Rejected-path property [Zhao 2026, Sect. "The Rejected-Path Property"]). A reader has the *rejected-path property* if it satisfies:

Betweenness
if two prefixes of a word end in Rej, every intermediate prefix also ends in Rej;

Admission cone
in a rejected word w, every admitted comparison (s, e) has $e \le J(w)$.

Lemma 2.14 (Rejected extension [Zhao 2026, Lemma "Rejected extension"]). *Let a reader have the rejected-path property. If v is a hole and w is a rejected word with $v \preceq w$, then w is a hole.*

Proof. The words v and w have the same states at the cuts of v, and v is rejected, so $J(w) \le |v|$. An admitted comparison (s, e) of w has $e \le J(w) \le |v|$ by the admission cone, so it is a comparison of v with the same states, contradicting the assumption that v is a hole. □

Theorem 2.15 (Bounded rejected tail [Zhao 2026, Theorem "Bounded rejected tail"]). *Let $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ be a reader with N states over a finite alphabet that has the rejected-path property. If it has a hole, then it has a hole w with $r(w) \le 4N - 2$.*

The theorem selects a hole; it bounds neither the tail of every hole nor a continuation of a given rejected prefix. Its proof distinguishes whether some reachable rejected state on a cycle of rejected states has two distinct letters that keep the run rejected; the bound holds for every alphabet size.

2.6 The Canonical Hierarchy, Depth, and Derivatives

Definition 2.16 (Canonical hierarchy, depth [Zhao 2026, Definition "Hierarchy and depth"]). Let v be a nonempty word over an alphabet Γ with last letter ℓ, the *marker*. Then $v = X_0\ell X_1\ell \cdots X_{k-1}\ell$ for unique *blocks* $X_i \in (\Gamma \setminus \{\ell\})^*$, and the *desubstitution* of v is the word $[X_0][X_1]\cdots[X_{k-1}]$ of length k over the alphabet $(\Gamma \setminus \{\ell\})^*$ of blocks, each block being one letter; v is its *expansion*. The *canonical hierarchy* of a nonempty word w is $w_0 = w$ and w_{j+1} = the desubstitution of w_j, whose marker ℓ_j is the last letter of w_j. A word is *unary* if all its letters are equal. The *depth* $\mathrm{depth}(w)$ is the least j such that w_j is unary.

A level that is not unary is strictly longer than the next one, so the depth is well defined. For example, for $n = 2$, $w = 01001101$ has $w_1 = [0][00][\varepsilon][0]$, then $w_2 = [\varepsilon][[00][\varepsilon]]$, and w_3 is a single letter, so $\mathrm{depth}(w) = 3$. Block alphabets can be infinite even when Γ is finite, and are infinite when $\Gamma \setminus \{\ell\} \neq \emptyset$; every word uses only finitely many of their letters. Over a one-letter alphabet, in particular for $n = 1$, every nonempty word has depth 0.

The hierarchy is compatible with comparisons: a comparison that does not end at the last cut is a comparison of the next level, refined by a common suffix of two blocks (Figure 3).

Lemma 2.17 (Alignment [Zhao 2026, Lemma "Alignment"]). *Let $v = X_0\ell X_1\ell \cdots X_{k-1}\ell$ be the expansion of a nonempty word $v' = [X_0]\cdots[X_{k-1}]$ of blocks avoiding ℓ, and let $a_i = |X_0\ell\cdots X_{i-1}\ell|$ be the cut at which X_i begins. For cuts $s < e < |v|$, the pair (s, e) is a comparison of v iff there are indices $i < i' < k$ and decompositions $X_i = xy$ and $X_{i'} = x'y$ with $s = a_i + |x|$, $e = a_{i'} + |x'|$, and $(i + 1, i' + 1)$ a comparison of v'.*

Proof. Every cut $h < |v|$ lies in exactly one interval $[a_i, a_i + |X_i|]$; write $X_i = xy$ with $h = a_i + |x|$, so that $v[h{:}] = y\ell X_{i+1}\ell\cdots X_{k-1}\ell$. Let s and e correspond in this way to $X_i = xy$ and $X_{i'} = x'y'$. Since y and y' avoid ℓ, the first marker of $v[s{:}]$ is at offset $|y|$ and that of $v[e{:}]$ at offset $|y'|$. If $v[e{:}] \preceq v[s{:}]$, then $y = y'$, and canceling $y\ell$ shows that the expansion of $[X_{i'+1}]\cdots[X_{k-1}]$ is a prefix of the expansion of $[X_{i+1}]\cdots[X_{k-1}]$. Expansions determine their block words, since the markers delimit the blocks, so $v'[i' + 1{:}] \preceq v'[i + 1{:}]$. Moreover $i < i'$: if $i = i'$, then $y = y'$ gives $s = e$, and if $i > i'$, then $s \ge a_i > a_{i'} + |X_{i'}| \ge e$. So $(i + 1, i' + 1)$ is a comparison of v'. The converse reverses these steps: $v[e{:}] = y\ell X_{i'+1}\ell\cdots \preceq y\ell X_{i+1}\ell\cdots = v[s{:}]$. □

For example, the comparison $(0, 6)$ of $w = 01001101$, with $w[6{:}] = 01$, starts in the blocks $X_0 = 0$ and $X_3 = 0$ of $w = (0)1(00)1(\varepsilon)1(0)1$, both at distance one from their markers and with the common

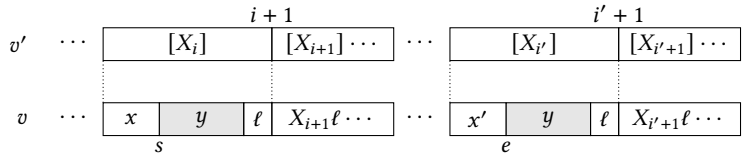


Fig. 3. The alignment lemma. Bottom: a comparison (s, e) of the expansion v with $e < |v|$ starts in two blocks $X_i = xy$ and $X_{i'} = x'y$ that have the common suffix y (shaded) after s and after e, so both cuts lie at the same distance from the markers that close their blocks. Top: what follows these two markers is the expansion of the block word v' after the cuts $i + 1$ and $i' + 1$, so (s, e) is a comparison of v iff, in addition, $(i + 1, i' + 1)$ is a comparison of v'.

suffix $y = 0$, and $(1, 4)$ is the terminal comparison of $w_1 = [0][00][\varepsilon][0]$ that ends at its last cut. A comparison of one level thus reappears at the next level as a comparison of blocks, together with a test at the same distance from the ends of two blocks. The derivatives of lettered monitors (Definition 2.23) check comparisons in exactly this form.

Depth measures how deeply such repetitions are nested. It matters because the compression behind Theorem 2.20 works one level of the hierarchy at a time, so its length bound grows with the depth. The depth of a rejected word is bounded in terms of its rejected tail, for every reader. The key step is the following lemma.

Lemma 2.18 (Return gap [Zhao 2026, Lemma "Return gap"]). *Let w be a rejected word of a reader. If (s, e) is a comparison of w and $q_s = q_e$, then $e - s \le r(w)$.*

Proof. Put $\varpi = e - s$. The comparison gives $w[:s]\, w[e:] = w[:|w| - \varpi]$. Since $q_s = q_e$, reading $w[e:]$ from q_s ends in $q_{|w|}$, so the state at the cut $|w| - \varpi$ is $q_{|w|} \in$ Rej. By minimality of $J(w)$, $|w| - \varpi \ge J(w)$, so $\varpi \le r(w)$. □

Theorem 2.19 (Depth from the rejected tail [Zhao 2026, Theorem "Depth from tail"]). *For every reader with N states, every nonempty rejected word w satisfies* $\mathrm{depth}(w) \le 2N(r(w) + 1) - 1$.

Theorem 2.20 (Bounded-depth search [Zhao 2026, Theorem "Bounded-depth search"]). *Let $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ be a reader with N states over a finite alphabet, and let $d \ge 0$. If it has a nonempty hole of depth at most d, then it has a nonempty hole of depth at most d and length at most* $\mathsf{Len}(N, d)$*, the explicit elementary function of Eq. (15). The same holds for every lettered monitor with N states over any alphabet (defined below, Definition 2.22).*

Theorem 2.19 applies to every nonempty rejected word, whether or not it is a hole, and neither theorem uses the rejected-path property or a property of subtype constraints; the alphabet size does not enter the bounds. Combined with Theorem 2.15, they decide the hole problem for readers with the rejected-path property.

Corollary 2.21 (Finite search for holes [Zhao 2026, Theorem "Finite search for holes", Lemma "Elementary estimate"]). *Let a reader with N states over a finite alphabet have the rejected-path property, and put $d_N = 2N(4N - 1) - 1$. It has a hole iff it has a hole of length at most* $\mathsf{Len}(N, d_N)$*, and* $\mathsf{Len}(N, d_N) \le 2^{2^{1280(N+1)^5}}$.

Proof. By Theorem 2.15, some hole w has $r(w) \le 4N - 2$. If w is empty, it meets the bound. Otherwise Theorem 2.19 gives $\mathrm{depth}(w) \le d_N$, and Theorem 2.20 gives a hole of length at most $\mathsf{Len}(N, d_N)$. The converse is trivial, and the estimate is Lemma A.12. □

Lettered monitors. The proof of Theorem 2.20 reads each level of the hierarchy of a hole with its own automaton, obtained from the automaton of the level below by a derivative. These automata evaluate their tests at positions inside blocks, so the tests see the letters as well as the states.

Definition 2.22 (Lettered monitor [Zhao 2026, Sect. "Lettered Monitors and Their Derivatives"]). A *lettered monitor* $\mathcal{M}$ over an alphabet Γ, possibly infinite, has a finite state set Q, a start state ζ_0, a transition function $\delta : Q \times \Gamma \to Q$, a *final-target* predicate $T^{\mathrm{fin}} \subseteq Q \times \Gamma$, and two relations Λ and Λ^{fin} on *labels*, the second for terminal comparisons. Labels are a symbol $\triangleright$, the *start label*, and the pairs $(\zeta, a) \in Q \times \Gamma$. Reading a word v gives the states $\zeta_{i+1} = \delta(\zeta_i, v[i])$ and the labels $L_0 = \triangleright$ and $L_i = (\zeta_{i-1}, v[i-1])$ for $1 \le i \le |v|$. A nonempty word v is a *hole* of $\mathcal{M}$ if $T^{\mathrm{fin}}(L_{|v|})$ holds, no terminal comparison $(s, |v|)$ has $\Lambda^{\mathrm{fin}}(L_s, L_{|v|})$, and no other comparison (s, e) has $\Lambda(L_s, L_e)$.

So the label of a cut $i \ge 1$ is the last transition before it: the state at the cut $i-1$ and the letter read from it. A reader $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ with states Q, start q_0, and transition δ determines the lettered monitor with the same states and transitions, $T^{\mathrm{fin}}(\zeta, a)$ iff $\delta(\zeta, a) \in \mathrm{Rej}$, and $\Lambda = \Lambda^{\mathrm{fin}}$ given by Adm on the *denoted* states, where $\mathrm{den}(\triangleright) = q_0$ and $\mathrm{den}(\zeta, a) = \delta(\zeta, a)$. Since $\mathrm{den}(L_i) = q_i$, the reader and this monitor have the same nonempty holes. The tests take labels, and not only the states they denote, because the monitors of higher levels evaluate them at positions inside blocks, which requires the letters.

Shifted blocks and derivatives. Fix a marker $\ell \in \Gamma$, a block $X \in (\Gamma \setminus \{\ell\})^*$, and $\zeta \in Q \sqcup \{\star\}$, where $\star$ is a fresh symbol. The *shifted block* of X read from $\zeta \in Q$ is the word ℓX read from ζ; read from $\star$, it is X read from ζ_0. It has a label $lab_g(\zeta, X)$ at each position $g = 0, \dots, |X|$: for $\zeta \in Q$, the label of the cut after $\ell X[:g]$, so that position 0 carries the *marker label* (ζ, ℓ); for $\zeta = \star$, the label $\triangleright$ at position 0 and the label of the cut after $X[:g]$ at positions $g \ge 1$. The *exit* $\delta_\ell(\zeta, X) \in Q$ is the state at the end of the shifted block. In a word of blocks read by the derivative below, the *entry state* of a position is the state at the cut before it, from which the shifted block of its letter is read.

Definition 2.23 (Derivative [Zhao 2026, Definition "Derivative"]). The *derivative* $\mathcal{M}_\ell$ of $\mathcal{M}$ along ℓ is the lettered monitor over $(\Gamma \setminus \{\ell\})^*$ with states $Q \sqcup \{\star\}$, start state $\star$, transitions δ_ℓ, and

$$\Lambda_\ell((\zeta, X), (\zeta', X')) \iff X = xy,\ X' = x'y,\ \Lambda\big(lab_{|x|}(\zeta, X), lab_{|x'|}(\zeta', X')\big) \text{ for some } x, x', y,$$

$$\Lambda^{\mathrm{fin}}_\ell((\zeta, X), (\zeta', X')) \iff \Lambda_\ell((\zeta, X), (\zeta', X')) \text{ or } \Lambda^{\mathrm{fin}}\big(lab_g(\zeta, X), (\delta_\ell(\zeta', X'), \ell)\big) \text{ for some } g \le |X|,$$

$$T^{\mathrm{fin}}_\ell(\zeta, X) \iff T^{\mathrm{fin}}(\delta_\ell(\zeta, X), \ell) \text{ and } \neg\Lambda^{\mathrm{fin}}\big(lab_g(\zeta, X), (\delta_\ell(\zeta, X), \ell)\big) \text{ for all } g \le |X|;$$

every relation with the start label $\triangleright$ of $\mathcal{M}_\ell$ as an argument is false.

In words: Λ_ℓ compares two positions at the same distance from the ends of two blocks, with equal letters after them, as in Lemma 2.17; the second disjunct of $\Lambda^{\mathrm{fin}}_\ell$ compares a position anywhere in the block of its first argument with the final cut, which carries the label of the closing marker; and T^{fin}_ℓ checks the original target and all terminal comparisons that start in the last block. Comparisons of the expansion that start at its cut 0 are represented inside the first shifted block, whose entry is $\star$, so setting the relations at the new start label to false discards none of them.

The label facts. Let $v = X_0\ell X_1\ell \cdots X_{k-1}\ell$ be the expansion of a nonempty word $v' = [X_0] \cdots [X_{k-1}]$ of blocks avoiding ℓ. Let $\zeta'_0 = \star$ and $\zeta'_{i+1} = \delta_\ell(\zeta'_i, X_i)$ be the run of v' in $\mathcal{M}_\ell$, put $a_i = |X_0\ell \cdots X_{i-1}\ell|$, and let $b_i = a_i + |X_i|$ be the position of the marker closing X_i. The intervals $[a_i, b_i]$ partition the cuts $0, \dots, |v| - 1$, and the labels L of v in $\mathcal{M}$ and L' of v' in $\mathcal{M}_\ell$ satisfy the *label facts* [Zhao 2026, proof of Lemma "Holes under desubstitution"]

$$L_{a_i+g} = lab_g(\zeta'_i, X_i) \quad (0 \le g \le |X_i|), \qquad L_{|v|} = (\zeta'_k, \ell), \qquad L'_{i+1} = (\zeta'_i, X_i). \tag{4}$$

In particular ζ'_{i+1} is the state of $\mathcal{M}$ at the cut b_i.

Lemma 2.24 (Holes under desubstitution [Zhao 2026, Lemma "Holes under desubstitution"]). *With v and v' as above, v is a hole of $\mathcal{M}$ iff v' is a hole of $\mathcal{M}_\ell$. Consequently, if $\mathcal{M}_0$ is a lettered monitor with N states and $\mathcal{M}_{j+1} = (\mathcal{M}_j)_{\ell_j}$ along the canonical hierarchy of a nonempty word w, then w is a hole of $\mathcal{M}_0$ iff w_j is a hole of $\mathcal{M}_j$, for every j, and $\mathcal{M}_j$ has $N + j$ states.*

Example 2.25 (A derivative). Let a lettered monitor $\mathcal{M}$ over $\{0, 1\}$ read $w = 01001101$, with the state ζ_h and the label L_h at the cut h. With the marker $\ell = 1$, the next level $w_1 = [0][00][\varepsilon][0]$ has the blocks $X_0 = 0$, $X_1 = 00$, $X_2 = \varepsilon$ and $X_3 = 0$, which begin at the cuts 0, 2, 5 and 6 of w:

i	X_i	cuts a_i–b_i	entry ζ'_i	label L'_{i+1}
0	0	0–1	$\star$	$(\star, 0)$
1	00	2–4	ζ_1	$(\zeta_1, 00)$
2	ε	5	ζ_4	(ζ_4, ε)
3	0	6–7	ζ_5	$(\zeta_5, 0)$

The marker is 1, so $\mathcal{M}_1$ is both the derivative along 1 and the monitor of level 1. By the label facts, the run of w_1 in $\mathcal{M}_1$ is $\star, \zeta_1, \zeta_4, \zeta_5, \zeta_7$, the states before the four markers, and the labels of w_1 are those in the last column; the shifted block of X_3, for instance, is the factor 10 of w between the cuts 5 and 7.

The comparisons of w_1 that start after its cut 0 are the terminal pairs $(1, 4)$, $(2, 4)$ and $(3, 4)$, which $\mathcal{M}_1$ tests with Λ_1^{fin}. For $(1, 4)$, the first disjunct $\Lambda_1(L'_1, L'_4)$ takes the two common suffixes 0 and ε of X_0 and X_3 and reads $\Lambda(L_0, L_6)$ and $\Lambda(L_1, L_7)$: the comparisons $(0, 6)$ and $(1, 7)$ of w (Figure 3). The second disjunct reads $\Lambda^{\mathrm{fin}}(L_0, L_8)$ and $\Lambda^{\mathrm{fin}}(L_1, L_8)$: the terminal comparisons of w that start in X_0. In the same way, $(2, 4)$ stands for the comparisons $(3, 6)$ and $(4, 7)$ and for the terminal comparisons from the cuts 2, 3 and 4, and $(3, 4)$ for $(5, 7)$ and the terminal comparison from the cut 5. Finally, $T_1^{\mathrm{fin}}(L'_4)$ reads the final target $T^{\mathrm{fin}}(L_8)$ and excludes the terminal comparisons from the cuts 6 and 7, which start in the last block. These are all the comparisons of w, so w is a hole of $\mathcal{M}$ iff w_1 is a hole of $\mathcal{M}_1$. The comparisons $(0, 3)$ and $(0, 4)$ of w_1 start at its start label, where every relation of $\mathcal{M}_1$ is false.

How the bounded-depth search works. Later sections extend the proof of Theorem 2.20, which has five steps; Appendix A.4 states them precisely. Let w be a nonempty hole of depth $D \le d$ with hierarchy (w_j, ℓ_j).

(1) *Derivatives.* Let $\mathcal{M}_0$ be the lettered monitor of the reader, and $\mathcal{M}_{j+1} = (\mathcal{M}_j)_{\ell_j}$. By Lemma 2.24, w_j is a hole of $\mathcal{M}_j$ for every j. The $N + j$ states of $\mathcal{M}_j$ are the N *ordinary* states of $\mathcal{M}_0$ and the *symbolic* start states $\star_1, \ldots, \star_j$ added by the derivatives; ordinary states lead to ordinary states (Appendix A.3).

(2) *Dictionary normal form.* At each level $1 \le i < D$, two isolated letters with the same exchange type are merged: the one with the longer expansion is replaced by the other, which keeps the comparisons and the state run of the level, and the level stays a hole (Lemma A.6). The result is again a hole of depth D, no longer. Repeating this bounds the dictionary at every level $i \ge 1$ by $\mathrm{G}(N, d)$. The terms are the following.

Dictionary. The *dictionary* of level i is the set of letters of w_i.

Protected and unprotected letters. The *protected* letters of level i are the marker ℓ_i and the letters of the *start prefix*, the first at most i positions of w_i, which are read from symbolic states; every other letter is *unprotected* and is read only from ordinary states.

Isolated letters. An unprotected letter is *isolated* if all its unequal comparison partners are protected: whenever it precedes one cut of a comparison and a different letter precedes the other cut, that letter is protected.

Exchange type. The *exchange type* of an unprotected letter is the finite table of how $\mathcal{M}_i$ treats it when it is read from an ordinary state: its transitions and final targets, and the values of Λ_i and Λ_i^{fin} when it is compared with itself (the *self table*), with a protected letter (the *protected rows*), and with the start label (the *initial row*).

(3) *The top level.* The unary top level w_D is shortened to at most $N + D$ letters by keeping a prefix with the same last label (Lemma A.8).

(4) *Simultaneous compression.* Going down from the top, the at most $\mathrm{G}(N, d)$ distinct blocks in use at each level are written as the tracks of one word of *columns*, aligned at their right ends, and shortened together by deleting the columns between two cuts with equal *summaries*. The summary of a tuple of blocks records, for every entry state, the exit of each block and the final marker labels to which Λ^{fin} relates its labels, which blocks are equal, and which pairs of blocks the derived admission relates; so it determines the derived monitor on the letters in use, and the shortened word is again a hole. Precisely, the new derived monitors *agree* with the old ones on the letters in use, the *support*: up to the renaming of each letter by its shortened version, they have the same states, transitions, targets and relations (Eq. (13)). The induction that produces the new markers level by level from the top is the *supported compression* of Lemma A.11.

(5) *The bound.* Each shortened block has fewer than $\mathrm{col}(N + d + 1, \mathrm{G}(N, d))$ letters, so each of the at most d expansions multiplies the length by at most this factor. With the top level this gives $\mathsf{Len}(N, d)$.

Shortening each block separately would fail, because comparisons relate different blocks through equal suffixes. The compression keeps a word a hole, that is, it preserves the absence of admitted comparisons, but it may destroy any particular comparison; positive conditions need the extension of Section 10.

3 Signed Leaf Cones

This section represents trees, and the solutions of systems, by sets of propositional atoms. In these coordinates every system over $\{\bot, f, \top\}$, of any arity and any variance, becomes a set of Horn clauses: implications with one premise, unit clauses (facts and forbidden roots), and one fixed family of binary exclusions. The representation has two consequences on which the rest of the paper rests. Least solutions are computed by closing a set of seeds, and every atom of a closure has a single seed (Section 3.3). Every pure move has a dual with both signs flipped, so two derivations that collide can be merged into one (Section 3.4).

We use the notation of Sections 2.2 and 2.3, in particular the directions $\Sigma = \{0, \dots, n-1\}$, the variance c, the polarity $\mathrm{par}(w)$, the order $\leq_c$, the normalizations ν_s, the dual t^∂ and the full constructor tree Ω, and we put $\Sigma^+ = \Sigma^* \setminus \{\varepsilon\}$. In this section and the next, all trees are trees over $\{\bot, f, \top\}$, and assignments range over arbitrary trees; Section 5, right after the characterization, shows that regular and finite trees and the signature $\{\bot, f\}$ reduce to this setting. Proofs omitted here are in Appendix B.

The positions of a tree t labeled $\bot$ or $\top$ are its *leaves*. We use the *chain property*: for every tree t and every word w, at most one prefix of w is a leaf of t, and if none is, then w is a position of t labeled f. Indeed, by induction on its prefixes, a prefix of w is a position as long as no shorter prefix is a leaf, and a position that is not a leaf has all its children.

3.1 Leaf Cones

Definition 3.1 (Signed leaf cones). For a tree T, the *top-leaf cone* A_T is the set of words w such that some prefix of w is a position of T labeled $\top$. For a tree t and a sign $s \in \{0, 1\}$, the *leaf cone of*

sign s of t is

$$L_t^s = A_{v_s t}.$$

We also write $L_t^+ = L_t^0$ and $L_t^- = L_t^1$, and call (L_t^0, L_t^1) the *leaf pair* of t.

The cones have a direct description. The *sign* of a leaf v of t is $b \oplus \mathrm{par}(v)$, where $b = 0$ if $t(v) = \top$ and $b = 1$ if $t(v) = \bot$. The tree $v_s t$ carries the label $\top$ at a leaf v iff v has sign s. By the chain property, therefore,

$$w \in L_t^s \iff \text{the path } w \text{ meets a leaf of } t \text{ of sign } s. \tag{5}$$

Since $v_1 t = (v_0 t)^\partial$ (Lemma 2.2(1)), the cone L_t^1 is the bottom-leaf cone of $v_0 t$: the two cones belong to the same normalized tree. The variance enters only through the sign of each leaf, and the parity that determines this sign accumulates along the path and stops at the leaf. There are two cones per tree, whatever the arity, not one per assignment of signs to the children. In terms of the signed correspondence (Lemma 2.5), $L^s_{\rho(v)}$ is the top-leaf cone of the covariant tree $\rho^\pm(v^s)$.

Theorem 3.2 (Leaf representation).

(1) *For every tree t, the sets L_t^0 and L_t^1 are disjoint, and each is* extension closed*: $w \in L_t^s$ implies $wi \in L_t^s$ for every letter i. Conversely, every pair (L^0, L^1) of disjoint extension-closed sets of words is the leaf pair of exactly one tree.*

(2) *For all trees t and u,*

$$t \le_c u \iff L_t^0 \subseteq L_u^0 \text{ and } L_u^1 \subseteq L_t^1.$$

(3) *$L_\top^0 = L_\bot^1 = \Sigma^*$, $L_\top^1 = L_\bot^0 = \varnothing$, and $L^s_{t^\partial} = L_t^{1-s}$. If $t = f(t_i)_i$, then neither cone of t contains ε, and for every letter i, word w, and sign s,*

$$iw \in L_t^s \iff w \in L_{t_i}^{s \oplus c(i)}.$$

In particular, $\varepsilon \in L_t^0$ iff $t = \top$, and $\varepsilon \in L_t^1$ iff $t = \bot$.

By (2), the order adds leaves of sign 0 and removes leaves of sign 1 as it goes up: after normalization, larger trees have more top leaves and fewer bottom leaves. The order is therefore not the inclusion of both cones. Inclusion of both cones is an *information order*, whose least element is the tree without leaves, Ω, with the leaf pair $(\varnothing, \varnothing)$. The least solutions of Section 3.3 are least in the information order.

Example 3.3 (Leaf cones of arrows). For the arrow, $n = 2$ and $f(a_0, a_1) = a_0 \to a_1$ with $c(0) = 1$ and $c(1) = 0$, so $\mathrm{par}(w)$ is the parity of the number of letters 0 in w. The leaf pairs of $\top$, $\bot$ and Ω are $(\Sigma^*, \varnothing)$, $(\varnothing, \Sigma^*)$ and $(\varnothing, \varnothing)$. In $\top \to \bot$, the leaf 0 is labeled $\top$ and has polarity 1, and the leaf 1 is labeled $\bot$ and has polarity 0; both have sign 1, so the leaf pair is $(\varnothing, \Sigma^+)$. Likewise $\bot \to \top$ has the leaf pair $(\Sigma^+, \varnothing)$, and $\bot \to \Omega$ has $(0\Sigma^*, \varnothing)$. By Theorem 3.2(3), both cones of every tree $f(a_0, a_1)$ lie in Σ^+, so (2) gives $\top \to \bot \le_c f(a_0, a_1) \le_c \bot \to \top$: these are the least and the greatest arrow types.

Figure 4 follows three arrow types up to the greatest one, $\bot \to \top$. The sign of each leaf is read off its label and its path, and each step up either removes a leaf of sign 1 or adds a leaf of sign 0, even where a subtree at a contravariant position becomes smaller.

3.2 Atoms and Clauses

Let V be a finite set of variables, and recall the signed variables v^s, their set $V^\pm$, and the flip $\overline{v^s} = v^{1-s}$ (Section 2.3); we put $\mathrm{sgn}(v^s) = s$. An *atom* is a pair $(p, w) \in C_V = V^\pm \times \Sigma^*$, and its *opposite* is $\overline{(p, w)} = (\bar{p}, w)$. An assignment ρ of trees to V determines the set of atoms

$$S_\rho = \{(v^s, w) : w \in L^s_{\rho(v)}\},$$

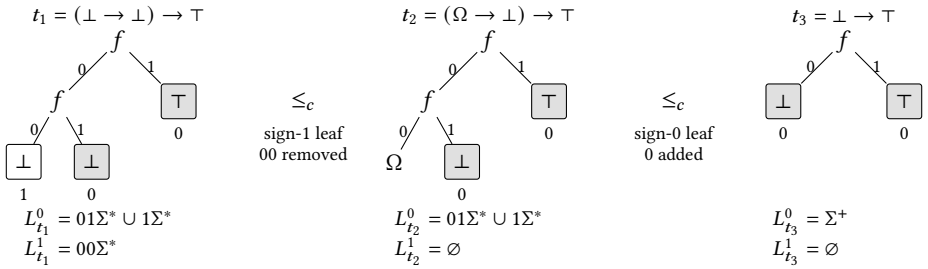


Fig. 4. Signed leaf cones of three arrow types, $c(0) = 1$ and $c(1) = 0$. The number below a leaf is its sign: 0 for $\top$ and 1 for $\bot$, plus the number of letters 0 on its path, modulo two. Leaves of sign 0 are shaded; the leaf $\bot$ at 01 has sign 0, and so does the leaf $\bot$ at 0 in t_3. Going up in the order removes leaves of sign 1 and adds leaves of sign 0 (Theorem 3.2(2)): $t_1 \le_c t_2$ shrinks L^1, and $t_2 \le_c t_3$ grows L^0 by replacing the subtree $\Omega \to \bot$ at the contravariant position 0 with the smaller tree $\bot$.

Table 3. The clause form of a literal, for all words w and all letters $i \in \Sigma$; here $\bar{c}(i) = 1 \oplus c(i)$. Every system also has the extensions $(p, w) \to (p, wi)$ for all $p \in V^{\pm}$, $w \in \Sigma^*$ and $i \in \Sigma$.

Literal	Pure moves	Root unit
$u \le v$	copies: $(u^0, w) \to (v^0, w)$ and $(v^1, w) \to (u^1, w)$	none
$f(a_i)_i \le u$	push_i: $(a_i^{c(i)}, w) \to (u^0, iw)$ pop_i: $(u^1, iw) \to (a_i^{\bar{c}(i)}, w)$	forbidden root (u^1, ε)
$u \le f(a_i)_i$	pop_i: $(u^0, iw) \to (a_i^{c(i)}, w)$ push_i: $(a_i^{\bar{c}(i)}, w) \to (u^1, iw)$	forbidden root (u^0, ε)
$u = \bot$	none	fact (u^1, ε)
$\top \le u$	none	fact (u^0, ε)

and an atom *holds* in ρ if it lies in S_ρ. An atom of sign 0 is *positive*, and an atom of sign 1 is *negative*. Opposite atoms are incompatible but not complementary: in Ω neither holds. For $U \subseteq V$ and $S \subseteq C_V$ we put $S|_U = S \cap C_U$. We read the word of an atom as a stack whose top is its first letter.

A set S of atoms is *extension closed* if $(p, w) \in S$ implies $(p, wi) \in S$ for every letter i, and an *exclusion pair* of S is a pair $a, \bar{a}$ of opposite atoms that both lie in S. By Theorem 3.2(1), applied to each variable, $\rho \mapsto S_\rho$ is a bijection from the assignments of trees to V onto the extension-closed subsets of C_V without exclusion pairs.

The literals of a system χ over V determine the *pure moves*, *facts* and *forbidden roots* listed in Table 3; facts and forbidden roots are the unit clauses. A push or a pop of direction i changes the sign by $c(i)$, and a copy keeps it. The *rules* of χ are its pure moves and the *extensions* $(p, w) \to (p, wi)$. We write Fact_χ and Neg_χ for the sets of facts and forbidden roots. Pure moves read and write only the top of the stack: push_i puts the letter i on top, pop_i removes it, and a copy leaves the stack unchanged. They form a pushdown system with control states $V^{\pm}$ and stack alphabet Σ, which Section 6 summarizes by finite automata. Extensions, in contrast, add a letter at the bottom of the stack.

The table is the signed system $\chi^{\pm}$ of Table 2 read in top-leaf cones. Every copy literal $p \le q$ of $\chi^{\pm}$ gives the pure moves $(p, w) \to (q, w)$, every lower constructor literal $f(p_i)_i \le q$ gives $(p_i, w) \to (q, iw)$, and every upper constructor literal $q \le f(p_i)_i$ gives $(q, iw) \to (p_i, w)$. The

forbidden roots are the atoms (q, ε) for the roots q of the upper constructor literals of $\chi^{\pm}$, and the facts are the atoms (q, ε) for its literals $\top \le q$.

Definition 3.4 (Models). A set $S \subseteq C_V$ is *consistent* for χ if it contains no forbidden root of χ and no exclusion pair. A *model* of χ is a consistent set of atoms that contains Fact_χ and is closed under all rules of χ. We write $\mathrm{Mod}(\chi)$ for the set of models of χ.

Lemma 3.5 (Clause form). *An assignment ρ satisfies a literal iff S_ρ is closed under the pure moves of the literal, contains its facts, and contains none of its forbidden roots. Consequently, $\rho \mapsto S_\rho$ is a bijection from the solutions of χ onto* $\mathrm{Mod}(\chi)$.

The proof (Appendix B) substitutes the identities of Theorem 3.2(3) into the inclusions of Theorem 3.2(2). Every literal thus contributes implications with one premise and units only; the only all-negative binary clauses are the exclusions $\neg(a \wedge \bar{a})$, which are the same for all systems, and no clause is a positive disjunction. A constructor literal contributes $2n$ schemes of pure moves and one forbidden root, and a copy literal two schemes. For $n = 0$, the literal $f() \le u$ says $u \neq \bot$ and $u \le f()$ says $u \neq \top$.

We write $a \to_\chi b$ if a pure move of χ leads from a to b.

Lemma 3.6 (Pure moves). *Let $(p, u) \to_\chi (q, t)$.*
(1) For every word σ, $(p, u\sigma) \to_\chi (q, t\sigma)$, by the same scheme of Table 3.
(2) $\mathrm{sgn}(q) \oplus \mathrm{par}(t) = \mathrm{sgn}(p) \oplus \mathrm{par}(u)$.

Item (1) holds because each scheme is stated for all words w, and (2) is a direct check of the six schemes (Appendix B). Both extend to sequences of pure moves. By (2), along such a sequence the sign of the current signed variable is determined by the first atom and the parity of the current word; we call (2) the *sign invariant*.

Example 3.7 (The system $x \le x \to x$). Let $\varphi = \{x \le x \to x\}$ for the arrow of Example 3.3, that is, the literal $x \le f(x, x)$. By Table 3, its pure moves are

$$\mathrm{pop}_0 : (x^0, 0w) \to (x^1, w), \quad \mathrm{push}_0 : (x^0, w) \to (x^1, 0w),$$
$$\mathrm{pop}_1 : (x^0, 1w) \to (x^0, w), \quad \mathrm{push}_1 : (x^1, w) \to (x^1, 1w),$$

it has the forbidden root (x^0, ε), and it has no facts. A move in the contravariant direction 0 changes the sign, and a move in the covariant direction 1 keeps it. In each row, the push is the pop read backwards with both signs reversed; Lemma 3.10 shows that this holds for every system. The tree $\top \to \bot$, with leaf pair $(\varnothing, \Sigma^+)$, is a solution: its atoms are the (x^1, w) with $w \neq \varepsilon$, and this set is closed under push_1 and under extensions.

3.3 Least Closures

A *derivation* of χ is a finite sequence of atoms $a_0, \ldots, a_k$, with $k \ge 0$, in which each a_{j+1} follows from a_j by a rule of χ; it is *pure* if all its rules are pure moves. We write $a \Rightarrow^*_\chi b$ if a derivation leads from a to b, and $a \to^*_\chi b$ if a pure derivation does; we drop the subscript when χ is clear. For an atom a and a set A of atoms put

$$\mathrm{Reach}_\chi(a) = \{b : a \Rightarrow^*_\chi b\}, \qquad \mu_\chi = \bigcup_{e \in \mathrm{Fact}_\chi} \mathrm{Reach}_\chi(e), \qquad H_\chi(A) = \mu_\chi \cup \bigcup_{a \in A} \mathrm{Reach}_\chi(a),$$

and $\mathrm{Cl}_\chi(a) = H_\chi(\{a\})$. An atom is *live* in χ if some model of χ contains it, and *dead* otherwise; Live_χ is the set of live atoms. Every rule has a single premise, so every atom b of $H_\chi(A)$ has a *seed*: an atom $x \in \mathrm{Fact}_\chi \cup A$ with $b \in \mathrm{Reach}_\chi(x)$.

Lemma 3.8 (Least closures). *Let χ be a system over V and $A \subseteq C_V$.*

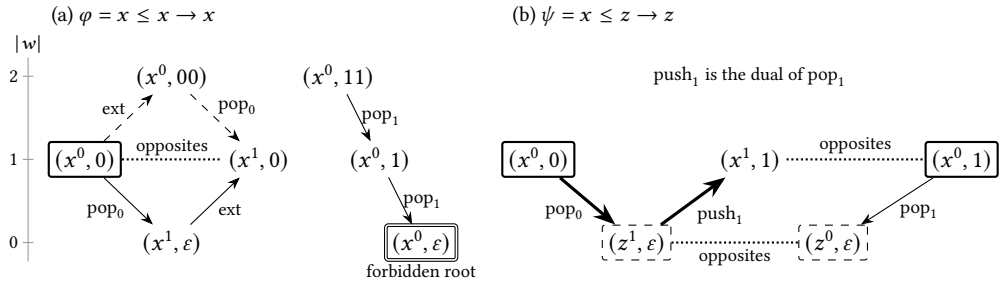


Fig. 5. Pure moves and extensions for the arrow, with each atom drawn at the height $|w|$ of its word: a pop lowers the stack, a push raises it, and an extension adds a letter at the bottom. Seeds are framed, hidden atoms dashed. (a) In $x \le x \to x$ (Examples 3.7 and 3.9), the seed $(x^0, 0)$ derives its own opposite $(x^1, 0)$, by a pop and an extension or, dashed, with the extension first (Lemma 3.11): a single-seed inconsistency without any unit. The atom $(x^0, 11)$ is dead for a different reason: it pops down to the forbidden root (x^0, ε). (b) In $x \le z \to z$ (Example 4.4), the seeds $(x^0, 0)$ and $(x^0, 1)$ derive the opposite atoms (z^1, ε) and (z^0, ε) of the hidden z. Reversing the second branch with flipped signs (Lemma 3.10) continues the first one (thick): the single seed $(x^0, 0)$ derives $(x^1, 1)$, the opposite of the second seed, which is collision return (Lemma 3.12).

(1) $H_\chi(A)$ *is the least set of atoms that contains* $\text{Fact}_\chi \cup A$ *and is closed under the rules of* χ.
(2) Some model of χ *contains* A *iff* $H_\chi(A)$ *is consistent, and then* $H_\chi(A)$ *is the least model that contains* A. *In particular,* χ *is satisfiable iff* μ_χ *is consistent, and then* μ_χ *is the least model of* χ*; and an atom* a *is live iff* $\text{Cl}_\chi(a)$ *is consistent, and then* $\text{Cl}_\chi(a)$ *is the least model that contains* a.
(3) If $H_\chi(A)$ *is not consistent, then there are seeds* $x, y \in \text{Fact}_\chi \cup A$*, possibly equal, such that* $\text{Reach}_\chi(x)$ *contains a forbidden root, or* $b \in \text{Reach}_\chi(x)$ *and* $\bar{b} \in \text{Reach}_\chi(y)$ *for some atom* b.
(4) For every atom (p, u) *and every word* α, $(p, u\alpha) \in \text{Reach}_\chi(p, u)$*; hence* $\text{Cl}_\chi(p, u\alpha) \subseteq \text{Cl}_\chi(p, u)$*, and if* (p, u) *is live, then so is* $(p, u\alpha)$.
(5) Let $V = X \cup Z$ *with* $X \cap Z = \varnothing$*, and let* $S \subseteq C_X$*. Then* $S = T|_X$ *for some model* T *of* χ *iff* $H_\chi(S)$ *is consistent and* $H_\chi(S)|_X = S$*; in this case* $H_\chi(S)$ *is the least such model.*

Item (3) says that every inconsistency has at most two seeds; it bounds neither the words of the seeds nor the lengths of the derivations. Item (5) reduces the extension of a given visible assignment to a single closure; Theorem 4.2 and the transfers of Section 5 use the lemma in this form.

Example 3.9 (Liveness in $x \le x \to x$*).* We continue Example 3.7. The system φ has no facts, so $\mu_\varphi = \varnothing$ is consistent: φ is satisfiable, and its least solution is $x = \Omega$. Every atom (x^1, u) is live, since the solution $x = \bot$ contains it. The atoms $(x^0, 1^k)$ are dead: k pops in direction 1 lead to the forbidden root (x^0, ε). The atom $(x^0, 0)$ is dead for a different reason. The pop $(x^0, 0) \to (x^1, \varepsilon)$ followed by an extension derives $(x^1, 0)$, the opposite of the seed, so $\text{Cl}_\varphi((x^0, 0))$ contains an exclusion pair. No forbidden root is derivable from $(x^0, 0)$: the only atoms (x^0, w) that it derives are its extensions $(x^0, 0w)$, and no move leads from a negative atom of x to a positive one. So this inconsistency has a single seed and uses no unit. Finally, $(x^0, 01)$ is live: its least model is the solution $((\Omega \to \bot) \to \bot) \to \bot$, whose atoms are the (x^0, w) with $w \in 01\Sigma^*$ and the (x^1, w) with $w \in 1^+\Sigma^* \cup 1^*001\Sigma^*$.

Figure 5(a) draws these derivations with each atom at the height of its word, so that the moves act as on a stack: pops go down, pushes go up, and an extension adds a letter at the bottom without changing the top.

3.4 Duality, Extensions First, and Collision Return

Lemma 3.10 (Duality). *If $a \to_\chi b$, then $\bar{b} \to_\chi \bar{a}$. Consequently, $a \to_\chi^* b$ implies $\bar{b} \to_\chi^* \bar{a}$.*

Proof. By Table 3: the two copies of a literal $u \le v$ are each other's duals, and so are push_i and pop_i of the same constructor literal, for each direction i separately; for instance, the dual of $(a_i^{c(i)}, w) \to (u^0, iw)$ is $(u^1, iw) \to (a_i^{\bar{c}(i)}, w)$. This holds also when variables are repeated among $u, a_0, \ldots, a_{n-1}$, since the pairing is by scheme. For derivations, reverse the sequence and dualize each move. □

Extensions have no duals: the reverse of an extension would delete the bottom letter of a stack, and no rule does that. The next lemma moves all extensions of a derivation to its beginning, so that duality applies to the rest.

Lemma 3.11 (Extensions first). *For all atoms (p, u) and b, $b \in \mathrm{Reach}_\chi(p, u)$ iff $(p, u\alpha) \to_\chi^* b$ for some word α.*

The proof (Appendix B) inducts on the derivation: an extension at the end appends a letter to the bottom of every stack of the pure derivation obtained so far, which Lemma 3.6(1) allows.

Lemma 3.12 (Collision return). *Let a and (q, v) be atoms. If $b \in \mathrm{Reach}_\chi(a)$ and $\bar{b} \in \mathrm{Reach}_\chi(q, v)$ for some atom b, then $(\bar{q}, v\delta) \in \mathrm{Reach}_\chi(a)$ for some word δ. If moreover $a = (p, u)$, then $(p, u\alpha) \to_\chi^* (\bar{q}, v\delta)$ for some words α and δ.*

Proof. By Lemma 3.11, $(q, v\delta) \to^* \bar{b}$ for some δ, and by Lemma 3.10, $b \to^* (\bar{q}, v\delta)$. Appending this pure derivation to a derivation from a to b gives $(\bar{q}, v\delta) \in \mathrm{Reach}_\chi(a)$. For the second claim apply Lemma 3.11 to this membership. □

Thus two derivations that end in an exclusion pair, possibly at a hidden variable, merge into one derivation from the first seed to the opposite of an extension of the second seed (Figure 5(b)): the first derivation continues along the second one, read backwards with flipped signs. The extension is necessary: in general the merged derivation does not reach the opposite of the second seed itself (Example 4.4). The first consequence concerns a single seed.

Corollary 3.13 (Deadness). *Let χ be satisfiable and let (p, u) be an atom. Then (p, u) is dead in χ iff at least one of the following holds:*

(a) $\mathrm{Reach}_\chi(p, u)$ *contains a forbidden root;*
(b) $(\bar{p}, u\delta) \in \mu_\chi$ *for some word* δ*;*
(c) $(\bar{p}, u\delta) \in \mathrm{Reach}_\chi(p, u)$ *for some word* δ*.*

Moreover, (a) holds iff $(p, u\alpha) \to_\chi^* e$ for some word α and some $e \in \mathrm{Neg}_\chi$, and (c) holds iff $(p, u\alpha) \to_\chi^* (\bar{p}, u\delta)$ for some words α and δ.

Causes (a) and (b) make the atom dead by support (Definition 6.9). Cause (c), a *self-collision*, may occur without such support: in Example 3.9, $(x^0, 0)$ is dead by $(x^0, 00) \to (x^1, 0)$, which is (c) with $\alpha = 0$ and $\delta = \varepsilon$. Intuitively, a self-collision has to change the sign, which the sign invariant ties to the polarities of the words, so it needs contravariant directions; in the covariant case every rule keeps the sign (Remark 4.5). The causes may overlap: for unary contravariant f and $\chi = \{x \le f(x)\}$, the atom (x^0, ε) is a forbidden root, and also $(x^0, 0) \to (x^1, \varepsilon)$. Section 6 expresses (a)–(c) by finite automata and reduces a self-collision of an atom that is not already dead by support to a *self event* (Definition 6.9): two cuts $i < j$ of the seed word, the suffix after j a prefix of the suffix after i, such that some signed variable that the seed reaches by popping the letters before i has its flip reached by popping the letters before j.

4 The Characterization

This section characterizes existential entailment over arbitrary trees by three conditions on single seeds. Let X and Z be disjoint finite sets of variables, let φ be a system over X, and let ψ be a system over $X \cup Z$ (Definition 2.4); every variable of ψ outside Z belongs to X. An atom (v^s, w) is *visible* if $v \in X$ and *hidden* if $v \in Z$. Restricting an assignment to X restricts its set of atoms to C_X, so by Lemma 3.5

$$\varphi \models \exists \bar{z}\, \psi \quad \Longleftrightarrow \quad \mathrm{Mod}(\varphi) \subseteq \mathcal{F}_\psi, \qquad \text{where } \mathcal{F}_\psi = \{T|_X : T \in \mathrm{Mod}(\psi)\}. \tag{6}$$

Here $\mathrm{Mod}(\varphi)$ consists of sets of visible atoms, and $\mathcal{F}_\psi$ is the *projected family* of ψ.

4.1 Three Conditions

Definition 4.1 (The three conditions). For φ and ψ as above, consider:

(I0) $\mu_\psi|_X \subseteq \mu_\varphi$;
(U) for every visible atom a that is live in φ, $\mathrm{Reach}_\psi(a)$ contains no forbidden root of ψ;
(II) for every visible atom a that is live in φ, $\mathrm{Reach}_\psi(a)|_X \subseteq \mathrm{Cl}_\varphi(a)$.

The names are mnemonic: (I0) is the inclusion for zero seeds, comparing the consequences of the facts alone; (U) concerns the unit clauses that are forbidden roots; and (II) is the inclusion for a single seed a.

When both systems are satisfiable, μ_ψ and μ_φ are their least models and $\mathrm{Cl}_\varphi(a)$ is the least model of φ that contains a (Lemma 3.8(2)). So (I0) says that every visible atom forced by ψ is forced by φ. Condition (U) says that ψ never derives a forbidden root from an atom that some solution of φ contains. Condition (II) says that every solution of φ that contains an atom a also contains every visible atom that ψ derives from a.

Theorem 4.2 (Characterization). *Let φ be a system over X and ψ a system over $X \cup Z$, for any arity and any variance, and let assignments range over arbitrary trees.*

(1) If φ is unsatisfiable, then $\varphi \models \exists \bar{z}\, \psi$. If φ is satisfiable and ψ is not, then $\varphi \not\models \exists \bar{z}\, \psi$.
(2) If φ and ψ are satisfiable, then $\varphi \models \exists \bar{z}\, \psi$ iff (I0), (U) *and* (II) *hold.*

Equivalently, $E_{\mathrm{all}}(\varphi, \psi) \iff \neg\, \mathrm{Sat}(\varphi) \vee \big(\mathrm{Sat}(\psi) \wedge (\mathrm{I0}) \wedge (\mathrm{U}) \wedge (\mathrm{II})\big)$.

Each condition concerns a single seed. An obstruction to extending a model S of φ may a priori involve two atoms of S whose closures under ψ collide in an exclusion pair, possibly at a hidden variable. The proof below uses collision return (Lemma 3.12) to turn every such collision into a violation of (I0) or of (II) at one seed. This step uses duality, which holds for every variance; no condition on pairs of seeds remains. When both systems are satisfiable, Figure 6 shows the four kinds of obstruction that the proof meets: three of them are the failures of the three conditions, and the fourth, a collision of two seeds, is reduced to the first or the third.

Proof. Item (1) is immediate from Definition 2.4.

For (2), assume that φ and ψ are satisfiable. We first show that the three conditions are necessary. Assume $\varphi \models \exists \bar{z}\, \psi$. The least model μ_φ of φ is $T|_X$ for some model T of ψ by Eq. (6), and $T \supseteq \mu_\psi$ by Lemma 3.8(2); hence $\mu_\psi|_X \subseteq T|_X = \mu_\varphi$, which is (I0). Let a be a visible atom that is live in φ. Its least model $\mathrm{Cl}_\varphi(a)$ is $T|_X$ for a model T of ψ. Since T contains a and is closed under the rules of ψ, it contains $\mathrm{Reach}_\psi(a)$. As T contains no forbidden root, this gives (U), and $\mathrm{Reach}_\psi(a)|_X \subseteq T|_X = \mathrm{Cl}_\varphi(a)$ gives (II).

Conversely, assume (I0), (U) and (II), let S be a model of φ, and put $T = H_\psi(S)$, where S is read as a set of atoms over $X \cup Z$. By Lemma 3.8(5) and Eq. (6), it suffices to show that $T|_X = S$ and that T

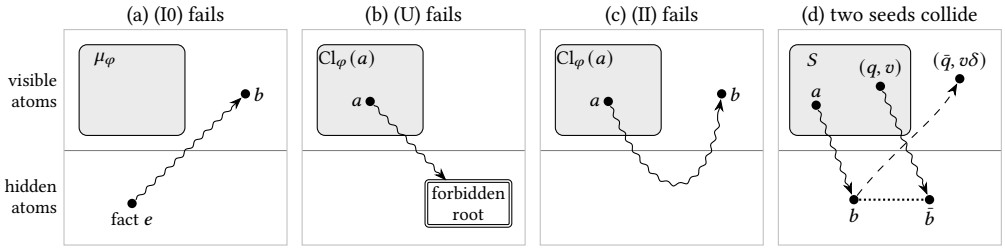


Fig. 6. When both systems are satisfiable, the four ways in which a model S of φ can fail to extend to a model of ψ, in the proof of Theorem 4.2. Upper halves hold visible atoms, lower halves hidden ones; wavy arrows are derivations of ψ, and the shaded box is a model of φ. (a) A fact of ψ derives a visible atom outside the least model μ_φ. (b) A seed a derives a forbidden root; then the least model $\mathrm{Cl}_\varphi(a)$ of φ that contains a cannot be extended. (c) A seed a derives a visible atom outside $\mathrm{Cl}_\varphi(a)$. (d) Two seeds of S derive opposite atoms b and $\bar{b}$, possibly hidden. Collision return (Lemma 3.12) reverses the second derivation by duality (dashed): the single seed a derives the opposite of the extension $(q, v\delta) \in S$ of the second seed, an atom outside $S \supseteq \mathrm{Cl}_\varphi(a)$, so this is case (c). If a is a fact of ψ instead, it is case (a).

is consistent. Every atom of S is live in φ, and $\mathrm{Cl}_\varphi(a) \subseteq S$ for every $a \in S$ by Lemma 3.8(1), since S is closed under the rules of φ and contains its facts.

Visible part. Clearly $S \subseteq T|_X$. A visible atom of T lies in $\mu_\psi|_X \subseteq \mu_\varphi \subseteq S$, by (I0), or in $\mathrm{Reach}_\psi(a)|_X$ for some $a \in S$, which lies in $\mathrm{Cl}_\varphi(a) \subseteq S$ by (II).

Consistency. Suppose that T is not consistent, and take seeds $x, y \in \mathrm{Fact}_\psi \cup S$ as in Lemma 3.8(3). If $\mathrm{Reach}_\psi(x)$ contains a forbidden root, then $x \notin \mathrm{Fact}_\psi$, because μ_ψ is consistent; so $x \in S$, contradicting (U). Otherwise $b \in \mathrm{Reach}_\psi(x)$ and $\bar{b} \in \mathrm{Reach}_\psi(y)$ for some atom b; since an exclusion pair is unordered, we may assume that $y \in S$ if one of x, y lies in S.

- If $x, y \in \mathrm{Fact}_\psi$, then $b, \bar{b} \in \mu_\psi$, contradicting the consistency of μ_ψ.
- If $x \in \mathrm{Fact}_\psi$ and $y = (q, v) \in S$, then Lemma 3.12 gives $(\bar{q}, v\delta) \in \mathrm{Reach}_\psi(x) \subseteq \mu_\psi$ for some δ. This atom is visible, so it lies in $\mu_\varphi \subseteq S$ by (I0). But $(q, v\delta) \in S$ because S is extension closed, and S has an exclusion pair.
- If $x \in S$ and $y = (q, v) \in S$, possibly $x = y$, then Lemma 3.12 gives $(\bar{q}, v\delta) \in \mathrm{Reach}_\psi(x)|_X$ for some δ, which lies in $\mathrm{Cl}_\varphi(x) \subseteq S$ by (II). Again $(q, v\delta) \in S$, and S has an exclusion pair.

Each case contradicts an assumption. Hence T is a model of ψ with $T|_X = S$, so $S \in \mathcal{F}_\psi$. □

The proof shows that, when the conditions hold, $H_\psi(S)$ is the least extension of every model S of φ; when a condition fails, the least model μ_φ or $\mathrm{Cl}_\varphi(a)$ that refutes it is a countermodel. The two qualifications of the theorem are needed.

- *The satisfiability dispatch.* Let $X = \{x\}$, φ empty, and $\psi = \{z = \bot,\ \top \leq z\}$ with z hidden. Then μ_ψ consists of atoms of z, and ψ has no rule that involves x other than its extensions, so the three conditions hold; but ψ is unsatisfiable, and the entailment fails.
- *Condition* (U). Let $n \geq 0$, $X = \{y\}$, φ empty, and $\psi = \{y \leq f(z_i)_i\}$ with distinct hidden variables z_i. Then ψ has no facts, and its only pure moves from visible atoms are the pops $(y^0, iw) \to (z_i^{c(i)}, w)$, after which no pure move applies. By Lemma 4.3 below, (I0) and (II) hold. But $y = \top$ has no extension: (U) fails at the live atom (y^0, ε), which is a forbidden root of ψ.

The theorem also depends on duality. For a set of one-premise rules without duals, say $(x^0, w) \to (z^0, w)$ and $(y^0, w) \to (z^1, w)$ with z hidden, the three conditions hold for the empty antecedent, but $x = y = \top$ produces an exclusion pair at z. No system has exactly these rules: by Lemma 3.10, a

system with these rules also has the duals $(z^1, w) \to (x^1, w)$ and $(z^0, w) \to (y^1, w)$, which expose the collision as the visible consequences $(x^0, w) \Rightarrow^* (y^1, w)$ and $(y^0, w) \Rightarrow^* (x^1, w)$, and (II) fails.

The conditions only need pure derivations of ψ, from extended seeds.

Lemma 4.3 (Pure forms).

(1) (U) *holds iff there are no visible atom (p, u) that is live in φ, word α and forbidden root e of ψ with $(p, u\alpha) \to_\psi^* e$.*

(2) (II) *holds iff $b \in \mathrm{Cl}_\varphi(p, u)$ for every visible atom (p, u) that is live in φ and every visible atom b with $(p, u) \to_\psi^* b$.*

Proof. If (p, u) is live in φ, so is $(p, u\alpha)$, and $\mathrm{Cl}_\varphi(p, u\alpha) \subseteq \mathrm{Cl}_\varphi(p, u)$ (Lemma 3.8(4)). By Lemma 3.11, $b \in \mathrm{Reach}_\psi(p, u)$ iff $(p, u\alpha) \to_\psi^* b$ for some α. Item (1) follows directly, since $(p, u\alpha)$ is again a live visible atom. For (2), the condition is clearly necessary. Conversely, if $b \in \mathrm{Reach}_\psi(p, u)|_X$ for a live (p, u), then $(p, u\alpha) \to_\psi^* b$ for some α, and the condition, applied to the live atom $(p, u\alpha)$, gives $b \in \mathrm{Cl}_\varphi(p, u\alpha) \subseteq \mathrm{Cl}_\varphi(p, u)$. □

Example 4.4 (The consequent $x \le z \to z$). For the arrow of Example 3.3, let $X = \{x\}$, $Z = \{z\}$ and $\psi = \{x \le z \to z\}$, the literal $x \le f(z, z)$. By Table 3, ψ has the forbidden root (x^0, ε), no facts, and the pure moves

$$(x^0, 0w) \to (z^1, w) \to (x^1, 1w), \qquad (x^0, 1w) \to (z^0, w) \to (x^1, 0w).$$

No pure move leaves a negative atom of x. So the pure derivations of ψ of positive length between visible atoms are exactly $(x^0, iw) \to^* (x^1, \bar{\imath}w)$, where $\bar{\imath} = 1 - i$: ψ exchanges the first letter and the sign. Since $\mu_\psi = \varnothing$, (I0) holds for every antecedent, and ψ is satisfiable, with $x = \bot$.

The empty antecedent. Let φ_0 be empty. Then (U) fails at (x^0, ε), with the countermodel $x = \top$. Condition (II) fails at $(x^0, 0)$: ψ derives $(x^1, 1)$, but $\mathrm{Cl}_{\varphi_0}((x^0, 0))$ consists of the atoms $(x^0, 0w)$, the leaf pair $(0\Sigma^*, \varnothing)$ of the countermodel $x = \bot \to \Omega$ (Example 3.3). Indeed $\bot \to \Omega \le_c z \to z$ would require $z \le_c \bot$ and $\Omega \le_c z$. The tree $\bot \to \top$, with leaf pair $(\Sigma^+, \varnothing)$, fails for a reason that a priori involves two seeds: the closures of $(x^0, 0)$ and $(x^0, 1)$ under ψ contain the exclusion pair (z^1, ε) and (z^0, ε). Collision return reverses the second branch $(x^0, 1) \to (z^0, \varepsilon)$ into $(z^1, \varepsilon) \to (x^1, 1)$, which continues the first branch: the single seed $(x^0, 0)$ derives $(x^1, 1)$, the opposite of the second seed, and (II) fails at $(x^0, 0)$. The extension in Lemma 3.12 is needed in general. The atom (x^0, ε) derives (z^0, ε) through its extension $(x^0, 1)$, which collides with the atom (z^1, ε) derived from $(x^0, 0)$; collision return gives $(x^1, 1)$, the opposite of an extension of (x^0, ε), whereas (x^1, ε) is not derivable from $(x^0, 0)$.

The antecedent $x \le x \to x$. Let $\varphi_1 = \{x \le x \to x\}$ be the system of Example 3.7. It is satisfiable. Condition (U) holds, because (x^0, ε) is a forbidden root of φ_1 as well, hence dead. For (II), by Lemma 4.3 it suffices that $(x^1, \bar{\imath}w) \in \mathrm{Cl}_{\varphi_1}((x^0, iw))$ for every live (x^0, iw), and φ_1 derives it: $(x^0, 0w) \to (x^1, w) \to (x^1, 1w)$ by pop_0 and push_1, and $(x^0, 1w) \to (x^0, w) \to (x^1, 0w)$ by pop_1 and push_0. By Theorem 4.2, $\varphi_1 \models \exists z\, \psi$; indeed $z = x$ is a witness.

4.2 Special Cases

The two remarks of this subsection are informal readings of the characterization in special cases; nothing later depends on them.

Remark 4.5 (Covariant and joint-parity instances). A *parity assignment* of a system χ is a map $\pi : V \to \{0, 1\}$ with $\pi(u) = \pi(v)$ for every literal $u \le v$ and $\pi(a_i) = \pi(u) \oplus c(i)$ for every i and every literal $f(a_i)_i \le u$ or $u \le f(a_i)_i$ of χ. Intuitively, the rules of such a system preserve the bit $s \oplus \pi(v)$ of an atom (v^s, w), which differs between opposite atoms; so self-collisions, cause (c) of

Corollary 3.13, do not arise, and renaming every atom (v^s, w) to $(v^{s\oplus\pi(v)}, w)$ turns the system into a covariant one. The covariant case itself is $\pi = 0$. The system $x \le x \to x$ has no parity assignment, since it would require $\pi(x) = \pi(x) \oplus c(0)$, and its self-collision $(x^0, 00) \to (x^1, 0)$ (Example 3.9) is exactly what a parity assignment rules out. Theorem 4.2 needs no parity assignment.

Remark 4.6 (One sign: the signature $\{\bot, f\}$). Let f be covariant, and consider trees over $\{\bot, f\}$ and systems without literals $\top \le u$ (Section 2.2). Every leaf of such a tree has sign 1, so $L_t^0 = \varnothing$, and L_t^1 is the complement of the set $K(t)$ of positions labeled f. Read through complements, a negative atom (v^1, w) says $w \notin K(v)$, and each rule on negative atoms is the contrapositive of a clause on constructor positions: $u \le v$ says $K(u) \subseteq K(v)$, $f(a_i)_i \le u$ says $\varepsilon \in K(u)$ and $iK(a_i) \subseteq K(u)$, $u \le f(a_i)_i$ says that $iw \in K(u)$ implies $w \in K(a_i)$, $u = \bot$ says $\varepsilon \notin K(u)$, and extensions say that $K(v)$ is prefix closed. Informally, over $\{\bot, f\}$ the three conditions then compare, for φ and for the projection of ψ, the unions of all solutions, the least solutions, and the closures of single positions, and no exclusions or self-collisions occur. The decision procedure does not treat $\{\bot, f\}$ separately; the next section reduces $\{\bot, f\}$ to $\{\bot, f, \top\}$ (Theorem 5.17).

A forbidden root has the empty word, so the pure derivations in Lemma 4.3(1) end with an empty stack. A failure of (II) with a live seed a and a consequence b is thus witnessed by a pure derivation of ψ from a live extension of a to b, with b outside the least model of φ that contains that extension.

The characterization concerns arbitrary trees over $\{\bot, f, \top\}$. Section 5 shows next that this setting suffices for the other tree classes and for the signature $\{\bot, f\}$ with f covariant. Sections 6 to 8 then turn the satisfiability tests, (I0), (U) and this form of (II) into problems about finite automata.

5 Tree Classes and the Signature without Top

The characterization of Section 4 concerns arbitrary trees over $\{\bot, f, \top\}$. This section shows that this one setting suffices: the problem in the other tree classes, for every arity $n \ge 0$ and every variance c, and over the signature $\{\bot, f\}$ with f covariant, reduces to it. Regular trees give the same answer as arbitrary trees, $E_{\text{reg}} = E_{\text{all}}$ (Theorem 5.12); finite trees give the same answer when both systems have finite solutions, and otherwise finite satisfiability alone decides (Theorem 5.13); and a gadget on every variable reduces $\{\bot, f\}$ to $\{\bot, f, \top\}$ in each tree class (Theorem 5.17). Nothing in this section depends on the later sections. In Sections 6 to 10, therefore, all trees are over $\{\bot, f, \top\}$ and assignments range over arbitrary trees, and Section 11 puts the pieces together.

The transfers to regular and to finite trees follow one pattern. A failed extension is caused by at most two seeds (Lemma 5.3), and the corresponding observations are expressed by finite ground bounds (Lemma 5.4). It then suffices that adding ground bounds to a system preserves the existence of solutions in the tree class: for regular trees and regular bounds by the graph equations of the bounds (Lemma 5.6), and for finite trees and finite bounds by a median argument, for every variance (Theorem 5.7). Least closures do not suffice for finite trees: they describe trees with as few leaves as possible, and these are infinite in general; the empty system has the empty least model, which assigns Ω to every variable. Proofs not given here are in Appendix C.

Throughout, trees are over $\{\bot, f, \top\}$ and ordered by $\le_c$ unless stated otherwise, φ and ψ are as in Definition 2.4, we use the notation of Sections 2 and 3 together with $f(t^n) = f(t, \dots, t)$ and $v_\pm$ for v_0 and v_1, and a model T *decodes* to the solution ρ with $S_\rho = T$ (Lemma 3.5).

5.1 Extensions and Two Seeds

An assignment ρ of X is *extensible* if it extends to a solution of ψ over arbitrary trees.

Lemma 5.1 (Extension criterion). *An assignment ρ of X is extensible iff $H_\psi(S_\rho)$ is consistent and $H_\psi(S_\rho)|_X = S_\rho$. In that case $H_\psi(S_\rho)$ decodes to an extension of ρ.*

This restates Lemma 3.8(5) for assignments.

Proof. By Lemma 3.8(5) and Lemma 3.5, applied to the set S_ρ, which is extension closed and has no exclusion pair, ρ is extensible exactly when $H_\psi(S_\rho)$ is consistent and has visible part S_ρ; then that closure decodes to an extension. □

Definition 5.2 (Observations). A *visible observation* is a visible atom $a = (x^\pm, w)$ together with one of the words *asserted* or *denied*. An asserted observation *holds* in an assignment ρ of X if $a \in S_\rho$, a denied one if $a \notin S_\rho$. The variable x and the word w are the variable and the word of the observation.

Lemma 5.3 (Two seeds). *Let ρ be an assignment of X that is not extensible. There are a set $A \subseteq S_\rho$ of visible atoms and a set C of visible atoms outside S_ρ, with $|A| + |C| \le 2$, such that no assignment ρ' of X with $A \subseteq S_{\rho'}$ and $C \cap S_{\rho'} = \varnothing$ is extensible. In particular, at most two visible observations hold in ρ and hold together in no extensible assignment.*

Proof. Every atom b of $H_\psi(S_\rho)$ has a seed in $\mathrm{Fact}_\psi \cup S_\rho$, so $b \in H_\psi(A_b)$ for a set $A_b \subseteq S_\rho$ with at most one element. By Lemma 5.1, $H_\psi(S_\rho)$ is not consistent or has a visible atom outside S_ρ. In the first case, Lemma 3.8(3) gives seeds $x, y \in \mathrm{Fact}_\psi \cup S_\rho$ such that $\mathrm{Reach}_\psi(x)$ contains a forbidden root, or $\mathrm{Reach}_\psi(x)$ and $\mathrm{Reach}_\psi(y)$ contain an exclusion pair; let A consist of those of x, y that lie in S_ρ, and $C = \varnothing$. In the second case, let $b \in H_\psi(S_\rho)|_X$ with $b \notin S_\rho$, and put $A = A_b$ and $C = \{b\}$. In both cases $|A| + |C| \le 2$. Let ρ' be an assignment of X with $A \subseteq S_{\rho'}$ and $C \cap S_{\rho'} = \varnothing$. Since H_ψ is monotone, $H_\psi(S_{\rho'}) \supseteq H_\psi(A)$, so $H_\psi(S_{\rho'})$ contains the same forbidden root or the same two opposite atoms, or the visible atom $b \notin S_{\rho'}$. By Lemma 5.1, ρ' is not extensible. The observations asserting the atoms of A and denying those of C prove the last claim. □

The lemma uses no satisfiability hypothesis. If ψ is unsatisfiable, then μ_ψ is inconsistent by Lemma 3.8(2), and $A = C = \varnothing$ will do. The two seeds of A may belong to the same variable.

5.2 Observation Pins

For a word w and trees b, g, let $\mathrm{emb}_b(w, g)$ be the tree with a pin path along w that carries g at its end and b at every child off the pin path: $\mathrm{emb}_b(\varepsilon, g) = g$, and $\mathrm{emb}_b(iw, g) = f(t_j)_j$ with $t_i = \mathrm{emb}_b(w, g)$ and $t_j = b$ for $j \neq i$.

Lemma 5.4 (Observation pins). *For every covariant tree t and every word w,*

$$\begin{aligned} w \in L^0_t &\iff \mathrm{emb}_\bot(w, \top) \le t, \quad w \notin L^0_t \iff t \le \mathrm{emb}_\top(w, f(\top^n)), \\ w \in L^1_t &\iff t \le \mathrm{emb}_\top(w, \bot), \quad w \notin L^1_t \iff \mathrm{emb}_\bot(w, f(\bot^n)) \le t, \end{aligned} \tag{7}$$

Consequently, for every variance c, every visible observation with variable x and word w is equivalent, for all assignments ρ of X, to a single inequality $g \le_c \rho(x)$ or $\rho(x) \le_c g$, where g is a finite ground tree.

For $n = 0$ the only word is ε, and the four trees are $\top$, $f()$, $\bot$ and $f()$.

5.3 Bound Pinning

Intersection of leaf cones. Over arbitrary trees, extensions can be intersected; over finite trees they cannot (item (2) below), which is why the transfer to finite trees pins observations by finite ground bounds instead. For trees t and u, let $t \sqcap u$ be the tree with the leaf pair $(L^0_t \cap L^0_u,\ L^1_t \cap L^1_u)$, which exists by Theorem 3.2(1); its signed leaf cones contain exactly the observations common to t and u. For assignments, $\sqcap$ acts variablewise.

Proposition 5.5 (Closure under intersection).

(1) *The intersection of a nonempty family of models of a system χ is a model of χ, and the intersection of a nonempty subfamily of the projected family $\mathcal{F}_\psi$ (Eq. (6)) belongs to $\mathcal{F}_\psi$. In particular, the solutions of χ and the assignments of X that extend to solutions of ψ are closed under $\sqcap$.*
(2) *For $n \geq 1$, $\bot \sqcap \top = \Omega$ is infinite, so the finite trees are not closed under $\sqcap$.*

The proof is in Appendix C. By (1), the model $H_\psi(S)$ of Lemma 3.8(5) is the intersection of all models T of ψ with $T|_X = S$, the least extension of S. Informally, $\sqcap$ is an information meet: it keeps the observations common to its arguments, it commutes with the constructor, and it is the median of t, u and Ω. So, over arbitrary trees, a relation that is the projection of the solutions of a system and contains two pairs (a, b) and (b, a) also contains $(a \sqcap b, a \sqcap b)$, and no system with hidden variables defines the disequality of two trees. The decision procedure does not use Proposition 5.5.

Ground bounds. A *ground bound* on a variable x is an inequality $g \leq_c x$, a *lower* bound, or $x \leq_c g$, an *upper* bound, where g is a ground tree; it is *finite* or *regular* if g is. We first treat regular trees.

Lemma 5.6 (Regular-bound pinning). *Let χ be a system over a finite set V of variables, with any variance, and let B be a finite set of regular ground bounds on variables of V. If $\chi \cup B$ has a solution, then it has a regular solution.*

Proof. Each tree g of a bound is the unfolding of a finite graph G_g whose constructor nodes have n ordered successors (Section 2.2). Give every node v of every such graph a fresh variable y_v, and let E consist of the literals $y_v \leq f(y_{v_i})_i$ and $f(y_{v_i})_i \leq y_v$ for every constructor node v with successors v_i, and of $y_v = \bot$ or $\top \leq y_v$ for every leaf v labeled $\bot$ or $\top$. By antisymmetry of $\leq_c$, a solution σ of E has $\sigma(y_v) = f(\sigma(y_{v_i}))_i$ at constructor nodes, so by induction on the length of a word w, w is a position of $\sigma(y_v)$ iff it is a position of the unfolding of G_g from v, with the same label: the graph equations E have exactly one solution, the unfoldings. Replace each bound $g \leq_c x$ by the copy literal $y_{r_g} \leq x$, where r_g is the root of G_g, and each bound $x \leq_c g$ by $x \leq y_{r_g}$; call the result B'. A solution of $\chi \cup B$, extended by the unfoldings, solves $\chi \cup E \cup B'$, so by Theorem 2.9(1) this system has a regular solution τ. By uniqueness, $\tau(y_{r_g}) = g$ for every bound, so the restriction of τ to V is a regular solution of $\chi \cup B$. □

For finite trees, the following theorem extends [Zhao 2026, Theorem "Finite countermodel transfer"] from the pin paths of a single word to arbitrary finite bounds; it needs a finite solution of the system without the bounds.

Theorem 5.7 (Finite-bound pinning). *Let χ be a system over a set V of variables, with any variance, that has a finite solution, and let B be a finite set of finite ground bounds on variables of V. If $\chi \cup B$ has a solution, then it has a finite solution.*

The proof reduces the theorem to covariant systems by the signed system $\chi^\pm$ of Table 2, and then symmetrizes a covariant solution by a median. A ground bound b is translated into a pair $b^\pm$ of covariant ground bounds on signed variables,

$$(g \leq_c x)^\pm = \{v_+ g \leq x^+,\ x^- \leq v_- g\}, \qquad (x \leq_c g)^\pm = \{x^+ \leq v_+ g,\ v_- g \leq x^-\}, \tag{8}$$

and $B^\pm$ is the union of these pairs. For an assignment ρ of V, $\rho^\pm$ is the assignment $v^\pm \mapsto v_\pm \rho(v)$ of Lemma 2.5. The *sign involution* ι maps an assignment σ of $V^\pm$ to the assignment

$$(\iota\sigma)(v^+) = \sigma(v^-)^\partial, \qquad (\iota\sigma)(v^-) = \sigma(v^+)^\partial.$$

The literals of Table 2 that contain $\bot$ or $\top$ belong to $\chi^\pm$; the proof needs them (Example 5.11).

Lemma 5.8 (Signed bounds). *Let χ be a system over V and B a finite set of finite ground bounds on V.*

(1) *An assignment ρ of V solves $\chi \cup B$ iff $\rho^{\pm}$ solves $\chi^{\pm} \cup B^{\pm}$.*
(2) *ι is an involution that maps the solutions of $\chi^{\pm} \cup B^{\pm}$ to solutions, and $\iota\rho^{\pm} = \rho^{\pm}$ for every assignment ρ of V.*
(3) *If $\iota\sigma = \sigma$, then $\sigma = \rho^{\pm}$ for the assignment $\rho(v) = v_{+}\sigma(v^{+})$; and $\rho \mapsto \rho^{\pm}$ and its inverse preserve finiteness.*

For covariant trees t_1, t_2, t_3, the *median* $\mathrm{med}(t_1, t_2, t_3)$ is the covariant tree whose cones consist of the words that lie in the corresponding cones of at least two of the three trees: for both signs,

$$L^{\pm}_{\mathrm{med}(t_1,t_2,t_3)} = \{w : w \in L^{\pm}_{t_j} \text{ for at least two } j\}.$$

It exists by Theorem 3.2(1): the sets on the right are extension closed, and they are disjoint, since a word in two of the cones $L^0_{t_j}$ and in two of the cones $L^1_{t_j}$ lies in both cones of one t_j. The median of assignments is taken variablewise. Informally, the median takes a majority vote of the three trees at every word; we use only the following properties.

Lemma 5.9 (Median).

(1) *The median is symmetric in its first two arguments and monotone in each argument, and $\mathrm{med}(t, t, u) = t$.*
(2) *$\mathrm{med}(f(a^1_i)_i, f(a^2_i)_i, f(a^3_i)_i) = f(\mathrm{med}(a^1_i, a^2_i, a^3_i))_i$ and $\mathrm{med}(t_1, t_2, t_3)^{\partial} = \mathrm{med}(t^{\partial}_1, t^{\partial}_2, t^{\partial}_3)$. In particular ι commutes with medians of assignments of signed variables.*
(3) *The median of three finite trees is finite.*
(4) *If three assignments solve a covariant system together with covariant ground bounds, so does their median.*

The covariant case of Theorem 5.7 is proved directly from the finite-satisfiability criterion of Zhao [2026] (Theorem A.1), in Appendix C.3. By that criterion, a system has a finite solution iff it has neither a label clash nor a cycle clash. A finite ground bound is equivalent to finitely many covariant observations, the *path pins* at its positions, and each path pin is a bound by one of the four trees of Lemma 5.4, which a chain of fresh variables along its word expresses. The fresh variables of a chain have ranks that strictly increase along the chain, so, by the ranked extensions of Zhao [2026], every cycle clash of the extended system lies among the original variables, where the finite solution of χ excludes it, and the solution of $\chi \cup B$ excludes label clashes. The pins are added one at a time.

Lemma 5.10 (Covariant finite-bound pinning). *Theorem 5.7 holds when f is covariant.*

Proof of Theorem 5.7. If $n = 0$, every tree is finite. Let $n \geq 1$, let δ be a finite solution of χ, and let ρ be a solution of $\chi \cup B$. By Lemma 5.8(1,3), $\delta^{\pm}$ is a finite solution of $\chi^{\pm}$, and $\rho^{\pm}$ is a solution of $\chi^{\pm} \cup B^{\pm}$. By Lemma 5.10, applied to the covariant system $\chi^{\pm}$ and the bounds $B^{\pm}$, the system $\chi^{\pm} \cup B^{\pm}$ has a finite solution α. By Lemma 5.8(2), $\iota\alpha$ is a solution of $\chi^{\pm} \cup B^{\pm}$, and it is finite because duality keeps the domain. Consider the median

$$\mathrm{med}(\alpha, \iota\alpha, \delta^{\pm}).$$

All three arguments solve $\chi^{\pm}$, so $\mathrm{med}(\alpha, \iota\alpha, \delta^{\pm})$ solves $\chi^{\pm}$ by Lemma 5.9(4), and it is finite by Lemma 5.9(3). The assignment $\delta^{\pm}$ need not satisfy the bounds; still $\mathrm{med}(\alpha, \iota\alpha, \delta^{\pm})$ satisfies $B^{\pm}$: for a lower bound $G \leq p$ in $B^{\pm}$, both $\alpha(p)$ and $(\iota\alpha)(p)$ lie above G, so $G = \mathrm{med}(G, G, \delta^{\pm}(p)) \leq \mathrm{med}(\alpha, \iota\alpha, \delta^{\pm})(p)$ by Lemma 5.9(1), and upper bounds are symmetric. By Lemma 5.9(2) and Lemma 5.8(2),

$$\iota\ \mathrm{med}(\alpha, \iota\alpha, \delta^{\pm}) = \mathrm{med}(\iota\alpha, \iota\iota\alpha, \iota\delta^{\pm}) = \mathrm{med}(\iota\alpha, \alpha, \delta^{\pm}) = \mathrm{med}(\alpha, \iota\alpha, \delta^{\pm}).$$

By Lemma 5.8(3), $\mathrm{med}(\alpha, \iota\alpha, \delta^{\pm}) = \tau^{\pm}$ for a finite assignment τ of V, which solves $\chi \cup B$ by Lemma 5.8(1). □

Example 5.11 (The extreme literals are needed). The signed literals of $u = \bot$ and $\top \le u$ in Table 2 cannot be dropped from $\chi^{\pm}$, although they contribute no rules on leaf cones, only facts (Table 3). Let $n \ge 1$, $\chi = \{\top \le x\}$, $B = \varnothing$, and $\delta(x) = \top$. Without its extreme literals the signed system is empty, and $\alpha(x^{+}) = \bot$, $\alpha(x^{-}) = \top$ is a finite solution with $\iota\alpha = \alpha$. Then $\mathrm{med}(\alpha, \iota\alpha, \delta^{\pm}) = \alpha$, which decodes to $x = \bot$ and violates χ. The third argument of the median must also be a finite solution of the unpinned system: Ω in every coordinate is fixed by ι and solves every signed system without extreme literals, but the median with it is infinite in general.

5.4 Regular and Finite Trees

Theorem 5.12 (Regular trees). *For every arity $n \ge 0$, every variance c, and all systems φ over a finite set X and ψ over the finite set $X \cup Z$:*

(1) a system is satisfiable iff it has a regular solution;
(2) every regular extensible assignment of X has a regular extension;
(3) if some solution of φ is not extensible, then some regular solution of φ is not extensible.

Consequently $E_{\mathrm{reg}}(\varphi, \psi) \iff E_{\mathrm{all}}(\varphi, \psi)$.

Proof. (1) This restates Theorem 2.9(1).

(2) Let ρ be regular and extensible, with an extension ρ' that solves ψ. Let B consist of the regular ground bounds $\rho(x) \le_c x$ and $x \le_c \rho(x)$ for $x \in X$. Then ρ' solves $\psi \cup B$, so by Lemma 5.6 the system $\psi \cup B$ has a regular solution τ. By antisymmetry of $\le_c$, τ agrees with ρ on X, so it is a regular extension of ρ.

(3) Let ρ be a solution of φ that is not extensible. By Lemma 5.3, at most two visible observations hold in ρ and hold together in no extensible assignment. By Lemma 5.4, each of them is equivalent to a ground bound on a visible variable by a finite, hence regular, tree; let B be the set of these bounds. Then ρ solves $\varphi \cup B$, so Lemma 5.6 gives a regular solution ρ_1 of $\varphi \cup B$. The observations hold in ρ_1, so ρ_1 is not extensible.

For the consequence, if E_{all} holds, every regular solution of φ is extensible and has a regular extension by (2); if E_{all} fails, (3) gives a regular solution of φ without any extension. □

Theorem 5.13 (Finite trees). *For every arity $n \ge 0$, every variance c, and all systems φ over X and ψ over $X \cup Z$,*

$$E_{\mathrm{fin}}(\varphi, \psi) \iff \neg\,\mathrm{Sat}_{\mathrm{fin}}(\varphi) \;\vee\; \big(\mathrm{Sat}_{\mathrm{fin}}(\psi) \wedge E_{\mathrm{all}}(\varphi, \psi)\big). \tag{9}$$

More precisely:

(1) if φ has a finite solution and some solution of φ is not extensible, then some finite solution of φ is not extensible;
(2) if ψ has a finite solution, then every finite extensible assignment of X has a finite extension.

Proof. (1) Let ρ be a solution of φ that is not extensible. By Lemma 5.3, at most two visible observations hold in ρ and hold together in no extensible assignment. By Lemma 5.4, each of them is equivalent to a ground bound on a visible variable; let B be the set of these bounds. Then ρ solves $\varphi \cup B$, and φ has a finite solution, so Theorem 5.7 gives a finite solution ρ_1 of $\varphi \cup B$. The observations hold in ρ_1, so ρ_1 is not extensible.

(2) Let ρ be a finite assignment of X with an extension ρ' that solves ψ. Let B consist of the bounds $\rho(x) \le_c x$ and $x \le_c \rho(x)$ for $x \in X$. Then ρ' solves $\psi \cup B$, and ψ has a finite solution, so Theorem 5.7 gives a finite solution τ of $\psi \cup B$. By antisymmetry of $\le_c$, τ agrees with ρ on X, so it is a finite extension of ρ.

For Eq. (9): if φ has no finite solution, both sides hold. If φ has a finite solution and ψ has none, that solution has no finite extension, and both sides fail. If both have finite solutions, (1) and (2) give $E_{\text{fin}}(\varphi,\psi) \iff E_{\text{all}}(\varphi,\psi)$. □

The bounds in the proof of (1) concern at most two visible variables. The proofs of Theorems 5.12 and 5.13 use no satisfiability hypothesis and no condition on the variance.

Corollary 5.14 (Decisions for regular and finite trees). *For fixed n and c, if existential entailment over $\{\bot, f, \top\}$ is decidable over arbitrary trees, then it is decidable over regular trees and over finite trees: $E_{\text{reg}} = E_{\text{all}}$, and E_{fin} is decided by Eq. (9), whose two finite-satisfiability tests are decidable by Theorem 2.9(2).*

Remark 5.15 (Both finite-satisfiability tests are needed). Let $n \geq 1$ and let c be any variance. For $X = \varnothing$, the empty antecedent and $\psi = \{z \leq f(z^n),\ f(z^n) \leq z\}$, the entailment holds over arbitrary and regular trees, with the unique witness $z = \Omega$, but fails over finite trees, since ψ has no finite solution. For $\varphi = \{x \leq f(x^n),\ f(x^n) \leq x\}$ and $\psi = \{x = \bot\}$, the entailment fails over arbitrary and regular trees, at $x = \Omega$, and holds vacuously over finite trees. Least models are not finite countermodels in general, and by Proposition 5.5(2) the finite trees are not closed under intersection of leaf cones; this is why Theorem 5.13 pins observations instead.

5.5 Trees without Top

In this subsection f is covariant, so $\leq_c$ is the covariant order $\leq$. A tree is $\top$-*free* if no position is labeled $\top$; the trees over $\{\bot, f\}$ are the $\top$-free trees (Section 2.2), and a $\top$-free tree is finite or regular in the same sense over $\{\bot, f\}$ and over $\{\bot, f, \top\}$. We write $\models_{\mathcal{D}}^{\{\bot,f\}}$ and $\models_{\mathcal{D}}^{\{\bot,f,\top\}}$ for existential entailment over the two signatures, with f covariant; over $\{\bot, f\}$ the systems have no literal $\top \leq u$ and all values are $\top$-free (Definition 2.4). We use the trees $T_0 = \bot$ and $T_{k+1} = f(T_k^n)$ of Section 2.3, now for every $n \geq 0$: the positions of T_k are the words of length at most k, labeled f below length k and $\bot$ at length k, and for $n = 0$ and $k \geq 1$ the tree T_k is $f()$. For a finite tree t, $h(t)$ is the maximal length of a position of t. The *gadget* on a variable v, with a fresh variable y_v, is the pair of literals

$$\mathsf{TF}(v, y_v) = \{v \leq y_v,\ y_v \leq f(y_v^n)\}.$$

Lemma 5.16 (Top-freeness gadget). *Let f be covariant.*

(1) If $y \leq f(y^n)$, then y is $\top$-free, and so is every tree $t \leq y$.

(2) For every $\top$-free tree t, the regular tree $y = \Omega$ satisfies $t \leq y \leq f(y^n)$. If t is finite, then so does the finite tree $y = T_{h(t)+1}$.

Proof. (1) For covariant trees L_t^0 is the set of words with a prefix labeled $\top$ in t, since $v_+ t = t$; so t is $\top$-free iff $L_t^0 = \varnothing$. By Theorem 3.2(2,3), $y \leq f(y^n)$ gives

$$L_y^0 \subseteq L_{f(y^n)}^0 = \bigcup_{i \in \Sigma} i\, L_y^0 = \Sigma L_y^0.$$

If L_y^0 were nonempty, a shortest word in it would have the form iv with $v \in L_y^0$, contradicting minimality. So $L_y^0 = \varnothing$; for $n = 0$ this holds because $\Sigma L_y^0 = \varnothing$. If $t \leq y$, then $L_t^0 \subseteq L_y^0 = \varnothing$ by Theorem 3.2(2). The argument concerns finite words only and applies to infinite trees y.

(2) Every position of Ω is labeled f, so $t(w) \leq \Omega(w)$ at every common position w since t is $\top$-free; and $\Omega = f(\Omega^n)$ has a single subtree. If t is finite, then $K(t)$ consists of words of length less than $h(t) + 1$, so $K(t) \subseteq K(T_{h(t)+1})$ and $t \leq T_{h(t)+1}$ by Lemma 2.3. By the same lemma, $T_{h(t)+1} \leq T_{h(t)+2} = f(T_{h(t)+1}^n)$. □

Theorem 5.17 (Reduction of $\{\bot, f\}$ to $\{\bot, f, \top\}$). *Let f be covariant, and let φ over X and ψ over $X \cup Z$ be systems over $\{\bot, f\}$. Take fresh variables y_v for all $v \in X \cup Z$, put $Y_X = \{y_x : x \in X\}$ and $Y_Z = \{y_z : z \in Z\}$, and let*

$$\varphi^\natural = \varphi \cup \bigcup_{x \in X} \mathsf{TF}(x, y_x), \qquad \psi^\natural = \psi \cup \bigcup_{z \in Z} \mathsf{TF}(z, y_z),$$

with visible variables $X \cup Y_X$ and hidden variables $Z \cup Y_Z$. For every tree class $\mathcal{D}$,

$$\varphi \models_{\mathcal{D}}^{\{\bot, f\}} \exists \bar{z}\, \psi \iff \varphi^\natural \models_{\mathcal{D}}^{\{\bot, f, \top\}} \exists \bar{z}\, \bar{y}\, \psi^\natural,$$

where $\bar{z}\, \bar{y}$ lists $Z \cup Y_Z$.

The gadget is placed on every visible variable $x \in X$, including those that do not occur in φ, and on every hidden variable $z \in Z$; the visible variables of ψ are covered by the gadgets of $\varphi^\natural$.

Proof. For a $\top$-free tree t in $\mathcal{D}$ let $y(t) = T_{h(t)+1}$ if $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$ and $y(t) = \Omega$ otherwise. By Lemma 5.16(2), $y(t)$ is a $\top$-free tree in $\mathcal{D}$ with $t \le y(t) \le f(y(t)^n)$. A $\top$-free assignment satisfies a literal over $\{\bot, f\}$ in the sense of $\{\bot, f\}$ iff it satisfies it in the sense of $\{\bot, f, \top\}$, since both signatures use the same trees, constructor and order on $\top$-free trees (Section 2.2).

Assume the left side, and let ρ be a solution of $\varphi^\natural$ with values in $\mathcal{D}$. By Lemma 5.16(1), the values $\rho(y_x)$ and $\rho(x)$, $x \in X$, are $\top$-free. So $\rho|_X$ is a solution of φ over $\{\bot, f\}$ with values in $\mathcal{D}$, and it has an extension σ to Z, with $\top$-free values in $\mathcal{D}$, that solves ψ. Put $\sigma(y_z) = y(\sigma(z))$ for $z \in Z$. Then ρ together with σ solves $\psi^\natural$, which does not constrain the variables y_x.

Assume the right side, and let ρ be a solution of φ over $\{\bot, f\}$ with values in $\mathcal{D}$. Extend it by $\rho(y_x) = y(\rho(x))$ for $x \in X$; this gives a solution of $\varphi^\natural$ over $\{\bot, f, \top\}$ with values in $\mathcal{D}$. It has an extension σ to $Z \cup Y_Z$, with values in $\mathcal{D}$, that solves $\psi^\natural$. By Lemma 5.16(1), every $\sigma(y_z)$ and every $\sigma(z)$ is $\top$-free, so $\sigma|_Z$ extends ρ to a solution of ψ over $\{\bot, f\}$ with values in $\mathcal{D}$. □

Corollary 5.18 (The signature $\{\bot, f\}$). *Let f be covariant, of any arity $n \ge 0$. In each tree class, existential entailment over $\{\bot, f\}$ reduces to existential entailment over $\{\bot, f, \top\}$ in the same tree class, at the same arity and with f covariant. The reduction doubles the numbers of visible and of hidden variables, so the size parameter s becomes $2s$, and it adds two literals per variable. Consequently, existential entailment over $\{\bot, f\}$ with f covariant is decidable over arbitrary, regular and finite trees whenever the covariant problem over $\{\bot, f, \top\}$ is, and $E_{\mathrm{reg}} = E_{\mathrm{all}}$ holds over $\{\bot, f\}$ as well.*

The last claim follows from Theorems 5.12 and 5.17. The finite witness $T_{h(t)+1}$ can be large, but it is used only in the proof; the reduction introduces variables and literals, not ground trees. Both requirements on the gadget are necessary.

Example 5.19 (Every visible variable, and covariance). The gadget cannot be omitted on a visible variable that does not occur in φ: with $X = \{x\}$, the entailment true $\models_{\mathcal{D}}^{\{\bot, f\}} \exists \bar{z}\, x \le f(z_i)_i$ holds in every tree class (take the z_i to be the children of x, or $\bot$ if $x = \bot$), whereas without a gadget on x its translation fails at $x = \top$. The gadget also needs covariance. If $c(j) = 1$ for some j, let $y = f(t_i)_i$ with $t_i = \top$ if $c(i) = 1$ and $t_i = \bot$ if $c(i) = 0$. Then $y \le_c f(y^n)$, since the covariant arguments require $\bot \le_c y$ and the contravariant ones $y \le_c \top$, but y has a label $\top$; for the arrow, $y = \top \to \bot$ satisfies $y \le_c y \to y$. We do not treat $\{\bot, f\}$ with a contravariant argument. That problem is not obtained from $\{\bot, f, \top\}$ by omitting the literals $\top \le u$: for every variance, the entailment above holds over $\{\bot, f\}$ but fails over $\{\bot, f, \top\}$ at $x = \top$.

5.6 Arities Zero and One

Remark 5.20 (Arity zero). Let $n = 0$. By Remark 2.8, all three tree classes coincide at arity zero, so $E_{\text{fin}} = E_{\text{reg}} = E_{\text{all}}$, and existential entailment is decidable by finite enumeration of the 3^s assignments. The route of Sections 3 and 4 also applies, with no nonempty words, but the enumeration is simpler. Copy literals cannot be eliminated at arity zero (Lemma 2.7 needs $n \geq 1$), which is why the proof of Theorem 5.7 treats $n = 0$ separately.

Remark 5.21 (Arity one). Informally, for $n = 1$ every tree is $f^k(\bot)$, $f^k(\top)$ or Ω, so every tree is regular, while finite and arbitrary trees still differ, as the system $\{x \leq f(x),\ f(x) \leq x\}$ shows. The variance $c(0) = 1$ is allowed, and then mixed polarity already occurs: $\{x \leq f(x)\}$ has no parity assignment (Remark 4.5). Later, the spine automaton of the two-word hole problem runs over $\Sigma = \{0\}$, so the branching case of Section 9 does not occur. The general argument applies without change.

6 Pushdown Summaries

The three conditions of Theorem 4.2 concern infinite sets of atoms: least closures, the sets $\text{Reach}_\chi(a)$, and live atoms. This section describes them with finite automata over the alphabet of directions. The pure moves of a system form a pushdown system whose control states are the signed variables and whose stack is the word of an atom (Section 3.2). They are the moves of the signed system $\chi^\pm$, a covariant system, so the upper and lower bound automata of covariant systems summarize the parts of a pure derivation that pop and push its stack, and every pure derivation splits at its lowest point into one part of each kind (Section 6.1). With them we describe the least closure of the facts (Section 6.2).

Two consequences of the signs have no counterpart for a covariant constructor. First, the lower automata are the upper automata with all signs flipped (Lemma 6.4). Second, an atom can be dead because the closure of its seed contains the opposite of an extension of the seed, the self-collision of Corollary 3.13(c). We show that a self-collision of an atom that is not already dead by support is found by comparing two suffixes of the seed word: it yields a strict *self event*, and deadness is the union of deadness by support, which is regular, and of the self events (Theorem 6.11). Deadness need not be regular (Example 6.12). The section ends with closure membership for a live seed (Lemma 6.14), the heads of pure derivations (Lemma 6.15), and the decision of satisfiability and of condition (I0) (Proposition 6.16). Proofs omitted here are in Appendix D.

Conventions. By Section 5, it suffices to work over $\{\bot, f, \top\}$ and arbitrary trees, for every arity and variance; we do so in this section and in Sections 7 to 10. Throughout, χ is a system over a finite set V of variables, we use the notation of Sections 2 and 3 and write $\bar{P} = \{\bar{p} : p \in P\}$ for $P \subseteq V^\pm$, and for $n = 0$, where ε is the only word, every statement below holds trivially.

6.1 Summaries and the Lowest Point

By Section 3.2, the pure moves of χ are the moves of the copy and constructor literals of $\chi^\pm$, read as a covariant system over $V^\pm$: a literal $p \leq q$ gives $(p, w) \to (q, w)$, a literal $f(p_i)_i \leq q$ gives $(p_i, w) \to (q, iw)$, and a literal $q \leq f(p_i)_i$ gives $(q, iw) \to (p_i, w)$; and the forbidden roots are the atoms (q, ε) for the roots q of the upper constructor literals of $\chi^\pm$. The extreme literals of $\chi^\pm$ give no moves. The *height* of an atom (p, w) is $|w|$. The automata below are the bound automata of a covariant system [Zhao 2026], for an arbitrary arity;[1] we write χ instead of $\chi^\pm$ in subscripts.

The closure D_χ summarizes the pure derivations that return to their starting height without going below it. Besides copies, such a derivation from a_i to b_i can push the letter i through a literal

[1] In the notation of Appendix A.1, D_χ is the closure $D^{\text{cov}}_{\chi^\pm}$, with the copy pairs included in the saturation.

$f(a_j)_j \le u$, continue with a derivation of the same kind from u to v, one level higher, and pop i again through a literal $v \le f(b_j)_j$; this is the decomposition rule below, and transitivity strings such pieces together. The upper automaton of p, after reading γ, holds the signed variables reached from $(p, \gamma\sigma)$ by popping γ, and the lower automaton of q, after reading β, those from which $(q, \beta\sigma)$ is reached by pushing β (Lemma 6.2).

Definition 6.1 (Closure and bound automata). The *closure* D_χ is the least reflexive and transitive relation on $V^\pm$ that contains every pair (p, q) with $p \le q$ in $\chi^\pm$ and satisfies the decomposition rule

$$\frac{f(a_j)_j \le u \in \chi^\pm \qquad D_\chi(u, v) \qquad v \le f(b_j)_j \in \chi^\pm}{D_\chi(a_i, b_i)} \qquad (i \in \Sigma).$$

Put $\uparrow_\chi p = \{r : D_\chi(p, r)\}$ and $\downarrow_\chi q = \{r : D_\chi(r, q)\}$. For $P, Q \subseteq V^\pm$ and $i \in \Sigma$, let

$$\delta_i^\uparrow(P) = \{b : \exists r \in P,\ \exists\, (r \le f(b_j)_j) \in \chi^\pm,\ D_\chi(b_i, b)\},$$
$$\delta_i^\downarrow(Q) = \{a : \exists r \in Q,\ \exists\, (f(a_j)_j \le r) \in \chi^\pm,\ D_\chi(a, a_i)\}.$$

The *upper automaton* of $p \in V^\pm$ has state set $2^{V^\pm}$, start state $\uparrow_\chi p$, and transitions $\delta_i^\uparrow$; its state after a word w is $P_p^\chi(w)$. The *lower automaton* of q has start state $\downarrow_\chi q$ and transitions $\delta_i^\downarrow$; its state after w is $Q_q^\chi(w)$. At cuts we write $P_{p,j}^\chi(w) = P_p^\chi(w[:j])$ and $Q_{q,j}^\chi(w) = Q_q^\chi(w[:j])$.

Both automata are deterministic and complete, and they read their word from left to right: $P_p^\chi(wi) = \delta_i^\uparrow(P_p^\chi(w))$. They have $4^{|V|}$ states each, and D_χ and every transition are computable. Every set $P_p^\chi(w)$ is upward closed under D_χ and every $Q_q^\chi(w)$ is downward closed, the empty set is absorbing in both automata, and $\delta_i^\uparrow$ and $\delta_i^\downarrow$ distribute over unions.

Lemma 6.2 (Pushdown summaries). *For all signed variables a, b, p, q and words β, γ:*

(a) $D_\chi(a, b)$ *iff* $(a, \varepsilon) \to_\chi^* (b, \varepsilon)$;
(b) $b \in P_p^\chi(\gamma)$ *iff* $(p, \gamma) \to_\chi^* (b, \varepsilon)$;
(c) $a \in Q_q^\chi(\beta)$ *iff* $(a, \varepsilon) \to_\chi^* (q, \beta)$.

By Lemma 3.6(1), the three derivations lift to every suffix: they give $(a, \sigma) \to_\chi^* (b, \sigma)$, $(p, \gamma\sigma) \to_\chi^* (b, \sigma)$ and $(a, \sigma) \to_\chi^* (q, \beta\sigma)$ for every word σ. So D_χ summarizes the derivations that return to their starting height, the upper automaton those that pop a word, and the lower automaton those that push one.

Lemma 6.3 (Lowest point). *For signed variables p, q and words w, v, $(p, w) \to_\chi^* (q, v)$ iff there are cuts $i \le |w|$ and $e \le |v|$ with $w[i:] = v[e:]$ and $P_{p,i}^\chi(w) \cap Q_{q,e}^\chi(v) \ne \varnothing$.*

The proof (Appendix D) follows the derivation move by move and keeps a split of the current atom (r', v') into a push head $v'[:e]$ above a suffix $v'[e:] = w[i:]$ of the start word, with a signed variable in $P_{p,i}^\chi(w) \cap Q_{r',e}^\chi(v')$. A copy keeps the split, a push extends the push head, and a pop shortens it or, if it is empty, moves the cut i one letter to the right. At the end, the cut i marks the first atom of least height of the derivation (Figure 7); the signed variable r there may belong to a visible or to a hidden variable, with either sign. In the notation of Figure 7, γ is the *pop head* and β the *push head* of the derivation.

Lemma 6.4 (Duality and sign coherence). *For all signed variables a, b, p, q and words w:*

(1) $D_\chi(a, b)$ *iff* $D_\chi(\bar{b}, \bar{a})$; *hence* $\downarrow_\chi \bar{q} = \overline{\uparrow_\chi q}$;
(2) $Q_{\bar{q}}^\chi(w) = \overline{P_q^\chi(w)}$;

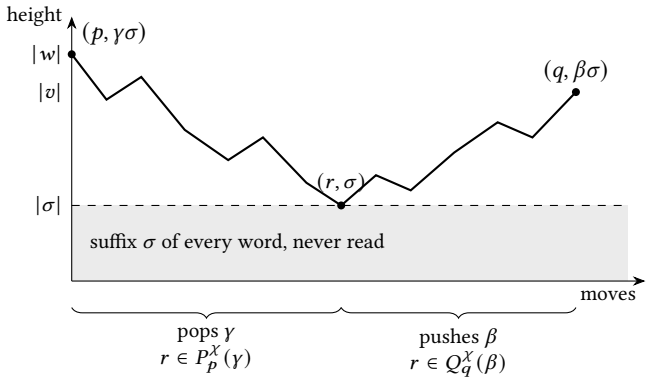


Fig. 7. The lowest point of a pure derivation from (p, w) to (q, v) (Lemma 6.3). No atom on the derivation goes below the least height $|\sigma|$, so the suffix σ is never read: $w = \gamma\sigma$ and $v = \beta\sigma$. The first atom (r, σ) of least height splits the derivation into a part summarized by the upper automaton of p and a part summarized by the lower automaton of q.

(3) $D_\chi(a, b)$ *implies* $\mathrm{sgn}(a) = \mathrm{sgn}(b)$*; every* $r \in P_p^{\chi}(w)$ *has* $\mathrm{sgn}(r) = \mathrm{sgn}(p) \oplus \mathrm{par}(w)$*; and every* $r \in Q_q^{\chi}(w)$ *has* $\mathrm{sgn}(r) = \mathrm{sgn}(q) \oplus \mathrm{par}(w)$.

Items (1) and (2) reverse the derivations of Lemma 6.2 by duality (Lemma 3.10). By item (2), Lemma 6.3 can be read with upper automata only. Item (3) is *sign coherence*, the sign invariant of Lemma 3.6(2) applied to the derivations of Lemma 6.2.

Example 6.5 (Shift closure: the signed system). Let $n = 2$ with f covariant, and let $\psi = \{x \le z,\ z \le y,\ z \le f(z, z)\}$ be the consequent of Example 1.1, now over $\{\bot, f, \top\}$, with z hidden. The copy and constructor literals of its signed system are

$$x^0 \le z^0, \quad z^0 \le y^0, \quad z^0 \le f(z^0, z^0), \quad y^1 \le z^1, \quad z^1 \le x^1, \quad f(z^1, z^1) \le z^1.$$

It has no facts, and its only forbidden root is (z^0, ε): the root of z is not $\top$. Since f is covariant, (x^0, w) says that ρx has a leaf $\top$ at a prefix of w, and (x^1, w) that it has a leaf $\bot$ at a prefix of w. The decomposition rule never applies, so D_ψ is the reflexive and transitive closure of the copies. Hence $\uparrow_\psi x^0 = \{x^0, z^0, y^0\}$ and $P_{x^0}^{\psi}(w) = \{z^0, y^0\}$ for $w \neq \varepsilon$; dually, $\downarrow_\psi x^1 = \{x^1, z^1, y^1\}$ and $Q_{x^1}^{\psi}(w) = \{z^1, y^1\}$ for $w \neq \varepsilon$, as Lemma 6.4(2) predicts. By Lemma 6.3, $(y^1, \sigma) \to_\psi^* (x^1, \beta\sigma)$ for all words β and σ: in every solution of ψ, if y has a leaf $\bot$ at a prefix of σ, then x has a leaf $\bot$ at a prefix of $\beta\sigma$. For trees over $\{\bot, f\}$ this says that every suffix of a constructor position of x is a constructor position of y. Similarly $(x^0, w) \to_\psi^* (z^0, \varepsilon)$ for every w, a pure derivation to the forbidden root: ψ forbids leaves $\top$ in x.

6.2 The Least Closure

Let

$$\mathrm{Fact}_\chi^{\uparrow} = \{r :\ r \in P_e^{\chi}(\gamma) \text{ for some } (e, \varepsilon) \in \mathrm{Fact}_\chi \text{ and some word } \gamma\}$$

be the set of signed variables that a fact reaches at the empty word after popping a word.

Lemma 6.6 (The least closure). *For every signed variable q and word v,*

$$(q, v) \in \mu_\chi \iff Q_{q,e}^{\chi}(v) \cap \mathrm{Fact}_\chi^{\uparrow} \neq \varnothing \text{ for some cut } e \le |v|.$$

In the definition of $\mathrm{Fact}_\chi^{\uparrow}$ *it suffices to consider words γ of length at most* $4^{|V|}$*, so* $\mathrm{Fact}_\chi^{\uparrow}$ *is computable.*

A fact (p, ε) derives (r, σ) for every σ: extend it to $(p, \gamma\sigma)$ and pop γ. So μ_χ contains every atom that a pure derivation pushing a word reaches from such an (r, σ), and Lemmas 3.11 and 6.3 show that there is nothing else. The bound on γ holds because the upper automaton of e has $4^{|V|}$ states.

6.3 Forbidden Roots and the Pop Bound

A pure derivation pops a letter only at the root of an upper constructor literal of $\chi^{\pm}$, and every such root is forbidden at the empty word. This gives the *pop bound*, the counterpart of the cone lemma of a covariant system [Zhao 2026], stated with the following notion and with $\mathsf{Neg}_\chi^{\downarrow} = \{r : D_\chi(r, m)$ for some $(m, \varepsilon) \in \mathsf{Neg}_\chi\}$.

Definition 6.7 (Root deadness). An atom (p, w) is *root-dead* in χ, written $\mathsf{DeadU}_\chi(p, w)$, if $\mathsf{Reach}_\chi(p, w)$ contains a forbidden root of χ.

Lemma 6.8 (Roots and the pop bound). *For every signed variable p and all words w, v:*

(1) if $P_p^\chi(wi) \neq \varnothing$ for some letter i, then some $r \in P_p^\chi(w)$ is the root of an upper constructor literal of $\chi^{\pm}$, $(p, w) \to_\chi^* (r, \varepsilon)$, and $(r, \varepsilon) \in \mathsf{Neg}_\chi$; in particular $\mathsf{DeadU}_\chi(p, w)$;
(2) (pop bound) *if $\mathsf{DeadU}_\chi(p, w[:k])$ fails, then $P_{p,j}^\chi(w) = \varnothing$ for every cut j with $k < j \leq |w|$;*
(3) $\mathsf{DeadU}_\chi(p, w)$ iff $P_p^\chi(wz) \cap \mathsf{Neg}_\chi^{\downarrow} \neq \varnothing$ for some word z;
(4) if $\mathsf{DeadU}_\chi(p, wv)$, then $\mathsf{DeadU}_\chi(p, w)$.

Proof. (1) If $b \in \delta_i^{\uparrow}(P_p^\chi(w))$, some $r \in P_p^\chi(w)$ is the root of an upper constructor literal $r \leq f(b_j)_j$ of $\chi^{\pm}$; then $(r, \varepsilon) \in \mathsf{Neg}_\chi$ by Section 3.2, and $(p, w) \to_\chi^* (r, \varepsilon)$ by Lemma 6.2(b). (2) If $P_{p,j}^\chi(w) \neq \varnothing$ with $j > k$, then $P_{p,k+1}^\chi(w) \neq \varnothing$, because the empty set is absorbing, and (1) applied to $w[:k]$ and the letter $w[k]$ gives $\mathsf{DeadU}_\chi(p, w[:k])$. (3) By Lemma 3.11, $(m, \varepsilon) \in \mathsf{Reach}_\chi(p, w)$ iff $(p, w\alpha) \to_\chi^* (m, \varepsilon)$ for some α; by Lemma 6.3 with the empty word $v = \varepsilon$, whose only cut is 0, this holds iff $P_p^\chi(w\alpha) \cap {\downarrow_\chi} m \neq \varnothing$. (4) By Lemma 3.8(4), $(p, wv) \in \mathsf{Reach}_\chi(p, w)$, so $\mathsf{Reach}_\chi(p, wv) \subseteq \mathsf{Reach}_\chi(p, w)$. □

6.4 Deadness

Corollary 3.13 lists three causes of deadness for a satisfiable system: (a) a forbidden root in $\mathsf{Reach}_\chi(p, w)$, which is $\mathsf{DeadU}_\chi(p, w)$; (b) the opposite of an extension of the seed in μ_χ; and (c) a self-collision, the opposite of an extension of the seed in $\mathsf{Reach}_\chi(p, w)$. The causes may overlap. Causes (a) and (b) together are deadness by support, which automata read. A self-collision of an atom that is not dead by support is read by a strict comparison of the seed word.

Definition 6.9 (Deadness by support, self events). An atom (p, w) is *dead by support* in χ, written $\mathsf{DeadT}_\chi(p, w)$, if $\mathsf{DeadU}_\chi(p, w)$ or $(\bar{p}, w\delta) \in \mu_\chi$ for some word δ. A *self event* of (p, w) in χ is a comparison (i, j) of w (Definition 2.10), that is, a pair of cuts $0 \leq i < j \leq |w|$ with $w[j:] \preceq w[i:]$, such that

$$P_{p,i}^\chi(w) \cap \overline{P_{p,j}^\chi(w)} \neq \varnothing.$$

Self events are strict: $i < j$. We write $\mathsf{SelfEv}_\chi(p, w, i, j)$ if (i, j) is a self event of (p, w).

Self events have a counterpart in the plain-entailment proof, whose readers already compare a root with its own signed copy [Zhao 2026]. Here both cuts belong to the word of one seed: the state of the upper automaton of p at the cut i and the flipped state at the cut j share a signed variable, which serves as the lowest point of a pure derivation from an extension of (p, w) to the opposite of an extension of (p, w) (Theorem 6.11).

Both kinds of deadness by support are read by the upper automaton of p together with one bit. Put

$$\mathsf{DeadU}^{\varepsilon}_{\chi} = \{r \in V^{\pm} : \mathsf{DeadU}_{\chi}(r, \varepsilon)\}, \qquad \mathsf{DeadT}^{\varepsilon}_{\chi} = \{r \in V^{\pm} : \mathsf{DeadT}_{\chi}(r, \varepsilon)\}.$$

Lemma 6.10 (Deadness from states). *For every signed variable p and all words w, v:*

(1) $\mathsf{DeadU}_{\chi}(p, w)$ *iff* $P^{\chi}_{p}(w) \cap \mathsf{DeadU}^{\varepsilon}_{\chi} \neq \varnothing$;

(2) $(\bar{p}, w) \in \mu_{\chi}$ *iff* $\overline{P^{\chi}_{p,e}(w)} \cap \mathsf{Fact}^{\uparrow}_{\chi} \neq \varnothing$ *for some cut* $e \leq |w|$;

(3) $\mathsf{DeadT}_{\chi}(p, w)$ *iff* $(\bar{p}, w) \in \mu_{\chi}$ *or* $P^{\chi}_{p}(w) \cap \mathsf{DeadT}^{\varepsilon}_{\chi} \neq \varnothing$;

(4) if $\mathsf{DeadT}_{\chi}(p, wv)$, *then* $\mathsf{DeadT}_{\chi}(p, w)$;

(5) the sets $\mathsf{DeadU}^{\varepsilon}_{\chi}$ *and* $\mathsf{DeadT}^{\varepsilon}_{\chi}$ *are computable.*

So the words w with $\mathsf{DeadU}_{\chi}(p, w)$, and those with $\mathsf{DeadT}_{\chi}(p, w)$, form regular languages that are closed under prefixes: the first is read by the upper automaton of p, and the second by that automaton together with a bit that records whether some state P read so far has $\bar{P} \cap \mathsf{Fact}^{\uparrow}_{\chi} \neq \varnothing$. Self events, in contrast, compare two cuts of one word. They are what remains of deadness.

Theorem 6.11 (Deadness). *Let χ be satisfiable. An atom (p, w) is dead iff* $\mathsf{DeadT}_{\chi}(p, w)$ *or (p, w) has a self event in χ. More precisely, every self event of (p, w) yields a self-collision of (p, w), and if (p, w) has a self-collision but* $\mathsf{DeadT}_{\chi}(p, w)$ *fails, then (p, w) has a self event.*

By Corollary 3.13, it remains to treat a self-collision, a pure derivation from an extension $(p, w\alpha)$ to an opposite atom $(\bar{p}, w\delta)$, of an atom (p, w) that is not dead by support. Split it at its lowest point. If its pop head or its push head is longer than w, then the upper automaton of p is nonempty after some proper extension of w (for the push head, by Lemma 6.4(2)), and the pop bound (Lemma 6.8) gives $\mathsf{DeadU}_{\chi}(p, w)$. Otherwise both heads end at cuts i, j of w, the suffixes $w[i{:}]$ and $w[j{:}]$ are comparable in the prefix order, and by Lemma 6.4(2) the lowest point lies in $P^{\chi}_{p,i}(w) \cap \overline{P^{\chi}_{p,j}(w)}$. Sign coherence excludes $i = j$, and the condition is symmetric in i and j, so the shorter suffix is a prefix of the longer one and (i, j) or (j, i) is a strict self event. Conversely, every self event yields a self-collision. The proof is in Appendix D.

Example 6.12 (A nonregular dead language). Let $\varphi = \{x \leq x \to x\}$ for the arrow, as in Examples 3.7 and 3.9. The copy and constructor literals of $\varphi^{\pm}$ are $x^0 \leq f(x^1, x^0)$ and $f(x^0, x^1) \leq x^1$, there are no facts, so $\mu_{\varphi} = \varnothing$, and the only forbidden root is (x^0, ε). The closure D_{φ} is the identity, and

$$P^{\varphi}_{x^0}(1^k) = \{x^0\}, \qquad P^{\varphi}_{x^0}(1^k 0) = \{x^1\}, \qquad P^{\varphi}_{x^0}(w) = \varnothing \text{ if } 1^k 0 \text{ is a proper prefix of } w.$$

Hence $\mathsf{DeadT}^{\varepsilon}_{\varphi} = \{x^0\}$, and $\mathsf{DeadT}_{\varphi}(x^0, w)$ holds iff $w \in 1^*$. The guard of a self event (i, j) requires $P^{\varphi}_{x^0,i}(w) = \{x^0\}$ and $P^{\varphi}_{x^0,j}(w) = \{x^1\}$, so $w = 1^k 0v$, $i \leq k$, and $j = k + 1$; the comparison requires $v \preceq 1^m 0v$ with $m = k - i$, that is, $v \preceq (1^m 0)^{\omega}$ (Lemma 2.1). By Theorem 6.11, the dead words of x^0 are

$$1^* \ \cup \ \{1^k 0v : \ v \preceq (1^m 0)^{\omega} \text{ for some } m \leq k\}.$$

This language is not regular: its intersection with 1^*01^*0 is $\{1^k 01^m 0 : m \leq k\}$. For example, $(x^0, 0)$ is dead by the self event $(0, 1)$ of the word 0, although $\mathsf{DeadT}_{\varphi}(x^0, 0)$ fails. This is the self event that the proof of Theorem 6.11 extracts from the self-collision $(x^0, 00) \to (x^1, 0)$ of Example 3.9: its lowest point $(x^1, 0)$ lies at the cut 1 of 00 and the cut 0 of 0. The atom $(x^0, 01)$ is live (Example 3.9).

Remark 6.13 (No self events without mixed parity). Informally, if χ has a parity assignment π (Remark 4.5), then the sets $P^{\chi}_{p,i}(w)$ consist of signed variables r with the same bit $\mathrm{sgn}(r) \oplus \pi(r)$, where $\pi(x^{\pm}) = \pi(x)$, and no such set meets a flipped one. Self events then do not arise, and deadness in a satisfiable χ is deadness by support; this is the automaton form of Remark 4.5, and it includes covariant systems, with $\pi = 0$.

6.5 Closure Membership

Lemma 6.14 (Closure membership). *Let (p, w) be live in χ, and let (q, v) be an atom. Then $(q, v) \in \mathrm{Cl}_\chi(p, w)$ iff $(q, v) \in \mu_\chi$ or there are cuts $i \le |w|$ and $e \le |v|$ with*

$$w[i{:}] \preceq v[e{:}] \qquad \text{and} \qquad P^\chi_{p,i}(w) \cap Q^\chi_{q,e}(v) \neq \varnothing.$$

Proof. By Lemmas 3.11 and 6.3, $(q, v) \in \mathrm{Reach}_\chi(p, w)$ iff there are a word α and cuts $i \le |w\alpha|$, $e \le |v|$ with $(w\alpha)[i{:}] = v[e{:}]$ and $P^\chi_p((w\alpha)[{:}i]) \cap Q^\chi_{q,e}(v) \neq \varnothing$. If $i > |w|$, then $P^\chi_p(w\alpha') \neq \varnothing$ for the nonempty prefix α' of α of length $i - |w|$, and Lemma 6.8(2) with $k = |w|$ gives $\mathrm{DeadU}_\chi(p, w)$, so $\mathrm{Cl}_\chi(p, w)$ contains a forbidden root, contradicting liveness by Lemma 3.8(2). Hence $i \le |w|$, and the equation reads $w[i{:}]\,\alpha = v[e{:}]$, which gives $w[i{:}] \preceq v[e{:}]$. Conversely, if $w[i{:}]\,\alpha = v[e{:}]$ and the intersection is nonempty, Lemma 6.3 gives $(p, w\alpha) \to^*_\chi (q, v)$, and $(p, w\alpha) \in \mathrm{Reach}_\chi(p, w)$. □

The two cuts belong to different words, so there is no restriction $i < e$. The prefix relation points from the seed to the consequence: the seed word may be continued to the word of the consequence, because extensions are rules.

6.6 Heads of Derivations

The pop head of a pure derivation passes only through root-dead atoms. Immediately before a pop, the signed variable of the current atom is the root of an upper constructor literal, whose empty-word atom is forbidden.

Lemma 6.15 (Head prefixes). *If $P^\chi_p(\gamma) \neq \varnothing$, then $\mathrm{DeadU}_\chi(p, \gamma[{:}i])$ for every $i < |\gamma|$.*

Proof. Since the empty set is absorbing, $P^\chi_p(\gamma[{:}i+1]) \neq \varnothing$, and Lemma 6.8(1) applies to $\gamma[{:}i]$ and the letter $\gamma[i]$. □

The lemma uses neither satisfiability nor any condition relating two systems. Applied to ψ, it says that every proper prefix γ' of the pop head γ of a pure derivation of ψ from a visible atom $(p, \gamma\sigma)$ has $\mathrm{DeadU}_\psi(p, \gamma')$; under condition (U), the atom (p, γ') is then dead in φ (Definitions 4.1 and 6.7). Section 8 uses this to end every pop head of ψ at or before the first rejected cut of the seed (Lemmas 8.4 and 8.6).

6.7 Satisfiability and Condition (I0)

Proposition 6.16 (Satisfiability and (I0)).

(1) A system χ is satisfiable iff $\downarrow_\chi m \cap \mathrm{Fact}^\uparrow_\chi = \varnothing$ for every forbidden root (m, ε) and there are no variable x and word w with $(x^+, w), (x^-, w) \in \mu_\chi$. If such x and w exist, then they exist with $|w| < 2^{4|V|+2}$.

(2) Let φ be a system over X and ψ a system over $X \cup Z$. Condition (I0)*, $\mu_\psi|_X \subseteq \mu_\varphi$, fails iff some $q \in X^\pm$ and some word v have $(q, v) \in \mu_\psi$ and $(q, v) \notin \mu_\varphi$, and then such q and v exist with $|v| < 2^{2|X|+2|X\cup Z|+2}$.*

Satisfiability and condition (I0) *are decidable, by searches over words of these lengths.*

Proof. (1) By Lemma 3.8(2), χ is satisfiable iff μ_χ is consistent, that is, contains no forbidden root and no exclusion pair. By Lemma 6.6 with $v = \varepsilon$, $(m, \varepsilon) \in \mu_\chi$ iff $\downarrow_\chi m \cap \mathrm{Fact}^\uparrow_\chi \neq \varnothing$. (2) is the definition of (I0). For the bounds, Lemma 6.6 reads membership in μ_χ at q by the lower automaton of q together with a bit that records whether some state read so far met $\mathrm{Fact}^\uparrow_\chi$; this automaton has $2^{2|V|+1}$ states. The second condition of (1) is reachability, for some $x \in V$, of a state with both bits set in the product of these automata for x^+ and x^-; and the condition of (2) is reachability, for some $q \in X^\pm$, of a state whose ψ-bit is set and whose φ-bit is not, in the product of the automata for q

in ψ and in φ. A shortest word that reaches a state visits no state twice, so it is shorter than the number of states of the product. The set $\mathsf{Fact}^{\uparrow}_{\chi}$ is computable by Lemma 6.6. □

Condition (U) needs more than reachability, because liveness involves self events; it is the subject of Section 7.

7 Condition (U)

From now on, φ is a system over the visible variables X and ψ a system over $X \cup Z$, as in Section 4. Condition (U) of Definition 4.1 asks that $\mathrm{Reach}_{\psi}(a)$ contain no forbidden root of ψ for every visible atom a that is live in φ. In the terms of Definition 6.7, this reads as follows.

Lemma 7.1 (Condition (U)). *Condition* (U) *holds iff no* $p \in X^{\pm}$ *and no word* w *satisfy* $\mathsf{DeadU}_{\psi}(p, w)$ *while* (p, w) *is live in* φ.

The words w with $\mathsf{DeadU}_{\psi}(p, w)$ form a regular language (Lemma 6.10), but the words w with (p, w) live in φ need not (Example 6.12), so (U) is not an inclusion of regular languages. When φ is satisfiable, however, Theorem 6.11 identifies liveness with the absence of DeadT_{φ}, which is regular, and of self events, which are comparisons of a single word whose two cuts are tested by a pair of automaton states. A failure of (U) is therefore a hole of a reader in the sense of Zhao [2026] (Definitions 2.11 and 2.12), and the reader has the rejected-path property (Definition 2.13). The hole search of that paper (Corollary 2.21) then decides (U).

7.1 The Source Automaton

One automaton reads the word of a visible seed for both systems. The two-word instances of Section 8 use it again, on the seed word.

Definition 7.2 (Source automaton). Let $p \in X^{\pm}$. The *source automaton* $\mathcal{B}_p$ is the deterministic automaton over Σ whose states are the triples $\zeta = (P, b, P')$ with $P \subseteq X^{\pm}$, $b \in \{0, 1\}$, and $P' \subseteq (X \cup Z)^{\pm}$, with start state $\big(\uparrow_{\varphi} p,\ [\overline{\uparrow_{\varphi} p} \cap \mathsf{Fact}^{\uparrow}_{\varphi} \neq \varnothing],\ \uparrow_{\psi} p\big)$ and transitions

$$\delta\big((P, b, P'), i\big) = \big(\delta^{\uparrow}_{i}(P),\ b \vee [\overline{\delta^{\uparrow}_{i}(P)} \cap \mathsf{Fact}^{\uparrow}_{\varphi} \neq \varnothing],\ \delta^{\uparrow}_{i}(P')\big),$$

where the first component uses the transitions of the upper automata of φ and the third those of ψ, and $[\,\cdot\,]$ is 1 for a true condition and 0 otherwise. For a state $\zeta = (P, b, P')$ put

$$\mathrm{dt}(\zeta) \iff b = 1 \text{ or } P \cap \mathsf{DeadT}^{\varepsilon}_{\varphi} \neq \varnothing, \qquad \mathrm{du}(\zeta) \iff P' \cap \mathsf{DeadU}^{\varepsilon}_{\psi} \neq \varnothing,$$

and let the *self guard* G_{s} consist of the pairs of states $((P_1, b_1, P'_1), (P_2, b_2, P'_2))$ with $P_1 \cap \overline{P_2} \neq \varnothing$.

Lemma 7.3 (Runs of the source automaton). *Let* $p \in X^{\pm}$, *let* w *be a word, and let* ζ_j *be the state of* $\mathcal{B}_p$ *after* $w[:j]$. *Then:*

(1) $\zeta_j = \big(P^{\varphi}_{p,j}(w),\ [(\bar{p}, w[:j]) \in \mu_{\varphi}],\ P^{\psi}_{p,j}(w)\big)$;
(2) $\mathrm{dt}(\zeta_j)$ *iff* $\mathsf{DeadT}_{\varphi}(p, w[:j])$, *and* $\mathrm{du}(\zeta_j)$ *iff* $\mathsf{DeadU}_{\psi}(p, w[:j])$; *if* $\mathrm{dt}(\zeta_j)$ *fails, it fails at every later cut, and if* $\mathrm{du}(\zeta_j)$ *holds, it holds at every earlier cut;*
(3) for $i < j \leq |w|$, *the pair* (i, j) *is a self event of* (p, w) *in* φ *iff it is a comparison of* w *and* $(\zeta_i, \zeta_j) \in G_{\mathrm{s}}$.

The automaton $\mathcal{B}_p$ *has* $N_{\mathrm{U}} = 2^{4|X|+2|Z|+1}$ *states.*

Item (1) holds by induction on j, using Lemma 6.10(2) for the bit; item (2) is Lemma 6.10(1,3) together with Lemma 6.8(4) and Lemma 6.10(4); and item (3) is Definition 6.9. The sets $\mathsf{Fact}^{\uparrow}_{\varphi}$, $\mathsf{DeadT}^{\varepsilon}_{\varphi}$, and $\mathsf{DeadU}^{\varepsilon}_{\psi}$ are computable (Lemmas 6.6 and 6.10), so $\mathcal{B}_p$, dt, du, and G_{s} are computable.

7.2 The Unit Reader

Definition 7.4 (Unit reader). For $p \in X^{\pm}$, the *unit reader* is the reader $\mathcal{R}_{\mathrm{U}}(p) = (\mathcal{B}_p, G_{\mathrm{s}}, \mathrm{Rej}_{\mathrm{U}})$ with the source automaton, the self guard as its admission relation, and the rejected states

$$\mathrm{Rej}_{\mathrm{U}} = \{\zeta : \ \mathrm{dt}(\zeta) \text{ fails and } \mathrm{du}(\zeta) \text{ holds}\}.$$

Lemma 7.5 (Holes of the unit reader). *Let φ be satisfiable and $p \in X^{\pm}$. A word w is a hole of $\mathcal{R}_{\mathrm{U}}(p)$ iff (p, w) is live in φ and $\mathrm{DeadU}_{\psi}(p, w)$. Consequently,* (U) *holds iff none of the $2|X|$ unit readers has a hole.*

Proof. By Lemma 7.3, w is rejected iff $\mathrm{DeadT}_{\varphi}(p, w)$ fails and $\mathrm{DeadU}_{\psi}(p, w)$ holds, and a comparison of w is admitted iff it is a self event of (p, w) in φ. So w is a hole iff $\mathrm{DeadU}_{\psi}(p, w)$ holds and (p, w) is neither dead by support nor has a self event in φ, that is, by Theorem 6.11, iff $\mathrm{DeadU}_{\psi}(p, w)$ holds and (p, w) is live in φ. The second claim follows by Lemma 7.1. □

Lemma 7.6 (Rejected paths). *Every unit reader $\mathcal{R}_{\mathrm{U}}(p)$ has the rejected-path property: it satisfies betweenness and the admission cone. No satisfiability assumption is needed.*

Proof. *Betweenness.* Let $w[:a]$ and $w[:b]$ be rejected, with $a < k < b$. Since dt fails at a, it fails at k, and since du holds at b, it holds at k (Lemma 7.3(2)). So $w[:k]$ is rejected.

Admission cone. Let w be rejected, with first rejected cut J, and let (i, j) be an admitted comparison of w. The guard $P^{\varphi}_{p,i}(w) \cap \overline{P^{\varphi}_{p,j}(w)} \neq \varnothing$ requires $P^{\varphi}_{p,j}(w) \neq \varnothing$. At the cut J, dt fails, so $\mathrm{DeadT}_{\varphi}(p, w[:J])$ fails, and in particular $\mathrm{DeadU}_{\varphi}(p, w[:J])$ fails. The pop bound (Lemma 6.8(2)) gives $P^{\varphi}_{p,j'}(w) = \varnothing$ for every $j' > J$. Hence $j \leq J$. □

The proof gives more than the admission cone: no admission, comparison or not, has its second cut after J.

Example 7.7 (Shift closure: condition (U)). Let $\psi = \{x \leq z,\ z \leq y,\ z \leq f(z, z)\}$ as in Example 6.5, with f binary and covariant, and let $\varphi_1 = \{x \leq y\}$. Over $\{\bot, f, \top\}$, condition (U) fails. Indeed, $\mathrm{DeadU}^{\varepsilon}_{\psi} = \{x^0, z^0\}$ and $z^0 \in P^{\psi}_{x^0}(w)$ for every w, so $\mathrm{DeadU}_{\psi}(x^0, w)$ holds for every word w. On the other side, φ_1 has neither facts nor forbidden roots, so $\mathrm{DeadT}_{\varphi_1}$ never holds, and it has no upper constructor literal, so $P^{\varphi_1}_{x^0}(w) = \varnothing$ for $w \neq \varepsilon$ and no comparison is admitted. Every word is a hole of $\mathcal{R}_{\mathrm{U}}(x^0)$, the empty word included. Its countermodel is the least model $\mathrm{Cl}_{\varphi_1}(x^0, \varepsilon)$, the solution $x = y = \top$: no z satisfies $\top \leq z \leq f(z, z)$.

Over $\{\bot, f\}$, the variables cannot take the value $\top$. Within $\{\bot, f, \top\}$, this can be expressed by a variable g with $y \leq g$ and $g \leq f(g, g)$, which allow exactly the trees y without leaves $\top$; this is the gadget of Lemma 5.16. The reduction of Theorem 5.17 puts such a gadget on every variable, but for this example one on y suffices: since $x \leq y$, it also keeps x free of $\top$, and $z \leq f(z, z)$ already keeps z free of $\top$. So let $\varphi_1' = \{x \leq y,\ y \leq g,\ g \leq f(g, g)\}$, with g visible. Its only forbidden root is (g^0, ε), and $x^0 \to y^0 \to g^0$ by copies, after which g^0 pops every word. So $\mathrm{DeadT}_{\varphi_1'}(x^0, w)$ holds for every w, and $\mathcal{R}_{\mathrm{U}}(x^0)$ rejects no word. For the other signed variables $p \in \{x^1, y^0, y^1, g^0, g^1\}$, the set $P^{\psi}_{p}(w)$ never meets $\mathrm{DeadU}^{\varepsilon}_{\psi}$, so $\mathcal{R}_{\mathrm{U}}(p)$ rejects no word either. Thus (U) holds for φ_1'; Example 8.11 shows that the entailment fails by condition (II).

Example 7.8 (Admitted self events are needed). Let $\varphi = \{x \leq x \to x\}$ as in Example 6.12, and let $\psi = \{x \leq a \to x,\ c \to y \leq a\}$ with a, c hidden. The antecedent is that of Example 4.4, but the consequent is new. For $x \leq z \to z$, (U) holds because its only root-dead atom, (x^0, ε), is a forbidden root of φ as well; this consequent is chosen so that (U) holds only because of admitted self events. The signed system of ψ has the upper constructor literals $x^0 \leq f(a^1, x^0)$ and $a^1 \leq f(c^0, y^1)$ and

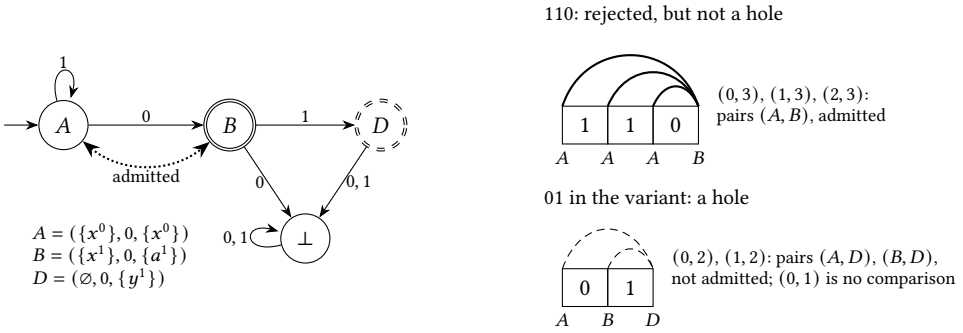


Fig. 8. The unit reader $\mathcal{R}_{\mathrm{U}}(x^0)$ of Example 7.8, after merging the states that cannot occur on a rejected word into a sink $\bot$ (Lemma 13.2). Rejected states are doubly circled; the only admitted pairs are (A, B) and (B, A), the self guard of the antecedent $x \le x \to x$. Every rejected word 1^k0, such as 110, ends in B and has the terminal comparisons $(i, k+1)$ with the admitted pair (A, B): it is rescued, because $(x^0, 1^k0)$ is dead in φ by a self event. In the example itself the state D, dashed, is not rejected and would be merged into $\bot$. In a variant in which y is hidden and ψ also contains $d \to e \le y$, with d, e hidden, the atom (y^1, ε) is a forbidden root, D is rejected, and the word 01 is a hole: none of its comparisons is admitted, so $(x^0, 01)$ is live in φ and root-dead in ψ, and (U) fails.

the lower ones $f(a^0, x^1) \le x^1$ and $f(c^1, y^0) \le a^0$; its closure is the identity, and $\mathrm{DeadU}^{\varepsilon}_{\psi} = \{x^0, a^1\}$. Reading from x^0,

$$P^{\psi}_{x^0}(1^k) = \{x^0\}, \quad P^{\psi}_{x^0}(1^k0) = \{a^1\}, \quad P^{\psi}_{x^0}(1^k00) = \{c^0\}, \quad P^{\psi}_{x^0}(1^k01) = \{y^1\},$$

and the empty set after longer words. So $\mathrm{DeadU}_{\psi}(x^0, w)$ holds iff $w \in 1^* \cup 1^*0$, and $\mathcal{R}_{\mathrm{U}}(x^0)$ rejects exactly the words 1^k0, which are not dead by support in φ. Each of them has the admitted comparison $(0, k+1)$: $P^{\varphi}_{x^0}(\varepsilon) = \{x^0\}$ meets $\overline{P^{\varphi}_{x^0}(1^k0)} = \{x^0\}$. So $\mathcal{R}_{\mathrm{U}}(x^0)$ has no hole. The other unit readers reject no word, because x^1, y^0, and y^1 have no root-dead atoms in ψ. Hence (U) holds. Without the admitted self events, the word 0 would be a hole: $(x^0, 0)$ is dead in φ only through its self event (Example 6.12).

Figure 8 draws this reader. The admission relation acts as a rescue: a rejected word, a candidate failure of (U), is discarded as soon as two of its cuts form a comparison with an admitted pair of states, because the seed is then dead in φ. A small change of the consequent, also shown there, leaves a rejected word that no comparison rescues, a hole.

7.3 Deciding Condition (U)

Zhao [2026] decides the hole problem for every reader with the rejected-path property (Corollary 2.21): a reader with N states over a finite alphabet that has this property has a hole iff it has a hole of length at most $\mathrm{Len}(N, d_N)$, where $d_N = 2N(4N-1) - 1$. The proof selects a hole with rejected tail at most $4N - 2$ (Theorem 2.15), bounds its depth (Theorem 2.19), and shortens it (Theorem 2.20). Applied to the unit readers, with their N_{U} states, this decides (U).

Theorem 7.9 (Condition (U) is decidable). *Let φ be satisfiable, and let $N_{\mathrm{U}} = 2^{4|X|+2|Z|+1}$ be the number of states of the source automaton. Condition* (U) *fails iff there are $p \in X^{\pm}$ and a word w with $|w| \le \mathrm{Len}(N_{\mathrm{U}}, d_{N_{\mathrm{U}}})$ such that (p, w) is live in φ and $\mathrm{DeadU}_{\psi}(p, w)$. Hence* (U) *is decidable. If $X \neq \emptyset$, then, with $v = |X|$, $s = |X| + |Z|$ and $E_3(x) = 2^{2^{2^x}}$,*

$$\mathrm{Len}(N_{\mathrm{U}}, d_{N_{\mathrm{U}}}) \le E_3(10v + 10s + 17) \le E_3(20s + 17).$$

The bound depends neither on the arity nor on the variance.

Proof. By Lemma 7.5, (U) fails iff some unit reader $\mathcal{R}_{\mathrm{U}}(p)$ has a hole. Each $\mathcal{R}_{\mathrm{U}}(p)$ has N_{U} states, the finite alphabet Σ, and the rejected-path property (Lemma 7.6), so by Corollary 2.21 it has a hole iff it has one of length at most $\mathsf{Len}(N_{\mathrm{U}}, d_{N_{\mathrm{U}}})$, and by Lemma 7.5 such a hole is a word as stated. If X is empty, there is no unit reader and (U) holds. For the decision, enumerate the $2|X|$ signed variables and the finitely many words up to this length, and test each pair by Lemma 7.3: the states of $\mathcal{B}_p$ along w are computable, and so are rejection and the admission of each of the finitely many comparisons of w.

For the estimate, let $v \geq 1$. Then $N_{\mathrm{U}} = 2^{2v+2s+1} \geq 8$, so $N_{\mathrm{U}} + 1 \leq \frac{9}{8} N_{\mathrm{U}}$, and $(\frac{9}{8})^5 < 2$. By Corollary 2.21, $\mathsf{Len}(N_{\mathrm{U}}, d_{N_{\mathrm{U}}}) \leq 2^{2^{1280(N_{\mathrm{U}}+1)^5}}$, and $1280(N_{\mathrm{U}} + 1)^5 < 2^{12} N_{\mathrm{U}}^5 = 2^{10v+10s+17}$; finally $v \leq s$. □

The bound of Theorem 7.9 is not the smallest one. A smaller reader gives a sharper bound (Proposition 13.4); it serves only the complexity analysis and is therefore placed in the part that is not mechanized (Section 13.1).

Informally, a different route would reduce (U) to plain entailment: the words w with $\mathsf{DeadU}_{\psi}(p, w)$ form a regular language W_p, and (U) asks that in every solution of φ the leaf cone of p avoid W_p, a comparison with a regular ground tree determined by W_p. We do not pursue this route; the unit reader gives an explicit witness bound without additional variables. Section 8 turns to condition (II).

8 Two-Word Holes with Self Events

Condition (II) fails when a visible atom that is live in φ has a visible ψ-consequence outside its closure in φ; by Lemma 4.3(2), the consequence can be taken to be derived by a pure derivation. By the lowest point (Lemma 6.3), the seed and the consequence then share a suffix, which follows a pop head in the seed and a push head in the consequence. Liveness of the seed is read by the source automaton of Section 7 together with the comparisons of the seed word that are self events (Theorem 6.11); membership in the closure is read by the lower automata of φ together with comparisons between the two words (Lemma 6.14); and the derivation is read by the bound automata of ψ. This section abstracts these tests into a hole problem for pairs of words read by four automata (Section 8.1), extracts its instances from φ and ψ (Section 8.2), and proves that, once satisfiability and (U) are known, (II) holds iff no extracted instance has a hole (Section 8.3). The extracted instances read the seed with a rejection test that already excludes the root-dead words of ψ. With this *enrichment* they have forced target heads by construction, the property on which Section 9 relies. Proofs omitted here are in Appendix D.

8.1 The Hole Problem

Definition 8.1 (Two-word hole problem with self events). An *instance* of the *two-word hole problem with self events* is a tuple

$$I = (A, \mathsf{Bad}, B, \mathsf{Rej}, G, G_{\mathrm{s}}, A', B', H)$$

that consists of complete deterministic automata A, B, A', B' over Σ, with state sets S_A, S_B, $S_{A'}$, $S_{B'}$ and a start state 0 each, together with sets $\mathsf{Bad} \subseteq S_A$, $\mathsf{Rej} \subseteq S_B$, $G \subseteq S_A \times S_B$, $G_{\mathrm{s}} \subseteq S_B \times S_B$, and $H \subseteq S_{A'} \times S_{B'}$. We call A the *spine automaton*, B the *target automaton*, A' and B' the *head automata*, G the *guard*, G_{s} the *self guard*, and H the *head relation*, and we put $N_A = |S_A|$, and similarly N_B, $N_{A'}$, $N_{B'}$. For words u and t, let $A_s(u)$ be the state of A after $u[:s]$ and $B_e(t)$ the state of B after $t[:e]$, define $A'_s(u)$ and $B'_e(t)$ in the same way, and write $M(w)$ for the state of an automaton M

after the whole word w. If some cut e of t has $B_e(t) \in \mathrm{Rej}$, the least one is the *first rejected cut* $J(t)$. A pair of words (u, t) is a *hole* of I iff

(a) $A_s(u) \notin \mathrm{Bad}$ for every cut $s \le |u|$;
(b) $J(t)$ is defined;
(c) there are no cuts $s \le |u|$ and $e \le J(t)$ with $t[e{:}] \preceq u[s{:}]$ and $(A_s(u), B_e(t)) \in G$;
(d) there are cuts $s \le |u|$ and $e \le |t|$ with $u[s{:}] = t[e{:}]$ and $(A'_s(u), B'_e(t)) \in H$;
(e) there are no cuts $i < j \le J(t)$ with $t[j{:}] \preceq t[i{:}]$ and $(B_i(t), B_j(t)) \in G_s$.

The *hole problem* asks whether a given instance has a hole.

We call u the *spine* and t the *target*. Unlike rejection in Definition 2.12, condition (b) requires a rejected prefix of t, not necessarily a rejected final state; for extracted instances these conditions coincide, as shown in Section 8.2. A state of A outside Bad is *safe*; a word is *safe from* a state r if A visits only safe states while reading it from r, including r and the last state; and the spine u is *safe* if it is safe from 0, which is condition (a). The other conditions are stated in terms of the following pairs of cuts.

Definition 8.2 (Comparisons and events). Let u and t be words. A *cross comparison* of (u, t) is a pair of cuts (s, e) with $0 \le s \le |u|$, $0 \le e \le |t|$, and $t[e{:}] \preceq u[s{:}]$. Its *offset* is $\Delta(s, e) = (|u|-s)-(|t|-e) \ge 0$, its *overhang* $o(s, e)$ is the suffix of u of length $\Delta(s, e)$, so that $u[s{:}] = t[e{:}]\, o(s, e)$, and it is *aligned* if $\Delta(s, e) = 0$. A *self comparison* of t is a comparison (i, j) of t (Definition 2.10): $i < j \le |t|$ and $t[j{:}] \preceq t[i{:}]$; its overhang is the word o of length $j - i$ with $t[i{:}] = t[j{:}]\, o$. For an instance I and a target t with $J(t)$ defined:

- an *event* of (u, t) is a cross comparison (s, e) with $e \le J(t)$ and $(A_s(u), B_e(t)) \in G$;
- a *self event* of t is a self comparison (i, j) of t with $j \le J(t)$ and $(B_i(t), B_j(t)) \in G_s$;
- a *head event* of (u, t) is an aligned cross comparison (s, e) with $(A'_s(u), B'_e(t)) \in H$.

Thus a hole is a pair with a safe spine, a target with $J(t)$ defined, no event, no self event, and a head event. A self comparison (i, j) of t is the cross comparison (i, j) of the pair (t, t), with offset $j - i > 0$; in particular, it is never aligned. Events relate two words; self events relate two cuts of the target. An instance with $G_s = \varnothing$ has no self events, and its holes are those of the suffix-sharing two-word problem of a covariant constructor; Remark 8.10 explains when the extracted instances have this form.

Head form. A head event (s, e) splits the two words as

$$u = \beta\sigma, \qquad t = \gamma\sigma, \qquad \beta = u[{:}s], \quad \gamma = t[{:}e], \quad \sigma = u[s{:}] = t[e{:}],$$

with $(A'(\beta), B'(\gamma)) \in H$. We call β the *spine head*, γ the *target head*, and σ the *common suffix*. In the extracted instances of Section 8.2, the target head is the pop head of Lemma 6.3 and the spine head its push head. Figure 9 draws the head form together with the two kinds of pairs that a hole must avoid. A cross comparison and a self comparison look alike: in both, a suffix of the target is laid at a cut of a word, the spine or the target itself, and agrees with that word letter by letter until the suffix ends.

Definition 8.3 (Forced target heads). An instance I has *forced target heads*, condition *(FH)*, if for every $(r, r') \in H$ with r reachable in A' and every word γ with $B'(\gamma) = r'$, no proper prefix of γ is rejected: $B_i(\gamma) \notin \mathrm{Rej}$ for all $i < |\gamma|$.

Lemma 8.4 (Heads end before the first rejected cut). *Let I satisfy (FH), and let (s, e) be a head event of (u, t), where $J(t)$ is defined. Then $e \le J(t)$: the target head ends at or before the first rejected cut, and $t[J(t){:}]$ is a suffix of the common suffix.*

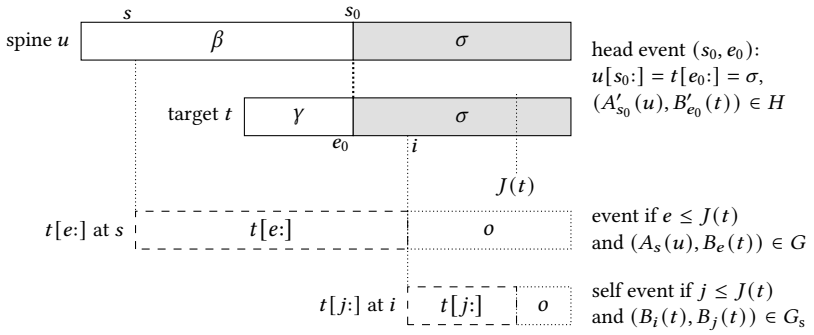


Fig. 9. The pairs of cuts that decide whether (u, t) is a hole (Definitions 8.1 and 8.2). Top: the spine and the target, aligned at their ends; a head event (s_0, e_0) splits them into the heads β and γ and the common suffix σ (shaded). Middle: a cross comparison (s, e) places $t[e{:}]$ at the cut s of u, where it must be a prefix of $u[s{:}]$, leaving the overhang o. Bottom: a self comparison (i, j) places $t[j{:}]$ at the cut i of t in the same way. A hole needs a head event, a safe spine and a first rejected cut $J(t)$ of the target, and it must have no event and no self event: every cross comparison (s, e) with $e \le J(t)$ must carry a state pair outside the guard G, and every self comparison (i, j) with $j \le J(t)$ a pair outside the self guard G_s.

Proof. The state $A'_s(u)$ is reachable, and $B'(t[{:}e]) = B'_e(t)$ is paired with it by H. By (FH), $B_i(t) = B_i(t[{:}e]) \notin \mathrm{Rej}$ for every $i < e$, so the first rejected cut is at least e. □

Section 9 uses Lemma 8.4 to split a hole at its first rejected cut. In the extracted instances, (FH) holds by construction (Lemma 8.6).

8.2 The Extracted Instances

The instance for a pair of visible signed variables $p, q \in X^{\pm}$ tests a pure derivation of ψ from a seed (p, t) to an atom (q, u). The names follow the readers of Zhao [2026]: the *target* is the word whose rejection is tested, here the seed word t, read by the source automaton $\mathcal{B}_p$ of Section 7; the *spine* is the word u of the derived atom. The derivation therefore runs from the target to the spine.

Definition 8.5 (Extracted instance). For $p, q \in X^{\pm}$, the *extracted instance* $I_{pq} = I_{pq}(\varphi, \psi)$ consists of:

- A, the lower automaton of q in φ, with $\mathrm{Bad} = \{Q : Q \cap \mathrm{Fact}^{\uparrow}_{\varphi} \neq \varnothing\}$;
- $B = \mathcal{B}_p$, the source automaton of p (Definition 7.2), with

$$\mathrm{Rej} = \{\zeta : \ \mathrm{dt}(\zeta) \text{ and } \mathrm{du}(\zeta) \text{ both fail}\};$$

- $G = \{(Q, (P, b, P')) : Q \cap P \neq \varnothing\}$, and G_s the self guard of Definition 7.2;
- A', the lower automaton of q in ψ, and B', the upper automaton of p in ψ, with $H = \{(Q', P') : Q' \cap P' \neq \varnothing\}$.

The start states $\downarrow_{\varphi} q$, the start state of $\mathcal{B}_p$, $\downarrow_{\psi} q$, and $\uparrow_{\psi} p$ are renamed 0.

Table 4 summarizes the instance. The automata have $N_A = 2^{2|X|}$, $N_B = 2^{4|X|+2|Z|+1}$, and $N_{A'} = N_{B'} = 2^{2(|X|+|Z|)}$ states, and the $4|X|^2$ instances are computable from φ and ψ.

Rejection is enriched: a cut of the seed is rejected only if it is neither dead by support in φ nor root-dead in ψ. Both properties are inherited by prefixes (Lemma 6.8(4), Lemma 6.10(4)), so a rejected cut stays rejected on every extension: $J(t)$ is defined iff t itself is rejected, that is, iff neither $\mathrm{DeadT}_{\varphi}(p, t)$ nor $\mathrm{DeadU}_{\psi}(p, t)$ holds.

Table 4. The extracted instance I_{pq} for a seed (p,t) and a derived atom (q,u). The state of B at a cut e of t is $(P^{\varphi}_{p,e}(t), [(\bar{p}, t[{:}e]) \in \mu_\varphi], P^{\psi}_{p,e}(t))$, by Lemma 7.3.

Part	Reads	Meaning
A, Bad	$Q^{\varphi}_{q,s}(u)$ on the spine u	safe spine: $(q,u) \notin \mu_\varphi$ (Lemma 6.6)
B, Rej	the source automaton on the target t	rejected: $\neg\mathsf{DeadT}_\varphi(p,t[{:}e])$ and $\neg\mathsf{DeadU}_\psi(p,t[{:}e])$
G	$Q^{\varphi}_{q,s}(u) \cap P^{\varphi}_{p,e}(t)$	event: $(q,u) \in \mathrm{Reach}_\varphi(p,t)$ (Lemma 6.14)
G_s	$P^{\varphi}_{p,i}(t) \cap \overline{P^{\varphi}_{p,j}(t)}$	self event: (p,t) is dead (Theorem 6.11)
A', B', H	$Q^{\psi}_{q,s}(u) \cap P^{\psi}_{p,e}(t)$	head event: $(p,t) \to^*_\psi (q,u)$ (Lemma 6.3)

Lemma 8.6 (Forced target heads). *Every extracted instance I_{pq} satisfies (FH). No hypothesis on φ or ψ is needed.*

Proof. Let $(Q', P') \in H$ and $B'(\gamma) = P'$. Then $P^{\psi}_{p}(\gamma) = P' \neq \varnothing$, and by Lemma 6.15, $\mathsf{DeadU}_\psi(p, \gamma[{:}i])$ for every $i < |\gamma|$. By Lemma 7.3(2), du holds at every such cut, so $B_i(\gamma) \notin$ Rej. □

Without the enrichment, that is, with Rej defined by the failure of dt alone, the target head can extend beyond the first rejected cut (Example 8.12).

8.3 Holes and Condition (II)

Theorem 8.7 (Holes and condition (II)). *Let φ be satisfiable, let* (U) *hold, let $p, q \in X^{\pm}$, and let u, t be words. Then (u,t) is a hole of I_{pq} iff (p,t) is live in φ, $(p,t) \to^*_\psi (q,u)$, and $(q,u) \notin \mathrm{Cl}_\varphi(p,t)$. Consequently:*

(1) condition (II) *holds iff no extracted instance I_{pq} has a hole;*
(2) for every hole (u,t) of I_{pq}, the least model $\mathrm{Cl}_\varphi(p,t)$ of φ that contains (p,t) is not the restriction to X of any model of ψ, so its solution has no extension to a solution of ψ.

The conditions of Definition 8.1 match the three requirements one by one. A head event is the lowest point of the pure derivation of ψ. A safe spine and the absence of events say that (q,u) lies neither in μ_φ nor in $\mathrm{Reach}_\varphi(p,t)$ (Lemma 6.14). A rejected target and the absence of self events say that (p,t) is neither dead by support nor has a self event, that is, that it is live (Theorem 6.11). Here (U) is used: a live seed must also be rejected, which requires that it not be root-dead in ψ. The bounds $e \le J(t)$ and $j \le J(t)$ in Definition 8.1 lose nothing: the guards require a nonempty set $P^{\varphi}_{p,e}(t)$, and the pop bound (Lemma 6.8(2)) empties these sets after $J(t)$. The proof is in Appendix D.

Corollary 8.8 (Entailment and holes). *Let φ and ψ be satisfiable. Then $\varphi \models \exists \bar{z}\, \psi$ iff condition* (I0) *holds, condition* (U) *holds, and no extracted instance I_{pq} with $p, q \in X^{\pm}$ has a hole.*

Proof. If (U) fails, the entailment fails by Theorem 4.2. Otherwise, by Theorem 4.2, the entailment holds iff (I0) and (II) hold, and by Theorem 8.7(1), (II) holds iff no I_{pq} has a hole. □

Conditions (I0) and (U) are decidable (Proposition 6.16 and Theorem 7.9), and every extracted instance satisfies (FH) (Lemma 8.6). It remains to decide the hole problem for instances with forced target heads, which Sections 9 and 10 do.

Example 8.9 (Condition (U) cannot be dropped). Without (U), a failure of (II) need not be a hole. Let φ be empty over $X = \{x, y\}$, and let $\psi = \{x \le y,\ x \le f(a,b)\}$ with a, b hidden and f binary and covariant. The seed (x^0, ε) is live in φ, the copy $(x^0, \varepsilon) \to (y^0, \varepsilon)$ is a pure move of ψ, and $(y^0, \varepsilon) \notin \mathrm{Cl}_\varphi(x^0, \varepsilon)$, so (II) fails at this seed. But x^0 is the root of the upper constructor literal $x^0 \le f(a^0, b^0)$ of $\psi^{\pm}$, so $\mathsf{DeadU}_\psi(x^0, \varepsilon)$ holds and the target ε of $I_{x^0 y^0}$ is not rejected: $(\varepsilon, \varepsilon)$ is not a

hole. Here (U) fails, with the countermodel $x = \top$. The enrichment trades (U) for (FH): with the rejection test given by the failure of dt alone, the holes still characterize the failures of (II) when φ is satisfiable, without assuming (U), but (FH) can fail (Example 8.12).

Remark 8.10 (Covariant and joint-parity instances). Informally, if φ has a parity assignment (Remark 4.5), in particular if f is covariant, the self guard never applies to reachable states of the target automaton (Remark 6.13), and the extracted instances behave like instances of the suffix-sharing two-word problem without self events.

Example 8.11 (Shift closure: a hole). Continue Example 7.7 with $\psi = \{x \le z,\ z \le y,\ z \le f(z,z)\}$ and $\varphi_1' = \{x \le y,\ y \le g,\ g \le f(g,g)\}$, both satisfiable. Condition (U) holds, and (I0) holds because ψ has no facts. Consider the seed variable y^1 and the derived variable x^1. The copy and constructor literals of $(\varphi_1')^{\pm}$ are $x^0 \le y^0$, $y^0 \le g^0$, $g^0 \le f(g^0,g^0)$, $y^1 \le x^1$, $g^1 \le y^1$, and $f(g^1,g^1) \le g^1$. In $I_{y^1x^1}$:

- A starts at $\downarrow_{\varphi_1'} x^1 = \{x^1, y^1, g^1\}$ and moves to $\{g^1\}$ on every letter; φ_1' has no facts, so every spine is safe; the state $\{g^1\}$ lies on a loop at which both letters are safe, so it is *branching* (Definition 9.5), the case in which Section 9 completes every pre-hole directly (Lemma 9.6);
- $P^{\varphi_1'}_{y^1}$ is $\{y^1, x^1\}$ at the empty word and empty afterwards, $\mu_{\varphi_1'} = \varnothing$, and $\mathsf{DeadT}^{\varepsilon}_{\varphi_1'} = \{x^0, y^0, g^0\}$; with Example 7.7, no cut of any target is dead by support or root-dead, so $J(t) = 0$ for every target t;
- hence the only possible event is $(0,0)$, a cross comparison iff $t \preceq u$, and there are no self events, since $P^{\varphi_1'}_{y^1,j}(t) = \varnothing$ for $j \ge 1$;
- $Q^{\psi}_{x^1}(\beta)$ always meets $\uparrow_{\psi} y^1 = \{y^1, z^1, x^1\}$, and $P^{\psi}_{y^1}(\gamma) = \varnothing$ for $\gamma \ne \varepsilon$, so the head events are the pairs $(s,0)$ with $u[s{:}] = t$.

So the holes of $I_{y^1x^1}$ are the pairs (u,t) in which t is a suffix but not a prefix of u, for example $(01,1)$, with head event $(1,0)$. By Theorem 8.7, ψ derives $(x^1, 01)$ from the live seed $(y^1, 1)$, and $(x^1, 01) \notin \mathrm{Cl}_{\varphi_1'}(y^1, 1)$. In words: if y has a leaf $\perp$ at a prefix of 1, then every z between x and y with $z \le f(z,z)$ has a leaf $\perp$ at a prefix of 1 and hence at a prefix of 01, and so does x. A finite countermodel is $x = y = f(f(\perp, f(\perp,\perp)),\perp)$, whose constructor positions are ε, 0, and 01, with g the complete binary tree whose constructor positions are the words of length at most 2. The least model $\mathrm{Cl}_{\varphi_1'}(y^1, 1)$ of Theorem 8.7(2) is the solution $x = y = f(\Omega,\perp)$ and $g = \Omega$. Since $J(t) = 0$, the target head is empty, and (FH) holds trivially for this instance.

Example 8.12 (A target head beyond the cuts dead by support). Continue Example 7.8 with $\varphi = \{x \le x \to x\}$ and $\psi = \{x \le a \to x,\ c \to y \le a\}$, both satisfiable: $x = \perp$ solves φ, and $x = y = \perp$, $a = c = \top$ solve ψ. Conditions (I0) and (U) hold. Consider the seed variable x^0, the derived variable y^1, and the pair $(u,t) = (\varepsilon, 01)$ in $I_{x^0y^1}$. Along the target $t = 01$, the upper automata give

$$P^{\varphi}_{x^0} :\ \{x^0\},\ \{x^1\},\ \varnothing, \qquad P^{\psi}_{x^0} :\ \{x^0\},\ \{a^1\},\ \{y^1\}$$

at the cuts 0, 1, 2. With $\mathsf{DeadT}^{\varepsilon}_{\varphi} = \{x^0\}$ and $\mathsf{DeadU}^{\varepsilon}_{\psi} = \{x^0, a^1\}$, cut 0 is dead by support, cut 1 is not dead by support but root-dead in ψ, and cut 2 is rejected: $J(01) = 2$. The head event is $(0,2)$, since $Q^{\psi}_{y^1}(\varepsilon) = \{y^1\} = P^{\psi}_{x^0}(01)$: the derivation pops 0 from x^0 to a^1 and then 1 from a^1 to y^1. The only cross comparison (s,e) with $e \le 2$ is $(0,2)$, whose guard $\{y^1\} \cap P^{\varphi}_{x^0}(01)$ is empty; the self comparisons of 01 ending at 2 have an empty self guard, and $(0,1)$ is not a self comparison. The spine is safe because φ has no facts. So $(\varepsilon, 01)$ is a hole: ψ derives (y^1, ε), that is $y = \perp$, from the live seed $(x^0, 01)$. A countermodel is $x = (\perp \to \perp) \to \perp$ and $y = \top$: from $x \le a \to x$ and $c \to y \le a$, any extension would have $c \to \top \le_c a \le_c \perp \to \perp$, hence $\top \le_c \perp$.

The target head is $\gamma = 01$. The rejection test given by the failure of dt alone would reject already at cut 1, inside the head: $(x^0, 0)$ is not dead by support, although it is dead by the self event $(0, 1)$ of the word 0 (Example 6.12). So the unenriched instance does not satisfy (FH). The enrichment rejects only at cut 2, because $(x^0, 0)$ is root-dead in ψ.

9 Non-Mortal Holes

Let I be an instance of the two-word hole problem with self events (Definition 8.1) that satisfies condition (FH) (Definition 8.3). This section shows that if I has a hole, then it has a hole whose length is polynomial in the sizes of its four automata, or it has a *mortal hole*: a hole (u, t) whose spine ends in a state without infinite safe continuations and whose target ends fewer than N_A letters after its first rejected cut (Theorem 9.17). Mortal holes are treated in Section 10.

The argument starts from a simple observation. Cutting a hole (u, t) at the first rejected cut $J(t)$ of its target splits off the *continuation* $\theta = t[J(t):]$. Condition (FH) places every target head before $J(t)$, so θ is a common suffix of both words, and what remains is a shorter pair (u_0, t_0) with $u = u_0\theta$ and $t = t_0\theta$, a *pre-hole* (Lemma 9.3 and Figure 10). Appending the same word to both words creates no new events and no new self events, and it keeps an existing one exactly when it continues the overhang of that event periodically (Lemma 9.1). The question is therefore whether a single continuation that keeps the spine automaton out of Bad can break every pending obligation of a pre-hole: its events, which are never aligned, and the self events of its target. The answer depends on the state $q = A(u_0)$ at the end of the spine of the pre-hole.

- If a cycle that offers a choice of two letters is reachable from q, a breaking continuation of length at most $\max(|u_0|, |t_0|) + 3N_A$ always exists (Lemma 9.6).
- If every reachable cycle is forced, then the long continuations follow finitely many ultimately periodic words, the *lassos* of q (Lemma 9.8). A lasso that is not purely periodic breaks every obligation (Lemma 9.13). A purely periodic lasso may let obligations survive, and then the pre-hole itself has to be shortened. The *pump* (Theorem 9.14) deletes factors of the two words while keeping the endpoint q and the absence of surviving obligations, down to a length polynomial in the sizes of the automata.
- In the remaining cases the continuation reaches a state without infinite safe continuations within fewer than N_A letters, and the hole is mortal.

Conventions. We use the instance, safety, comparison, event, and forced-head conventions of Definitions 8.1 to 8.3. For an automaton M, let $M(r, w)$ denote its state after reading w from r, and let $M_s(w) = M(0, w[:s])$ be its state at the cut s of w. We abbreviate the state counts as

$$a = N_A, \qquad b = N_B, \qquad a' = N_{A'}, \qquad b' = N_{B'};$$

all four are at least 1.

An *infinite word* $\xi' = \xi'[0]\xi'[1]\cdots$ over Σ has prefixes $\xi'[:k]$ and suffixes $\xi'[k:]$ as for finite words, and $v\xi'$ is the infinite word that starts with the finite word v. A number $\varpi \geq 1$ is a *period* of ξ' if $\xi'[i + \varpi] = \xi'[i]$ for all $i \geq 0$; ξ' is *purely periodic* if it has a period, and its *least period* is then the least of its periods. For a nonempty finite word o, $o^\omega = ooo\cdots$ is the infinite word with period $|o|$ that begins with o; a finite word is a prefix of o^ω exactly when it is a prefix of some o^k, in agreement with the notation $\theta \preceq o^\omega$ of Section 2.1. Holes are written (u, t), pre-holes (u_0, t_0), continuations θ, and infinite words ξ.

9.1 The Continuation Identity

For a cross comparison (s, e) of (u, t) and a word θ, we say that θ *keeps* (s, e) if $t[e:]\,\theta \preceq u[s:]\,\theta$, and that θ *breaks* it otherwise; for a self comparison (i, j) of t, θ keeps it if $t[j:]\,\theta \preceq t[i:]\,\theta$. Non-aligned

cross comparisons can be unboundedly many: if $u = \ell^k$ and $t = \ell^m$ for a letter ℓ, every pair of cuts with a nonnegative offset is one. We therefore do not count them but control them through common continuations.

Lemma 9.1 (Continuation identity). *Let u, t and θ be words such that $J(t)$ is defined.*

(a) $J(t\theta) = J(t)$, and each of the four automata has the same state at every cut $s \le |u|$ of $u\theta$ as at the cut s of u, and at every cut $e \le |t|$ of $t\theta$ as at the cut e of t.

(b) Let $s \le |u\theta|$ and $e \le J(t)$. Then (s, e) is a cross comparison of $(u\theta, t\theta)$ if and only if $s \le |u|$, (s, e) is a cross comparison of (u, t), and θ keeps it. Likewise, for $i < j \le J(t)$, the pair (i, j) is a self comparison of $t\theta$ if and only if it is a self comparison of t and θ keeps it.

(c) A word θ keeps a cross or self comparison with overhang o if and only if $\theta \preceq o\theta$. If $o \neq \varepsilon$, this holds if and only if $\theta \preceq o^\omega$. Every word keeps every aligned cross comparison.

In words: a common continuation creates no new comparison among the cuts up to $J(t)$, and it keeps an existing one exactly when it continues the overhang periodically. The proof is in Appendix E.2. We use the lemma through the following *continuation facts*, which hold for all words u and t with $J(t)$ defined and every word θ:

(K1) the events of $(u\theta, t\theta)$ are exactly the events of (u, t) that θ keeps, and the self events of $t\theta$ are exactly the self events of t that θ keeps;
(K2) every word keeps every aligned cross comparison, so aligned events can never be removed;
(K3) if θ breaks a cross or self comparison, so does every word with prefix θ;
(K4) every head event of (u, t) is a head event of $(u\theta, t\theta)$;
(K5) if u is safe, then $u\theta$ is safe if and only if θ is safe from $A(u)$.

A non-aligned event or a self event with overhang o is thus a periodicity obligation: a continuation removes it exactly when it departs from o^ω.

9.2 Pre-Holes and the Decomposition at the First Rejected Cut

Definition 9.2 (Pre-hole). A *pre-hole* of I is a pair (u_0, t_0) such that u_0 is safe, $J(t_0) = |t_0|$, (u_0, t_0) has a head event, and (u_0, t_0) has no aligned event. Its *obligations* are its events and the self events of t_0. A word θ *completes* (u_0, t_0) if θ is safe from $A(u_0)$ and breaks every obligation of (u_0, t_0).

No obligation of a pre-hole is aligned, so every obligation has a nonempty overhang. The overhang of an event has length at most $|u_0|$ and that of a self event at most $|t_0|$, so every obligation has an overhang of length at most

$$W = \max(|u_0|, |t_0|).$$

A pre-hole without obligations is a hole, completed by the empty word.

Lemma 9.3 (Decomposition at the first rejected cut). *If (u_0, t_0) is a pre-hole and θ completes it, then $(u_0\theta, t_0\theta)$ is a hole. Conversely, let I satisfy* (FH) *and let (u, t) be a hole. Then $\theta = t[J(t):]$ is a suffix of u, and if $u = u_0\theta$ and $t = t_0\theta$, then (u_0, t_0) is a pre-hole and θ completes it.*

In one direction, appending a completing word keeps the spine safe, keeps the head event, and leaves no event and no self event, by (K5), (K4) and (K1). In the other, (FH) places the target head of a head event before $J(t)$ (Lemma 8.4). Hence θ is no longer than the common suffix and can be split off both words, and (K1) and (K2) show that θ breaks every obligation of (u_0, t_0) and that (u_0, t_0) has no aligned event. The proof is in Appendix E.2. Condition (FH) cannot be dropped (Example 9.20).

Example 9.4 (Shift closure: decomposition). In the instance $I_{y^1x^1}$ of Example 8.11, every target t has $J(t) = 0$. So the hole $(01, 1)$ splits as $\theta = t[0:] = 1$ and $(u_0, t_0) = (0, \varepsilon)$. The pre-hole $(0, \varepsilon)$ has the head event $(1, 0)$, and this aligned pair is not an event, because $A_1(0) = \{g^1\}$ misses

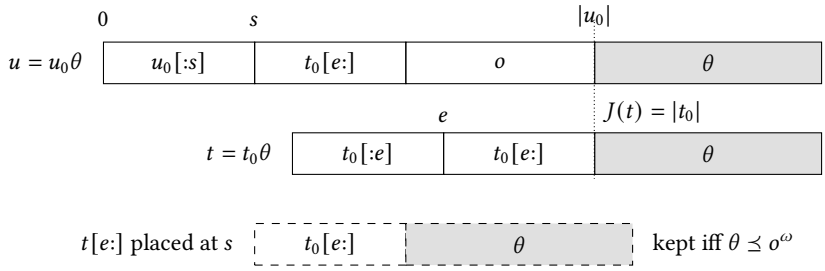


Fig. 10. A hole $(u, t) = (u_0\theta, t_0\theta)$ cut at the first rejected cut $J(t) = |t_0|$ of its target (Lemma 9.3); the continuation θ is shaded. An event (s, e) of the pre-hole (u_0, t_0) places $t_0[e:]$ at the cut s of u_0, followed by the overhang $o = o(s, e)$. After θ is appended, (s, e) is still a cross comparison if and only if the shifted copy of θ (bottom row) agrees with $o\theta$, that is, if and only if $\theta \preceq o^\omega$ (Lemma 9.1). A self event (i, j) of t_0 behaves in the same way, with t_0 in the top row. A hole requires θ to break every event and every self event.

$P^{\varphi'_i}_{y^i}(\varepsilon) = \{y^1, x^1\}$. Its only obligation is the event $(0, 0)$, with overhang 0; the target ε has no self comparison. By Lemma 9.1(c), a continuation keeps this obligation iff it is a prefix of 0^ω. So the continuation 0 keeps it, and indeed $(00, 0)$ has the event $(0, 0)$, while every continuation that contains the letter 1 breaks it; the shortest one, $\theta = 1$, gives back the hole $(01, 1)$.

9.3 Kinds of State

Let $\mathcal{G}_A$ be the *safe graph* of A: its vertices are the safe states of A, with an edge from r to $A(r, \ell)$, labeled by the letter ℓ, whenever $A(r, \ell)$ is safe. The labels of the paths from r are exactly the words safe from r. A safe state is *cyclic* if it lies on a cycle of $\mathcal{G}_A$, and a letter ℓ is *safe at* a safe state r if $A(r, \ell)$ is safe.

Definition 9.5 (Kinds of state). A safe state q of A is

- *mortal* if no infinite word is safe from q, equivalently, if no cyclic state is reachable from q in $\mathcal{G}_A$;
- *branching* if some cyclic state r reachable from q in $\mathcal{G}_A$ has two distinct safe letters $\ell \neq \ell'$;
- *thin* if it is neither mortal nor branching.

From a mortal state every safe run dies within N_A letters. From a branching state some reachable cycle offers a choice of letters, so a safe continuation can depart from any given period. From a thin state infinite safe runs exist, but once they reach a cycle they are forced, letter by letter. The two descriptions of mortality agree because an infinite path in the finite graph $\mathcal{G}_A$ repeats a vertex, and a reachable cyclic state can be visited forever. A branching state reaches a cyclic state, so it is not mortal; the three kinds are therefore pairwise disjoint, and every safe state has exactly one of them. Each kind is decided by reachability in $\mathcal{G}_A$.

Branching asks only for two distinct safe letters at a reachable cyclic state; the states they lead to may coincide, and the letter that leaves the cycle need not admit any further safe continuation, since the breaker stops right after it. Requiring all letters, or two specific letters, to be safe would misclassify states over larger alphabets. For instance, for arity $n \leq 1$ no state has two distinct safe letters, and for $n = 0$ there are no letters at all.

9.4 The Breaker

Lemma 9.6 (Breaker). *Let q be a branching state and $W \geq 0$. There is a word θ, safe from q, with $|\theta| \leq W + 3N_A$, that is not a prefix of o^ω for any word o with $1 \leq |o| \leq W$. In particular, every pre-hole (u_0, t_0) with $A(u_0)$ branching is completed by a word of length at most $\max(|u_0|, |t_0|) + 3N_A$.*

The idea is to steer the spine into a cycle that offers a choice of letters, to run around it for longer than any overhang, and then to leave it. The word $\theta = \alpha\lambda^k\ell$ follows a shortest path α to a cyclic state r with two distinct safe letters, runs k times around a shortest cycle λ through r, with $k|\lambda| \geq W + |\lambda|$, and leaves the cycle by a safe letter $\ell \neq \lambda[0]$; one of the two safe letters at r differs from $\lambda[0]$. If θ were a prefix of o^ω, then over the stretch λ^k, of length at least $|o| + |\lambda|$, it would have both periods $|o|$ and $|\lambda|$, hence the period $\gcd(|o|, |\lambda|)$ by the theorem of Fine and Wilf [Fine and Wilf 1965]. So after λ^k it would have to continue with $\lambda[0]$, not with ℓ. This uses only equality of letters and holds over every alphabet. Every obligation of a pre-hole has a nonempty overhang of length at most $W = \max(|u_0|, |t_0|)$, so θ breaks it by Lemma 9.1(c). The proof is in Appendix E.3.

In Example 9.4, the spine state $A(0) = \{g^1\}$ lies on a loop at which both letters are safe, so it is branching, and the breaker also completes the pre-hole $(u_0, t_0) = (0, \varepsilon)$. Here $W = 1$, and the construction may take the cyclic state $r = \{g^1\}$ itself, $\alpha = \varepsilon$, the cycle $\lambda = 0$ and $k = 2$; this gives $\theta = \alpha\lambda^k\ell = 001$ with the breaker letter $\ell = 1$ and the hole $(0001, 001)$, while the shortest completion $\theta = 1$ gives back $(01, 1)$.

The bound uses $\max(|u_0|, |t_0|)$ because the overhang of a self event of t_0 can be as long as t_0, which may be longer than u_0. The maximum cannot be replaced by $|u_0|$, as the following example shows.

Example 9.7 (Self events need the longer word). Let $\Sigma = \{0, 1\}$ and $t_0 = 0001011100$. Let A and A' have a single state 0, with Bad $= \varnothing$, so that $N_A = 1$ and the state 0 of A is branching, and let $G = \varnothing$. Let B and B' both be the automaton whose states are the prefixes of t_0, with start state ε, and one sink; it moves from v to $v\ell$ while $v\ell$ is a prefix of t_0, and to the sink otherwise. Let Rej $= \{t_0\}$, let H consist of the pair of the state of A' and the state t_0 of B', and let G_s consist of the pairs $(t_0[:i], t_0)$ of states of B with $0 \leq i \leq 7$. Condition (FH) holds, since t_0 is the only word that leads B' to the state t_0, and its proper prefixes are not rejected. The pair (ε, t_0) is a pre-hole, and its obligations are the self events $(i, 10)$ with $0 \leq i \leq 7$, whose overhangs are the suffixes $t_0[i:]$, of length at least 3. The eight factors $t_0[i:i+3]$, $0 \leq i \leq 7$, are the eight words of length 3 over Σ. So every word θ of length at most 3 is a prefix of some $t_0[i:]$ with $i \leq 7$ and keeps the self event $(i, 10)$. Hence no word of length at most $|u_0| + 3N_A = 3$ completes (ε, t_0), whereas Lemma 9.6 provides a completion of length at most $|t_0| + 3N_A = 13$.

9.5 Thin States and Lassos

A *lasso* of a safe state q is an infinite word $\alpha\lambda^\omega$ with $\lambda \neq \varepsilon$, $|\alpha| + |\lambda| \leq N_A$, $\alpha\lambda$ safe from q, and $A(q, \alpha\lambda) = A(q, \alpha)$. Every lasso of q is safe from q, and there are finitely many.

Lemma 9.8 (Thin states). *Let q be a safe state of A.*

(a) If q is mortal, every word safe from q has length less than N_A and leads to a mortal state.
(b) If q is thin, every cyclic state reachable from q has exactly one safe letter, and the edge along it leads to a cyclic state.
(c) If q is thin, every infinite word safe from q is a lasso of q, and q has at least one lasso.
(d) If q is thin, θ is safe from q, and $A(q, \theta)$ is mortal, then $|\theta| < N_A$.

Thus a thin region consists of finitely many lassos together with acyclic paths into mortal states. Each part follows a safe run until it repeats a state; the proof is in Appendix E.3.

Definition 9.9 (Survival). A cross or self comparison *survives* an infinite word ξ if every finite prefix of ξ keeps it. An infinite word ξ is *survivor-free* for a pre-hole (u_0, t_0) if no obligation of (u_0, t_0) survives ξ.

Lemma 9.10 (Survival). *Let u and t be words and ξ an infinite word.*

(a) A cross or self comparison with nonempty overhang o survives ξ if and only if $\xi = o^\omega$. Every aligned cross comparison survives ξ.

(b) Let $s \le |u|$ and $e \le |t|$ with $|t| - e \le |u| - s$. Then (s, e) is a cross comparison of (u, t) that survives ξ if and only if $u[s{:}]\,\xi = t[e{:}]\,\xi$. Likewise, for $i < j \le |t|$, the pair (i, j) is a self comparison of t that survives ξ if and only if $t[i{:}]\,\xi = t[j{:}]\,\xi$.

(c) Let $\xi = \alpha\lambda^\omega$ with $\lambda \neq \varepsilon$. If a comparison with nonempty overhang o does not survive ξ, then the prefix of ξ of length $|\alpha| + |\lambda| + |o| - 1$ breaks it.

Part (a) is Lemma 9.1(c) applied to every prefix of ξ. Part (b) turns survival into an equality of infinite words, which is the form used by the pump; it holds because two finite words are prefixes of the same infinite word exactly when the shorter is a prefix of the longer. Part (c) compares two ultimately periodic words by the theorem of Fine and Wilf. The proofs are in Appendix E.4.

Corollary 9.11 (Lasso completion). *Let (u_0, t_0) be a pre-hole, and let ξ be a lasso of $A(u_0)$ that is survivor-free for (u_0, t_0). Then the prefix of ξ of length $\max(|u_0|, |t_0|) + N_A - 1$ completes (u_0, t_0).*

Proof. The prefix is safe from $A(u_0)$, since ξ is. Write $\xi = \alpha\lambda^\omega$ with $|\alpha|+|\lambda| \le N_A$. Each obligation has a nonempty overhang o with $|o| \le \max(|u_0|, |t_0|)$ and does not survive ξ. By Lemma 9.10(c) the prefix of ξ of length $|\alpha| + |\lambda| + |o| - 1 \le \max(|u_0|, |t_0|) + N_A - 1$ breaks it, and by (K3) so does the longer prefix. □

9.6 Short Pre-Holes

Pre-holes form a recognizable set of pairs. A pre-hole asks for a safe spine, a target rejected exactly at its end, and conditions on the aligned pairs of cuts only, and the aligned pairs are the pairs at equal distance from the ends within the longest common suffix. A finite graph that reads both words forward, aligned at their right ends, can therefore track the four automata and two bits: whether some aligned pair seen so far carries H, and whether some carries G. Both bits are reset whenever the two words read different letters. Its paths spell exactly the pre-holes, and a shortest path visits no node twice.

Proposition 9.12 (Short pre-holes). *Let E be a set of safe states of A. If some pre-hole (u_0, t_0) has $A(u_0) \in E$, then some pre-hole (u_1, t_1) has $A(u_1) \in E$ and $|u_1|, |t_1| < 12aba'b'$. In particular, it is decidable whether a pre-hole with $A(u_0) \in E$ exists.*

The graph has the nodes $(r_A, r_B, r_{A'}, r_{B'}, h, g, \phi)$, with a state of each automaton, two bits h and g, and a phase $\phi \in \{0, 1, 2\}$ that records whether only the spine, only the target, or both words are being read; hence $12aba'b'$ nodes. The construction and its correctness are in Appendix E.5. The proposition does not preserve survivor-freeness for a given lasso; it is used where every pre-hole with the given endpoint is good enough, namely for branching endpoints and for lassos that are not purely periodic.

9.7 Aperiodic Lassos

Lemma 9.13 (Aperiodic lassos). *If an infinite word ξ is not purely periodic, then no cross or self comparison with a nonempty overhang survives ξ. In particular, ξ is survivor-free for every pre-hole.*

Proof. By Lemma 9.10(a), a comparison with nonempty overhang o survives ξ only if $\xi = o^\omega$, which is purely periodic. Every obligation of a pre-hole has a nonempty overhang. □

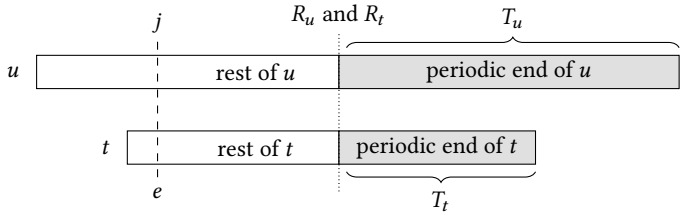


Fig. 11. The rest and the periodic end of the spine and of the target for a purely periodic ξ with least period ϖ, drawn with the two boundaries R_u and R_t on the dotted line. Two cuts in the periodic ends have equal ξ-suffixes if and only if their end distances agree modulo ϖ. A rest cut j of u has an equal ξ-suffix with a cut e of t only if e is a rest cut at the same distance before the boundary, $R_u - j = R_t - e$ (dashed), and a rest cut never matches a cut of a periodic end (Lemma E.7).

9.8 The Pump

The remaining non-mortal case is a pre-hole whose spine ends in a thin state q and a purely periodic lasso of q that is survivor-free for it. Here survivor-freeness depends on the pre-hole, and it can fail for short pre-holes and hold for long ones, so Proposition 9.12 does not apply. The pump shortens the pre-hole directly, keeping its endpoint and its survivor-freeness. Put

$$\mathrm{P}(a, b, a', b') = 2aba'b' + a^2a' + ab + aa' + bb'.$$

Theorem 9.14 (Pump). *Let ξ be a purely periodic infinite word with least period ϖ, and let (u, t) be a pre-hole for which ξ is survivor-free. Then there is a pre-hole (u', t') for which ξ is survivor-free, with $A(u') = A(u)$ and*

$$|u'|, |t'| < 2aba'b' + aa'\varpi + b\varpi + aa' + bb'.$$

In particular, if $\varpi \le N_A$, then $|u'|, |t'| < \mathrm{P}(a, b, a', b')$.

The theorem uses no property of the state $A(u)$, and a lasso enters only through the bound on its period: if a lasso $\xi = \alpha\lambda^\omega$ is purely periodic, then its least period is at most $|\lambda| \le N_A$ (Lemma E.1(b)), so the second bound applies. The proof rests on three observations and a count; the details are in Appendix E.6.

Survival as an equality of infinite words. By Lemma 9.10(b), a pre-hole (u, t) is survivor-free for ξ if and only if no pair of cuts (s, e) with $|t| - e \le |u| - s$ and $u[s{:}]\,\xi = t[e{:}]\,\xi$ carries G, and no pair of cuts $i < j$ of t with $t[i{:}]\,\xi = t[j{:}]\,\xi$ carries G_s; the aligned pairs are among the former. Call such pairs *ξ-matched*. The pump only has to make sure that a deletion creates no new ξ-matched pair.

Rests and periodic ends. Call a cut s of a word w *periodic* if $w[s{:}]\,\xi$ has period ϖ. The periodic cuts form an interval $R_w \le s \le |w|$; the factor $w[{:}R_w]$ is the *rest* of w and $w[R_w{:}]$, of length $T_w = |w| - R_w$, its *periodic end*. The ξ-matched pairs have a rigid structure (Figure 11). Two periodic cuts have equal ξ-suffixes exactly when their *end distances*, the distances to the ends of their words, agree modulo ϖ. A periodic cut never matches a rest cut. A rest cut j of u matches at most one cut of t, its *partner*, which is then a rest cut at the same distance before the boundary of the rest: $R_u - j = R_t - e$. Finally, the infinite word $\xi' = u[j{:}]\,\xi$ at a rest cut is *non-self-similar*: $z\xi' \ne \xi'$ for every nonempty finite word z. Indeed, $z\xi' = \xi'$ would give ξ' the period $|z|$, and then also the period ϖ of its suffix ξ, which would make j a periodic cut. Moreover, the periodic end of a target t with $J(t) = |t|$ is short, $T_t < N_B\,\varpi$ (Lemma E.8): two of its cuts with the same state of B and end distances that agree modulo ϖ are followed by the same letters up to the end of t, so the earlier one would lead B to the rejected state $B(t)$ before the end.

Table 5. The four deletions of the pump (Lemma E.12). Each applies to a pre-hole (u, t) for which ξ is survivor-free when the condition in the second column holds; it deletes the factors in the third column, which are found by pigeonhole on the signature in the fourth column, of size N_i.

Deletion	Condition	Deleted factors	Signature	Count
1, spine end	$T_u \geq T_t + N_1$	a factor of the periodic end of u at end distance at least T_t	states of A, A'; end distance mod ϖ	$N_1 = aa'\varpi$
2, synchronous rest	$\lvert C_u\rvert > N_2$	the factor between two partnered rest cuts of u, and the one between their partners	four states, head bit	$N_2 = 2aba'b'$
3, spine rest	$\lvert U_u\rvert > N_3$	the factor between two unpartnered rest cuts of u	states of A, A'	$N_3 = aa'$
4, target rest	$\lvert U_t\rvert > N_4$	the factor between two unpartnered rest cuts of t	states of B, B'	$N_4 = bb'$

Deletions. Delete a factor w from $u = xwy$ and a factor w' from $t = x'w'y'$, at least one of them nonempty, such that each automaton is in the same state at both ends of each deleted factor. Then the states at the remaining cuts are unchanged, so safety, rejection and the endpoint $A(u)$ are kept. The shortened pair is again a pre-hole that is survivor-free for ξ, provided some head event of (u, t) survives the deletion and every ξ-matched pair of the shortened words comes from a ξ-matched pair of the original words (Lemma E.10). Two mechanisms provide this lifting. A factor of the periodic end of u whose length is a multiple of ϖ is *absorbed*: $w\,(y\xi) = y\xi$, so inserting it back changes no infinite suffix. *Rigidity*, the non-self-similarity of the suffix after a rest cut, handles two specific configurations: the synchronous deletion of the factor between two partnered rest cuts of u together with the factor between their partners, and the deletion of a factor that ends at an unpartnered rest cut (Lemmas E.11 and E.12). A factor that merely ends at a rest cut can create new matches: for $\xi = 0^\omega$, $u = 1010$ and $t = 110$, deleting the factor $u[1{:}2] = 0$, which ends at the rest cut 2, turns u into t and creates the matched pair $(0, 0)$. Four deletions, listed in Table 5, are found by pigeonhole on the states at the ends of the factors, the end distance modulo ϖ, and one bit recording whether a head event lies beyond the cut. Write $N_1, \ldots, N_4$ for their signature counts and C_u, U_u and U_t for the sets of partnered rest cuts of u, unpartnered rest cuts of u, and unpartnered rest cuts of t.

If no deletion applies, then the rests are short, since partners inject the partnered rest cuts of t into those of u, so that $R_u \leq N_2 + N_3$ and $R_t \leq N_2 + N_4$; the periodic end of t is shorter than $b\varpi$; and $T_u < T_t + N_1$. Hence both words are shorter than $N_1 + N_2 + N_3 + N_4 + b\varpi = 2aba'b' + aa'\varpi + b\varpi + aa' + bb'$ (Lemma E.13). Induction on $|u| + |t|$ proves the theorem. No deletion inside the periodic end of t is needed, and in particular no factors of both periodic ends are deleted together.

Remark 9.15 (An automaton instead of the pump). Informally, one could instead recognize the survivor-free pre-holes of a periodic lasso by a two-tape automaton that records a set of spine states for every residue class of end distances modulo the period. Such an automaton would have exponentially many states, whereas the pump needs no automaton and gives a polynomial bound.

9.9 The Three Cases

Definition 9.16 (Mortal hole). A *mortal hole* of I is a hole (u, t) such that $A(u)$ is mortal and $|t| - J(t) < N_A$.

Put

$$\mathsf{L}_{\mathrm{nm}}(a, b, a', b') = 4 \max\big(12aba'b',\ \mathsf{P}(a, b, a', b')\big) + 6a.$$

Since $a, b, a', b' \geq 1$, this is at most $54a^2ba'b'$, a polynomial in the four state counts (Appendix E.7).

Theorem 9.17 (Three cases). *Let I satisfy* (FH). *Then I has a hole if and only if at least one of the following holds:*

(1) some pre-hole (u_0, t_0) has $A(u_0)$ branching;
(2) some pre-hole (u_0, t_0) has $A(u_0)$ thin, and some infinite word safe from $A(u_0)$ is survivor-free for (u_0, t_0);
(3) I has a mortal hole.

More precisely, let $(u, t) = (u_0\theta, t_0\theta)$ be a hole decomposed as in Lemma 9.3. If $A(u_0)$ is branching, (u_0, t_0) satisfies (1). *If $A(u_0)$ is thin and $A(u)$ is not mortal, then (u_0, t_0) satisfies* (2) *with an infinite word that begins with θ. Otherwise (u, t) is a mortal hole. Moreover, if* (1) *or* (2) *holds, then I has a hole (u, t) with $|u| + |t| \leq \mathsf{L}_{\mathrm{nm}}(a, b, a', b')$; this part does not need* (FH).

A pre-hole as in (1) is shortened by Proposition 9.12, keeping its endpoint, and completed by Lemma 9.6. In case (2) the infinite word is a lasso ξ of the thin state $q = A(u_0)$ (Lemma 9.8(c)). If ξ is not purely periodic, every pre-hole with endpoint q is survivor-free for ξ (Lemma 9.13), so Proposition 9.12 shortens the pre-hole; if ξ is purely periodic, its least period is at most N_A and the pump shortens it. In both cases Corollary 9.11 completes the short pre-hole. Conversely, Lemma 9.3 decomposes a hole as $(u_0\theta, t_0\theta)$. If $A(u_0)$ is branching, case (1) holds. Otherwise, if $A(u_0)$ is thin and $A(u)$ has an infinite safe continuation ξ', then $\theta\xi'$ is safe from $A(u_0)$ and survivor-free, because its prefix θ already breaks every obligation. In the remaining cases $A(u)$ is mortal, and Lemma 9.8(a) and (d) give $|t| - J(t) = |\theta| < N_A$. The proofs are in Appendix E.7.

Corollary 9.18 (Short holes outside the mortal case). *Let I satisfy* (FH). *Then I has a hole if and only if it has a hole (u, t) with $|u| + |t| \leq \mathsf{L}_{\mathrm{nm}}(a, b, a', b')$ or it has a mortal hole.*

Proof. A hole of either kind is a hole. Conversely, if I has a hole, then by Theorem 9.17 one of (1)–(3) holds; in cases (1) and (2) the theorem provides a hole with $|u| + |t| \leq \mathsf{L}_{\mathrm{nm}}(a, b, a', b')$. □

The bound L_{nm} depends on neither the alphabet nor the self guard. The alphabet enters only the number of candidate pairs, and the self guard is absorbed by the use of $\max(|u_0|, |t_0|)$ in Lemma 9.6 and Corollary 9.11 and by the pump, whose signatures already contain the states of B.

Remark 9.19 (When mortal holes cannot occur). Informally, if no state reachable from 0 in $\mathcal{G}_A$ is mortal, for example if $\mathrm{Bad} = \varnothing$ and $|\Sigma| \geq 1$, then case (3) does not arise, and only the short holes of Corollary 9.18 matter.

Example 9.20 (Condition (FH) is needed). Let $\Sigma = \{\ell\}$, let A, A' and B have the single state 0, with $\mathrm{Bad} = \varnothing$, $\mathrm{Rej} = \{0\}$ and $G = G_{\mathrm{s}} = \varnothing$, let B' have the states 0 and 1, with $B'(0, \ell) = B'(1, \ell) = 1$, and let $H = \{(0, 1)\}$. The pair (ε, ℓ) is a hole, with the head event $(0, 1)$, but I has no pre-hole: every target has $J(t) = 0$, so a pre-hole has $t_0 = \varepsilon$, whose only aligned pair $(|u_0|, 0)$ carries the state 0 of B'. Since the only state of A is thin, none of the cases (1)–(3) holds. Condition (FH) fails: the head $\gamma = \ell$ has $B'(\gamma) = 1$, and its proper prefix ε is rejected.

10 Mortal Holes

Let $I = (A, \mathrm{Bad}, B, \mathrm{Rej}, G, G_s, A', B', H)$ be an instance of the two-word hole problem with self events (Definition 8.1), with the notation of Definitions 8.1 and 8.2. A *mortal hole* of I is a hole (u, t) whose spine ends in a mortal state and whose target ends fewer than N_A letters after its first rejected cut (Definition 9.16). By Theorem 9.17, an instance with forced target heads has a hole iff it has a short hole or a mortal hole. This section bounds the length of a shortest mortal hole and thereby decides the hole problem under (FH) (Theorem 10.17).

Where mortal holes come from. In an extracted instance I_{pq}, mortality depends on φ alone. The spine automaton is the lower automaton of q in φ, and its bad states are those that meet $\mathrm{Fact}^{\uparrow}_{\varphi}$, so a spine u is safe iff $(q, u) \notin \mu_\varphi$ (Table 4): the derived atom is not one of the atoms that every solution of φ contains. Facts come only from extreme literals (Table 3), so, informally, mortal holes arise only when φ forces atoms (Remark 9.19): a spine that ends in a mortal state derives an atom that is not forced, although all its long extensions are. A mortal hole therefore admits no arbitrarily long safe continuation. The argument of Section 9, which replaces the continuation of a hole by a new one taken from the infinite safe continuations of the spine, does not apply, and the pair has to be shortened as it stands.

Example 10.1 (A mortal hole). Let $n = 2$ with f covariant, let $X = \{x, y, a, b, c\}$ and $Z = \varnothing$, and

$$\varphi = \{x \le f(x, b),\ b \le f(a, a),\ a = \bot,\ f(y, c) \le y\}, \qquad \psi = \{x \le f(y, y)\}.$$

Both systems are satisfiable. Condition (I0) holds, since ψ has no facts, and so does (U): the only atom that is root-dead in ψ is its forbidden root (x^0, ε), which is also a forbidden root of φ and hence dead in φ. Consider $I_{y^1 x^1}$.

- The only fact of φ is (a^1, ε), and $\mathrm{Fact}^{\uparrow}_{\varphi} = \{a^1\}$. The spine automaton starts at $\downarrow_\varphi x^1 = \{x^1\}$, loops there on 0, moves to $\{b^1\}$ on 1, and from $\{b^1\}$ to the bad state $\{a^1\}$ on either letter; so $\{b^1\}$ is mortal. In terms of trees, $x \le f(x, b)$, $b \le f(a, a)$ and $a = \bot$ force a leaf $\bot$ of x at a prefix of every word $0^k 1 i$.
- The φ-component of the target automaton is $\{y^1\}$ along 0^*, $\{c^1\}$ after the first letter 1, and empty afterwards. The atoms $(y^1, 0^k)$ are dead by support, because (y^1, ε) is a forbidden root of φ and $(y^1, 0^k)$ pops to it, while no atom (y^1, w) is root-dead in ψ. So $J(t)$ is defined iff t contains the letter 1, and it is then the cut after the first 1.
- The guard and the self guard never apply, because the sets that they intersect are disjoint.
- The pop head is empty: ψ derives $(x^1, 0\sigma)$ from (y^1, σ) by a single push, and the head events are the pairs $(1, 0)$ with $u[1{:}] = t$.

Hence the holes of $I_{y^1 x^1}$ are exactly the pairs $(0^{j+1}1, 0^j 1)$ with $j \ge 0$, all of them mortal, with $J(t) = |t|$; the spine and the common suffix $0^j 1$ are unbounded. These pairs are also the only pre-holes, and their spines end in the mortal state, so the instance has holes, but only in case (3) of Theorem 9.17. In the corresponding failure of (II), ψ derives $(x^1, 0^{j+1}1)$ from the live seed $(y^1, 0^j 1)$, but the least model of φ that contains the seed does not contain $(x^1, 0^{j+1}1)$. For $j = 0$, the solution $a = c = \bot$, $b = f(\bot, \bot)$, $x = f(f(\bot, b), b)$, $y = f(\top, \bot)$ of φ is not a solution of ψ: $x \le f(y, y)$ would require $f(\bot, b) \le f(\top, \bot)$, hence $b \le \bot$.

In a mortal hole the target ends shortly after its first rejected cut, but the spine and the common suffix of a head event can be arbitrarily long, as in the example. They cannot be shortened independently: events relate cuts of the spine to cuts of the target, self events relate two cuts of the target, and a head event demands that two suffixes be *equal*. We proceed in four steps.

(1) We pack the pair into the *separator word* $u\sharp t$ over $\Sigma \sqcup \{\sharp\}$. Events and self events become comparisons of this word, and a head event becomes an *anchored* comparison, one whose matched factor is followed by the separator. Mortal holes are then exactly the Π-*holes* of one reader, its holes with an anchored comparison whose state pair lies in a relation Π. Mortality and the short rejected tail enter through the reader itself, and self events enter through its admission relation (Sections 10.1 and 10.2).
(2) The short rejected tail bounds the depth of the separator word, by the depth theorem of Zhao [2026] (Section 10.3).
(3) We attach to each level of the hierarchy a *positive channel*, which records the anchored witness in the form in which it appears at that level, and show that channels have derivatives exactly as lettered monitors do (Section 10.4). The bounded-depth search of that paper does not look at the channel and can lose the witness (Example 10.13); once its exchange types, its shortening of the top level and its column summaries record the channel, it gives the anchored compression theorem (Section 10.5).
(4) Combining the three steps bounds the length of a shortest mortal hole by a doubly exponential function of the automaton state counts, hence a triply exponential function of s for extracted instances, which decides the hole problem (Section 10.6).

Condition (FH) is used only in the last theorem (Remark 10.20). The proofs are in Appendix F.

10.1 Anchored Comparisons

Fix a letter $\sharp$, the *separator*. Comparisons in this section are comparisons of one word, in the end the separator word $u\sharp t$ of Section 10.2, which is a different word from the pair (u, t); we therefore write them (x, y) (here x and y are cuts, not variables), whereas the cross comparisons (s, e) of Definition 8.2 relate cuts of the two words of the pair. For a comparison (x, y) of a word w, the factor $w[x{:}x + |w| - y] = w[y{:}]$ is its *matched factor*, and the letter $w[x + |w| - y]$ that follows the matched factor inside $w[x{:}]$ is its *continuation letter*; it exists because $x < y$. For a terminal comparison $(x, |w|)$ the matched factor is empty and the continuation letter is $w[x]$.

Definition 10.2 (Anchored comparison, Π-hole). A comparison of a word w is *anchored* if its continuation letter is $\sharp$. Let $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ be a reader with states Q over a finite alphabet $\Gamma \ni \sharp$ (Definition 2.11), and let $\Pi \subseteq Q \times Q$. A Π-*hole* of the reader is a nonempty hole w (Definition 2.12) that has an anchored comparison (x, y) with $(q_x, q_y) \in \Pi$.

If w contains $\sharp$ exactly once, at the position m, then a pair of cuts (x, y) is an anchored comparison iff

$$x \le m, \qquad y = x + |w| - m, \qquad w[y{:}] = w[x{:}m].$$

Indeed, the continuation letter of a comparison (x, y) is $\sharp$ iff $x + |w| - y = m$, and then the matched factor $w[y{:}]$ has length $m - x$, so the prefix relation $w[y{:}] \preceq w[x{:}]$ says $w[y{:}] = w[x{:}m]$; conversely these three conditions make (x, y) a comparison, since $x < y$ and $w[x{:}m]$ is a prefix of $w[x{:}]$. The terminal pair $(m, |w|)$ is always anchored, with empty matched factor. Every other anchored comparison matches a nonempty factor that ends immediately before the separator with a suffix of the whole word.

The equation $y = x + |w| - m$ is what matters. A comparison (x, y) with $y > m$ only says that the suffix after y is a *prefix* of the suffix after x; the equation forces the two matched factors to have the same length, and the prefix relation becomes an equality. In the separator word this is exactly the difference between an event, a prefix relation $t[e{:}] \preceq u[x{:}]$, and a head event, an equality $t[e{:}] = u[x{:}]$ (Lemma 10.4). A compression that preserves only "some comparison with a state pair in Π" is therefore not enough.

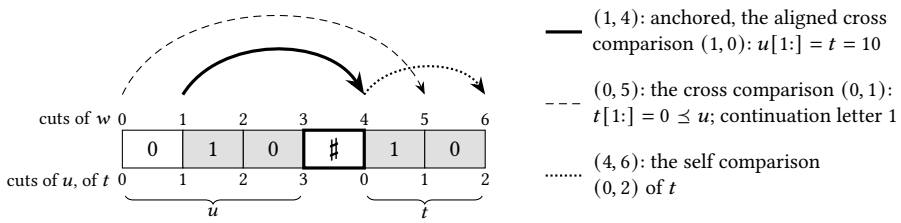


Fig. 12. The separator word $w = u\sharp t$ for $u = 010$ and $t = 10$, the word of Example 10.12; each arrow points from the first cut of a comparison to its second. The cut $x \le 3$ of w is the cut x of u, and the cut $4 + e$ is the cut e of t. Every comparison of w joins a cut of u to a cut of t, as the cross comparison $(0,5)$ does, or two cuts of t, as $(4,6)$ does (Lemma 10.4). The comparison $(1,4)$ is anchored: its matched factor $w[1{:}3] = 10$ (shaded) ends right before the separator and equals the suffix $w[4{:}]$. This is how a head event, an equality of two suffixes, appears in w; a cross comparison that is not aligned, such as $(0,5)$, has a continuation letter inside u. The anchored comparisons of w are $(1,4)$, $(2,5)$ and $(3,6)$.

Example 10.3 (An unanchored positive comparison). Over $\Gamma = \{0, 1, \sharp\}$ take the reader whose states $0, \dots, 4$ form the prefix chain of $01\sharp 0$, that is, $\delta(0,0) = 1$, $\delta(1,1) = 2$, $\delta(2,\sharp) = 3$ and $\delta(3,0) = 4$, with all other transitions into a sink 5. Let $\mathrm{Rej} = \{4\}$, $\mathrm{Adm} = \varnothing$ and $\Pi = \{(0,3)\}$. The only rejected word is $w = 01\sharp 0$, so it is the only hole. It has the comparison $(0,3)$, since $w[3{:}] = 0 \preceq w$, with state pair $(0,3) \in \Pi$, but the continuation letter of $(0,3)$ is $w[1] = 1$. The only anchored comparison of w is the terminal pair $(2,4)$, whose state pair $(2,4)$ is not in Π. Hence the reader has no Π-hole, although its only hole has a comparison labeled by Π.

10.2 The Separator Word

For words $u, t \in \Sigma^*$ put $w = u\sharp t$, a word over $\Gamma = \Sigma \sqcup \{\sharp\}$, and $m = |u|$, so that $|w| = |u| + |t| + 1$. The cuts $x \le m$ of w are the cuts of u, and the cut $m + 1 + e$ of w is the cut e of t.

Lemma 10.4 (Comparisons of the separator word). *Let $u, t \in \Sigma^*$, $w = u\sharp t$ and $m = |u|$.*

(i) No pair (x, y) with $y \le m$ is a comparison of w.
(ii) For $x \le m$ and $e \le |t|$, the pair $(x, m + 1 + e)$ is a comparison of w iff (x, e) is a cross comparison of (u, t).
(iii) For $i < j \le |t|$, the pair $(m + 1 + i, m + 1 + j)$ is a comparison of w iff (i, j) is a comparison of t.
(iv) For $x \le m$ and $e \le |t|$, the pair $(x, m + 1 + e)$ is an anchored comparison of w iff (x, e) is an aligned cross comparison of (u, t), that is, iff $u[x{:}] = t[e{:}]$.

So every comparison of w either joins a cut of the spine to a cut of the target, and is then a cross comparison, or joins two cuts of the target (Figure 12). The separator keeps the two words apart: every matched factor is a suffix of t, so none runs across the separator, and a matched factor that starts in u stops either inside u or right before the separator; the comparisons of the second kind are the anchored ones. The proof is in Appendix F.1. All conditions of a mortal hole, mortality included, are therefore read by one reader, which we now construct.

The reader $\mathcal{R}_I^\Pi$. Mortality of a safe state is decided by reachability in the safe graph of A (Definition 9.5). The reader $\mathcal{R}_I^\Pi$ over $\Gamma = \Sigma \sqcup \{\sharp\}$ reads $u\sharp t$ in three phases.

- On the spine, its states $\mathrm{pre}(q, q')$, with $q \in S_A$ and $q' \in S_{A'}$, follow A and A'; only safe states q are reached.
- On the target up to its first rejected cut, its states $\mathrm{act}(r, r')$, with $r \in S_B$ and $r' \in S_{B'}$, follow B and B'.

- After the first rejected cut, its states $\mathrm{done}(r', k)$, with $r' \in S_{B'}$ and $1 \le k \le N_A$, follow B' and count the letters read, capped at N_A.

A state sink absorbs every failure. The automata A, B, A', B' start in their state 0 (Definition 8.1); the start state of $\mathcal{R}_I^\Pi$ is $\mathrm{pre}(0,0)$, or sink if $0 \in \mathsf{Bad}$. For a letter $i \in \Sigma$ the transitions are

$$\mathrm{pre}(q,q') \xrightarrow{i} \mathrm{pre}(A(q,i), A'(q',i)) \text{ if } A(q,i) \notin \mathsf{Bad}, \text{ else sink},$$

$$\mathrm{pre}(q,q') \xrightarrow{\sharp} \mathrm{act}(0,0) \text{ if } q \text{ is mortal, else sink},$$

$$\mathrm{act}(r,r') \xrightarrow{i} \mathrm{done}(B'(r',i),1) \text{ if } r \in \mathsf{Rej}, \text{ else } \mathrm{act}(B(r,i), B'(r',i)),$$

$$\mathrm{done}(r',k) \xrightarrow{i} \mathrm{done}(B'(r',i), \min(k+1, N_A)),$$

and every other transition, in particular every transition on $\sharp$ after the separator, leads to sink. The rejected states are $\mathrm{act}(r,r')$ with $r \in \mathsf{Rej}$ and $\mathrm{done}(r',k)$ with $k \le N_A - 1$. The admission relation consists of the pairs

$$\big(\mathrm{pre}(q,q'), \mathrm{act}(r,r')\big) \text{ with } (q,r) \in G, \qquad \big(\mathrm{act}(r_1,r_1'), \mathrm{act}(r_2,r_2')\big) \text{ with } (r_1,r_2) \in G_{\mathrm{s}},$$

and no others. The relation Π relates $\mathrm{pre}(q,q')$ to $\mathrm{act}(r,r')$ and to $\mathrm{done}(r',k)$ iff $(q',r') \in H$, and relates no other pairs.

Thus $\mathcal{R}_I^\Pi$ reads the separator only from a mortal state, rejects a word only if fewer than N_A letters follow the first rejected cut of the target, and lets Π compare the head automata on the two sides of the separator. The target automaton is kept up to and including the first rejected cut, because events and self events may end there; after it, events and self events no longer exist (Definition 8.2), so only B' and the counter remain. The admissions from the self guard G_{s} are the only difference from a reader for instances without self events; the compression below allows an arbitrary admission relation.

We count all states of $\mathcal{R}_I^\Pi$, reachable or not, and put

$$\bar{N}_I = 1 + N_A N_{A'} + N_B N_{B'} + N_A N_{B'}, \qquad d_I = 2\bar{N}_I N_A - 1.$$

Lemma 10.5 (The Π-reader). *The reader $\mathcal{R}_I^\Pi$ and the relation Π are computable from I, and:*

(i) $\mathcal{R}_I^\Pi$ has exactly $\bar{N}_I$ states;

(ii) a word w over $\Sigma \sqcup \{\sharp\}$ is a Π-hole of $\mathcal{R}_I^\Pi$ iff $w = u\sharp t$ with $u, t \in \Sigma^*$ and (u,t) a mortal hole of I;

(iii) for such w the first rejected cut is $|u| + 1 + J(t)$, so the rejected tail of w is $|t| - J(t)$.

On $u\sharp t$, the state at a cut $x \le |u|$ carries $A_x(u)$ and $A'_x(u)$ while u is safe, and the state at the cut $|u| + 1 + e$ carries $B_e(t)$ up to $J(t)$ and $B'_e(t)$ throughout. By Lemma 10.4, the admitted comparisons are then exactly the events and the self events, and Π on the anchored comparisons is exactly the head condition. The proof is in Appendix F.1.

10.3 The Depth of the Separator Word

The length bound of the bounded-depth search grows with the depth of the word (Theorem 2.20). For a mortal hole, the depth of the separator word is bounded at once.

Lemma 10.6 (Depth of mortal holes). *Every mortal hole (u,t) of I satisfies* $\mathrm{depth}(u\sharp t) \le d_I$.

Proof. By Lemma 10.5, $u\sharp t$ is a nonempty rejected word of $\mathcal{R}_I^\Pi$, a reader with $\bar{N}_I$ states, and its rejected tail is $|t| - J(t) \le N_A - 1$. Theorem 2.19 gives $\mathrm{depth}(u\sharp t) \le 2\bar{N}_I((N_A - 1) + 1) - 1 = d_I$. □

The bound is coarse, since it involves the spine and head automata through $\bar{N}_I$, but it enters the final estimate only through a polynomial in $\bar{N}_I$ and N_A, and changes only the constant in the exponent (Lemma 10.18).

10.4 Positive Channels

The bounded-depth search sees each level of the hierarchy through a lettered monitor, the derivative of the monitor one level below (Definitions 2.22 and 2.23); the derivatives carry the negative conditions, because a comparison of one level reappears at the next as a comparison of blocks or as a test in the final target. We carry the positive condition in the same way. At level 0 it is an anchored comparison in Π, a comparison together with a test of its continuation letter; one level up, the continuation letter is the first letter of the expansion of a block, or a letter inside the last block, and the comparison may have become a test of the final label. Five components cover all the forms that arise.

Definition 10.7 (Positive channel). Let $\mathcal{M}$ be a lettered monitor over an alphabet Γ with state set Q. A *positive channel* $\Theta = (F, R, S, S', \Phi)$ on $\mathcal{M}$ consists of a predicate $F \subseteq Q \times \Gamma$, three relations R, S and S' on labels, and a set $\Phi \subseteq \Gamma$ of letters. For a nonempty word v with labels $L_0, \dots, L_{|v|}$, the *positive condition* $\mathrm{Pos}_\Theta(v)$ holds iff one of the following clauses holds:

(F) $F(L_{|v|})$;
(R) some comparison (x, y) of v with $y < |v|$ has $R(L_x, L_y)$ and $v[x + |v| - y] \in \Phi$;
(S) some cut $x < |v|$ has $S(L_x, L_{|v|})$;
(S′) some cut $x < |v|$ has $S'(L_x, L_{|v|})$ and $v[x] \in \Phi$.

A *positive hole* of $(\mathcal{M}, \Theta)$ is a hole v of $\mathcal{M}$ with $\mathrm{Pos}_\Theta(v)$.

Clauses (R) and (S′) together say that some comparison has its labels in a relation, R if it is not terminal and S' if it is, and its continuation letter in Φ; the continuation letter of the terminal comparison $(x, |v|)$ is $v[x]$. Clauses (F) and (S) are tests without a continuation letter. They are empty at level 0, but derivatives move witnesses into them, as the derivatives of lettered monitors move admissions into the terminal relation and the final target.

For a reader $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ over $\Gamma \ni \sharp$ with start q_0 and $\Pi \subseteq Q \times Q$, the *base channel* on the lettered monitor of the reader (Definition 2.22) is

$$\Theta_\Pi = (\varnothing, \widehat{\Pi}, \varnothing, \widehat{\Pi}, \{\sharp\}), \qquad \widehat{\Pi}(L, L') \iff (\mathrm{den}\, L, \mathrm{den}\, L') \in \Pi.$$

Lemma 10.8 (The base channel). *A nonempty word w satisfies $\mathrm{Pos}_{\Theta_\Pi}(w)$ iff it has an anchored comparison (x, y) with $(q_x, q_y) \in \Pi$. Hence the Π-holes of the reader are exactly the positive holes of its lettered monitor with the base channel.*

Proof. Since $\mathrm{den}\, L_i = q_i$, clause (R) is an anchored comparison (x, y) with $y < |w|$ and $(q_x, q_y) \in \Pi$, and clause (S′) is the same for a terminal comparison $(x, |w|)$, whose continuation letter is $w[x]$; clauses (F) and (S) are empty. The reader and its lettered monitor have the same nonempty holes. □

The derivative of a channel uses the shifted blocks of Definition 2.23. The letter that follows the position g of the shifted block of X is $(X\ell)[g]$: the letter $X[g]$ for $g < |X|$, and the marker that opens the next shifted block for $g = |X|$.

Definition 10.9 (Derivative of a channel). Let $\Theta = (F, R, S, S', \Phi)$ be a positive channel on $\mathcal{M}$ and $\ell \in \Gamma$. The *derivative* $\Theta_\ell = (F_\ell, R_\ell, S_\ell, S'_\ell, \Phi_\ell)$ is the positive channel on $\mathcal{M}_\ell$ given by

$$\begin{aligned}
\Phi_\ell &= \{X : (X\ell)[0] \in \Phi\},\\
R_\ell((\zeta, X), (\zeta', X')) &\iff X = xy,\ X' = x'y,\ R\big(lab_{|x|}(\zeta, X), lab_{|x'|}(\zeta', X')\big) \text{ for some } x, x', y,\\
S'_\ell &= R_\ell,\\
S_\ell((\zeta, X), (\zeta', X')) &\iff pt(\zeta, X; \delta_\ell(\zeta', X')),\\
F_\ell(\zeta, X) &\iff F(\delta_\ell(\zeta, X), \ell) \text{ or } pt(\zeta, X; \delta_\ell(\zeta, X)),
\end{aligned}$$

where the *positive terminal test* is

$$pt(\zeta, X; \zeta'') \iff \text{for some } g \le |X| :\ S\big(lab_g(\zeta, X), (\zeta'', \ell)\big),\\ \text{or } S'\big(lab_g(\zeta, X), (\zeta'', \ell)\big) \text{ and } (X\ell)[g] \in \Phi,$$

and every relation with the start label $\triangleright$ of $\mathcal{M}_\ell$ as an argument is false.

The relation R_ℓ is built from R exactly as the derived admission Λ_ℓ is built from Λ, and the positive terminal test is the analog of the terminal test in $\Lambda_\ell^{\mathrm{fin}}$ and T_ℓ^{fin}: it relates a position of a block, possibly the closing marker, to the final label, which is (ζ'', ℓ) when the final marker is read from the state ζ''. The predicate F concerns only the final label, so it enters F_ℓ but not S_ℓ. The set Φ_ℓ tests the first letter of the expansion of a block, because the continuation letter of a comparison of the expansion is the first letter of the expansion of the continuation letter of the corresponding comparison of blocks (Lemma F.1). This is how the anchoring equation travels up the hierarchy: "followed by $\sharp$" at level 0 becomes "followed by a block whose expansion starts with $\sharp$" at level 1, and so on.

Lemma 10.10 (Positive desubstitution). *Let Θ be a positive channel on $\mathcal{M}$, let $\ell \in \Gamma$, and let v be the expansion of a nonempty word v' of blocks avoiding ℓ. Then* $\mathrm{Pos}_\Theta(v) \iff \mathrm{Pos}_{\Theta_\ell}(v')$. *Consequently v is a positive hole of $(\mathcal{M}, \Theta)$ iff v' is a positive hole of $(\mathcal{M}_\ell, \Theta_\ell)$.*

The second claim follows from the first and Lemma 2.24. The proof of the first, in Appendix F.2, translates each clause by the label facts Eq. (4) and the alignment lemma (Lemma 2.17): a nonterminal comparison of v becomes a comparison of blocks, read by R_ℓ or, if it ends at the last cut, by S'_ℓ; a terminal comparison of v that starts in the block X_i becomes a test of S_ℓ or, if X_i is the last block, of F_ℓ.

Corollary 10.11 (Channel towers). *Let $\mathcal{M}_0$ be a lettered monitor with N states, Θ_0 a positive channel on it, and w a nonempty word with canonical hierarchy (w_j, ℓ_j). Put $\mathcal{M}_{j+1} = (\mathcal{M}_j)_{\ell_j}$ and $\Theta_{j+1} = (\Theta_j)_{\ell_j}$. Then w is a positive hole of $(\mathcal{M}_0, \Theta_0)$ iff w_j is a positive hole of $(\mathcal{M}_j, \Theta_j)$, for every j.*

Proof. Each w_{j+1} is a nonempty word of blocks avoiding ℓ_j, with expansion w_j. Apply Lemma 10.10 at each level. □

Neither the lemma nor the corollary assumes that $\sharp$ occurs once, or at all, and the reader may have an arbitrary admission relation.

Example 10.12 (A witness travels up the tower). Take $\Gamma = \{0, 1, \sharp\}$, the five states $0, \dots, 4$ with start 0, the transitions

state	0	1	2	3	4
on 0	3	0	0	3	0
on 1	1	1	4	0	2
on $\sharp$	4	3	2	4	0

Adm $= \varnothing$, Rej $= \{0, 1\}$ and $\Pi = \{(3, 4), (4, 2)\}$. The word $w = 010\sharp10$ has the run $0, 3, 0, 3, 4, 2, 0$ and ends in a rejected state, so it is a hole. Its anchored comparisons are $(1, 4)$, $(2, 5)$ and $(3, 6)$, with state pairs $(3, 4)$, $(0, 2)$ and $(3, 0)$. Only the first, which matches the factor 10 before the separator with the final 10, is in Π, so w is a Π-hole; for the base channel, its witness $(1, 4)$ is clause (R). The hierarchy of w is $w_1 = abc$ with $a = [\varepsilon]$, $b = [1]$ and $c = [\sharp1]$, and $w_2 = [ab]$, a single letter. At level 1, with the marker 0, the cut 1 lies in the block $X_1 = 1$ and the cut 4 in the last block $X_2 = \sharp1$, both with the common suffix 1; so the comparison of blocks is the terminal pair $(2, 3)$ of w_1, and the witness becomes clause (S′) at the cut 2, with continuation letter c, which lies in Φ_1 because its expansion starts with $\sharp$. At level 2, with the marker c, the cut 2 of w_1 is the end of the only block

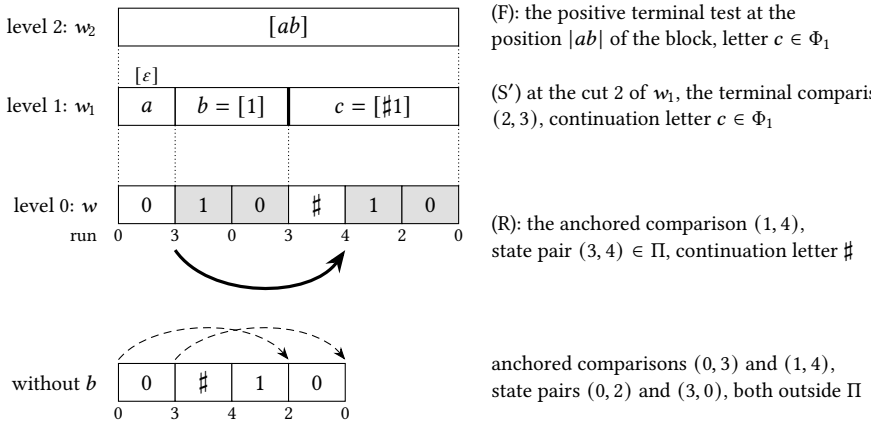


Fig. 13. A positive witness travels up the hierarchy (Examples 10.12 and 10.13). Bottom up: the hole $w = 010\sharp10$ of the reader of Example 10.12 with its run, the word $w_1 = abc$ of blocks for the marker 0, each letter drawn over its block and the marker that closes it, and the one-letter word $w_2 = [ab]$ for the marker c. The anchored witness $(1, 4)$ in Π, whose matched factor 10 is shaded, is clause (R) of the base channel at level 0, clause (S′) of the derived channel at level 1, and clause (F) at level 2, where it is a test of the final label (Definition 10.7 and Lemma 10.10). The compression for plain entailment sees no difference between the prefixes a and ab of the block and deletes the column of b, which carries the matched factor 10; the result $0\sharp10$ (last row) is still a hole, but no longer a Π-hole. Recording the channel in the summaries keeps b.

ab of w_2; so the witness becomes the second disjunct of the final target F_2, through the positive terminal test at the position $|ab|$, whose letter is the marker $c \in \Phi_1$. Clause (R) at level 0 has become clause (S′) at level 1 and clause (F) at level 2.

Figure 13 draws the three levels of this example. Each level sees the witness only through its own letters: once the matched factor 10 has become the single letter b and the separator the first letter of c, the comparison turns into a test at the boundary between b and c, and one level higher into a test of the final label.

10.5 Anchored Compression

What the compression for plain entailment misses. Steps 2 and 4 of the bounded-depth search (Section 2.6, and precisely Appendix A.4) replace the word by another one: step 2 merges isolated letters with the same exchange type, and step 4 deletes the columns between two cuts of a column word with equal summaries. Each of them only has to keep the word a hole. A Π-hole must also *contain* an anchored comparison in Π. The exchange types and summaries record nothing about Π, so a column deletion, and similarly a merge, can destroy every such comparison.

Example 10.13 (The compression for plain entailment loses the anchored witness). Continue Example 10.12. Step 4 shortens the block ab over the level-1 monitor with the marker c. In this monitor c leads every state to 2, and the exits of the shifted blocks of the prefixes a and ab are 0 from every entry state; since Adm is empty, all terminal and admission bits are false. So the summaries after a and after ab coincide, and step 4 deletes the column holding b. Expanding $[a]$ gives $0\sharp10$ (Figure 13, last row), with run $0, 3, 4, 2, 0$: a hole, whose anchored comparisons $(0, 3)$ and $(1, 4)$ have the state pairs $(0, 2)$ and $(3, 0)$, both outside Π. The deleted letter b is exactly the letter whose expansion is the matched factor 10 before the separator.

The bound of the next theorem is an explicit function $\mathsf{Len}^\sharp(N,d)$. Like $\mathsf{Len}(N,d)$ of Zhao [2026], it is built from constants $\mathsf{C}^\sharp$, $\mathsf{A}^\sharp$, $\mathsf{G}^\sharp$ and pcol, which bound the anchored exchange types, the protected and isolated letters, the dictionaries and the positive column summaries. They are defined in Eq. (17) and do not depend on the alphabet; the theorem states their closed form.

Theorem 10.14 (Anchored compression). *Let $(\mathcal{M}, \mathsf{Adm}, \mathsf{Rej})$ be a reader with $N \geq 1$ states over a finite alphabet $\Gamma \ni \sharp$, let $\Pi \subseteq Q \times Q$, and let $d \geq 0$. If the reader has a Π-hole of depth at most d, then it has a Π-hole of depth at most d and length at most $\mathsf{Len}^\sharp(N,d)$. More generally, for every lettered monitor $\mathcal{M}_0$ with N states over any alphabet and every positive channel Θ_0 on it, if $(\mathcal{M}_0, \Theta_0)$ has a positive hole of depth at most d, then it has one of depth at most d and length at most $\mathsf{Len}^\sharp(N,d)$. Moreover, with $\Xi^\sharp(N,d) = 6N^2 + 10N(N+d)(d+1) + 7N + 3d + 4$,*

$$\mathsf{G}^\sharp(N,d) \leq 2^{\Xi^\sharp(N,d)} \qquad \text{and} \qquad \mathsf{Len}^\sharp(N,d) \leq 2^{2^{5\Xi^\sharp(N,d)+6}}.$$

The theorem is the analog of Theorem 2.20 for Π-holes. It preserves the *existence* of an anchored comparison in Π, not its coordinates or its state pair; to keep a particular state pair, apply it to the singleton relation of that pair. It needs no uniqueness of the separator. The first claim is the second one for the lettered monitor of the reader and the base channel, by Lemma 10.8.

Why recording the channel suffices. The positive condition is a property of the labels at the cuts of a witness and of one letter, the continuation letter. The shortenings of Zhao [2026] (steps 2–4 in Section 2.6) preserve such data once they record the channel. The merges preserve the state run, the comparison positions, and the relevant channel tests; the top shortening preserves the labels of the cuts of the witness, which it never deletes, and the letter, which is constant on a unary word; and the summaries determine the derived channel, so that agreement preserves labels and letters up to renaming. This is why recording the channel suffices, with no further condition on the separator.

Proof outline. The complete proof is in Appendix F.8. The proof of Theorem 2.20 merges letters with equal exchange types and deletes columns between equal summaries. We add the five components of the channel to both, so that each shortening keeps some anchored witness, though not necessarily the original one; every argument about the negative conditions is reused unchanged. Let w be a positive hole of depth $D \leq d$, with hierarchy (w_j, ℓ_j); we follow the five steps of that proof (Section 2.6).

(1) *Towers.* By Corollary 10.11, w_j is a positive hole of $(\mathcal{M}_j, \Theta_j)$ for every j; the clause of the witness may change from level to level (Example 10.12).

(2) *Merging letters.* In the terms of step 2 in Section 2.6, a letter of w_i is unprotected if it is neither the marker nor a letter of the start prefix, and isolated if, moreover, all its unequal comparison partners are protected; its exchange type is the table of how $\mathcal{M}_i$ treats it when it is read from an ordinary state. The *anchored exchange type* of an unprotected letter a at a level $1 \leq i < D$ adds the same table for the channel: the row of F_i at a, the self table, the protected rows and the initial row of each of R_i, S_i and S_i', and the bit $[a \in \Phi_i]$ (Definition F.2). There are at most $\mathsf{C}^\sharp(N,d)$ anchored types (Lemma F.3). Merging two isolated letters with the same anchored type keeps the comparisons and the run of the level, so it keeps every value of Λ_i, Λ_i^{fin}, R_i, S_i and S_i' at the comparisons of the level, exactly as in the argument of Zhao [2026] for Λ_i and Λ_i^{fin}; it keeps the final label, because the marker is protected; and it keeps the position of every continuation letter and changes that letter at most from a to b, which have the same bit. So the merged level is again positive, and expanding it gives a positive hole of $(\mathcal{M}_0, \Theta_0)$ of the same depth, no longer, with a smaller dictionary (Lemma F.4). Repeated merges bound the dictionary,

the set of letters, of every level $i \geq 1$ by $\mathsf{G}^\sharp(N, d)$ (Lemma F.5). A merge is not injective and gives no agreement; after each merge the hierarchy and the towers are recomputed.

(3) *The top level.* The top level w_D is unary. Deleting a loop between two cuts with equal states keeps a hole, since every pair of cuts of a unary word is a comparison, and it keeps the positive witness if no cut of the witness is deleted. Protecting the at most three cuts of one witness leaves at most three stretches of pairwise distinct states, so $|w_D| \leq 3(N + D)$ after shortening (Lemma F.6).

(4) *Deleting columns.* The *positive summary* of a tuple of blocks consists of the summary of Appendix A.4 for the monitor, a second such summary for the monitor that takes F, R and S as its final target and its two relations, the S'-tests with a known next letter, the last pending S'-test (a test at the end of a block, whose letter is the next one to be read), and, for each block, whether it is empty and otherwise the bit Φ of its first letter. These fields are updated column by column and determine the derived channel on the blocks, in particular S_ℓ without an F-term and F_ℓ with one (Lemma F.8). Deleting the columns between two cuts with equal positive summaries therefore keeps both the derived monitor and the derived channel (Lemma F.10). In Example 10.13, the pending S'-test is false after a and true after ab, where it records the witness $(1, 4)$; so the positive summaries after a and after ab differ, and the column holding b is kept (Example F.9).

(5) *Agreement and the bound.* Agreement of monitors on the letters in use (step 4 in Section 2.6) is extended to the five components of the channel, which must also take equal values at corresponding letters and labels. Extended agreement preserves positive holes and is preserved under derivation (Lemma F.7). The supported compression of Zhao [2026], which shortens the letters in use level by level from the top, then runs unchanged with the positive summaries (Lemma F.11). Each of the at most d expansions costs a factor $\mathrm{pcol}(N + d + 1, \mathsf{G}^\sharp(N, d))$, and the top level costs $3(N + d + 1)$, which gives $\mathsf{Len}^\sharp(N, d)$. The new top word is a positive hole of the new top monitor, and Lemmas 2.24 and 10.10 reconstruct a positive hole of $(\mathcal{M}_0, \Theta_0)$ of depth at most D.

The closed-form estimate is Lemma F.12. □

10.6 Short Mortal Holes and Decidability

Put

$$\mathsf{Len}_I^\sharp = \mathsf{Len}^\sharp(\bar{N}_I, d_I),$$

with $\bar{N}_I$ and d_I as in Section 10.2, and recall the bound L_{nm} of Theorem 9.17, evaluated at the state counts $(N_A, N_B, N_{A'}, N_{B'})$.

Theorem 10.15 (Short mortal holes). *If I has a mortal hole, then it has a mortal hole (u, t) with $|u| + 1 + |t| \leq \mathsf{Len}_I^\sharp$. No assumption on I is needed; in particular (FH) is not used.*

Proof. Let (u, t) be a mortal hole. By Lemma 10.5, $w = u\sharp t$ is a Π-hole of $\mathcal{R}_I^\Pi$, a reader with $\bar{N}_I$ states over the finite alphabet $\Sigma \sqcup \{\sharp\}$. By Lemma 10.6, $\mathrm{depth}(w) \leq d_I$. By Theorem 10.14, $\mathcal{R}_I^\Pi$ has a Π-hole $\tilde{w}$ of length at most $\mathsf{Len}_I^\sharp$. By Lemma 10.5(ii), $\tilde{w} = \tilde{u}\sharp\tilde{t}$ with $(\tilde{u}, \tilde{t})$ a mortal hole, and $|\tilde{u}| + 1 + |\tilde{t}| = |\tilde{w}|$. □

Corollary 10.16 (Mortal holes are decidable). *It is decidable whether an instance of the two-word hole problem with self events has a mortal hole.*

Theorem 10.17 (Short holes). *Let I satisfy (FH). Then I has a hole iff it has a hole (u, t) with*

$$|u| + |t| \leq \max\big(\mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'}),\ \mathsf{Len}_I^\sharp\big).$$

In particular, it is decidable whether an instance with forced target heads has a hole.

Proof. By Corollary 9.18, I has a hole iff it has a hole with $|u| + |t| \le \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'})$ or a mortal hole. In the second case Theorem 10.15 gives a mortal hole, which is a hole, with $|u| + |t| < \mathsf{Len}^{\sharp}_I$. The bound is computable from I, and the alphabet Σ is finite, so there are finitely many candidate pairs; for each, the conditions of Definition 8.1 are checked over finitely many cuts. □

The proof of Corollary 10.16 is the same enumeration with the bound of Theorem 10.15 and the additional tests that $A(u)$ is mortal and $|t| - J(t) < N_A$ (Appendix F.9). Both bounds are independent of the alphabet, and hence of the arity; the alphabet enters only the number of candidate pairs. For $n = 0$ the only candidate is $(\varepsilon, \varepsilon)$.

The size of the bound. Write $E_1(x) = 2^x$ and $E_{k+1}(x) = 2^{E_k(x)}$.

Lemma 10.18 (Mortal bound). *Let $\alpha \ge 3$ and $\beta \ge 0$ be integers with $\bar{N}_I \le 2^\alpha$ and $N_A \le 2^\beta$. Then $\mathsf{Len}^{\sharp}_I \le E_3(3\alpha + 2\beta + 9)$.*

Corollary 10.19 (Mortal bound for extracted instances). *Let φ be a flat system over X and ψ a flat system over $X \cup Z$, with $v = |X|$, $h = |Z|$ and $s = v + h \ge 1$. Every extracted instance I_{pq} (Definition 8.5) satisfies*

$$\mathsf{Len}^{\sharp}_{I_{pq}} \le E_3(10v + 12s + 15) \le E_3(22s + 15),$$

for every arity n and every variance. So if I_{pq} has a mortal hole, it has one whose separator word has length at most $E_3(22s + 15)$.

Proof. By Section 8.2, $N_A = 2^{2v}$, $N_B = 2^{4v+2h+1} = 2^{2v+2s+1}$ and $N_{A'} = N_{B'} = 2^{2s}$. Hence $N_A N_{A'} = N_A N_{B'} = 2^{2v+2s}$ and $N_B N_{B'} = 2^{2v+4s+1}$, so $\bar{N}_I = 1 + 2^{2v+2s+1} + 2^{2v+4s+1} \le 2^{2v+4s+2}$, using $s \ge 1$. Lemma 10.18 with $\alpha = 2v+4s+2$ and $\beta = 2v$ gives $E_3(3(2v+4s+2)+4v+9) = E_3(10v+12s+15)$, and $v \le s$. The second claim is Theorem 10.15. None of these sizes depends on n. □

Remark 10.20 (Where (FH) is used). Condition (FH) is used only in Theorem 10.17, through Theorem 9.17. Every other result of this section, in particular Theorem 10.15 and Corollary 10.16, holds for every instance.

11 The Main Theorem

This section states the main theorem and proves it by assembling the results of Sections 2 to 10. Apart from bookkeeping, only two things are proved here: how a counterexample word yields a countermodel (Lemma 11.3), and two short estimates that turn the length bounds of the earlier sections into closed forms (Proposition 11.5).

Throughout, the arity $n \ge 0$ and the variance c are fixed. For a flat system φ over X and a flat system ψ over $X \cup Z$ we write

$$v = |X|, \qquad h = |Z|, \qquad s = v + h,$$

so that s is the size parameter of Section 2.3, counted after flattening. As in Section 10.6, $E_1(x) = 2^x$ and $E_{k+1}(x) = 2^{E_k(x)}$.

11.1 Statement

When the entailment fails although both systems are satisfiable, the failure is certified by a single word, which comes from one of the three conditions of Theorem 4.2.

Definition 11.1 (Counterexample words). Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, both satisfiable. A *counterexample word* of (φ, ψ) is a word of one of the following three kinds, together with the signed variables named in it.

(C1) An (I0)-*word*: a word w with $(q, w) \in \mu_\psi$ and $(q, w) \notin \mu_\varphi$ for some $q \in X^{\pm}$.
(C2) A (U)-*word*: a word w such that (p, w) is live in φ and $\mathrm{DeadU}_\psi(p, w)$ holds (Definition 6.7), for some $p \in X^{\pm}$.
(C3) A *hole word*, used only when (U) holds: the separator word $u\sharp t$ of a hole (u, t) of an extracted instance I_{pq} with $p, q \in X^{\pm}$ (Definition 8.5).

Its *length* is the length of the word, so a hole word has length $|u| + |t| + 1$.

By Proposition 6.16(2), an (I0)-word exists iff (I0) fails. By Lemma 7.1, a (U)-word exists iff (U) fails. If (U) holds, then by Theorem 8.7(1) a hole word exists iff (II) fails.

Theorem 11.2 (Main theorem). *Let $n \geq 0$ be an arity and c a variance of f.*

(1) Existential entailment $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ over $\{\bot, f, \top\}$ is decidable for each of the tree classes $\mathcal{D} = \mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$ and $\mathcal{T}_{\mathrm{fin}}$.
(2) If f is covariant, existential entailment over $\{\bot, f\}$ is decidable for each of the three tree classes.
(3) Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, with $v = |X|$ and $s = |X| + |Z|$, and let $\mathcal{D}$ be a tree class. If φ and ψ are $\mathcal{D}$-satisfiable and $\varphi \not\models_{\mathcal{D}} \exists \bar{z}\, \psi$, then (φ, ψ) has a counterexample word of length at most $E_3(10v + 12s + 15) \leq E_3(22s + 15)$. Every counterexample word determines at most two visible observations such that φ has solutions with values in $\mathcal{D}$ in which they hold, and no such solution has an extension with values in $\mathcal{D}$ to a solution of ψ.
(4) Let f be covariant, let φ and ψ be flat systems over $\{\bot, f\}$ with $v = |X|$ and $s = |X| + |Z|$, and let $\mathcal{D}$ be a tree class. If φ and ψ are $\mathcal{D}$-satisfiable over $\{\bot, f\}$ and the entailment fails over $\{\bot, f\}$ in $\mathcal{D}$, then the pair $(\varphi^\natural, \psi^\natural)$ of Theorem 5.17 has a counterexample word of length at most $E_3(20v + 24s + 15) \leq E_3(44s + 15)$, and every counterexample word of this pair determines a solution of φ over $\{\bot, f\}$ in $\mathcal{D}$ without an extension over $\{\bot, f\}$ in $\mathcal{D}$ to a solution of ψ.

The bounds in (3) and (4) depend neither on n nor on c.

Unflattened inputs are first flattened (Lemma 2.6), and s then counts the variables after flattening, as the bounds require (Section 2.3). The satisfiability hypotheses in (3) and (4) are needed only to make the certificate a word: if φ is not $\mathcal{D}$-satisfiable the entailment holds, and if φ is $\mathcal{D}$-satisfiable and ψ is not, every $\mathcal{D}$-solution of φ is a countermodel. The bounds concern certificates, not countermodels or witnesses: a countermodel is in general an infinite regular tree assignment (Lemma 11.3), and when the entailment holds the witnesses for the hidden variables may have to be infinite (Remark 5.15). Part (2) is not claimed for $\{\bot, f\}$ with a contravariant argument; the reduction of Theorem 5.17 needs covariance (Example 5.19).

11.2 The Decision Procedure

The input consists of the arity n, the variance c, the signature ($\{\bot, f, \top\}$, or $\{\bot, f\}$ with f covariant), a tree class $\mathcal{D}$, and two finite conjunctions φ and ψ of inequalities and equalities between type expressions, with visible variables X and hidden variables Z. The output is whether $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ holds and, if it fails, a certificate: a counterexample word when both systems are $\mathcal{D}$-satisfiable, and otherwise the failure of the $\mathcal{D}$-satisfiability test for ψ. Each step cites the result that justifies it; Table 6 lists the sections involved.

Step 1. *Flatten.* Replace φ and ψ by the flat systems of Lemma 2.6. The answer is unchanged in every tree class and for both signatures. From now on s counts the variables after flattening.
Step 2. *Arity zero.* If $n = 0$, the trees are $\bot < f() < \top$, all finite, and the three classes coincide (Remarks 2.8 and 5.20); over $\{\bot, f\}$ only $\bot$ and $f()$ remain. Enumerate the at most 3^s assignments of $X \cup Z$, check φ and ψ for each, and answer by Definition 2.4. Stop.

Table 6. The steps of the decision procedure, the results that justify them, and the sections where these are proved. The characterization (Theorem 4.2), built on the leaf cones of Section 3, underlies Steps 4 to 7.

Step	Task	Results used	Sections
Step 1	flatten	Lemma 2.6	Section 2
Step 2	arity zero	Remarks 2.8 and 5.20	Sections 2 and 5
Step 3	$\{\bot, f\}$	Lemma 5.16, Theorem 5.17, and Corollary 5.18	Section 5
Step 4	satisfiability	Proposition 6.16(1), Theorems A.1 and 2.9, Theorem 4.2(1)	Sections 2, 4 and 6, Appendix A
Step 5	(I0)	Lemma 6.6, Proposition 6.16(2)	Section 6
Step 6	(U)	Lemmas 7.5 and 7.6 and Theorem 7.9; Corollary 2.21	Sections 2 and 7
Step 7	(II)	Theorem 8.7, Corollary 8.8, and Lemma 8.6; Corollary 9.18; Theorems 10.15 and 10.17 and Corollary 10.19	Sections 8 to 10
Step 8	tree classes	Theorems 5.12 and 5.13 and Corollary 5.14	Section 5

Step 3. *The signature* $\{\bot, f\}$. If the signature is $\{\bot, f\}$, replace (φ, ψ) by $(\varphi^{\natural}, \psi^{\natural})$, with visible variables $X \cup Y_X$ and hidden variables $Z \cup Y_Z$, and continue over $\{\bot, f, \top\}$. The answer is unchanged in every tree class (Theorem 5.17); the size parameter doubles (Corollary 5.18).

Step 4. *Satisfiability in* $\mathcal{D}$. Decide whether φ and ψ are $\mathcal{D}$-satisfiable: for $\mathcal{D} = \mathcal{T}$ by Proposition 6.16(1), for $\mathcal{D} = \mathcal{T}_{\mathrm{reg}}$ by the same test, since $\mathrm{Sat}_{\mathrm{reg}} = \mathrm{Sat}$ (Theorem 2.9(1)), and for $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$ by Theorem 2.9(2). If φ is not $\mathcal{D}$-satisfiable, answer *holds*. If φ is $\mathcal{D}$-satisfiable and ψ is not, answer *fails*: every solution of φ in $\mathcal{D}$ is a countermodel. Otherwise both systems are satisfiable over arbitrary trees; continue.

Step 5. *Condition* (I0). For each $q \in X^{\pm}$, decide whether some word w has $(q, w) \in \mu_\psi$ and $(q, w) \notin \mu_\varphi$ (Proposition 6.16(2)). If so, answer *fails*, with the (I0)-word w.

Step 6. *Condition* (U). For each $p \in X^{\pm}$, test the words w of length at most $\mathsf{Len}(N_{\mathrm{U}}, d_{N_{\mathrm{U}}})$, with $N_{\mathrm{U}} = 2^{4|X|+2|Z|+1}$ as in Theorem 7.9, for (p, w) live in φ and $\mathsf{DeadU}_\psi(p, w)$, using the source automaton $\mathcal{B}_p$ and its self guard (Theorem 7.9 and Lemmas 7.3 and 7.5). If one passes, answer *fails*, with the (U)-word w.

Step 7. *Condition* (II). Now (I0) and (U) hold. For each $p, q \in X^{\pm}$, compute the extracted instance I_{pq} (Definition 8.5) and

$$K_{pq} = \max\big(\mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'}),\ \mathsf{Len}^{\sharp}_{I_{pq}}\big),$$

with the state counts of I_{pq}, and test the pairs (u, t) with $|u| + |t| \le K_{pq}$ for conditions (a)–(e) of Definition 8.1. This search finds a hole iff I_{pq} has one: by Lemma 8.6, I_{pq} has forced target heads, so Corollary 9.18 leaves only holes shorter than L_{nm} and mortal holes, and Theorem 10.15 shortens the latter (Theorem 10.17; numerically, Corollary 10.19). If some I_{pq} has a hole (u, t), answer *fails*, with the hole word $u\sharp t$; otherwise answer *holds* (Corollary 8.8).

Step 8. *Tree classes.* Return the answer. Steps 5 to 7 decide entailment over arbitrary trees, and their answer is the answer for $\mathcal{D}$: $E_{\mathrm{reg}} = E_{\mathrm{all}}$ (Theorem 5.12), and $E_{\mathrm{fin}} = E_{\mathrm{all}}$ when both systems have finite solutions (Theorem 5.13). A countermodel in $\mathcal{D}$ is obtained from a counterexample word by Lemma 11.3. Over $\{\bot, f\}$, a countermodel of $(\varphi^{\natural}, \psi^{\natural})$, including one returned in Step 4, is restricted to the original visible variables (Lemma 11.3(4)).

Condition (U) is decided in Step 6 before the holes of the extracted instances are used in Step 7. This order matters: Theorem 8.7 assumes (U), and without it a failure of (II) need not be a hole (Example 8.9). The bounded search of Step 7 itself needs only forced target heads, condition (FH), which every extracted instance satisfies by construction (Lemma 8.6); without (FH) the search would be incomplete (Example 9.20). Equivalently, after Step 4 one may test all candidate words of the three kinds of Definition 11.1 up to the bound of Theorem 11.2(3): a candidate hole of some I_{pq} found while (U) fails still gives the correct answer *fails*, and if no candidate passes, all three conditions hold by Proposition 11.5.

11.3 Countermodels

A counterexample word determines a countermodel through least closures: the least model of φ that contains the seed of the word is a solution of φ that no solution of ψ extends. The word also gives at most two observations, which ground bounds pin in every tree class.

Lemma 11.3 (Countermodels). *Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, both satisfiable, and let a counterexample word of (φ, ψ) be given. Define a set A of seeds and a set O of visible observations (Definition 5.2) by*

- *for an* (I0)*-word w with q: $A = \varnothing$, and O denies (q, w);*
- *for a* (U)*-word w with p: $A = \{(p, w)\}$, and O asserts (p, w);*
- *for a hole word $u\sharp t$ of I_{pq}: $A = \{(p, t)\}$, and O asserts (p, t) and denies (q, u).*

(1) $H_\varphi(A)$ is a model of φ, namely μ_φ, $\mathrm{Cl}_\varphi(p, w)$ or $\mathrm{Cl}_\varphi(p, t)$. It decodes to a solution ρ_A of φ in which the observations of O hold.

(2) No assignment of X in which the observations of O hold extends to a solution of ψ. In particular, ρ_A is a countermodel over arbitrary trees.

(3) Let $\mathcal{D}$ be a tree class in which φ is satisfiable, and let B be the set of finite ground bounds equivalent to the observations of O by Lemma 5.4. Then $\varphi \cup B$ has a solution in $\mathcal{D}$, and every such solution is a solution of φ in $\mathcal{D}$ without any extension to a solution of ψ.

(4) Suppose that $(\varphi, \psi) = (\varphi_0^\natural, \psi_0^\natural)$ for systems φ_0 over X_0 and ψ_0 over $X_0 \cup Z_0$ over $\{\bot, f\}$, with f covariant, as in Theorem 5.17. If ρ is a solution of φ with values in a tree class $\mathcal{D}$ that has no extension with values in $\mathcal{D}$ to a solution of ψ, then $\rho|_{X_0}$ is a solution of φ_0 over $\{\bot, f\}$ in $\mathcal{D}$ without an extension over $\{\bot, f\}$ in $\mathcal{D}$ to a solution of ψ_0.

Proof. (1) Since φ is satisfiable, μ_φ is its least model (Lemma 3.8(2)). A (U)-word has (p, w) live in φ by definition, and a hole (u, t) of I_{pq} has (p, t) live in φ by Theorem 8.7, which applies because (U) holds; so $\mathrm{Cl}_\varphi(p, w)$, respectively $\mathrm{Cl}_\varphi(p, t)$, is the least model of φ that contains the seed (Lemma 3.8(2)). The observations hold in $H_\varphi(A)$: $(q, w) \notin \mu_\varphi$ by definition, the seed lies in its own closure, and $(q, u) \notin \mathrm{Cl}_\varphi(p, t)$ by Theorem 8.7. By Lemma 3.5, $H_\varphi(A)$ is S_{ρ_A} for a unique solution ρ_A of φ.

(2) Let ρ' be an assignment of X in which the observations hold, and suppose that it extends to a solution of ψ. By Lemma 3.5, $S_{\rho'} = T|_X$ for a model T of ψ. Since T contains Fact_ψ and is closed under the rules of ψ, it contains μ_ψ, and it contains $\mathrm{Reach}_\psi(a)$ for each of its atoms a (Lemma 3.8(1)). For an (I0)-word, $(q, w) \in \mu_\psi \subseteq T$ is visible, so $(q, w) \in S_{\rho'}$, against the denial. For a (U)-word, $(p, w) \in T$, so $T \supseteq \mathrm{Reach}_\psi(p, w)$ contains a forbidden root of ψ, against the consistency of T. For a hole word, $(p, t) \in T$ and $(p, t) \to_\psi^* (q, u)$ by Theorem 8.7, so $(q, u) \in T|_X = S_{\rho'}$, against the denial. Applied to ρ_A, this proves the second claim; Theorem 8.7(2) is the case of a hole word.

(3) By Lemma 5.4, an observation holds in an assignment iff the corresponding ground bound does, so ρ_A solves $\varphi \cup B$. For $\mathcal{D} = \mathcal{T}$ take ρ_A; for $\mathcal{D} = \mathcal{T}_{\mathrm{reg}}$, Lemma 5.6 gives a regular solution, the

bounds being finite and hence regular; and for $\mathcal{D} = \mathcal{T}_{\text{fin}}$, Theorem 5.7 gives a finite one, since φ has a finite solution. Every solution of $\varphi \cup B$ satisfies the observations, so it has no extension by (2).

(4) The values $\rho(x)$, $x \in X_0$, lie below the values $\rho(y_x)$, which satisfy $y_x \le f(y_x^n)$, so they are $\top$-free by Lemma 5.16(1), and $\rho|_{X_0}$ solves φ_0 over $\{\bot, f\}$. Suppose that σ, with values in $\mathcal{D}$, extends $\rho|_{X_0}$ to a solution of ψ_0 over $\{\bot, f\}$. Put $\sigma(y_z) = T_{h(\sigma(z))+1}$ for $z \in Z_0$ if $\mathcal{D} = \mathcal{T}_{\text{fin}}$, and $\sigma(y_z) = \Omega$ otherwise. By Lemma 5.16(2), ρ together with σ solves ψ, which does not constrain the variables y_x; this contradicts the choice of ρ. □

Informally, a countermodel in $\mathcal{D}$ can then be found by enumerating the finite assignments, or finite graph presentations, of X and testing $\varphi \cup B$; we do not analyze this search.

Example 11.4 (Shift closure: the procedure end to end). Run the procedure on Example 1.1 with $\varphi_1 = \{x \le y\}$ and $\psi = \{x \le z,\ z \le y,\ z \le f(z,z)\}$, over $\{\bot, f\}$ with f binary and covariant, in the class of finite trees. Both systems are flat. Step 3 adds $\mathsf{TF}(x, y_x)$ and $\mathsf{TF}(y, y_y)$ to φ_1 and $\mathsf{TF}(z, y_z)$ to ψ; the assignment $\bot$ to every variable solves both new systems, so Step 4 continues.

- Step 5: $\psi^\natural$ has no facts, so $\mu_{\psi^\natural} = \varnothing$ and (I0) holds.
- Step 6: every atom (v^0, w) with $v \in \{x, y, y_x, y_y\}$ is dead by support in $\varphi_1^\natural$, since for $v \in \{x, y\}$ the copy $v^0 \to y_v^0$, and then pops of y_v^0, lead to the forbidden root (y_v^0, ε). Since f is covariant, pure moves preserve signs (Lemma 3.6(2)), and the forbidden roots (z^0, ε) and (y_z^0, ε) of $\psi^\natural$ have sign 0, so no atom of sign 1 is root-dead in $\psi^\natural$. Hence no $\mathcal{R}_{\mathrm{U}}(p)$ rejects a word, and (U) holds, as in Example 7.7.
- Step 7: the computation of Example 8.11 carries over, with y_y in the role of g. The spine automaton of x^1 moves from $\{x^1, y^1, y_x^1, y_y^1\}$ to $\{y_x^1, y_y^1\}$ on every letter and is safe; $J(t) = 0$ for every target t; the only possible event is $(0,0)$, present iff t is a prefix of u; and the head events are the pairs $(s, 0)$ with $u[s{:}] = t$. So $(01, 1)$ is a hole of $I_{y^1 x^1}$, and the search may answer *fails* with the hole word $01\sharp 1$, of length 4.

By Lemma 11.3, the word gives $A = \{(y^1, 1)\}$, and $\mathcal{O}$ asserts $(y^1, 1)$ and denies $(x^1, 01)$. The least model $\mathrm{Cl}_{\varphi_1^\natural}(y^1, 1)$ is the regular countermodel $x = y = f(\Omega, \bot)$, $y_x = y_y = \Omega$: the closure of the seed consists of the atoms $(y^1, 1\sigma)$ and $(x^1, 1\sigma)$. For finite trees, the pins of Lemma 5.4 for the two observations are

$$y \le \mathrm{emb}_\top(1, \bot) = f(\top, \bot), \qquad f(f(\bot, f(\bot, \bot)), \bot) = \mathrm{emb}_\bot(01, f(\bot, \bot)) \le x.$$

Every finite solution of $\varphi_1^\natural$ with these two bounds is a countermodel, for instance $x = y = f(f(\bot, f(\bot, \bot)), \bot)$ with $y_x = y_y = T_3$, and its restriction to $\{x, y\}$ (Lemma 11.3(4)) is the countermodel of Example 1.1. On trees without $\top$, the two pins say $1 \notin K(y)$ and $01 \in K(x)$, which is exactly the failure of $\mathrm{Suf}(K(x)) \subseteq K(y)$ found there.

11.4 Counterexample Bounds

The automata of the extracted instances have

$$N_A = 2^{2v}, \qquad N_B = 2^{4v+2h+1} = 2^{2v+2s+1}, \qquad N_{A'} = N_{B'} = 2^{2s}$$

states (Section 8.2), and the unit readers of condition (U) have $N_{\mathrm{U}} = N_B$ states (Theorem 7.9).

Proposition 11.5 (Counterexample bounds). *Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, both satisfiable, with $v = |X|$ and $s = |X| + |Z|$.*

(1) If (I0) *fails, there is an* (I0)*-word of length less than $2^{2v+2s+2}$.*

(2) If (U) *fails, there is a* (U)*-word of length at most $\mathsf{Len}(N_{\mathrm{U}}, d_{N_{\mathrm{U}}}) \le E_3(10v + 10s + 17)$.*

Table 7. Length bounds for shortest counterexample words (Definition 11.1), with $v = |X|$ and $s = |X| + |Z|$, and their growth in s: singly or triply exponential. Over $\{\bot, f\}$ the reduction of Theorem 5.17 replaces v and s by $2v$ and $2s$.

Failing condition	Word	Length bound	Growth	Source
(I0)	w	$< 2^{2v+2s+2}$	single	Proposition 11.5(1)
(U)	w	$\le E_3(10v + 10s + 17)$	triple	Theorem 7.9
(II), not mortal	$u\sharp t$	$\le \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'}) + 1$	single	Corollary 9.18
(II), mortal	$u\sharp t$	$\le E_3(10v + 12s + 15)$	triple	Theorem 10.15 and Corollary 10.19

(3) If (I0) *and* (U) *hold and* (II) *fails, there are* $p, q \in X^{\pm}$ *and a hole* (u, t) *of* I_{pq} *such that either* $|u| + |t| \le \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'})$*, a bound singly exponential in* s*, or* (u, t) *is a mortal hole and* $|u\sharp t| \le \mathsf{Len}^{\sharp}_{I_{pq}} \le E_3(10v + 12s + 15)$.

If $v \ge 1$*, each of these bounds is at most* $E_3(10v + 12s + 15) \le E_3(22s + 15)$*, and the mortal closed form is the largest. If* $v = 0$*, there is no counterexample word.*

Proof. (1) This is Proposition 6.16(2), with $|X| = v$ and $|X \cup Z| = s$.

(2) This is Theorem 7.9.

(3) By Theorem 8.7(1), which applies because φ is satisfiable and (U) holds, some I_{pq} has a hole. It satisfies (FH) by Lemma 8.6, so by Corollary 9.18 it has a hole with $|u| + |t| \le \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'})$ or a mortal hole. The function L_{nm} is a polynomial in the four state counts (Section 9.9), each of which is at most $2^{2v+2s+1}$, so the first bound is singly exponential in s. In the second case, Theorem 10.15 gives a mortal hole whose separator word has length at most $\mathsf{Len}^{\sharp}_{I_{pq}}$, and Corollary 10.19 bounds this by $E_3(10v + 12s + 15)$.

For the comparison, let $v \ge 1$, so that $s \ge 1$. The bound of (1) is at most 2^{4s+2}, and the non-mortal hole word has length at most $\mathsf{L}_{\mathrm{nm}} + 1 \le 54N_A^2 N_B N_{A'} N_{B'} + 1 < 2^{6v+6s+7}$ (Section 9.9); both are far below the triply exponential bounds. Of these, $E_3(10v + 10s + 17) \le E_3(10v + 12s + 15)$ since $s \ge 1$, so the mortal closed form is the largest, and $E_3(10v + 12s + 15) \le E_3(22s + 15)$ since $v \le s$. If $v = 0$, then $X^{\pm}$ is empty, and Definition 11.1 admits no word. □

Table 7 summarizes the cases. Both triply exponential terms come from the bounded-depth search of Zhao [2026] and its anchored extension, and in both cases the depth is bounded through a short rejected tail (Theorem 2.19). A mortal hole has a rejected tail shorter than $N_A = 2^{2v}$, which involves only visible variables; its reader has $\bar{N}_I \le 2^{2v+4s+2}$ states, and its depth is at most $d_I = 2\bar{N}_I N_A - 1 < 2^{4v+4s+3}$ (Lemma 10.6). A unit reader has $N_{\mathrm{U}} = 2^{2v+2s+1}$ states, and the search of Corollary 2.21 bounds the depth of a suitable hole by $d_{N_{\mathrm{U}}} < 2^{4v+4s+5}$. The mortal closed form is the largest of the closed forms in the table. This does not order the exact bounds: for $s \le 2$ and $v \ge 1$, the exact (U) bound exceeds the exact mortal bound, because $d_{N_{\mathrm{U}}} \approx 8N_{\mathrm{U}}^2$ is much larger than d_I. In the part that is not mechanized, the smaller reader of Section 13.1 gives an even smaller bound for (U) (Proposition 13.4), and Section 13 analyzes the complexity of the procedure.

11.5 Proof of the Main Theorem

Proof of Theorem 11.2. *Parts (1) and (2).* Each step of the procedure of Section 11.2 terminates and is correct by the result it cites (Table 6); three points need a remark. First, if the procedure passes Step 4, both systems are satisfiable in $\mathcal{D}$, hence over arbitrary trees, so Steps 5 to 7 decide entailment over arbitrary trees by Theorem 4.2(2), and Step 8 transfers the answer to $\mathcal{D}$. Second, Step 6 runs before Step 7, so (U) holds when Step 7 starts, and Theorems 8.7 and 10.17 apply to every

I_{pq}, which has forced target heads by Lemma 8.6. Third, Steps 2, 6, and 7 are finite searches: there are finitely many assignments at arity zero, the alphabet Σ is finite, the bounds are computable, and each candidate is tested over finitely many cuts. This proves (1), and (2) follows by Step 3, which preserves the answer by Theorem 5.17.

Part (3). Let φ and ψ be $\mathcal{D}$-satisfiable with $\varphi \not\models_{\mathcal{D}} \exists \bar{z}\, \psi$. Then both are satisfiable over arbitrary trees, and $\varphi \not\models \exists \bar{z}\, \psi$: for $\mathcal{D} = \mathcal{T}_{\mathrm{reg}}$ by Theorem 5.12, and for $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$ by Eq. (9), since both systems have finite solutions. By Theorem 4.2(2), (I0), (U) or (II) fails. In the first two cases Proposition 11.5(1,2) gives an (I0)-word or a (U)-word; otherwise (I0) and (U) hold and Proposition 11.5(3) gives a hole word. A counterexample word exists only if $v \geq 1$, and then its length is at most $E_3(10v + 12s + 15) \leq E_3(22s + 15)$ by Proposition 11.5. This argument uses no hypothesis on n; for $n = 0$ the only word is ε, and every counterexample word has length at most 1. By Lemma 11.3(3), every counterexample word yields a countermodel in $\mathcal{D}$, since φ is $\mathcal{D}$-satisfiable. None of the bounds involves n or c.

Part (4). Let f be covariant, and let φ and ψ over $\{\bot, f\}$ be $\mathcal{D}$-satisfiable over $\{\bot, f\}$, with failing entailment over $\{\bot, f\}$ in $\mathcal{D}$. For a $\top$-free tree t in $\mathcal{D}$, let $y(t)$ be the tree of Lemma 5.16(2) in $\mathcal{D}$ with $t \leq y(t) \leq f(y(t)^n)$. Extending a $\{\bot, f\}$-solution of φ by $y_x = y(\rho(x))$, and a $\{\bot, f\}$-solution σ of ψ by $y_z = y(\sigma(z))$, shows that $\varphi^\natural$ and $\psi^\natural$ are $\mathcal{D}$-satisfiable over $\{\bot, f, \top\}$, and by Theorem 5.17 the entailment of $(\varphi^\natural, \psi^\natural)$ fails in $\mathcal{D}$. This pair has $2|X|$ visible and $2|Z|$ hidden variables (Corollary 5.18), so (3), applied with $2v$ and $2s$, gives a counterexample word of length at most $E_3(10 \cdot 2v + 12 \cdot 2s + 15) = E_3(20v + 24s + 15) \leq E_3(44s + 15)$. It yields a countermodel for $(\varphi^\natural, \psi^\natural)$ in $\mathcal{D}$, which restricts to a countermodel over $\{\bot, f\}$ in $\mathcal{D}$ by Lemma 11.3(4). □

11.6 Plain Entailment

Plain entailment is the special case of existential entailment without hidden variables and with the consequent $x \leq y$ (Definition 2.4). Its decidability, proved by Zhao [2026], is therefore a corollary, with the same procedure.

Corollary 11.6 (Plain entailment). *For every arity $n \geq 0$ and every variance c, plain entailment $\varphi \models_{\mathcal{D}} x \leq y$ over $\{\bot, f, \top\}$ is decidable for each tree class $\mathcal{D}$, and*

$$\varphi \models_{\mathrm{reg}} x \leq y \iff \varphi \models x \leq y, \qquad \varphi \models_{\mathrm{fin}} x \leq y \iff \neg\, \mathrm{Sat}_{\mathrm{fin}}(\varphi) \ \vee\ \varphi \models x \leq y.$$

If f is covariant, plain entailment over $\{\bot, f\}$ is decidable for each tree class.

Proof. With $Z = \varnothing$ and $\psi = \{x \leq y\}$, Theorem 11.2(1,2) decides the entailment. The consequent has the solution $x = y = \bot$ in every tree class, so Theorem 5.12 gives the first equivalence and Eq. (9) the second. □

12 Constrained Types and Simplification

This section states the subsumption of constrained type schemes as an existential entailment, discusses the correctness of simplifications, and ends with examples, all of which involve variables that occur both co- and contravariantly. The proof of the subsumption formula is in Appendix G.

The signature is $\{\bot, f, \top\}$ with an arity n and a variance c, or $\{\bot, f\}$ with f covariant, and $\mathcal{D}$ is a tree class. For arrow types, $n = 2$, $c(0) = 1$, $c(1) = 0$, and we write $a \to b$ for $f(a, b)$. Constraints are inequalities between finite terms; by Lemma 2.6 they are flattened, with the names of the subterms of the antecedent visible and those of the consequent hidden. Plain entailments are taken in the same signature and tree class.

12.1 Subsumption

A *type scheme* $S = \forall \bar{a}.\, C \Rightarrow (\Gamma \Rightarrow \tau)$ consists of a finite set $\bar{a}$ of bound type variables, a finite conjunction C of inequalities between terms, a *context* Γ, that is, a finite map from program variables to terms, and a term τ. Its other type variables are *free*. For a valuation η of the free variables in $\mathcal{D}$, the *instances* of S under η are the pairs $(\rho \circ \Gamma, \rho(\tau))$, where ρ extends η to $\bar{a}$ with values in $\mathcal{D}$, satisfies C, and $\rho \circ \Gamma$ maps each v in $\operatorname{Dom}\Gamma$ to $\rho(\Gamma(v))$. Following Trifonov and Smith [1996, Section 4.2], an instance (Δ, T) is below an instance (Δ', T') if $T \leq_c T'$, $\operatorname{Dom}\Delta \subseteq \operatorname{Dom}\Delta'$, and $\Delta'(v) \leq_c \Delta(v)$ for every $v \in \operatorname{Dom}\Delta$; contexts are compared contravariantly. Then $S_1 \leq S_2$ if, under every valuation of the free variables, every instance of S_2 is above some instance of S_1.

Proposition 12.1 (Subsumption formula). *Let $S_i = \forall \bar{a}_i.\, C_i \Rightarrow (\Gamma_i \Rightarrow \tau_i)$ for $i = 1, 2$, where the bound variables are renamed so that $\bar{a}_1$ and $\bar{a}_2$ are disjoint from each other and from the free variables of both schemes. Let the visible variables be $\bar{a}_2$ and the free variables, and let $\bar{a}_1$ be hidden.*

(1) If $\operatorname{Dom}\Gamma_1 \subseteq \operatorname{Dom}\Gamma_2$, then

$$S_1 \leq S_2 \iff C_2 \models_{\mathcal{D}} \exists \bar{a}_1\, (C_1 \wedge R_{12}), \qquad R_{12} = (\tau_1 \leq \tau_2) \wedge \bigwedge_{v \in \operatorname{Dom}\Gamma_1} \Gamma_2(v) \leq \Gamma_1(v).$$

(2) If $\operatorname{Dom}\Gamma_1 \not\subseteq \operatorname{Dom}\Gamma_2$, then $S_1 \leq S_2$ iff C_2 has no solution in $\mathcal{D}$.

Both renamings are needed. For $S_1 = b$, with b free, and $S_2 = \forall b.\, b$, with empty contexts, the formula read without renaming $\bar{a}_2$ would be the valid entailment true $\models b \leq b$; but at $b = \top$ the instance $\bot$ of S_2 lies above no instance of S_1, so $S_1 \not\leq S_2$.

This is Eq. (1) with contexts. The formula has no restriction on the occurrences of variables: in a scheme for a higher-order program, a type variable typically occurs in the domain and in the codomain of arrows, and then at both polarities.

Corollary 12.2 (Deciding subsumption). *For $\{\bot, f, \top\}$ with any arity and variance, in particular for arrow types over $\bot$, $\to$ and $\top$, and for $\{\bot, f\}$ with f covariant, subsumption of type schemes is decidable in each tree class. Over regular trees, the setting of Trifonov and Smith [1996], it agrees with subsumption over arbitrary trees.*

Proof. By Proposition 12.1 and Lemma 2.6, subsumption is an existential entailment between flat systems, or the test that C_2 is unsatisfiable. The latter is the entailment $C_2 \models_{\mathcal{D}} f(\bot, \ldots, \bot) \leq \bot$, since its consequent is false in every tree class. After flattening, both entailments are decided by Theorem 11.2. Regular and arbitrary trees agree for both questions (Theorem 5.12). □

12.2 Simplification

A system $C(\bar{x}, \bar{h})$ may be replaced by $C'(\bar{x}, \bar{k})$, where $\bar{x}$ remain observable, if $\exists \bar{h}\, C \iff \exists \bar{k}\, C'$. The forward implication is the existential entailment $C \models \exists \bar{k}\, C'$ with $\bar{h}$ visible, and the reverse one exchanges the systems; equivalence of type schemes is subsumption in both directions (Proposition 12.1) and need not preserve the solutions on every type variable. These are the correctness conditions of constraint simplification and minimization [Eifrig et al. 1995; Pottier 1996, 2001; Trifonov and Smith 1996], and by Theorem 11.2 they are decidable, for arrow types without any restriction on polarity.

Informally, some simplifications need only plain entailment, such as deleting a literal that the remaining literals entail. Existential entailment is needed when a transformation changes the hidden variables: eliminating or merging internal variables, or replacing a scheme by one with different bound variables (Example 12.5).

12.3 Examples

Example 12.3 (Two identity schemes). Let $S_1 = \forall\alpha.\, \alpha \to \alpha$ and $S_2 = \forall\beta\gamma.\, (\beta \le \gamma) \Rightarrow \beta \to \gamma$, with empty contexts. By Proposition 12.1, $S_1 \le S_2$ iff $\beta \le \gamma \models \exists\alpha\, (\alpha \to \alpha \le \beta \to \gamma)$. Both sides of the inequality are arrows, so it says $\beta \le_c \alpha$ and $\alpha \le_c \gamma$, and $\alpha = \beta$ is a witness. Conversely, $S_2 \le S_1$ iff $\models \exists\beta\gamma\, (\beta \le \gamma \wedge \beta \to \gamma \le \alpha \to \alpha)$, with α visible, that is, iff some β, γ satisfy $\alpha \le \beta \le \gamma \le \alpha$, and $\beta = \gamma = \alpha$ is a witness. So the two schemes are equivalent. The hidden variable α of the first entailment occurs in the domain and in the codomain of the same arrow, so it has both polarities, and after flattening the literal $f(\alpha, \alpha) \le u$ admits no parity assignment (Remark 4.5), which would need $\pi(\alpha) = \pi(u) \oplus 1$ and $\pi(\alpha) = \pi(u)$.

Trifonov and Smith [1996] decide subsumption by a sound approximation through canonical bounds, and simplifiers prove entailments by syntactic reasoning about the bounds of variables; such tests are incomplete [Pottier 2001]. The following example of Pottier [1998b, Section 8.5] shows a correct simplification that reasoning through lower bounds misses.

Example 12.4 (A missed redundant constraint). For arrow types put

$$g = \bot \to \top, \qquad k = g \to \bot, \qquad C(a) = (a \to \bot \le a).$$

Then $C \models k \le a$ in each tree class. Indeed $a \neq \bot$, since $a \to \bot \le_c \bot$ is false. If $a = \top$, then $k \le_c a$. Otherwise a is an arrow $a_0 \to a_1$, so $a \le_c g$, because g is the greatest arrow (Example 3.3); by contravariance $k = g \to \bot \le_c a \to \bot \le_c a$. Thus adding or deleting the literal $k \le a$ is an exact simplification, and equivalently $C \models \exists z\, (k \le z \wedge z \le a)$, with the witness $z = k$. The variable a has the single lower bound $a \to \bot$ and the upper bound $\top$ in C. Proving $k \le a$ through this lower bound requires $k \le a \to \bot$, that is, $a \le g$, and this is not entailed: $a = \top$ satisfies C. A test that proves an inequality $k \le a$ only through a lower bound of a therefore fails on this true simplification.

The antecedent mentions a at both polarities, and its flattening admits no parity assignment. The example also shows why the signed system cannot be read on its own. The signed literals of C (Table 2) are $f(a^-, \bot) \le a^+$ and $a^- \le f(a^+, \top)$, read covariantly. Put $F = f(\bot, \bot)$. The covariant values $a^+ = F$ and $a^- = \bot$ satisfy them, but they violate $a^- = (a^+)^\partial$, the condition of Lemma 2.5. The signed form of $k \le a$ requires $v_0 k \le a^+$ (Lemma 2.2(2)), and $v_0 k = f(v_1 g, \bot) = f(F, \bot)$ by Lemma 2.2(1). Since $f(F, \bot) \not\le F$, a test that drops the coupling between a^+ and a^- reports a countermodel of a true entailment. The leaf cones of Section 3 keep this coupling: the two cones of a are the cones of one tree, and the exclusions and duality of Section 3.4 enforce it.

The last example concerns a merge of type variables, which changes the bound variables of a scheme and so needs existential entailment.

Example 12.5 (Merging the variables of self-application). The term $\lambda x.\, x\, x$ has the inferred scheme

$$S = \forall\alpha\beta.\, (\alpha \le \alpha \to \beta) \Rightarrow \alpha \to \beta,$$

in which α is both a function and its own argument. A simplifier might merge β into α and propose $S' = \forall\gamma.\, (\gamma \le \gamma \to \gamma) \Rightarrow \gamma \to \gamma$. By Proposition 12.1, $S \le S'$ iff

$$\gamma \le \gamma \to \gamma \;\models\; \exists\alpha\beta\, (\alpha \le \alpha \to \beta \;\wedge\; \alpha \to \beta \le \gamma \to \gamma),$$

which holds with the witness $\alpha = \beta = \gamma$. But $S' \le S$ iff

$$\alpha \le \alpha \to \beta \;\models\; \exists\gamma\, (\gamma \le \gamma \to \gamma \;\wedge\; \gamma \to \gamma \le \alpha \to \beta),$$

and the second literal says $\alpha \le_c \gamma \le_c \beta$. The values $\alpha = \top \to \bot$ and $\beta = \bot$ satisfy the antecedent: the domain $\top$ of α lies above α, and its codomain $\bot$ below β. No γ lies between the arrow α and $\bot$, so the entailment fails, and the instance $(\top \to \bot) \to \bot$ of S, the type of $\lambda x.\, x\, x$ applied to arguments that accept everything and return $\bot$, lies above no instance of S'. The merge loses

generality. In both entailments every hidden variable has both occurrence signs in the consequent, and a hidden variable is bounded by an arrow in which it occurs itself, $\alpha \le \alpha \to \beta$ and $\gamma \le \gamma \to \gamma$, and Theorem 11.2 decides both.

13 Complexity and Sharper Bounds (Not Mechanized)

This section is not mechanized. Every result before it, up to and including the main theorem (Theorem 11.2) and its applications (Section 12), is mechanized (Section 14), and nothing before it uses a result of this section. The section collects the complexity of the problem and bounds sharper than those of the main theorem. Its proofs use the mechanized results freely; those omitted here are in Appendix H, which is not mechanized either.

Throughout, $s = |X| + |Z|$ is the number of variables after flattening (Section 2.3), $v = |X|$, and $E_1(x) = 2^x$, $E_{k+1}(x) = 2^{E_k(x)}$. Section 13.1 gives a smaller reader for condition (U), with a sharper bound, and Section 13.2 records two facts on tree classes and signed systems. The upper bounds of Section 13.3 hold for every arity n and every variance c, with constants that do not depend on n; the arity enters only the size of the alphabet and of the input. The lower bounds of Section 13.4 are proved for a covariant constructor of arity at least two.

Remark 13.1 (Complexity). The procedure of Section 11.2 is written to make its correctness evident: Steps 6 and 7 enumerate all words up to a triply exponential length. The applications of Section 12 need only the statement of the main theorem. This section gives sharper procedures, the complexity of the problem, and lower bounds.

13.1 A Smaller Reader for Condition (U)

The bound of Theorem 7.9 is not the smallest one. Two observations give a much smaller bound. First, most states of the source automaton (Definition 7.2) can never occur on a rejected word, so they can be merged into one sink. Second, the depth bound needs only the first rejected cut, and on the words that the unit reader rejects this cut is determined by the antecedent alone, which is read by a much smaller automaton. These results serve the complexity bounds of Section 13.3; the decision procedure and its correctness do not use them.

Lemma 13.2 (The reduced unit reader). *Let $p \in X^{\pm}$, $v = |X|$ and $s = |X| + |Z|$. Let $\widehat{\mathcal{R}}_p$ arise from the reachable part of $\mathcal{R}_{\mathrm{U}}(p)$ by replacing every state $\zeta = (P, b, P')$ with $b = 1$ or with $\mathrm{du}(\zeta)$ false by one new absorbing state $\perp \notin \mathsf{Rej}$: every transition into a replaced state is redirected to $\perp$, and no pair that involves $\perp$ is admitted. Then:*

(1) a word is rejected by $\widehat{\mathcal{R}}_p$ iff it is rejected by $\mathcal{R}_{\mathrm{U}}(p)$, and then both readers have the same state at every cut, so they have the same holes;
(2) $\widehat{\mathcal{R}}_p$ has the rejected-path property;
(3) $\widehat{\mathcal{R}}_p$ has at most $1 + 2^{v+1}(2^s - 1) < 2^{v+s+1}$ states.

No satisfiability assumption is needed.

The replaced states are closed under transitions, because the bit b never returns to 0 and du, once false, stays false (Lemma 7.3(2)); so a rejected word never meets them. The count uses sign coherence: on a kept state, P' is nonempty and fixes the common sign of all elements of P and P'. The proof is in Appendix H.

Lemma 13.3 (Antecedent rejection clock). *Let $p \in X^{\pm}$ and $v = |X|$, and let $\mathcal{A}_p$ be the reader whose automaton is the reachable part of the upper automaton of p in φ, whose admission relation is empty, and whose rejected states are the sets P with $P \cap \mathrm{DeadT}^{\varepsilon}_{\varphi} = \varnothing$. It has at most $2^{v+1} - 1$ states.*

If w is rejected by $\mathcal{R}_{\mathrm{U}}(p)$, then w is rejected by $\mathcal{A}_p$ with the same first rejected cut, and if moreover $w \neq \varepsilon$, then $\mathrm{depth}(w) \leq 2(2^{v+1}-1)(r(w)+1)-1$.

On a word rejected by $\mathcal{R}_{\mathrm{U}}(p)$, the bit and du are constant, so only the first component of the source automaton decides rejection; the depth bound is then Theorem 2.19 for $\mathcal{A}_p$, which uses neither admissions nor holes. The proof is in Appendix H.

Proposition 13.4 (A sharper bound for (U)). *Let φ be satisfiable and $X \neq \varnothing$, let $v = |X|$ and $s = |X| + |Z|$, and put*

$$A_{\mathrm{U}} = 2^{v+1}, \qquad K_{\mathrm{U}} = 2^{v+s+1}, \qquad D_{\mathrm{U}} = 2A_{\mathrm{U}}(4K_{\mathrm{U}}-1)-1.$$

If (U) *fails, there are $p \in X^{\pm}$ and a word w with $|w| \leq \mathsf{Len}(K_{\mathrm{U}}, D_{\mathrm{U}})$ such that (p, w) is live in φ and* $\mathsf{DeadU}_{\psi}(p, w)$. *Moreover* $\mathsf{Len}(K_{\mathrm{U}}, D_{\mathrm{U}}) \leq E_3(5v+3s+16)$.

Proof. If (U) fails, some $\mathcal{R}_{\mathrm{U}}(p)$ has a hole (Lemma 7.5), hence so does $\widehat{\mathcal{R}}_p$, which has $m \leq K_{\mathrm{U}}$ states and the rejected-path property (Lemma 13.2). By Theorem 2.15 it has a hole w_0 with $r(w_0) \leq 4m-2$; if $w_0 = \varepsilon$, we are done. Otherwise w_0 is rejected by $\mathcal{R}_{\mathrm{U}}(p)$, and Lemma 13.3 gives $\mathrm{depth}(w_0) \leq 2(2^{v+1}-1)(4m-1)-1 \leq D_{\mathrm{U}}$. Theorem 2.20 for $\widehat{\mathcal{R}}_p$ with $d = D_{\mathrm{U}}$ gives a hole of length at most $\mathsf{Len}(m, D_{\mathrm{U}}) \leq \mathsf{Len}(K_{\mathrm{U}}, D_{\mathrm{U}})$, since Len is nondecreasing in each argument (Appendix A.5). This is a hole of $\mathcal{R}_{\mathrm{U}}(p)$ by Lemma 13.2(1), hence a word as stated by Lemma 7.5.

For the estimate, write $A = A_{\mathrm{U}}$, $K = K_{\mathrm{U}}$ and $D = D_{\mathrm{U}}$. Then $D+1 \leq 8AK$ and $K + D \leq 9AK$, so the quantity Ξ of Lemma A.12 satisfies $\Xi(K, D) \leq 3K^2 + 288A^2K^3 + 3K + 24AK + 4 \leq 322A^2K^3$, and $5\Xi(K, D) + 3 \leq 1613A^2K^3 < 2^{11}A^2K^3 = 2^{5v+3s+16}$. By Lemma A.12, $\mathsf{Len}(K, D) \leq 2^{2^{5\Xi(K,D)+3}} \leq E_3(5v+3s+16)$. □

Remark 13.5 (Singly exponential cases). In two cases the witnesses of Proposition 13.4 are much shorter. Let φ be satisfiable and $p \in X^{\pm}$.

- *No self events.* If φ has a parity assignment (Remark 4.5), in particular if f is covariant, then no pair of reachable states of $\mathcal{B}_p$ is admitted (Remark 6.13). The holes of $\mathcal{R}_{\mathrm{U}}(p)$ are then exactly its rejected words, and a shortest one visits no state of $\widehat{\mathcal{R}}_p$ twice, so it is shorter than 2^{v+s+1}.
- *Arity one.* If $n = 1$, every pair of cuts $i < j$ of a word is a comparison. If w is a hole of $\mathcal{R}_{\mathrm{U}}(p)$, then so is its prefix $w_J = w[:J(w)]$, because the comparisons of w_J are comparisons of w with the same states. By Lemma 13.3, w_J is rejected by $\mathcal{A}_p$ with the first rejected cut $J(w)$, and no two of its cuts $i < j \leq J(w)$ have the same state of $\mathcal{A}_p$: the unary word would then repeat the run of $\mathcal{A}_p$ with period $j-i$, and the cut $J(w)-(j-i)$ would already be rejected. Hence $J(w) < 2^{v+1}-1$, and (U), if it fails, fails at a word of length at most $2^{v+1}-2$.

13.2 Tree Classes and Signed Systems

Two facts used below concern the tree classes. Like the rest of this section, neither is mechanized, and no result before this section uses them. The first, Lemma 13.6, describes finite and regular trees through leaf cones (Section 3); the lower bounds of Section 13.4 use it, and the decision procedure and the transfers of Section 5 do not. The second, Lemma 13.7, extends the satisfiability criteria of Zhao [2026] (Theorem A.1(2,3)) to copy literals and to all three tree classes; the upper bounds of Section 13.3 use it, and the decision procedure decides satisfiability by Proposition 6.16 and Theorem 2.9 instead. Both proofs are in Appendix H.

Lemma 13.6 (Cones and tree classes). *For every tree t, the positions of t labeled f are exactly the words outside $L_t^0 \cup L_t^1$. Hence t is finite iff $\Sigma^* \setminus (L_t^0 \cup L_t^1)$ is finite, and t is regular iff L_t^0 and L_t^1 are regular languages.*

Lemma 13.7 (Signed satisfiability). *For every arity $n \geq 0$, every variance c, every system χ over $\{\bot, f, \top\}$, copy literals included, and every tree class $\mathcal{D}$,*

$$\mathrm{Sat}_{\mathcal{D},c}(\chi) \iff \mathrm{Sat}_{\mathcal{D}}(\chi^{\pm}), \qquad \textit{and} \qquad \mathrm{Sat}_{c}(\chi) \iff \mathrm{Sat}_{\mathrm{reg},c}(\chi).$$

13.3 Upper Bounds

Witness lengths. When both systems are satisfiable in the tree class considered and the entailment fails, a counterexample word (Definition 11.1) of length at most $E_3(10v + 12s + 15) \leq E_3(22s + 15)$ exists over $\{\bot, f, \top\}$, and of length at most $E_3(20v + 24s + 15) \leq E_3(44s + 15)$ over $\{\bot, f\}$ with f covariant (Theorem 11.2); every counterexample word yields a countermodel (Lemma 11.3). Proposition 11.5 and Table 7 bound the kinds separately. Words for (I0) and holes that are not mortal are singly exponential in s; for the latter the pump keeps the purely periodic case polynomial in the sizes of the automata (Theorem 9.14), for every instance of the two-word problem with self events that satisfies (FH). The words for (U) and the mortal holes need the triply exponential estimates of compression. All these bounds are existential: they bound a shortest counterexample, not every one.

Searching without storing the word. Enumerating all words up to the bound of Theorem 11.2 decides the problem in triply exponential space. The words need not be stored. The compression of Theorem 10.14 only ever uses a dictionary of boundedly many blocks per level, and a level is described, for the purpose of the levels above it, by the derived monitor and channel on these blocks, which the column summaries determine (Lemma F.8). A nondeterministic search can therefore build the hierarchy from the bottom, one level at a time, keeping only the current derived tables.

Proposition 13.8 (Summary search). *Let $\mathcal{M}_0$ be a lettered monitor with $N \geq 1$ states over a finite alphabet Γ, let Θ_0 be a positive channel on it, both given by their tables, and let $d \geq 0$. Whether $(\mathcal{M}_0, \Theta_0)$ has a positive hole of depth at most d is decidable in nondeterministic space polynomial in $N + d + 1$, $\mathrm{G}^{\sharp}(N, d)$ and $|\Gamma|$, hence in deterministic space $2^{O((N+d+1)^3)} \cdot \mathrm{poly}(|\Gamma|)$.*

Proof. Put $g = \mathrm{G}^{\sharp}(N, d)$. We use the summaries, column words and agreement of Appendix A.4 and their extensions by channels in Appendix F. The search keeps a level $j \leq d$ and a *current pair* $(\mathcal{W}, \Theta)$: a lettered monitor with the N states of $\mathcal{M}_0$ and j symbolic states, and a positive channel on it, over a current alphabet. Initially $j = 0$ and $(\mathcal{W}, \Theta) = (\mathcal{M}_0, \Theta_0)$, over Γ; for $j \geq 1$ the current alphabet is $\{1, \ldots, h\}$ with $h \leq g$. The search repeats one of two moves.

- *Accept* if some letter z and some $m \leq 3(N + j)$ make z^m a positive hole of $(\mathcal{W}, \Theta)$. This is checked by computing the labels of z^m and testing every pair of cuts.
- *Derive*, if $j < d$: choose a marker ℓ and a number $1 \leq h \leq g$, and guess a valid column word with h tracks over the current alphabet without ℓ, one column at a time, while computing the positive summary of the tracks read so far by the automaton of Lemma F.8, starting from the summary of h empty blocks; a binary counter stops the guess before $\mathrm{pcol}(N + j, h)$ columns. Require that the final equality fields declare the h blocks pairwise distinct. Replace $(\mathcal{W}, \Theta)$ by the restriction of $(\mathcal{W}_\ell, \Theta_\ell)$ to the h blocks, named $1, \ldots, h$, which the final summary determines at every state (Appendix A.4, step 4, and Lemma F.8(2)), and increase j.

Space. A current pair over $h \leq \max(g, |\Gamma|)$ letters with $M \leq N + d + 1$ states is given by its transitions, final targets, the relations Λ, Λ^{fin}, R, S, S' on labels, the predicate F and the set Φ, in $O(M^2h^2 + Mh \log M)$ bits. A positive summary has at most $\mathrm{pcol}(M, h)$ values, hence $O(M^2h^2 + Mh \log M)$ bits, and so does the counter; a column has h entries. Only the current pair, one summary, one column, the counter and j are stored. Since $g \leq 2^{\Xi^{\sharp}(N,d)}$ by Theorem 10.14 and $\Xi^{\sharp}(N, d) \leq$

$30(N+d+1)^3$, the nondeterministic space is polynomial in $N+d+1$, g and $|\Gamma|$, and Savitch's theorem squares it, which keeps the stated form.

Soundness. We show by induction on the number of moves that there are markers $\widetilde{\ell_0}, \ldots, \widetilde{\ell_{j-1}}$ and an injective map ρ_j from the current alphabet into the letters of level j, with a valid image family for these markers (Appendix A.4), such that $(\mathcal{W}, \Theta) \equiv_{E,\rho_j} (\mathcal{M}_j^{\widetilde{\ell}}, \Theta_j^{\widetilde{\ell}})$, where E is the current alphabet. Initially ρ_0 is the identity. In a derive move, the guessed column word spells blocks $Y_1, \ldots, Y_h$ over E that avoid ℓ; by Lemma F.8(1) the computed summary is theirs, so the new pair is the restriction of $(\mathcal{W}_\ell, \Theta_\ell)$ to $\{Y_1, \ldots, Y_h\}$. By Lemma F.7, $(\mathcal{W}_\ell, \Theta_\ell) \equiv_{\{Y_a\},\rho_j^*} ((\mathcal{M}_j^{\widetilde{\ell}})_{\rho_j(\ell)}, (\Theta_j^{\widetilde{\ell}})_{\rho_j(\ell)})$, and the images avoid $\rho_j(\ell)$. Put $\widetilde{\ell_j} = \rho_j(\ell)$ and $\rho_{j+1}(a) = \rho_j^*(Y_a)$, which is injective because the Y_a are distinct and ρ_j is injective. If the search accepts with z^m, then $\rho_j(z)^m$ is a positive hole of $(\mathcal{M}_j^{\widetilde{\ell}}, \Theta_j^{\widetilde{\ell}})$, since agreement preserves positive holes (Appendix F.5). Expanding it along the markers gives, by Lemmas 2.24 and 10.10, a positive hole w of $(\mathcal{M}_0, \Theta_0)$. Each expansion ends with its marker and its blocks avoid it, so the canonical hierarchy of w recovers the expanded levels and reaches the unary word $\rho_j(z)^m$ at level j, and $\mathrm{depth}(w) \le j \le d$.

Completeness. Suppose a positive hole of depth at most d exists. By Lemma F.5 there is one, w, of depth $D \le d$ whose dictionaries satisfy $g_i \le g$ for $1 \le i \le D$; let (w_j, ℓ_j) be its hierarchy, $\mathcal{A}_j$ its dictionaries and $(\mathcal{M}_j, \Theta_j)$ its towers. We follow a run in which, for $1 \le j \le D$, the current pair is the restriction of $(\mathcal{M}_j, \Theta_j)$ to $\mathcal{A}_j$, up to the names of the letters. At level $j < D$, choose $\ell = \ell_j$ and $h = g_{j+1}$. The letters of $\mathcal{A}_{j+1}$ are h distinct blocks over $\mathcal{A}_j$ that avoid ℓ_j, and their positive summary depends only on the tables at the letters of $\mathcal{A}_j$, so it is also their summary for the current pair. By the deletion argument in the proof of Lemma F.10, some valid column word with fewer than $\mathrm{pcol}(N+j, h)$ columns has the same final summary; the search guesses it, and the new pair is the restriction of $(\mathcal{M}_{j+1}, \Theta_{j+1})$ to $\mathcal{A}_{j+1}$, because the summary determines it. At level D, $w_D = z^m$ is a positive hole of $(\mathcal{M}_D, \Theta_D)$ by Corollary 10.11, hence of the current pair, and Lemma F.6 shortens it to $m \le 3(N+D)$. The search accepts. □

The *trivial channel* $(Q \times \Gamma, \varnothing, \varnothing, \varnothing, \varnothing)$ makes every nonempty word positive, so its positive holes are the holes of the monitor, and Proposition 13.8 also decides whether a monitor has a nonempty hole of depth at most d.

Theorem 13.9 (Complexity of the decision). *Existential entailment over $\{\bot, f, \top\}$, for every arity and variance, and over $\{\bot, f\}$ with f covariant, is decidable in each of the three tree classes in deterministic space $E_2(O(s)) \cdot \mathrm{poly}(m)$, where m is the size of the input, with the arity, the variance and the systems given explicitly. It is therefore in 2-EXPSPACE and in 3-EXPTIME.*

Proof. By Lemma 2.6, flattening is linear, so let the systems be flat. Over $\{\bot, f\}$, apply the reduction of Theorem 5.17, which is polynomial and doubles s (Corollary 5.18). For $n = 0$ enumerate the 3^s assignments (Remark 5.20), in polynomial space. Let $n \ge 1$.

Satisfiability. For each class $\mathcal{D}$, $\mathrm{Sat}_{\mathcal{D}}$ of a system χ is decided in polynomial time: by Lemma 13.7, it is satisfiability of the signed system of the system $\chi^\flat$ of Lemma 2.7, and by Theorem A.1(1) this is the absence of a label clash, and over finite trees also of a cycle clash, which are closure and graph tests of polynomial size; regular satisfiability is satisfiability. If φ is not $\mathcal{D}$-satisfiable, the entailment holds; if φ is and ψ is not, it fails. Otherwise both are satisfiable, and Theorems 5.12 and 5.13 reduce the problem to arbitrary trees.

The three conditions. By Corollary 8.8, the entailment holds iff (I0) and (U) hold and no extracted instance has a hole. Condition (I0) is decided in polynomial space: the closure D_χ of Definition 6.1 is computed by saturation in polynomial time, so is every transition of the bound automata, whose states are sets of signed variables, and the tests in the proof of Proposition 6.16, together with the

computation of $\mathrm{Fact}^{\uparrow}_{\chi}$ (Lemma 6.6), are reachability questions in these automata or in products of two of them, each with one bit, and are decided in nondeterministic, hence deterministic, polynomial space. For (U), compute the tables of the reduced unit readers $\widehat{\mathcal{R}}_p$ of Lemma 13.2, with fewer than $K_{\mathrm{U}} = 2^{v+s+1}$ states each, in space $2^{O(s)}\mathrm{poly}(m)$ (Lemma 7.3). By Lemmas 7.5 and 13.2, (U) fails iff some $\widehat{\mathcal{R}}_p$ has a hole. The empty word is tested directly. As in the proof of Proposition 13.4, by Theorem 2.15 and Lemma 13.3, a reduced unit reader with a hole has the empty hole or a nonempty hole of depth at most $D_{\mathrm{U}} = 2^{v+2}(4K_{\mathrm{U}} - 1) - 1$, and its nonempty holes are those of its lettered monitor (Definition 2.22). So Proposition 13.8 with the trivial channel decides (U) in space $2^{O((K_{\mathrm{U}}+D_{\mathrm{U}}+1)^3)}\mathrm{poly}(n) = E_2(O(s))\,\mathrm{poly}(n)$.

For (Π), fix $p, q \in X^{\pm}$. The instance I_{pq} satisfies (FH), so by Corollary 9.18 it has a hole iff it has a hole with $|u| + |t| < \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'})$, a bound singly exponential in s (Proposition 11.5), or a mortal hole. The first alternative is decided by enumerating such pairs and checking the conditions of Definition 8.1 cut by cut, in space $2^{O(s)}\mathrm{poly}(m)$. For the second, by Lemmas 10.5 and 10.6, I_{pq} has a mortal hole iff $\mathcal{R}^{\Pi}_I$ has a Π-hole of depth at most d_I; this reader has $\bar{N}_I \le 2^{2v+4s+2}$ states (proof of Corollary 10.19), so $d_I < 2^{4v+4s+3}$, and its tables, including the mortality of the states of A, are computed by reachability in space $2^{O(s)}\mathrm{poly}(m)$. Its Π-holes are the positive holes of its lettered monitor with the base channel (Lemma 10.8), and Proposition 13.8 decides their existence in space $E_2(O(s))\,\mathrm{poly}(n)$.

Total. All tests together use deterministic space $E_2(O(s))\,\mathrm{poly}(m)$, and $s \le m$. A deterministic machine in this space has at most $2^{E_2(O(s))\mathrm{poly}(m)}$ configurations, which gives the time bound. □

Remark 13.10 (Plain entailment). Plain entailment $\varphi \models_{\mathcal{D}} x \le y$ is the case $\psi = \{x \le y\}$, $Z = \varnothing$. Theorem 13.9 therefore also places plain entailment, for every arity and variance and in each tree class, in 2-EXPSPACE. The earlier procedure for plain entailment places non-entailment in N3EXPTIME and entailment in coN3EXPTIME, hence in 3EXPSPACE [Zhao 2026, Corollary "Complexity upper bounds"]; the summary search improves the deterministic space upper bound to 2-EXPSPACE, one exponential lower.

Arity one. For $n = 1$ there is a single letter, words are determined by their lengths, and a suffix comparison is a comparison of lengths. Here a direct shortening gives singly exponential witnesses for every instance, mortal or not.

Theorem 13.11 (Arity one). *Let $n = 1$, with either variance.*

(1) Let I be an instance of the two-word problem with self events that satisfies (FH), and let $r_I \le N_A N_{A'}$ be the number of states of $A \times A'$ reachable from the start. If I has a hole, it has a hole (u, t) with $|t| \le N_B - 1$ and $|u| + 1 + |t| \le r_I + 2N_B - 2$.

(2) Every reader with N states over a one-letter alphabet that has a hole has a hole of length at most $N - 1$.

(3) Let φ and ψ be flat systems over $\{\bot, f, \top\}$ with s variables. If they are $\mathcal{D}$-satisfiable and $\varphi \not\models_{\mathcal{D}} \exists \bar{z}\, \psi$, a counterexample word of length less than 2^{4s+3} exists. Over $\{\bot, f\}$ with f covariant, if φ and ψ are $\mathcal{D}$-satisfiable over $\{\bot, f\}$ and the entailment fails, the pair $(\varphi^{\natural}, \psi^{\natural})$ of Theorem 5.17, which has $2s$ variables, has a counterexample word of length less than 2^{8s+3}. Existential entailment at arity one is in PSPACE, for both variances and in each tree class, over $\{\bot, f, \top\}$ and, for covariant f, over $\{\bot, f\}$.

Proof. Write the letter as 0. For words 0^a and 0^b and cuts $x \le a$, $y \le b$, the pair (x, y) is a cross comparison of $(0^a, 0^b)$ iff $b - y \le a - x$, and it is aligned iff $b - y = a - x$; every pair of cuts $i < j$ of a word is a comparison.

(1) Let (u, t) be a hole, $J = J(t)$, and let (x, e) be a head event; by (FH), $e \le J$ (Lemma 8.4). Since $|u| - x = |t| - e \ge |t| - J$, we may delete $|t| - J$ letters from the end of both words: let u' be the prefix of u of length $L = |u| - (|t| - J)$ and t' the prefix of t of length J. The cuts of u' and t' are cuts of u and t with the same states, and $J(t') = J$. For $x \le L$ and $y \le J$, the pair (x, y) is a cross comparison of (u', t') iff $J - y \le L - x$, iff $|t| - y \le |u| - x$, iff it is one of (u, t); so (u', t') has no event, t' has no self event, and the head event (x, e) is a head event of (u', t'), with $x \le L$. The spine u' is safe. So (u', t') is a hole whose target is first rejected at its end.

The states $B_0(t'), \dots, B_J(t')$ are pairwise distinct: if $B_i = B_j$ with $i < j \le J$, the run of the single letter repeats with period $j - i$ from i on, so $B_{J-(j-i)}(t') = B_J(t') \in \mathrm{Rej}$ with $J - (j - i) < J$, contradicting the choice of J. Hence $|t'| = J \le N_B - 1$.

If $L < J$, then $|u'| + 1 + |t'| \le 2J \le 2N_B - 2$. Otherwise let m be the least cut of u' at which $A \times A'$ is in the state it has at the cut $L - J$. The states of $A \times A'$ at the cuts $0, \dots, m$ are pairwise distinct, so $m \le r_I - 1$: if two of them coincided, the run would be periodic from the first of them, and the state at the cut $L - J$ would occur before m, contrary to the choice of m. Let $u'' = 0^{m+J}$. Its states at the cuts $x \le m$ are those of u', and at the cut $m + y$, $y \le J$, those of u' at $L - J + y$. The pair (u'', t') is a hole. Its spine is safe, since all its states occur on u'. The head event (x, e) has $x = L - J + e$ and becomes the aligned pair $(m + e, e)$, with the same states. A cross comparison (x, y) of (u'', t') with $x \ge m$ corresponds to the cross comparison $(L - J + x - m, y)$ of (u', t'), with the same states, and one with $x < m$ has the same states as the pair (x, y) of (u', t'), which is a cross comparison because $L - x > L - m \ge J \ge J - y$; neither is an event. The target and its self events are unchanged. Finally $|u''| + 1 + |t'| \le r_I - 1 + 2J + 1 \le r_I + 2N_B - 2$.

(2) Let w be a hole with first rejected cut J. Its prefix of length J is rejected, and its comparisons are pairs of cuts $i < j \le J$, which are comparisons of w with the same states; so it is a hole. Its states at the cuts $0, \dots, J$ are pairwise distinct, as in (1), so $J \le N - 1$.

(3) By Proposition 11.5(1), a failure of (I0) has a word of length less than $2^{2v+2s+2}$. A failure of (U) has a (U)-word of length at most $2^{v+1} - 2$ (Remark 13.5). A hole of an extracted instance has one with $|u| + 1 + |t| \le r_I + 2N_B - 2$ by (1), where $r_I \le 2^{2v+2s}$ and $N_B = 2^{2v+2s+1}$. Each bound is less than 2^{4s+3}. Over $\{\bot, f\}$, the pair of Theorem 5.17 is $\mathcal{D}$-satisfiable over $\{\bot, f, \top\}$ and its entailment fails (proof of Theorem 11.2), and it has $2s$ variables, which gives 2^{8s+3}. For membership in PSPACE, follow the proof of Theorem 13.9, which handles satisfiability, the tree classes, $\{\bot, f\}$ and (I0) in polynomial space, and replace the searches for (U) and (II) by an enumeration of the candidates up to these bounds. A unary word is stored by its length, in $O(s)$ bits. The state of an automaton at a cut is a set of signed variables, recomputed from the start in polynomial space, and each condition of Definitions 2.12 and 8.1 quantifies over cuts, which binary counters enumerate; comparisons are tests of lengths. So all tests use polynomial space. For $\{\bot, f\}$ the gadget doubles s, which keeps the bounds singly exponential. □

Remark 13.12 (Arity zero). For $n = 0$ every tree is $\bot$, $f()$ or $\top$ (Remark 2.8), and the entailment says that every assignment of X that satisfies φ has an assignment of Z that satisfies ψ (Remark 5.20). Assignments have polynomial size and are checked in linear time, so the problem is in Π_2^p, hence in PSPACE, in each tree class, which coincide.

13.4 Lower Bounds

Long witnesses. Shortest counterexamples can be exponentially long, already for plain entailment over $\{\bot, f\}$ with a binary covariant constructor and antecedents without literals $u = \bot$. The construction uses a small automaton for the set of words that are not extensions of one exponentially long word, the record of a binary counter.

Proposition 13.13 (Exponential witnesses). *For every $k \geq 1$ there are a word $W_k \in \{0,1\}^*$ of length $L_k = 2(k+1)2^k$ and a satisfiable flat system φ_k over $\{\bot, f\}$, with f binary and covariant, without hidden variables and without literals $u = \bot$, over $9k + 30$ variables, such that $\varphi_k \models x \leq y$ fails over arbitrary and over regular trees, and:*

(1) every word of $K(\rho x) \setminus K(\rho y)$, for every solution ρ of φ_k, has the prefix W_k, and some regular solution has $W_k \in K(\rho x) \setminus K(\rho y)$;
(2) for the instance of Theorem 5.17 over $\{\bot, f, \top\}$, conditions (I0) *and* (U) *hold, and every hole (u, t) of an extracted instance has $|t| \geq L_k$; so every counterexample word of that instance (Definition 11.1) is a hole word longer than L_k.*

For every arity $n \geq 3$, statements (1) and (2) hold for a system over $\{\bot, f\}$ with f covariant of arity n and $18k + 61$ variables, which may contain literals $u = \bot$. Hence shortest counterexample words can have length $2^{\Omega(s)}$, for every fixed arity $n \geq 2$.

Proof. *A counter.* Over the alphabet $\Gamma = \{0, 1, \#, \$\}$ let $H = C_0 \# C_1 \# \cdots \# C_{2^k-1}\$$, where C_j is the binary expansion of j with k bits, least significant bit first; $|H| = (k+1)2^k$. Consider the nondeterministic automaton $\mathcal{N}$ over Γ, all of whose states are final, with the following states and transitions; a transition that is not listed does not exist.

- E loops on every letter.
- $\mathsf{F}_0, \ldots, \mathsf{F}_k$ check the format: F_i goes on a bit to F_{i+1} if $i < k$ and to E if $i = k$; on # to F_0 if $i = k$ and to E otherwise; on \$ to E if $i < k$.
- I checks that C_0 is zero: it goes on 0 to I and on 1 to E.
- $\mathsf{C}(c)$, $c \in \{0,1\}$, tracks the carry of an increment and spawns checkers: on a bit b it goes to both $\mathsf{C}(c \wedge b)$ and $\mathsf{W}(b \oplus c, k)$, and on # to $\mathsf{C}(1)$.
- $\mathsf{W}(b, j)$, $0 \leq j \leq k$, waits for the corresponding bit of the next block: for $j > 0$ it goes on every letter other than \$ to $\mathsf{W}(b, j-1)$; $\mathsf{W}(b, 0)$ goes on the bit $1 - b$ to E.
- $\mathsf{O}(o)$, $o \in \{0,1\}$, records whether the current block is all ones so far: on a bit b it goes to $\mathsf{O}(o \wedge b)$; on # to E if $o = 1$ and to $\mathsf{O}(1)$ if $o = 0$; $\mathsf{O}(0)$ goes on \$ to E.

The initial states are F_0, I, $\mathsf{C}(1)$ and $\mathsf{O}(1)$, and there are $3k + 9$ states. We claim that the words with a run of $\mathcal{N}$ are exactly the words in $\Gamma^* \setminus H\Gamma^*$.

No branch reaches E on a prefix of H: the format is right; C_0 is zero; a checker spawned at the bit of offset p in block C_i reads the k letters between it and the bit of offset p in C_{i+1}, which is $b \oplus c$ for the bit b and the incoming carry c, because $C_{i+1} = C_i + 1$; a # follows only blocks that are not all ones; and \$ follows the all-ones block C_{2^k-1}. After H no state remains: F_k, $\mathsf{C}(\cdot)$, $\mathsf{O}(1)$ and the checkers have no transition on \$, and I has none on #. So no word in $H\Gamma^*$ has a run. Every proper prefix of H has a run through the format states. Let w first differ from H at position $i < |H|$. If $H[i]$ is a bit of C_j and $w[i] \in \{\#, \$\}$, the format branch, at some F_p with $p < k$, reaches E. If $w[i]$ is the other bit, then I reaches E if $j = 0$, and otherwise the checker spawned at the same offset of C_{j-1} is at $\mathsf{W}(H[i], 0)$ and reaches E. If $H[i] = \#$, a bit leads F_k to E, and \$ leads $\mathsf{O}(0)$ to E, since C_j is not all ones. If $H[i] = \$$, a bit leads F_k to E and # leads $\mathsf{O}(1)$ to E. Since E loops, w has a run, which proves the claim.

Binary words. Code the tokens by $0 \mapsto 00$, $1 \mapsto 01$, $\# \mapsto 10$, $\$ \mapsto 11$, and let W_k be the code of H, of length L_k. Let $\mathcal{L}$ have the states q, $(q, 0)$ and $(q, 1)$ for every state q of $\mathcal{N}$, all final, the initial states of $\mathcal{N}$, a transition on the bit b from q to (q, b), and a transition on the bit b' from (q, b) to every successor of q on the token coded by bb'. It has $9k + 27$ states. By induction on the number of complete codes, the base states reachable on the code of a token word are the states reachable on that word in $\mathcal{N}$, and after one further bit the reachable set is nonempty iff that set is. So a binary

word has a run of $\mathcal{L}$ iff its longest prefix of even length is the code of a token word with a run of $\mathcal{N}$, that is, iff it does not begin with W_k. The language of $\mathcal{L}$ is therefore $\{0,1\}^* \setminus W_k\{0,1\}^*$.

The system. Let φ_k have the variables x, y, d and Y_q for the states q of $\mathcal{L}$, and the literals

$$f(d,d) \le Y_q, \qquad f(Y_r,d) \le Y_q \ \ (q \xrightarrow{0} r), \qquad f(d,Y_r) \le Y_q \ \ (q \xrightarrow{1} r), \qquad Y_q \le y \ \ (q \text{ initial}).$$

In every solution, $K(Y_q)$ contains every word that has a run of $\mathcal{L}$ from q, by induction on its length, using Lemma 2.3; hence $K(y) \supseteq \{0,1\}^* \setminus W_k\{0,1\}^*$, and every word of $K(x) \setminus K(y)$ begins with W_k. Conversely, let $K(Y_q)$ be the set of words with a run from q, $d = \bot$, $K(y)$ the language of $\mathcal{L}$, and $K(x)$ the set of prefixes of W_k. These sets are regular and prefix closed, since all states are final, and they satisfy every literal, so this is a regular solution with $W_k \in K(x) \setminus K(y)$. This proves (1); the antecedent is satisfiable and the entailment fails over arbitrary and regular trees (Theorem 5.12).

(2) The consequent is $\psi = \{x \le y\}$ and $Z = \varnothing$, so $\psi^\natural = \psi$ has no facts and no forbidden roots: (I0) and (U) hold. Its pure moves are the copies $(x^0, w) \to (y^0, w)$ and $(y^1, w) \to (x^1, w)$. Let (u,t) be a hole of an extracted instance I_{pq} of $\varphi^\natural$ and ψ. By Theorem 8.7, $(p,t) \to^* (q,u)$ by pure moves of ψ, and the least solution ρ of $\varphi^\natural$ that contains (p,t) omits (q,u); so $u = t$ and (p,q) is (x^0, y^0) or (y^1, x^1). By Lemma 5.16(1), ρx and ρy are $\top$-free, so their positive cones are empty, and $p = y^1$, $q = x^1$. For $\top$-free covariant trees the negative cone is the complement of the constructor positions (Remark 4.6), so $t \in K(\rho x) \setminus K(\rho y)$. The restriction of ρ to X solves φ_k over $\{\bot, f\}$, and (1) gives $|t| \ge L_k$.

For $n \ge 3$, apply Proposition 13.15 below: the solutions of the translated antecedent are, on the original variables, the images of the solutions of φ_k, with the same sets of constructor positions, and the translation has $2(9k + 30) + 1$ variables; it contains the literal $o = \bot$. Statement (2) carries over by the same argument. Finally $s \le 18k + 61$ and $\log_2 L_k \ge k + 1$, so $L_k \ge 2^{(s-61)/18}$. □

The antecedents have no finite solutions, since $K(y)$ is infinite, so the proposition gives no lower bound over finite trees.

Larger arities. A lower bound for a binary constructor does not transfer to a constructor of larger arity by *bare padding*, that is, by filling the new arguments of every constructor with $\bot$. The translated antecedent constrains values that are not translations of binary trees, and such values may refute the translated entailment.

Example 13.14 (Bare padding fails). Over $\{\bot, f\}$ with f_2 binary and covariant, true $\models \exists a\, b \ \ x \le f_2(a,b)$ holds in every tree class: take the children of x, or $a = b = \bot$ if $x = \bot$. With f_3 ternary and the new argument filled by $\bot$, the entailment true $\models \exists a\, b \ \ x \le f_3(a,b,\bot)$ fails at the finite tree $x = f_3(\bot, \bot, f_3(\bot,\bot,\bot))$. In the same way, padding the antecedent of Proposition 13.13 leaves x unconstrained, and a value of x with a constructor at the one-letter word 2, which lies outside $K(y)$, is a countermodel with a witness of length one.

The repair restricts every variable to the image of the binary trees, by a variant of the gadget of Section 5.5.

Proposition 13.15 (Embedding binary trees). *Let $n \ge 2$, let f be covariant of arity n, and let f_2 be binary and covariant. For a tree t over $\{\bot, f_2\}$ let $E(t)$ be the tree over $\{\bot, f\}$ with $K(Et) = K(t)$. For a system χ over $\{\bot, f_2\}$, let χ^E replace every literal $f_2(a_0,a_1) \le u$ by $f(a_0,a_1,o,\dots,o) \le u$ and every literal $u \le f_2(a_0,a_1)$ by $u \le f(a_0,a_1,o,\dots,o)$, for a fresh variable o. For φ over X and ψ over $X \cup Z$, put*

$$\varphi^{(n)} = \varphi^E \cup \{o = \bot\} \cup \bigcup_{x \in X} B(x, m_x), \qquad \psi^{(n)} = \psi^E \cup \bigcup_{z \in Z} B(z, m_z),$$

where $B(v,m) = \{v \le m,\ m \le f(m,m,o,\dots,o)\}$, with the visible variables $X \cup \{o\} \cup \{m_x\}_{x\in X}$ and the hidden variables $Z \cup \{m_z\}_{z\in Z}$. Then:

(1) E is a bijection from the trees over $\{\bot, f_2\}$ onto the trees over $\{\bot, f\}$ whose constructor positions lie in $\{0,1\}^*$; it preserves and reflects the order, maps each tree class onto its part of the image, and $E(f_2(a,b)) = f(Ea, Eb, \bot, \dots, \bot)$;

(2) for every tree class $\mathcal{D}$, $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ over $\{\bot, f_2\}$ iff $\varphi^{(n)} \models_{\mathcal{D}} \exists \bar{z}\, \bar{m}\, \psi^{(n)}$ over $\{\bot, f\}$, and the restrictions to X of the solutions of $\varphi^{(n)}$ are exactly the images under E of the solutions of φ.

The translation has $2s+1$ variables and size linear in the input for fixed n.

Proof. (1) Prefix-closed subsets of $\{0,1\}^*$ are exactly the sets of constructor positions of binary trees and of the trees in the image, and the order on both sides is inclusion of these sets (Lemma 2.3). A tree over $\{\bot, f\}$ is finite iff this set is finite, and regular iff it is a regular language (Lemma 13.6), on both sides.

(2) Let m be a tree over $\{\bot, f\}$ with $m \le f(m,m,\bot,\dots,\bot)$. Then $K(m) \subseteq \{\varepsilon\} \cup 0K(m) \cup 1K(m)$, and by induction on length every word of $K(m)$ lies in $\{0,1\}^*$. Conversely, the regular tree Θ with $K(\Theta) = \{0,1\}^*$ and the finite trees Θ_h with $K(\Theta_h) = \{0,1\}^{<h}$ satisfy this literal, and every tree in the image of E lies below Θ, and below Θ_h if it is finite with all positions shorter than h. So, with $o = \bot$, a value of v admits a value of m with $B(v,m)$ in the class $\mathcal{D}$ iff it lies in the image of E. On this image, a literal of χ^E with $o = \bot$ holds for $E \circ \sigma$ iff the corresponding literal of χ holds for σ, by (1). Both claims of (2) now follow as in the proof of Theorem 5.17, with E in place of the identity on $\top$-free trees. □

Hardness. Henglein and Rehof [1998, Theorems 1 and 2] proved PSPACE-hardness of plain entailment over $\{\bot, f, \top\}$ for a binary covariant constructor. We prove PSPACE-hardness for every fixed arity $n \ge 2$, in each tree class, by explicit reductions from the universality and the inclusion problems of nondeterministic automata over $\{0,1\}$, which are PSPACE-complete [Stockmeyer and Meyer 1973]. For a word $w \in \{0,1\}^*$ let $\mathrm{code}(w) = e(w)1$, where e replaces each letter 0 by 00 and each 1 by 01. The code is prefix free: if $\mathrm{code}(w') \preceq \mathrm{code}(w)$ and $|w'| < |w|$, then position $2|w'|$ carries the final 1 of $\mathrm{code}(w')$ but the letter 0 in $\mathrm{code}(w)$. From an automaton for L one obtains in linear time an automaton for $\mathrm{code}(L)$: replace each transition by two, the first labeled 0, and add a transition labeled 1 from every final state to one new final state.

Theorem 13.16 (PSPACE-hardness). *Let $n \ge 2$ be fixed and f covariant. In each of the three tree classes:*

(1) existential entailment over $\{\bot, f\}$ is PSPACE-hard;

(2) plain entailment over $\{\bot, f, \top\}$ is PSPACE-hard;

(3) consequently existential entailment over $\{\bot, f, \top\}$ is PSPACE-hard, and so is the problem with the variance as part of the input, already on covariant instances.

We claim no hardness for a fixed variance with a contravariant argument, and none for $n = 1$, where the problem is in PSPACE by Theorem 13.11. The reductions use the extra arguments of f either for an unconstrained filler, whose positions every condition ignores, or for $\bot$, under literals that already confine the relevant values to the letters 0 and 1; neither is bare padding.

Proof. Let $\Sigma = \{0,\dots,n-1\}$. For an automaton $\mathcal{N}$ over $\{0,1\}$ with states Q, initial state q_0 and final states F, write L_q for the words accepted from q.

(1) Let $\mathcal{F}(\mathcal{N})$ have the variables x, d and X_q for $q \in Q$, and the literals

$$x \le X_{q_0}, \quad X_q \le f(X_{q'}, d, \dots, d)\ (q \xrightarrow{0} q'), \quad X_q \le f(d, X_{q'}, d, \dots, d)\ (q \xrightarrow{1} q'), \quad X_q = \bot\ (q \in F).$$

We claim: a tree t over $\{\bot, f\}$ in a class $\mathcal{D}$ is the value of x in some $\mathcal{D}$-solution iff no word of L_{q_0} is a constructor position of t. If $\pi \in K(t)$ and $q_0 \xrightarrow{\pi} q$, then by induction on π the subtree of t at π lies below X_q: an upper literal at a constructor node passes to the child in the direction of the transition. For $\pi \in L_{q_0}$ with $q \in F$ this contradicts $X_q = \bot$. Conversely, let h bound the lengths of the positions of t if t is finite, and put $h = \infty$ otherwise, with $\Sigma^{\leq \infty} = \Sigma^*$. Let $K(d) = \Sigma^{\leq h}$ and let $K(X_q)$ be the words of $\Sigma^{\leq h}$ with no prefix in L_q. These sets are prefix closed, finite if $h < \infty$ and regular otherwise. If $0w \in K(X_q)$ and $q \xrightarrow{0} q'$, then no prefix of w lies in $L_{q'}$, so $w \in K(X_{q'})$, and every $iw \in K(X_q)$ with $i \neq 0$ has $w \in K(d)$; so the upper literals hold by Lemma 2.3, and symmetrically for the letter 1. For $q \in F$, $\varepsilon \in L_q$, so $X_q = \bot$. Finally $K(t) \subseteq K(X_{q_0})$, since $K(t)$ is prefix closed and avoids L_{q_0}.

Given automata $\mathcal{N}_1, \mathcal{N}_2$, let $\mathcal{N}_i'$ accept $\mathrm{code}(L(\mathcal{N}_i))$. Let $\varphi = \mathcal{F}(\mathcal{N}_2')$, with all its variables visible, and let $\psi = \mathcal{F}(\mathcal{N}_1')$ with fresh variables except x, all of them hidden. Both systems are satisfiable, by the assignment $\bot$. By the claim, $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ iff every $\top$-free $t \in \mathcal{D}$ whose constructor positions avoid $\mathrm{code}(L(\mathcal{N}_2))$ also avoids $\mathrm{code}(L(\mathcal{N}_1))$. If $L(\mathcal{N}_1) \subseteq L(\mathcal{N}_2)$, this holds. If $w \in L(\mathcal{N}_1) \setminus L(\mathcal{N}_2)$, let t be the finite tree whose constructor positions are the prefixes of $\mathrm{code}(w)$; a prefix of $\mathrm{code}(w)$ in $\mathrm{code}(L(\mathcal{N}_2))$ would equal $\mathrm{code}(w)$ by prefix freeness, so t is a countermodel. Inclusion of automata is PSPACE-hard, and the reduction is polynomial.

(2) Let $\mathcal{N}$ accept $\mathrm{code}(L(\mathcal{N}_0))$ for a given automaton $\mathcal{N}_0$, and let $C = (00 \mid 01)^*1$ be the set of all codes and P the set of proper prefixes of codes, the words $e(w)$ and $e(w)0$. Let χ have the variables x, d, b, r, r' and X_q for $q \in Q$, and the literals

$$x \leq X_{q_0}, \quad X_q \leq f(X_{q'}, d, b, \dots, b)\ (q \xrightarrow{0} q'), \quad X_q \leq f(d, X_{q'}, b, \dots, b)\ (q \xrightarrow{1} q'), \quad X_q = \bot\ (q \in F),$$
$$b = \bot, \qquad f(r', b, \dots, b) \leq r, \qquad f(r, r, b, \dots, b) \leq r'.$$

We show that $\chi \models_{\mathcal{D}} x \leq r$ iff $L(\mathcal{N}_0) = \{0,1\}^*$.

Suppose $\mathcal{N}_0$ is universal, so that $\mathcal{N}$ accepts every code, and let ρ be a solution. *The values of r.* Call r the state of the words $e(w)$ and r' that of the words $e(w)0$. By induction along P, if $\pi \in P$ is a position of ρr, then its subtree lies above the value of its state: from r, the literal $f(r', b, \dots, b) \leq r$ gives a node f or $\top$ whose child 0, if it exists, lies above r', and from r' the other literal gives children 0 and 1 above r. In particular ρr is not $\bot$ at any position in P. *The values of x.* Every $\pi \in P$ is a proper prefix of an accepted code, so an accepting run reaches, on π, a state q with an outgoing transition; as in (1), if π is a position of ρx, its subtree lies below X_q, which lies below a constructor tree, so ρx is not $\top$ at π, and the children of that subtree in the directions $i \geq 2$ are $\bot$. Every code c is accepted, so if c is a position of ρx, its subtree lies below $X_q = \bot$ for a final q. The children of a word of P are words of P, codes, and words πi with $\pi \in P$ and $i \geq 2$, and ρx is $\bot$ at the last two kinds; so every position of ρx is of one of these three kinds. Hence at every common position either ρx is $\bot$, or the position lies in P, where ρx has a label in $\{\bot, f\}$ and ρr one in $\{f, \top\}$. So $\rho x \leq \rho r$.

Suppose $w \notin L(\mathcal{N}_0)$, and let $c = \mathrm{code}(w)$, so that no prefix of c is accepted by $\mathcal{N}$, by prefix freeness. Let x be the finite tree whose constructor positions are the prefixes of c, c included. For $q \in Q$ let X_q be the join, the union of constructor positions, of the subtrees of x at the prefixes π of c with $q_0 \xrightarrow{\pi} q$; each is the path tree along the rest of c. Then $x \leq X_{q_0}$, $X_q = \bot$ for $q \in F$, and the upper literals hold with $b = \bot$ and d the finite tree with $K(d) = \{0,1\}^{\leq |c|}$, as in (1). Let $r_0 = r_0' = \top$, $r_{h+1} = f(r_h', \bot, \dots, \bot)$ and $r_{h+1}' = f(r_h, r_h, \bot, \dots, \bot)$. By induction $r_{h+1} \leq r_h$ and $r_{h+1}' \leq r_h'$, so (r_h, r_h') satisfies the two lower literals, and for $h > |c|$ the tree r_h has the label $\bot$ at c, where x has f. This finite solution refutes $x \leq r$.

Universality is PSPACE-hard and the reduction is polynomial; since PSPACE is closed under complement, this proves (2). Item (3) follows from (1) by Theorem 5.17, or from (2), since plain entailment is the case $Z = \varnothing$ and the covariant instances have the variance $c = 0$. □

The reduction of (2) reproves, for $n = 2$, the theorem of Henglein and Rehof [1998], in all three tree classes, and extends it to every fixed arity at least two. Together with Theorems 13.9 and 13.11, the problem is PSPACE-hard and in 2-EXPSPACE for every fixed $n \geq 2$ and covariant f, and in PSPACE for $n \leq 1$; whether it is PSPACE-hard for $n = 1$ is open.

14 Mechanization

The Lean 4 development [de Moura and Ullrich 2021], using Mathlib [The mathlib Community 2020], mechanizes every numbered result before Section 13 and the supporting results in the appendices. This includes flattening, the transfers between tree classes, the reduction of covariant $\{\bot, f\}$ to $\{\bot, f, \top\}$, the hole bounds and their closed forms in the number of variables, countermodels, and the applications to constrained types. The paper and Lean follow the same proof routes. The complexity results and sharper bounds of Section 13 and Appendix H are outside the mechanized part.

The correspondence is recorded in `DeciENSSE/Paper/*.lean`. Each result has a declaration in the namespace `DeciENSSE.Paper`, with one entry per numbered item for multipart results. The contract described in `scripts/README-contract.md` checks the paper labels, item numbers, result kinds, and declaration names. It rejects dangling or duplicate entries; strict mode also rejects missing required entries. The generated table in Appendix I.2 lists the declarations and their statement-form notes. This check enforces the correspondence of names and items; reviewing the meaning of a Lean statement against the mathematical statement remains necessary.

The public existential-entailment procedures cover both signatures in all three tree classes, including arity zero. Each is a computable definition with a correctness theorem, without satisfiability premises. Decidability comes from bounded searches, rather than from a classical choice of answers. Plain entailment is obtained by taking no hidden variables. Flattening extends the procedures to finite type expressions. Classical reasoning is used in proofs of correctness and bounds; it does not replace these searches.

The trusted statements are the tree semantics and order in `DeciNSSE/Semantics/`, the syntax, satisfaction, and entailment relations in `DeciENSSE/Semantics.lean`, the statements of the public deciders, and the contract statements themselves. The kernel checks their proofs. The only axioms used are `propext`, `Classical.choice`, and `Quot.sound`. Executing compiled procedures also relies on the compiler. To build the library and check coverage and axioms, run from the repository root:

```
lake build
python3 scripts/contract.py --strict --axioms
```

The second command checks every contract declaration against this axiom list.

The representation conventions are few: finite variable carriers and sum types for visible and hidden variables, lists of literals, finite graphs for regular trees and finite terms for finite trees, optional subtrees, and default letters for total hierarchy constructions. Appendix I describes these conventions and the module layout. They change the representation of the statements, not the proof routes. The artifact is included in the supplementary material.

15 Related Work

Table 8 summarizes selected results on entailment questions that involve existential quantifiers or projections.

Table 8. Selected results related to existential subtype entailment. "Open alphabet" means that a feature not occurring in the formula is always available.

Problem	Trees	Result	Reference
Structural subtyping, first-order theory (hence existential entailment)	finite	decidable	Kuncak and Rinard [2003]
Non-structural subtyping over $\{\bot, f\}$, f binary and covariant, first-order theory	finite, regular, arbitrary	undecidable, with six quantifier blocks	Su et al. [2002]
Feature-tree ordering constraints, existential entailment, open alphabet	finite, arbitrary	PSPACE-complete	Müller et al. [1998, 2001]
Restricted entailment (projection inclusion) in a set-based selector domain	complete infinite	exponential time; PSPACE-hard	Flanagan and Felleisen [1999]
Subtyping of constrained types with $\to$, $\bot$, $\top$	regular	posed as open; sound decidable approximation	Trifonov and Smith [1996]
Subsumption of polar type schemes	extensible algebra	decidable	Dolan [2016]
Plain NSSE, one constructor of any arity and variance	finite, regular, arbitrary	decidable	Zhao [2026]
Existential entailment, one constructor of any arity and variance over $\{\bot, f, \top\}$, and over $\{\bot, f\}$ with f covariant	finite, regular, arbitrary	decidable (mechanized); in 2-EXPSPACE and PSPACE-hard for each fixed arity ≥ 2, already with f covariant (not mechanized)	this paper

Constrained Types and Their Simplification. Constrained types with recursive constraints were used for type inference for objects by Eifrig et al. [1995]. Trifonov and Smith [1996] defined subtyping between constrained types semantically, over regular trees with arrows, $\bot$, and $\top$; Eq. (1) is their definition, for types without assumptions on program variables, written as an existential entailment. Their decision procedure is a sound approximation based on canonical constraint maps and primitive derivations, and Corollary 12.2 decides their relation exactly. Pottier [1996] described an incomplete entailment algorithm, together with the removal of unreachable constraints and simplifying substitutions. Pottier [2001] developed polarity-based garbage collection, canonization, and minimization, and proved completeness in the equality setting; completeness is lost for subtyping. Pottier [1998b, Section 8.5] gives a correct simplification that canonical bounds miss; Example 12.4 recalls it. Aiken et al. [1999] eliminate quantified variables and obtain optimal representations for unconstrained polymorphic types with subtyping; their extension to constrained types is sound but not complete. Fähndrich and Aiken [1996] use polarity-based minimization, removal of intermediate variables, and cycle collapse in an implemented analysis, and they define the comparison of constrained types by a universal-existential condition. For constrained types, all these works use sound transformations or approximations; Theorem 11.2 gives a complete test.

Flanagan and Felleisen [1999] decide a closely related problem. Their restricted entailment $S_1 \models_E S_2$ is the inclusion $\mathrm{Sol}(S_1)|_E \subseteq \mathrm{Sol}(S_2)|_E$ of projections onto a set E of external variables, which after renaming is the existential entailment $S_1 \models \exists \bar{z}\, S_2$ with $\bar{z}$ the variables of S_2 outside E. They decide it completely, in exponential time, and show PSPACE-hardness. Their domain, however,

is the set-based domain $D = \mathcal{P}(\text{Const}) \times D \times D$ of complete infinite trees with total selectors and a contravariant domain component, not the ground trees of non-structural subtyping, so their decision procedure does not apply to our setting.

Feature Trees and Set Constraints. For ordering constraints over feature trees, the full first-order theory is undecidable, while existential entailment is PSPACE-complete over finite and over possibly infinite trees [Müller et al. 1998, 2001]. Their procedure saturates the constraints, builds polynomial-size automata that recognize consequences expressed by paths, hides the existential variables while keeping their states, and decides $\varphi \models \exists \bar{z}\, \psi$ by an inclusion of the resulting languages. Our three conditions and the pure derivations through hidden variables play the role of their consequence languages and hidden states. Their setting differs from ours in one essential respect: the feature alphabet is open, so a feature that does not occur in the formula is always available. Their PSPACE-hardness proof, for instance, builds countermodels with such a fresh feature [Müller et al. 2001, Lemma 5.5], and Müller et al. [2000] use fresh features to construct countermodels for ordering constraints. A ranked signature has no fresh direction: a constructor node has exactly its prescribed children, and $\bot$ and $\top$ have none.

The alphabet matters already for feature trees, as the following example over arbitrary trees shows. Over the features $\{0, 1\}$, the constraints $a(x) \wedge x[0]x \wedge x[1]x \wedge x \leq y$ make x the complete infinite tree labeled a everywhere, so every y above x equals x, and the constraints entail $y \leq x$. Once a third feature is available, y may extend x along it, and the entailment fails. The open-alphabet and the fixed-alphabet problems thus have different answers on this instance, and a decision procedure for the first does not decide the second. The same phenomenon appears in our setting when the arity grows: an entailment for a binary constructor need not survive bare padding to a larger arity (Example 13.14). In set-constraint languages, entailment of atomic set constraints is PSPACE-complete over infinite signatures [Niehren et al. 1999]; these constraints have no existential conclusions, and sets of trees are a different domain from trees under the subtype order.

The First-Order Theory. The undecidability proof of Su et al. [2002], recalled in Section 1, reduces the Post correspondence problem to a sentence whose matrix uses implications, disjunctions, and negations; for example, it expresses that the subtypes of a tree form a chain. Our problem is the validity of $\forall \bar{x}\, (\varphi \Longrightarrow \exists \bar{z}\, \psi)$ with conjunctive φ and ψ, a proper restriction of the $\forall^* \exists^*$ fragment. The same paper decides the first-order theory over signatures with only unary and nullary constructors. Su [2002] and the extended version of the paper [Su et al. 2006] reduce entailment to emptiness of constrained automata on tuples of trees, which is undecidable in general. The remark of Su [2002, Section 7.2.2] quoted in Section 1 concerns an adaptation of that undecidability proof; no formula with a single alternation is displayed, and undecidability of a fragment with one alternation but an unrestricted matrix would still not settle the case of conjunctions. Kuncak and Rinard [2003] showed that the first-order theory of structural subtyping of non-recursive types is decidable, by a term-power construction. This covers existential entailment for structural subtyping, which compares only trees of the same shape, over finite trees.

The Automata Route to Plain Entailment. Niehren and Priesnitz [1999] proved PSPACE-completeness of plain non-structural entailment for the fragment without literals $x = \bot$ and $x = \top$, by a reduction to universality of finite automata. Niehren and Priesnitz [2001, 2003] introduced cap automata, decided the unary case, and identified the role of word equations in the general problem. Henglein and Rehof [1998] proved PSPACE-hardness of non-structural entailment for a binary covariant constructor, which Theorem 13.16, in the part of this paper that is not mechanized, extends to every fixed arity at least two, and the theses of Priesnitz [Priesnitz 2000, 2004] develop

the automata approach further. Niehren et al. [2005] classify the complexity of satisfiability over posets and note that their technique does not extend beyond satisfiability. Our reduction keeps the spirit of this line: the obstructions are periodicity conditions between suffixes of paths. The new features are that two words with a common suffix are compared, that a word may be compared with itself through a self event, and that the separator encoding returns the two words to the one-word readers of Zhao [2026].

Pushdown Saturation Through Internal Variables. The type inference of Noonan et al. [2016] for machine code uses derived type variables, which extend a variable by a word of field labels, and observes that "the full set of constraints entailed by C has a finite encoding by an automaton". It builds a transducer for derivations through uninteresting variables by a modification of the saturation procedure for prefix rewriting [Büchi 1964; Caucal 1992], and normalizes derivations to pops followed by pushes. This is a close precedent for our pure derivations through hidden variables and their lowest-point decomposition (Lemma 6.3). Their entailment is derivability in a proof system, while ours is semantic entailment over ground trees. Completeness for the semantic notion is what needs the leaf-cone characterization and the analysis of two-word holes, which have no counterpart there. The saturation in our setting is the computation of the closure D_χ of the signed system; no further saturation is needed, because the pushdown summaries are read off from it (Definition 6.1 and Lemma 6.2).

Algebraic and Semantic Subtyping. Algebraic subtyping [Dolan 2016; Dolan and Mycroft 2017] interprets types in an extensible algebra rather than as ground trees and infers principal types by biunification. Subsumption between its polar type schemes is decided in the thesis [Dolan 2016]; the conference paper does not discuss deciding subsumption [Dolan and Mycroft 2017]. Simple-sub [Parreaux 2020] gives a compact account of this inference; MLstruct extends it to a Boolean algebra of structural types and does not formally describe its subsumption check [Parreaux and Chau 2022]; and Parreaux et al. [2024] keep subtyping constraints in first-class polymorphic types. Because the semantics differ, decidability there does not decide the ground-tree problem. Semantic subtyping interprets types as sets of values. Castagna and Xu [2011] decide subtyping between polymorphic regular types in EXPTIME, and Gesbert et al. [2015] decide semantic subtyping through a logic. For polymorphic application, Castagna et al. [2015] decide tallying, but inferring the type substitutions in general uses sound and complete semi-decision procedures whose termination is left open. These are questions about types, not about the solutions of constraint systems over ground trees.

Representations of Solution Spaces. Kozen [1995] represents the solution spaces of set constraints as rational spaces and relates logical entailment to rational embeddings between these spaces. Our leaf representation (Theorem 3.2) is a representation theorem of a different kind, in the tradition of rings of sets [Birkhoff 1937] and Alexandrov topologies [Alexandroff 1937]: a tree of any variance is a pair of disjoint sets of words that are closed under extension, and the order of trees is inclusion of the first set and reverse inclusion of the second. This elementary fact becomes useful because every literal then becomes clauses with a single premise. For a covariant constructor over $\{\bot, f\}$, solution sets are closed under unions and intersections (Remark 4.6); in general they are closed under intersections of leaf cones (Proposition 5.5).

Decidability of Plain Entailment. Zhao [2026] decides plain entailment $\varphi \models x \leq y$ for one constructor of any arity and variance, by reducing non-entailment to holes of finite readers and bounding a shortest hole through its rejected tail, its depth, and a compression argument. It lists existential entailment among the natural extensions that are not decided by testing component inequalities independently. We use the results recalled in Section 2 and Appendix A. Condition (U) is a hole problem of exactly the kind solved there, decided by its hole search applied to the unit

reader (Section 7); the readers of that paper already compare a root with its own signed copy, the plain counterpart of our self events. For the two-word holes of condition (II), the rejected tail of a mortal hole is short by definition, so its depth theorem already bounds the depth of the separator word (Lemma 10.6). Our main technical additions to that machinery are the pump for holes that are not mortal (Theorem 9.14) and the extension of the bounded-depth search to holes that must contain an anchored comparison (Theorem 10.14).

16 Conclusion

We have shown that existential subtype entailment $\varphi \models \exists \bar{z}\, \psi$ for one constructor is decidable over arbitrary, regular, and finite trees, for every arity and every variance over $\{\bot, f, \top\}$, and over $\{\bot, f\}$ when the constructor is covariant (Theorem 11.2). Signed leaf cones turn every system into a Horn system with unary rules, and collision return reduces the entailment to three conditions on single seeds. Two of them become hole problems for finite readers. Condition (U) is a hole problem of the kind solved for plain entailment [Zhao 2026]; condition (II) is a hole problem for two words that share a suffix, in which a word can also obstruct itself through a self event. Holes that are not mortal are found through breakers, lassos and a pump, and are singly exponential. Mortal holes are packed around a separator into one word of bounded depth, and a compression that carries a positive channel up the hierarchy keeps the anchored comparison that a mortal hole must contain. The transfers to regular and finite trees hold for every variance, and a gadget reduces covariant $\{\bot, f\}$ to $\{\bot, f, \top\}$; both come before the hole problems, which are therefore solved in a single setting.

The Signature without Top and a Contravariant Argument. We do not know whether existential entailment over $\{\bot, f\}$ is decidable when f has a contravariant argument. Neither omitting the literals $\top \le u$ nor the gadget of Theorem 5.17 reduces this case to $\{\bot, f, \top\}$ (Example 5.19): the gadget excludes leaves $\top$ only because each covariant child of its variable lies below the variable itself. A decision for this case needs either a gadget that bounds the leaves of both polarities without $\top$ literals, or a direct treatment of the cones of trees without $\top$, in which every leaf is $\bot$ but its sign still depends on the polarity of its path.

Complexity. By the results of the part that is not mechanized (Section 13), the problem is PSPACE-hard for every fixed arity $n \ge 2$ with f covariant, and in 2-EXPSPACE (Theorems 13.9 and 13.16). For $n \le 1$ it is in PSPACE (Theorem 13.11 and Remark 13.12), but we know no lower bound for $n = 1$: the reductions of Theorem 13.16 need two independent directions, and a unary constructor offers only one. Closing the gap between PSPACE and 2-EXPSPACE for $n \ge 2$ is open, as is the complexity for a fixed variance with a contravariant argument, for which we claim no lower bound.

Witness Lengths. Shortest counterexample words are at most triply exponential (Theorem 11.2), and they can be exponentially long (Proposition 13.13, not mechanized). Words for (I0) and holes that are not mortal are singly exponential, and the closed form for (U) is at most the mortal closed form (Proposition 11.5), which therefore gives the uniform bound of Theorem 11.2. Whether mortal holes have a doubly exponential bound is open. The length bound of anchored compression is exponential in the depth and in the square of the dictionary size, and the dictionaries are exponential in the number of states and the depth; dictionaries of polynomial size would give a doubly exponential bound. Bounded depth and the subset structure of the automata do not give such dictionaries by themselves, and a different compression, or a direct analysis of the long common suffix of a mortal hole, may be needed. The words for (U) come from the compression of Zhao [2026], whose bound has the same form, so an improvement of the overall bound would need it there as well.

The exponential lower bound holds over arbitrary and regular trees; we have no superpolynomial lower bound over finite trees.

Richer Signatures and Conclusions. Several constructors, possibly together with nullary base types, change the order on labels: incomparable constructors clash, and the leaf cones would have to record which constructor occurs at each position, while the polarity of a path would depend on the constructors along it. Entailment of disjunctions, $\varphi \models \psi_1 \vee \psi_2$, is not covered either: a disjunction is not a Horn clause, and the single-seed characterization of Section 4 rests on the Horn form.

Mechanization. The mechanized proof follows the paper's routes and covers every numbered result before Section 13, including flattening, the bounds in the number of variables, countermodels, and the applications. Its decision procedures are computable definitions, and the paper–Lean contract checks the correspondence (Section 14 and Appendix I). Mechanizing Section 13, in particular the smaller unit reader of Section 13.1, the summary search of Proposition 13.8 and the reductions of Theorem 13.16, would make the complexity bounds machine checked as well.

Implementation in Type Inference. The procedure of Theorem 11.2 is not meant to be run as stated. A practical subsumption or simplification checker could test satisfiability and condition (I0) by bounded searches, and then search for short holes of the unit readers and of the extracted instances. Each counterexample found can be checked directly and yields a countermodel (Lemma 11.3); only the absence of counterexamples depends on the bounds. Which of these cases arise in the constrained types produced by inference, and how often mortal holes or long words for (U) are needed, are empirical questions that we leave open.

Use of LLM-Based Tools

Tools based on large language models (LLMs) assisted proof attempts, checks of arguments, exploratory programs, and the formalization. Kernel-checked proofs support the decision procedures (Section 14). Search results were treated as evidence, not as proofs of universal claims.

References

Alexander Aiken and Edward L. Wimmers. 1993. Type Inclusion Constraints and Type Inference. In *Proceedings of the Conference on Functional Programming Languages and Computer Architecture (FPCA)*. 31–41.

Alexander Aiken, Edward L. Wimmers, and Jens Palsberg. 1999. Optimal Representations of Polymorphic Types with Subtyping. *Higher-Order and Symbolic Computation* 12 (1999), 237–282. doi:10.1023/A:1010056315933

Paul Alexandroff. 1937. Diskrete Räume. *Matematicheskii Sbornik (Recueil Mathématique), Nouvelle Série* 2(44), 3 (1937), 501–519.

Garrett Birkhoff. 1937. Rings of Sets. *Duke Mathematical Journal* 3, 3 (1937), 443–454.

J. Richard Büchi. 1964. Regular Canonical Systems. *Archiv für mathematische Logik und Grundlagenforschung* 6 (1964), 91–111.

Giuseppe Castagna, Kim Nguyễn, Zhiwu Xu, and Pietro Abate. 2015. Polymorphic Functions with Set-Theoretic Types. Part 2: Local Type Inference and Type Reconstruction. In *Proceedings of the 42nd Annual ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*. doi:10.1145/2676726.2676991

Giuseppe Castagna and Zhiwu Xu. 2011. Set-Theoretic Foundation of Parametric Polymorphism and Subtyping. In *Proceedings of the 16th ACM SIGPLAN International Conference on Functional Programming (ICFP)*. 94–106. doi:10.1145/2034773.2034788

Didier Caucal. 1992. On the Regular Structure of Prefix Rewriting. *Theoretical Computer Science* 106, 1 (1992), 61–86.

Leonardo de Moura and Sebastian Ullrich. 2021. The Lean 4 Theorem Prover and Programming Language. In *Automated Deduction – CADE 28 (Lecture Notes in Computer Science, Vol. 12699)*. Springer, 625–635. doi:10.1007/978-3-030-79876-5_37

Stephen Dolan. 2016. *Algebraic Subtyping*. Ph. D. Dissertation. University of Cambridge.

Stephen Dolan and Alan Mycroft. 2017. Polymorphism, Subtyping, and Type Inference in MLsub. In *Proceedings of the 44th ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*. 60–72. doi:10.1145/3009837.3009882

Jonathan Eifrig, Scott Smith, and Valery Trifonov. 1995. Sound Polymorphic Type Inference for Objects. In *Proceedings of the Conference on Object-Oriented Programming Systems, Languages, and Applications (OOPSLA)*. 169–184.

Manuel Fähndrich and Alexander Aiken. 1996. *Making Set-Constraint Program Analyses Scale.* Technical Report UCB/CSD-96-917. University of California, Berkeley.

Nathan J. Fine and Herbert S. Wilf. 1965. Uniqueness Theorems for Periodic Functions. *Proc. Amer. Math. Soc.* 16 (1965), 109–114. doi:10.1090/S0002-9939-1965-0174934-9

Cormac Flanagan and Matthias Felleisen. 1999. Componential Set-Based Analysis. *ACM Transactions on Programming Languages and Systems* 21, 2 (1999). doi:10.1145/316686.316703

You-Chin Fuh and Prateek Mishra. 1990. Type Inference with Subtypes. *Theoretical Computer Science* 73, 2 (1990), 155–175.

Nils Gesbert, Pierre Genevès, and Nabil Layaïda. 2015. A Logical Approach to Deciding Semantic Subtyping. *ACM Transactions on Programming Languages and Systems* 38, 1 (2015), 3:1–3:31. doi:10.1145/2812805

Fritz Henglein and Jakob Rehof. 1998. Constraint Automata and the Complexity of Recursive Subtype Entailment. In *Automata, Languages and Programming (ICALP 1998) (Lecture Notes in Computer Science, Vol. 1443)*. Springer, 616–627. doi:10.1007/BFb0055089

Dexter Kozen. 1995. Rational Spaces and Set Constraints. In *TAPSOFT '95: Theory and Practice of Software Development (LNCS, Vol. 915)*. 42–61. doi:10.1007/3-540-59293-8_186

Viktor Kuncak and Martin Rinard. 2003. Structural Subtyping of Non-Recursive Types Is Decidable. In *Proceedings of the 18th Annual IEEE Symposium on Logic in Computer Science (LICS)*. doi:10.1109/LICS.2003.1210049

M. Lothaire. 2002. *Algebraic Combinatorics on Words.* Encyclopedia of Mathematics and its Applications, Vol. 90. Cambridge University Press.

John C. Mitchell. 1991. Type Inference with Simple Subtypes. *Journal of Functional Programming* 1, 3 (1991), 245–285.

Martin Müller, Joachim Niehren, and Andreas Podelski. 2000. Ordering Constraints over Feature Trees. *Constraints* 5, 1–2 (2000), 7–42.

Martin Müller, Joachim Niehren, and Ralf Treinen. 1998. The First-Order Theory of Ordering Constraints over Feature Trees. In *Proceedings of the 13th Annual IEEE Symposium on Logic in Computer Science (LICS)*. 432–443. doi:10.1109/LICS.1998.705677

Martin Müller, Joachim Niehren, and Ralf Treinen. 2001. The First-Order Theory of Ordering Constraints over Feature Trees. *Discrete Mathematics and Theoretical Computer Science* 4, 2 (2001), 193–234. doi:10.46298/dmtcs.267

Joachim Niehren, Martin Müller, and Jean-Marc Talbot. 1999. Entailment of Atomic Set Constraints Is PSPACE-Complete. In *Proceedings of the 14th Annual IEEE Symposium on Logic in Computer Science (LICS)*. 285–294. doi:10.1109/LICS.1999.782623

Joachim Niehren and Tim Priesnitz. 1999. Entailment of Non-structural Subtype Constraints. In *Advances in Computing Science – ASIAN'99 (Lecture Notes in Computer Science, Vol. 1742)*. Springer, 251–265. doi:10.1007/3-540-46674-6_22

Joachim Niehren and Tim Priesnitz. 2001. Non-structural Subtype Entailment in Automata Theory. In *Theoretical Aspects of Computer Software (TACS 2001) (Lecture Notes in Computer Science, Vol. 2215)*. Springer, 360–384.

Joachim Niehren and Tim Priesnitz. 2003. Non-structural Subtype Entailment in Automata Theory. *Information and Computation* 186, 2 (2003), 319–354. doi:10.1016/S0890-5401(03)00140-8

Joachim Niehren, Tim Priesnitz, and Zhendong Su. 2005. Complexity of Subtype Satisfiability over Posets. In *Programming Languages and Systems (ESOP 2005) (Lecture Notes in Computer Science, Vol. 3444)*. Springer, 357–373. doi:10.1007/978-3-540-31987-0_25

Matt Noonan, Alexey Loginov, and David Cok. 2016. Polymorphic Type Inference for Machine Code. In *Proceedings of the 37th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*. 27–41. doi:10.1145/2908080.2908119

Lionel Parreaux. 2020. The Simple Essence of Algebraic Subtyping: Principal Type Inference with Subtyping Made Easy (Functional Pearl). *Proceedings of the ACM on Programming Languages* 4, ICFP (2020), 124:1–124:28. doi:10.1145/3409006

Lionel Parreaux, Aleksander Boruch-Gruszecki, Andong Fan, and Chun Yin Chau. 2024. When Subtyping Constraints Liberate: A Novel Type Inference Approach for First-Class Polymorphism. *Proceedings of the ACM on Programming Languages* 8, POPL (2024). doi:10.1145/3632890

Lionel Parreaux and Chun Yin Chau. 2022. MLstruct: Principal Type Inference in a Boolean Algebra of Structural Types. *Proceedings of the ACM on Programming Languages* 6, OOPSLA2 (2022). doi:10.1145/3563304

François Pottier. 1996. Simplifying Subtyping Constraints. In *Proceedings of the First ACM SIGPLAN International Conference on Functional Programming (ICFP)*. 122–133. doi:10.1145/232627.232642

François Pottier. 1998a. A Framework for Type Inference with Subtyping. In *Proceedings of the Third ACM SIGPLAN International Conference on Functional Programming (ICFP)*. 228–238. doi:10.1145/289423.289448

François Pottier. 1998b. *Type Inference in the Presence of Subtyping: From Theory to Practice.* Ph. D. Dissertation. Université Paris 7.

François Pottier. 2001. Simplifying Subtyping Constraints: A Theory. *Information and Computation* 170, 2 (2001), 153–183. doi:10.1006/inco.2001.2963

Tim Priesnitz. 2000. *Entailment von nicht-strukturellen Teiltyp-Constraints.* Diplomarbeit. Universität des Saarlandes.

Tim Priesnitz. 2004. *Subtype Satisfiability and Entailment.* Ph. D. Dissertation. Saarland University. doi:10.22028/D291-25888

Jakob Rehof. 1996. Problem #16: Is the Subtype Entailment Problem Decidable? TLCA List of Open Problems. https://tlca.di.unito.it/opltlca/opltlcasu23.html

Larry J. Stockmeyer and Albert R. Meyer. 1973. Word Problems Requiring Exponential Time: Preliminary Report. In *Proceedings of the 5th Annual ACM Symposium on Theory of Computing (STOC)*. 1–9. doi:10.1145/800125.804029

Zhendong Su. 2002. *Algorithms for and the Complexity of Constraint Entailment.* Ph. D. Dissertation. University of California, Berkeley.

Zhendong Su, Alexander Aiken, Joachim Niehren, Tim Priesnitz, and Ralf Treinen. 2002. The First-Order Theory of Subtyping Constraints. In *Proceedings of the 29th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*. 203–216. doi:10.1145/503272.503292

Zhendong Su, Alexander Aiken, Joachim Niehren, Tim Priesnitz, and Ralf Treinen. 2006. The First-Order Theory of Subtype Constraints. Extended version of the POPL 2002 paper. https://www.ps.uni-saarland.de/Publications/documents/fot_04.pdf

The mathlib Community. 2020. The Lean Mathematical Library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP)*. 367–381. doi:10.1145/3372885.3373824

Valery Trifonov and Scott Smith. 1996. Subtyping Constrained Types. In *Static Analysis Symposium (SAS) (Lecture Notes in Computer Science, Vol. 1145)*. Springer, 349–365. doi:10.1007/3-540-61739-6_52

Yaoyu Zhao. 2026. Non-Structural Subtype Entailment Is Decidable. Manuscript.

A Results Used from the Plain-Entailment Proof

This appendix collects what we use from the decidability proof for plain entailment of Zhao [2026] beyond Section 2: the satisfiability criteria (Appendix A.1); the cores of the canonical hierarchy (Appendix A.2); the ordinary and symbolic states of derived monitors (Appendix A.3); the structure of the proof of Theorem 2.20 in the precise terms that later sections extend (Appendix A.4); and the constants of its length bound (Appendix A.5). The results are quoted with the hypotheses of that paper and proved there, for every arity and every variance; we give proofs only where we extend a result. Except in Appendix A.1, nothing here uses a property of subtype constraints, and every alphabet may be infinite unless it is said to be finite.

A.1 Satisfiability

In the syntax of Zhao [2026] a system has no copy literals (Section 2.4). For a covariant system χ without copy literals over a finite set V, the *closure* D^{cov}_{χ} is the least reflexive and transitive relation on V such that, for every $i \in \Sigma$,

$$\frac{f(a_j)_j \le u \in \chi \qquad D^{\mathrm{cov}}_{\chi}(u,v) \qquad v \le f(b_j)_j \in \chi}{D^{\mathrm{cov}}_{\chi}(a_i,b_i)}. \tag{10}$$

It is computed by saturation, which terminates since there are $|V|^2$ pairs. The *path bounds* $\mathrm{Up}(\eta,u,v)$ and $\mathrm{Lo}(\eta,u,v)$, for words $\eta \in \Sigma^*$ and $u,v \in V$, are defined by $\mathrm{Up}(\varepsilon,u,v) \iff \mathrm{Lo}(\varepsilon,u,v) \iff D^{\mathrm{cov}}_{\chi}(u,v)$ and

$$\begin{aligned} \mathrm{Up}(i\eta,u,v) &\iff \exists z,\bar{b}\,\left[D^{\mathrm{cov}}_{\chi}(u,z) \wedge (z \le f(b_j)_j) \in \chi \wedge \mathrm{Up}(\eta,b_i,v)\right], \\ \mathrm{Lo}(i\eta,u,v) &\iff \exists \bar{a},z\,\left[(f(a_j)_j \le z) \in \chi \wedge D^{\mathrm{cov}}_{\chi}(z,v) \wedge \mathrm{Lo}(\eta,u,a_i)\right]. \end{aligned} \tag{11}$$

In every solution, $\mathrm{Up}(\eta,u,v)$ says that the subtree of u at η, when it exists, is below v, and $\mathrm{Lo}(\eta,u,v)$ says that u is below the subtree of v at η, when it exists [Zhao 2026, Lemma "Path rules"]. A *label clash* is a literal $f(\bar{a}) \le u$ or $\top \le u$ of χ, with *root label* f or $\top$, and a literal $v \le f(\bar{b})$ or $v = \bot$, with root label f or $\bot$, such that $D^{\mathrm{cov}}_{\chi}(u,v)$ and the first root label is greater than the second; that is, every such pair except two constructor literals. A *cycle clash* is a nonempty word η and variables a,b with

$$\mathrm{Lo}(\eta,a,a), \qquad D^{\mathrm{cov}}_{\chi}(a,b), \qquad \mathrm{Up}(\eta,b,b). \tag{12}$$

Theorem A.1 (Satisfiability criteria [Zhao 2026, Sect. "Closure and Supported Path Bounds"; Theorems "Least shape and satisfiability", "Covariant finite satisfiability" and "Regular countermodel transfer"; Lemma "Common shape and selection"; Lemma "Finite signed correspondence"]). *For every arity and every variance:*

(1) Let χ be a covariant system without copy literals. In every solution σ of χ, $D_\chi^{\mathrm{cov}}(u,v)$ implies $\sigma(u) \le \sigma(v)$. The system χ has a solution iff it has no label clash, and a finite solution iff it has neither a label clash nor a cycle clash. If χ has no label clash, then it has a cycle clash iff it has one whose word η has length at most $|V|^2$.

(2) For every system χ without copy literals and every variance c, $\mathrm{Sat}_c(\chi) \iff \mathrm{Sat}(\chi^{\pm})$ and $\mathrm{Sat}_{\mathrm{fin},c}(\chi) \iff \mathrm{Sat}_{\mathrm{fin}}(\chi^{\pm})$.

(3) Every system without copy literals that has a solution has a regular solution.

The signed system of a system without copy literals again has no copy literals, so part (1) decides both satisfiability problems of part (2). The extension of parts (2) and (3) to copy literals and to all three tree classes, Lemma 13.7, serves only the complexity bounds of the part of the paper that is not mechanized (Section 13); the decision procedure does not need it.

A.2 Cores

Let w be a nonempty word with canonical hierarchy (w_j, ℓ_j). Every position of w_j carries a *core*, which is either a cut of w, an *ordinary core*, or a symbolic *start core*. At level 0, position i has core i. At level $j+1$, the first position has a new start core $\star_{j+1}$, and position $i \ge 1$ inherits the core of the $(i-1)$st occurrence of ℓ_j in w_j, counting occurrences from 0; this is the marker that closes the block X_{i-1}, and the block X_i is read from it. Let C_j be the set of ordinary cores of w_j.

Lemma A.2 (Cores [Zhao 2026, Lemma "Cores"]).

(1) $C_{j+1} \subseteq C_j$. The start cores of w_j occupy an initial segment of its positions, are pairwise distinct, and number at most j; the ordinary cores increase from left to right.

(2) If $C_j \neq \varnothing$ and $e_j = \max C_j$, then

$$C_{j+1} = \{p : p < e_j \text{ and } w[e_j{:}] \preceq w[p{:}]\}.$$

In particular, (p, e_j) is a comparison of w for every $p \in C_{j+1}$. If $C_j = \varnothing$, then $C_{j+1} = \varnothing$.

The cut e_j is the *terminal core* of level j, the core of the last position of w_j. Thus each level collects the left ends of the comparisons whose right end is the terminal core of the level below. The proof of Theorem 2.19 combines this lemma with Lemma 2.18. Both lemmas hold for every word, respectively every rejected word, and every reader.

A.3 Lettered Monitors and Derivatives

Lettered monitors (Definition 2.22), the lettered monitor of a reader, which has the same nonempty holes as the reader, shifted blocks, derivatives (Definition 2.23), the label facts Eq. (4) and the desubstitution lemma (Lemma 2.24) are stated in Section 2.6. We add the structure of the states along a hierarchy.

Ordinary and symbolic states. Along a hierarchy, the states of $\mathcal{M}_i$ are the N states of $\mathcal{M}_0$, the *ordinary* states, and the i *symbolic* states $\star_1, \ldots, \star_i$ added by the derivatives. The ordinary states form a closed copy: a derived transition from an ordinary state runs the previous monitor over ℓX, so it stays ordinary. The *entry state* of a position x of w_i is the state of $\mathcal{M}_i$ at the cut x, from which its letter is read. By the label facts and the core recursion, a position with an ordinary core p has as entry state the state of $\mathcal{M}_0$ at the cut p of w, and a position with a start core has a symbolic entry state [Zhao 2026, Appendix "Details of the Bounded-Depth Compression"].

A.4 Structure of the Bounded-Depth Search

Let $\mathcal{M}_0$ be a lettered monitor with $N \geq 1$ states, for instance the monitor of a reader, and let w be a nonempty hole of depth $D \leq d$, with hierarchy (w_i, ℓ_i) and derived monitors $\mathcal{M}_i$. The proof of Theorem 2.20 has the five steps sketched in Section 2.6.

Step 1: derivatives. By Lemma 2.24, w_i is a hole of $\mathcal{M}_i$, which has $N + i$ states.

Step 2: dictionary normal form. For a word v let $\mathrm{alph}(v)$ be its set of letters. The level-i *dictionary* is $\mathcal{A}_i = \mathrm{alph}(w_i)$, and $g_i = |\mathcal{A}_i|$. The *start count* $s_i \leq i$ is the number of positions of w_i with a start core; by Lemma A.2 they form the *start prefix* $w_i[:s_i]$. The set U_i of *entry states* is the set of ordinary states together with the symbolic states that are entry states of positions in the start prefix, so $|U_i| \leq N + s_i$. Every entry state occurring in w_i lies in U_i. The *protected* letters are

$$P_i = \{\ell_i\} \cup \mathrm{alph}(w_i[:s_i]), \qquad |P_i| \leq s_i + 1.$$

An unprotected letter occurs only outside the start prefix, so all its occurrences have ordinary entry states. A comparison (s, e) of w_i is *left maximal* if $s \geq 1$ and its predecessor letters $w_i[s-1]$ and $w_i[e-1]$ differ. The *boundary pairs* BP_i are the ordered pairs $(w_i[s-1], w_i[e-1])$ over the left-maximal comparisons with $e < |w_i|$, and the *endpoint pairs* EP_i the same pairs over those with $e = |w_i|$; each pair is counted once, and $b_i = |\mathrm{BP}_i|$. Endpoint pairs have ℓ_i as second component. A letter $a \in \mathcal{A}_i \setminus P_i$ is *isolated* if no pair in BP_i has a as one component and a letter outside P_i as the other; then all unequal comparison partners of a, endpoint partners included, are protected. The *exchange type* of an unprotected letter a consists of the following tables of $\mathcal{M}_i$:

- the transition $\zeta \mapsto \delta_i(\zeta, a)$ and the target $T_i^{\mathrm{fin}}(\zeta, a)$ for every ordinary state ζ;
- for each $R \in \{\Lambda_i, \Lambda_i^{\mathrm{fin}}\}$, the *self table* $R((\zeta, a), (\zeta', a))$ for ordinary ζ, ζ';
- for each such R, the *protected rows* $R((\zeta, a), (\vartheta, z))$ and $R((\vartheta, z), (\zeta, a))$ for ordinary ζ, $\vartheta \in U_i$, and $z \in P_i$;
- for each such R, the *initial row* $R(\triangleright, (\zeta, a))$ for ordinary ζ, which is false at every level $i \geq 1$.

Lemma A.3 (Exchange types [Zhao 2026, Lemma "Exchange types"]). *At every level $1 \leq i < D \leq d$ there are at most $\mathrm{C}(N, d)$ exchange types, with C from Eq. (14).*

Lemma A.4 (Comparison-preserving maps [Zhao 2026, Lemma "Comparison-preserving maps"]). *Let v be a nonempty word with last letter ℓ, and h a letter map that* identifies no unequal predecessor pair*: for every comparison (s, e) of v with $s \geq 1$, terminal ones included, $h(v[s-1]) = h(v[e-1])$ implies $v[s-1] = v[e-1]$. Then v and $h(v)$ have exactly the same comparisons, and on the letters of v, $h(a) = h(\ell)$ implies $a = \ell$.*

A *level map* at level i is a map h from $\mathcal{A}_i$ into itself. It acts letterwise on w_i, and $h(w_i)$ is expanded with the original lower markers $\ell_{i-1}, \ldots, \ell_0$ to a new word w'. The induced maps on the higher levels are $h_0 = h$ and $h_{r+1}([X]) = [h_r^*(X)]$, where h_r^* applies h_r letterwise to a block.

Lemma A.5 (Shape under a level map [Zhao 2026, Lemma "Shape under a level map"]). *Let h be a level map at level i that identifies no unequal predecessor pair of w_i, fixes the start prefix $w_i[:s_i]$ letterwise, preserves the level-i state run, and makes $h(w_i)$ a hole of $\mathcal{M}_i$. Then w' is a hole of $\mathcal{M}_0$, its markers below i are unchanged, and its dictionaries below i are subsets of the old ones. For every $r \geq 0$, $w'_{i+r} = h_r(w_{i+r})$, $\ell'_{i+r} = h_r(\ell_{i+r})$, and h_r identifies no unequal predecessor pair of w_{i+r}. Every start count is unchanged, and exactly the same levels are unary; in particular w' has depth D and no dictionary of w' is larger than the corresponding one of w.*

Lemma A.6 (Isolated exchange [Zhao 2026, Lemma "Isolated exchange"]). *Let $1 \leq i < D$, and let distinct isolated letters a, b of w_i have the same exchange type, where the level-zero expansion of*

b is not longer than that of a. The level map that replaces a by b satisfies the hypotheses of Lemma A.5, so the resulting word is a hole of depth D, no longer than w, with every start count unchanged and no dictionary larger; its level-i dictionary is strictly smaller.

Of two isolated letters with the same exchange type, one always has the shorter expansion, so the lemma applies in one of the two directions.

A hole is *normal* if, at every level $1 \le i < D$, distinct isolated letters have distinct exchange types, computed in its own hierarchy.

Lemma A.7 (Dictionary normal form and bounded dictionaries [Zhao 2026, Lemmas "Dictionary normal form", "Dictionary counting and propagation" and "Bounded dictionaries"]).

(1) Every nonempty hole has a normal hole of the same depth, no greater length, the same start counts, and no larger dictionary at any level.
(2) For a normal hole of depth $D \le d$ and $1 \le i < D$, $g_i \le \mathrm{A}(N,d) + 2b_i$. For every nonempty word and every level i, $b_i \le b_{i+1} + g_{i+1} - 1$.
(3) For a normal hole of depth $D \le d$, with $\mathrm{A} = \mathrm{A}(N,d)$, at every $1 \le i < D$,

$$2b_i \le (\mathrm{A}-1)(3^{D-1-i}-1), \qquad g_i \le 1 + (\mathrm{A}-1)3^{D-1-i}.$$

Consequently every nonempty hole of depth $D \le d$ has a hole of depth D and no greater length with $g_i \le \mathrm{G}(N,d)$ for every $1 \le i \le D$.

The counting in part (2) partitions $\mathcal{A}_i$ into the at most $d+1$ protected letters, the isolated letters, at most one per exchange type, and the remaining letters, each of which is a component of a boundary pair with both components unprotected. For the propagation, Lemma 2.17 shows that every boundary pair of level i consists of the first unequal letters, scanning from the right, of the two blocks of a boundary or endpoint pair of level $i+1$, with ℓ_i standing for an exhausted block. No bound on g_0 is needed.

Step 3: the top level.

Lemma A.8 (Unary words [Zhao 2026, Appendix "Details of the Bounded-Depth Compression"]). *Let z^m be a hole of a lettered monitor with state set Q. Then $z^{m'}$ is a hole for some $1 \le m' \le |Q|$.*

Proof. Let ζ_k be the state at the cut k of z^m, so that the label of the cut $k \ge 1$ is (ζ_{k-1}, z). The states $\zeta_0, \zeta_1, \ldots$ are the orbit of the start state under the letter z, so every state of the orbit occurs first at an index below $|Q|$; let t be the least index with $\zeta_t = \zeta_{m-1}$, so that $t < |Q|$ and $t \le m-1$. The prefix z^{t+1} has the labels of z^m at its cuts, and its last label (ζ_t, z) is the last label of z^m. Every pair of cuts of a unary word is a comparison, so a nonterminal comparison of z^{t+1} is one of z^m with the same labels, and a terminal one $(s, t+1)$ has the labels of the terminal comparison (s, m) of z^m. Hence z^{t+1} is a hole. □

Applied to w_D in $\mathcal{M}_D$, the lemma gives at most $N + D$ letters.

Step 4: simultaneous compression. Let $\mathcal{M}$ be a lettered monitor with state set Q and start ζ_0, and ℓ a marker. Write $lab_0(\zeta)$ for the initial shifted-block label: (ζ, ℓ) for $\zeta \in Q$ and $\triangleright$ for $\star$. For blocks

$X_1, \dots, X_k$ avoiding ℓ, their *summary* is the tuple of the four functions

$$\begin{aligned} exit_a(\zeta) &\in Q, && \text{the exit of the shifted block of } X_a \text{ read from } \zeta,\\ term_a(\zeta,\zeta'') &\in \{0,1\}, && \text{whether some label of it is related by } \Lambda^{\mathrm{fin}} \text{ to } (\zeta'',\ell),\\ eq(a,b) &\in \{0,1\}, && \text{whether } X_a = X_b,\\ adm(a,b,\zeta,\zeta') &\in \{0,1\}, && \text{whether } \Lambda_\ell \text{ relates } (\zeta, X_a) \text{ and } (\zeta', X_b), \end{aligned}$$

for blocks a, b, entry states $\zeta, \zeta' \in Q \sqcup \{\star\}$, and states $\zeta'' \in Q$. It determines the restriction of $\mathcal{M}_\ell$ to the letters $X_1, \dots, X_k$, together with their equalities, and it has at most $\mathrm{col}(|Q|, k)$ values Eq. (14).

The *column word* of the tuple aligns the blocks at their right ends, pads them on the left, and reads the resulting word of k-tuples of optional letters, the *columns*, from left to right. It is *valid* if every track consists of absent entries followed by letters; a subsequence of a valid column word is valid. A deterministic automaton computes the summary along the column word. The tuple of k empty blocks has $exit_a(\zeta) = \delta(\zeta, \ell)$ for $\zeta \in Q$, $exit_a(\star) = \zeta_0$, $term_a(\zeta,\zeta'') = \Lambda^{\mathrm{fin}}(lab_0(\zeta), (\zeta'', \ell))$, all blocks equal, and $adm(a,b,\zeta,\zeta') = \Lambda(lab_0(\zeta), lab_0(\zeta'))$. Reading a column γ updates all components simultaneously, from the old values. If γ_a is absent, the components of block a are unchanged. If $\gamma_a = z$, then

$$exit'_a(\zeta) = \delta(exit_a(\zeta), z), \qquad term'_a(\zeta,\zeta'') = term_a(\zeta,\zeta'') \vee \Lambda^{\mathrm{fin}}\big((exit_a(\zeta), z), (\zeta'', \ell)\big).$$

Equality becomes $eq(a,b) \wedge \gamma_a = \gamma_b$. If $\gamma_a = z$ and $\gamma_b = z'$ are both present,

$$adm'(a,b,\zeta,\zeta') = \Lambda\big((exit_a(\zeta), z), (exit_b(\zeta'), z')\big) \vee \big(z = z' \wedge adm(a,b,\zeta,\zeta')\big);$$

if only $\gamma_a = z$ is present, $adm'(a,b,\zeta,\zeta') = \Lambda((exit_a(\zeta), z), lab_0(\zeta'))$; if only $\gamma_b = z'$ is present, $adm'(a,b,\zeta,\zeta') = \Lambda(lab_0(\zeta), (exit_b(\zeta'), z'))$; and if both are absent, the bit is unchanged. The update is correct because Λ_ℓ compares positions at the same distance from the ends of two blocks, with equal letters after them: appending a column either relates the two new last positions, or extends the common suffix of a pair related earlier by an equal letter.

Lemma A.9 (Column shortening [Zhao 2026, Appendix "Details of the Bounded-Depth Compression"]). *Deleting the columns between two cuts of a valid column word with equal summaries gives a valid column word with the same final summary. Hence every tuple of k blocks avoiding ℓ can be replaced by subsequences, one per block, with the same summary and all lengths below $\mathrm{col}(|Q|, k)$. The replacement is injective on the letters, preserves marker avoidance, maps holes of $\mathcal{M}_\ell$ over these letters to holes, and keeps a unary level unary.*

For lettered monitors $\mathcal{U}, \mathcal{V}$ with the same state set, a finite set E of letters, and a map ρ from E into the alphabet of $\mathcal{V}$, *agreement* $\mathcal{U} \equiv_{E,\rho} \mathcal{V}$ means: their start states coincide, ρ is injective on E, and for every $a \in E$ and every state ζ,

$$\delta_{\mathcal{V}}(\zeta, \rho(a)) = \delta_{\mathcal{U}}(\zeta, a), \quad T^{\mathrm{fin}}_{\mathcal{V}}(\zeta, \rho(a)) \iff T^{\mathrm{fin}}_{\mathcal{U}}(\zeta, a), \quad R_{\mathcal{V}}(\widehat{\rho}L, \widehat{\rho}L') \iff R_{\mathcal{U}}(L, L') \tag{13}$$

for $R \in \{\Lambda, \Lambda^{\mathrm{fin}}\}$ and all labels L, L' whose letters lie in E, the start label included; here $\widehat{\rho}(\triangleright) = \triangleright$ and $\widehat{\rho}(\zeta, a) = (\zeta, \rho(a))$, and states are not renamed. Agreement preserves runs, labels up to $\widehat{\rho}$, and holes of words over E, and agreements compose.

Lemma A.10 (Agreement under derivation [Zhao 2026, Lemma "Agreement under derivation"]). *If $\mathcal{U} \equiv_{E,\rho} \mathcal{V}$ and $\ell \in E$, then for every finite family F of blocks over E avoiding ℓ, the letterwise map ρ^* gives $\mathcal{U}_\ell \equiv_{F,\rho^*} \mathcal{V}_{\rho(\ell)}$, and the image blocks avoid $\rho(\ell)$.*

For markers $\boldsymbol{\ell} = (\ell_0, \dots, \ell_{j-1})$ write $\mathcal{M}^{\boldsymbol{\ell}}_0 = \mathcal{M}_0$ and $\mathcal{M}^{\boldsymbol{\ell}}_{h+1} = (\mathcal{M}^{\boldsymbol{\ell}}_h)_{\ell_h}$. A finite family of level-j letters is *valid* for the markers if every block avoids the marker of its level and its letters are,

recursively, valid letters of the level below; $\mathrm{Exp}_{\ell,j}(a)$ is the level-zero word obtained by expanding a valid level-j letter a, each block followed by its marker.

Lemma A.11 (Supported simultaneous compression [Zhao 2026, Lemma "Supported simultaneous compression"]). *Fix a nonempty word w with hierarchy markers ℓ and dictionaries $\mathcal{A}_h$, and $j \le d$. Suppose $|\mathcal{A}_h| \le g$ for $1 \le h \le j$, and put $\kappa = \mathrm{col}(N+d+1, g)$. For every set $E_j \subseteq \mathcal{A}_j$ there are new markers $\widetilde{\ell}$ and a map ρ_j, injective on E_j, such that*

$$\mathcal{M}_j^{\ell} \equiv_{E_j,\rho_j} \mathcal{M}_j^{\widetilde{\ell}}, \qquad |\mathrm{Exp}_{\widetilde{\ell},j}(\rho_j(a))| \le \kappa^j \quad (a \in E_j),$$

and the image family is valid for the new markers. The recursion keeps at each level $h < j$ the support

$$E_h = \{\ell_h\} \cup \bigcup_{X \in E_{h+1}} \mathrm{alph}(\tau_h(X)) \;\subseteq\; \mathcal{A}_h,$$

where $\tau_h(X)$ is the subsequence of X produced by Lemma A.9; at level zero no cardinality bound is needed. Agreement holds at every state, not only at the states visited by w.

Step 5: the bound. Normalize the given hole by Lemma A.7, so that $g_i \le \mathrm{G} = \mathrm{G}(N,d)$ for $1 \le i \le D$, and keep it fixed as the original hierarchy; put $\kappa = \mathrm{col}(N+d+1, \mathrm{G})$. Lemma A.8 gives a hole z^m of $\mathcal{M}_D$ with $1 \le m \le N+D$. Compressing $E_D = \{z\}$ by Lemma A.11, agreement makes $\rho_D(z)^m$ a hole of the new level-D monitor, and validity allows its exact expansion by Lemma 2.24 to a nonempty hole of $\mathcal{M}_0$. Cutting that expansion recovers the unary word $\rho_D(z)^m$ at level D, so the depth is at most D, and the length is at most

$$m\kappa^D \le (N+D)\kappa^D \le (N+d+1)\kappa^d = \mathsf{Len}(N,d).$$

No step uses finiteness of the alphabet, and the alphabet size does not enter the bound.

A.5 Constants

For integers $N \ge 1$, $d \ge 0$, and $k \ge 0$ put

$$\begin{aligned} \mathrm{col}(N,k) &= N^{(N+1)k} \cdot 2^{N(N+1)k} \cdot 2^{k^2} \cdot 2^{(N+1)^2k^2}, & \mathrm{C}(N,d) &= N^N\, 2^{2N^2+4N(N+d)(d+1)+3N}, \\ \mathrm{A}(N,d) &= d+1+\mathrm{C}(N,d), & \mathrm{G}(N,d) &= 2+\mathrm{A}(N,d)\,3^d, \end{aligned} \tag{14}$$

and

$$\mathsf{Len}(N,d) = (N+d+1)\cdot \mathrm{col}\big(N+d+1, \mathrm{G}(N,d)\big)^d. \tag{15}$$

All five functions are nondecreasing in each argument. Their meaning is as follows; $D \le d$ is the depth of the hole being shortened.

- $\mathrm{col}(N,k)$ bounds the number of summary values of a k-tuple of blocks for a lettered monitor with N states. It is the product of the numbers of values of the summary fields:

Factor	Summary field
$N^{(N+1)k}$	exits: k blocks, $N+1$ entry states
$2^{N(N+1)k}$	terminal-comparison bits, also for N final-marker states
2^{k^2}	equality bits for ordered pairs of blocks
$2^{(N+1)^2k^2}$	derived-admission bits for pairs of blocks and entry states

- $\mathrm{C}(N,d)$ bounds the number of exchange types at a level $1 \le i < D$ (Lemma A.3). The transition table of the ordinary states has N^N values; the target table uses N bits, the two self tables $2N^2$ bits, the protected rows $4N|U_i||P_i|$ bits, where $|P_i| \le d+1$ and $|U_i| \le N+d$, and the initial rows $2N$ bits.

- $\mathsf{A}(N,d)$ bounds the protected letters, at most $d+1$, together with one isolated letter for each exchange type.
- $\mathsf{G}(N,d)$ bounds the dictionary at every level $1 \le i \le D$ of a suitable normal hole (Lemma A.7).
- $\mathsf{Len}(N,d)$ is the length bound. The top level of the shortened hole has at most $N+D \le N+d+1$ letters, and each of the at most d expansion steps multiplies the length by at most $\mathsf{col}(N+d+1, \mathsf{G}(N,d))$, which bounds the summary values of the level-j monitors, with $N+j$ states, for every $j \le d$. For $d=0$ the bound is $N+1$: a hole of depth 0 is unary and is its own top level.

Lemma A.12 (Elementary estimate [Zhao 2026, Lemma "Elementary estimate"]). *Put* $\Xi(N,d) = 3N^2+4N(N+d)(d+1)+3N+3d+4$. *Then* $\mathsf{G}(N,d) \le 2^{\Xi(N,d)}$ *and* $\mathsf{Len}(N,d) \le 2^{2^{5\Xi(N,d)+3}}$. *For* $N \ge 1$ *and* $d_N = 2N(4N-1)-1$,

$$\mathsf{Len}(N,d_N) \le 2^{2^{1280(N+1)^5}}.$$

Proof. Write $\Xi = \Xi(N,d)$ and $\Xi_0 = 3N^2+4N(N+d)(d+1)+3N$. From $N \le 2^N$ we get $\mathsf{C}(N,d) \le 2^{\Xi_0}$; a sum of two quantities bounded by 2^p and 2^q is at most 2^{p+q+1}, and $d+1 \le 2^{d+1}$, so $\mathsf{A}(N,d) \le 2^{\Xi_0+d+2}$; with $3^d \le 2^{2d}$ this gives $\mathsf{G}(N,d) \le 2^{\Xi_0+3d+3} \le 2^{\Xi}$. Put $M = N+d+1$, $g = \mathsf{G}(N,d)$, and $P = 2^{\Xi}$. Then $M+1 \le \Xi$, so $M, M+1, d, g \le P$. Replacing M by 2^M in the first factor of Eq. (14) yields $\mathsf{col}(M,g) \le 2^{2M(M+1)g+g^2+(M+1)^2g^2} \le 2^{4P^4} = 2^{2^{4\Xi+2}}$. Therefore $\mathsf{Len}(N,d) \le 2^{\Xi+d2^{4\Xi+2}} \le 2^{2^{5\Xi+3}}$, because $d \le 2^{\Xi}$ and $\Xi \le 2^{5\Xi+2}$. For $d_N = 8N^2-2N-1$ we have d_N, $N+d_N$, $d_N+1 \le 8N^2$, so $4N(N+d_N)(d_N+1) \le 256N^5$ and $\Xi(N,d_N) \le 256N^5+27N^2+3N+4$. Hence $5\Xi(N,d_N)+3 \le 1280N^5+135N^2+15N+23 \le 1280(N+1)^5$. □

B Proofs for Leaf Cones and the Characterization

This appendix proves the results of Sections 3 and 4 whose proofs were omitted there. We keep their notation. For a tree t, a word w *meets* a leaf v of t if $v \preceq w$. By the chain property, a word meets at most one leaf, a word that meets no leaf is a position labeled f, and a proper prefix of a position is never a leaf. We use Eq. (5) throughout: $w \in L_t^s$ iff w meets a leaf of t of sign s.

Proof of Theorem 3.2. (1) A word meets at most one leaf, and every extension wi of w meets the leaf that w meets. By Eq. (5), the two cones are disjoint and extension closed.

Conversely, let (L^0, L^1) be a pair of disjoint extension-closed sets, and put $L = L^0 \cup L^1$. Let D be the set of words none of whose proper prefixes lies in L. For $w \in D$ put $t(w) = f$ if $w \notin L$, and, if $w \in L^s$, put $t(w) = \top$ if $s \oplus \mathrm{par}(w) = 0$ and $t(w) = \bot$ if $s \oplus \mathrm{par}(w) = 1$; this is well defined because $L^0 \cap L^1 = \varnothing$. The set D contains ε and is prefix closed, and $wi \in D$ iff $w \in D$ and $w \notin L$, that is, iff $w \in D$ and $t(w) = f$. So t is a tree. Its leaves are the words of $D \cap L$, and by construction a leaf $w \in L^s$ has sign s. We show $L_t^s = L^s$. If $w \in L_t^s$, then w meets a leaf v of sign s, so $v \in L^s$, and $w \in L^s$ by extension closure. If $w \in L^s$, let v be the shortest prefix of w that lies in L. Then $v \in D$, so v is a leaf of t. If $v \in L^{s'}$, then $w \in L^{s'}$ by extension closure, so $s' = s$ by disjointness; thus w meets the leaf v of sign s, and $w \in L_t^s$.

For uniqueness, we show that the leaf pair determines the tree. A word w is a position of t iff no proper prefix of w is a leaf, iff no proper prefix of w lies in $L_t^0 \cup L_t^1$: a proper prefix in this union meets a leaf, which is again a proper prefix of w. A position w is labeled f iff it is not a leaf, iff $w \notin L_t^0 \cup L_t^1$, since the only leaf that a position can meet is itself. A leaf $w \in L_t^s$ has sign s, so its label is $\top$ if $s \oplus \mathrm{par}(w) = 0$ and $\bot$ otherwise.

(2) For trees t and u put $T = v_0 t$ and $U = v_0 u$; they have the domains of t and u. By Lemma 2.2(2), $t \le_c u$ iff $T \le U$ in the covariant order, that is, by Eq. (3) for the covariant variance,

$$t \le_c u \quad \Longleftrightarrow \quad T(w) \le U(w) \ \text{ at every common position } w \text{ of } t \text{ and } u.$$

Moreover, a leaf v of t has sign 0 iff $T(v) = \top$, and sign 1 iff $T(v) = \bot$, because ν_0 exchanges the label of v iff $\mathrm{par}(v) = 1$.

Assume $t \leq_c u$, and let $w \in L^0_t$. Then w meets a leaf v of t with $T(v) = \top$, and every proper prefix of v is a position of t labeled f. If u has a leaf at a proper prefix v' of v, then v' is a common position, $f = T(v') \leq U(v')$ forces $U(v') = \top$, and w meets the leaf v' of u of sign 0. Otherwise v is a position of u, by the chain property, and $\top = T(v) \leq U(v)$ forces $U(v) = \top$; again $w \in L^0_u$. Symmetrically, let $w \in L^1_u$, so that w meets a leaf v of u with $U(v) = \bot$. If t has a leaf at a proper prefix v' of v, then $T(v') \leq U(v') = f$ forces $T(v') = \bot$; otherwise v is a position of t, and $T(v) \leq \bot$ forces $T(v) = \bot$. In both cases $w \in L^1_t$.

Conversely, assume $L^0_t \subseteq L^0_u$ and $L^1_u \subseteq L^1_t$, and let w be a common position. If $T(w) = \top$, then w is a leaf of t of sign 0, so $w \in L^0_u$. As w is a position of u, the leaf of u that w meets is w itself, so $U(w) = \top$. If $U(w) = \bot$, then likewise $T(w) = \bot$. In all remaining cases $T(w) \in \{\bot, f\}$ and $U(w) \in \{f, \top\}$. So $T(w) \leq U(w)$ in every case, and $t \leq_c u$.

(3) The tree $\top$ consists of the leaf ε, of polarity 0 and hence of sign 0; every word meets it, so $L^0_\top = \Sigma^*$ and $L^1_\top = \varnothing$. Likewise the leaf ε of $\bot$ has sign 1. The dual t^∂ has the leaves of t, with $\bot$ and $\top$ exchanged, so every leaf changes its sign, and $L^s_{t^\partial} = L^{1-s}_t$; equivalently, $\nu_s t^\partial = \nu_{1-s} t$ by Lemma 2.2(1). Let $t = f(t_i)_i$. Its root is labeled f, so ε meets no leaf and lies in neither cone. The positions of t other than ε are the words iv with v a position of t_i, with $t(iv) = t_i(v)$, and $\mathrm{par}(iv) = c(i) \oplus \mathrm{par}(v)$. Hence iv is a leaf of t iff v is a leaf of t_i, and then the sign of iv in t is the sign of v in t_i plus $c(i)$. So iw meets a leaf of t of sign s iff w meets a leaf of t_i of sign $s \oplus c(i)$. Finally, ε lies in a cone iff it meets a leaf, that is, iff ε is a leaf, and the leaf ε has sign 0 iff it is labeled $\top$. □

Proof of Lemma 3.5. Let ρ be an assignment and write $t_v = \rho(v)$. By Theorem 3.2(2), $u \leq v$ holds iff $L^0_{t_u} \subseteq L^0_{t_v}$ and $L^1_{t_v} \subseteq L^1_{t_u}$, that is, iff S_ρ is closed under the two copies of $u \leq v$.

Let $t = f(t_{a_i})_i$. Every nonempty word has the form iw, so by Theorem 3.2(3), $L^s_t = \bigcup_{i \in \Sigma} i L^{s \oplus c(i)}_{t_{a_i}}$. The literal $f(a_i)_i \leq u$ holds iff $t \leq_c t_u$, iff $L^0_t \subseteq L^0_{t_u}$ and $L^1_{t_u} \subseteq L^1_t$. The first inclusion says that $w \in L^{c(i)}_{t_{a_i}}$ implies $iw \in L^0_{t_u}$, for all i and w, which is closure under push_i. The second says that $\varepsilon \notin L^1_{t_u}$, which is the avoidance of the forbidden root (u^1, ε), and that $iw \in L^1_{t_u}$ implies $w \in L^{\bar{c}(i)}_{t_{a_i}}$, which is closure under pop_i. The literal $u \leq f(a_i)_i$ holds iff $L^0_{t_u} \subseteq L^0_t$ and $L^1_t \subseteq L^1_{t_u}$. The first inclusion says that $\varepsilon \notin L^0_{t_u}$ and that $iw \in L^0_{t_u}$ implies $w \in L^{c(i)}_{t_{a_i}}$, which are the forbidden root (u^0, ε) and closure under pop_i; the second says that $w \in L^{\bar{c}(i)}_{t_{a_i}}$ implies $iw \in L^1_{t_u}$, which is closure under push_i.

The literal $u = \bot$ holds iff $t_u = \bot$, that is, by Theorem 3.2(1,3), iff the leaf pair of t_u is $(\varnothing, \Sigma^*)$. This holds iff $\varepsilon \in L^1_{t_u}$: if so, then $L^1_{t_u} = \Sigma^*$ by extension closure, and $L^0_{t_u} = \varnothing$ by disjointness. So $u = \bot$ holds iff S_ρ contains the fact (u^1, ε). Likewise $\top \leq u$ holds iff $t_u = \top$, iff $(u^0, \varepsilon) \in S_\rho$. This proves the first claim.

For the second, $\rho \mapsto S_\rho$ is a bijection from the assignments onto the extension-closed sets without exclusion pairs, as noted before Table 3. Every model has these two properties, and by the first claim ρ is a solution of χ iff S_ρ is a model of χ. So the bijection restricts to a bijection from the solutions onto $\mathrm{Mod}(\chi)$. □

Proof of Lemma 3.6. (1) Every scheme of Table 3 holds for all words w; replacing w by $w\sigma$ turns the move $(p, u) \to (q, t)$ into $(p, u\sigma) \to (q, t\sigma)$.

(2) For a copy, $t = u$ and $\mathrm{sgn}(q) = \mathrm{sgn}(p)$. For the other moves we use $\mathrm{par}(iw) = c(i) \oplus \mathrm{par}(w)$. The move $(a_i^{c(i)}, w) \to (u^0, iw)$ has $c(i) \oplus \mathrm{par}(w) = 0 \oplus \mathrm{par}(iw)$; the move $(u^1, iw) \to (a_i^{\bar{c}(i)}, w)$

has $1 \oplus \mathrm{par}(iw) = 1 \oplus c(i) \oplus \mathrm{par}(w)$; the move $(u^0, iw) \to (a_i^{c(i)}, w)$ has $\mathrm{par}(iw) = c(i) \oplus \mathrm{par}(w)$; and the move $(a_i^{\bar{c}(i)}, w) \to (u^1, iw)$ has $1 \oplus c(i) \oplus \mathrm{par}(w) = 1 \oplus \mathrm{par}(iw)$. □

Proof of Lemma 3.8. (1) Every set that contains $\mathsf{Fact}_\chi \cup A$ and is closed under the rules contains $H_\chi(A)$, by induction along derivations. Conversely, $H_\chi(A)$ contains $\mathsf{Fact}_\chi \cup A$, since derivations may have length zero, and it is closed under the rules: each rule has a single premise, so applying it to an atom of $\mathrm{Reach}_\chi(x)$ yields an atom of $\mathrm{Reach}_\chi(x)$.

(2) A model M that contains A contains Fact_χ and is closed under the rules, so $H_\chi(A) \subseteq M$ by (1); since M is consistent, so is every subset of M. Conversely, if $H_\chi(A)$ is consistent, then it is a model by (1), and by the first part it is the least model containing A. The remaining claims are the cases $A = \varnothing$ and $A = \{a\}$; for the first, χ is satisfiable iff it has a model, by Lemma 3.5.

(3) An inconsistent set contains a forbidden root or an exclusion pair, and every atom of $H_\chi(A)$ has a seed in $\mathsf{Fact}_\chi \cup A$.

(4) The extensions by the letters of α derive $(p, u\alpha)$ from (p, u), so $\mathrm{Reach}_\chi(p, u\alpha) \subseteq \mathrm{Reach}_\chi(p, u)$ and $\mathrm{Cl}_\chi(p, u\alpha) \subseteq \mathrm{Cl}_\chi(p, u)$. A model that contains (p, u) is closed under extensions, so it contains $(p, u\alpha)$.

(5) If $S = T|_X$ for a model T of χ, then T contains S and Fact_χ and is closed under the rules, so $H_\chi(S) \subseteq T$ by (1). Hence $H_\chi(S)$ is consistent, and $S \subseteq H_\chi(S)|_X \subseteq T|_X = S$. Conversely, if $H_\chi(S)$ is consistent and $H_\chi(S)|_X = S$, then $H_\chi(S)$ is a model by (1), with visible part S, and it is contained in every model that contains S. □

Proof of Lemma 3.11. We show by induction on k that if a derivation of length k leads from (p, u) to b, then $(p, u\alpha) \to^* b$ for some α. For $k = 0$ take $\alpha = \varepsilon$. For $k > 0$, let the derivation end with a rule from b' to b, and by induction let $(p, u\alpha') \to^* b'$. If the last rule is a pure move, append it, with $\alpha = \alpha'$. If it is an extension from $b' = (q, t)$ to $b = (q, ti)$, apply Lemma 3.6(1) with $\sigma = i$ to each move of the pure derivation from $(p, u\alpha')$ to (q, t); this gives $(p, u\alpha' i) \to^* (q, ti)$, and $\alpha = \alpha' i$. Conversely, if $(p, u\alpha) \to^* b$, then the extensions by the letters of α, followed by this pure derivation, form a derivation from (p, u) to b. □

Proof of Corollary 3.13. Put $a = (p, u)$. Suppose that (a), (b) or (c) holds. In case (a), $\mathrm{Cl}_\chi(a) \supseteq \mathrm{Reach}_\chi(a)$ contains a forbidden root. In cases (b) and (c), $\mathrm{Cl}_\chi(a)$ contains $(\bar{p}, u\delta)$, and it contains $(p, u\delta)$ by Lemma 3.8(4), an exclusion pair. In each case $\mathrm{Cl}_\chi(a)$ is inconsistent, and a is dead by Lemma 3.8(2).

Conversely, let a be dead. Then $\mathrm{Cl}_\chi(a)$ is inconsistent; take seeds $x, y \in \mathsf{Fact}_\chi \cup \{a\}$ as in Lemma 3.8(3). Since χ is satisfiable, μ_χ is consistent. If $\mathrm{Reach}_\chi(x)$ contains a forbidden root, then $x \notin \mathsf{Fact}_\chi$, so $x = a$, and (a) holds. Otherwise $b \in \mathrm{Reach}_\chi(x)$ and $\bar{b} \in \mathrm{Reach}_\chi(y)$ for some atom b, and x and y are not both facts. Since $\{b, \bar{b}\}$ is unordered, we may assume $y = a$. If x is a fact e, then Lemma 3.12 gives $(\bar{p}, u\delta) \in \mathrm{Reach}_\chi(e) \subseteq \mu_\chi$ for some δ, which is (b). If $x = a$, it gives $(\bar{p}, u\delta) \in \mathrm{Reach}_\chi(a)$, which is (c). The last two claims follow from Lemma 3.11. □

C Proofs for Tree Classes

This appendix proves the results of Section 5 whose proofs were deferred, in its notation. For a covariant tree t, $\nu_0 t = t$ and $\nu_1 t = t^\partial$, so L_t^0 is the set of words with a prefix labeled $\top$ in t, and L_t^1 the set of words with a prefix labeled $\bot$. By Lemma 2.2(3), duality reverses the covariant order, and it commutes with the constructor, $f(t_i)_i^\partial = f(t_i^\partial)_i$, since it exchanges labels position by position.

C.1 Observation Pins

Proof of Lemma 5.4. Let $\mathrm{Pref}(w)$ be the set of prefixes of w, including w, and let $\mathrm{Off}(w) = \Sigma^* \setminus (\mathrm{Pref}(w) \cup w\Sigma^*)$ be the set of words that leave the pin path, that is, the words $w[{:}k]\, j\, v$ with

$k < |w|$ and $j \neq w[k]$. The positions of $\mathrm{emb}_b(w, g)$ are the words of $\mathrm{Pref}(w)$, of which the proper prefixes of w are labeled f; the children $w[{:}k]\ j$ off the pin path, labeled b; and w followed by the positions of g. A word of $\mathrm{Off}(w)$ passes a child off the pin path, a proper prefix of w passes no leaf, and an extension of w follows g. Hence the four trees of Eq. (7) have the following cones:

tree	L^0	L^1
$g_1 = \mathrm{emb}_\bot(w, \top)$	$w\Sigma^*$	$\mathrm{Off}(w)$
$g_2 = \mathrm{emb}_\top(w, f(\top^n))$	$\Sigma^* \setminus \mathrm{Pref}(w)$	$\varnothing$
$g_3 = \mathrm{emb}_\top(w, \bot)$	$\mathrm{Off}(w)$	$w\Sigma^*$
$g_4 = \mathrm{emb}_\bot(w, f(\bot^n))$	$\varnothing$	$\Sigma^* \setminus \mathrm{Pref}(w)$

We use Theorem 3.2(1,2): the cones of t are extension closed and disjoint, and $t \leq u$ iff $L^0_t \subseteq L^0_u$ and $L^1_u \subseteq L^1_t$.

- $g_1 \leq t$ iff $w\Sigma^* \subseteq L^0_t$ and $L^1_t \subseteq \mathrm{Off}(w)$. If $w \in L^0_t$, then $w\Sigma^* \subseteq L^0_t$ by extension closure; L^1_t contains no extension of w, by disjointness, and no proper prefix of w, since otherwise it would contain w; so $L^1_t \subseteq \mathrm{Off}(w)$. Conversely, $g_1 \leq t$ gives $w \in L^0_t$.
- $t \leq g_2$ iff $L^0_t \subseteq \Sigma^* \setminus \mathrm{Pref}(w)$, since the condition on the cones of sign 1 is vacuous. By extension closure, L^0_t meets $\mathrm{Pref}(w)$ iff $w \in L^0_t$.
- $t \leq g_3$ iff $L^0_t \subseteq \mathrm{Off}(w)$ and $w\Sigma^* \subseteq L^1_t$; this is the first case with the two cones exchanged.
- $g_4 \leq t$ iff $L^1_t \subseteq \Sigma^* \setminus \mathrm{Pref}(w)$, iff $w \notin L^1_t$, as in the second case.

For the consequence, let x be a visible variable and ρ an assignment. The cones of $\rho(x)$ are the covariant cones of $t = \nu_0\rho(x)$, since $\nu_1\rho(x) = t^\partial$ by Lemma 2.2(1). So an observation with variable x and word w is equivalent to one of the four inequalities between t and a tree g_k of Eq. (7). By Lemma 2.2(1,2), $g_k \leq \nu_0\rho(x)$ iff $\nu_0 g_k \leq_c \rho(x)$, and $\nu_0\rho(x) \leq g_k$ iff $\rho(x) \leq_c \nu_0 g_k$. The tree $\nu_0 g_k$ has the domain of g_k. □

C.2 Signed Bounds and Medians

Proof of Lemma 5.8. (1) By Lemma 2.5, ρ solves χ iff $\rho^\pm$ solves $\chi^\pm$. For a lower bound, Lemma 2.2(2) gives $g \leq_c \rho(x)$ iff $\nu_+ g \leq \nu_+\rho(x)$ iff $\nu_-\rho(x) \leq \nu_- g$; these are the two bounds of Eq. (8) at $\rho^\pm$. Upper bounds are symmetric.

(2) Since duality is an involution, so is ι. Let σ be an assignment of $V^\pm$. In every row of Table 2, and for the two bounds of each pair in Eq. (8), $\iota\sigma$ satisfies one entry iff σ satisfies the other, because duality reverses the covariant order, commutes with the constructor, exchanges $\bot$ and $\top$, and maps $\nu_+ g$ to $\nu_- g$. For instance, $\iota\sigma$ satisfies $u^+ \leq v^+$ iff $\sigma(u^-)^\partial \leq \sigma(v^-)^\partial$, iff $\sigma(v^-) \leq \sigma(u^-)$; it satisfies $f(a_i^{c(i)})_i \leq u^+$ iff $f(\sigma(a_i^{\bar{c}(i)})^\partial)_i \leq \sigma(u^-)^\partial$, iff $\sigma(u^-) \leq f(\sigma(a_i^{\bar{c}(i)}))_i$; it satisfies $u^+ = \bot$ iff $\sigma(u^-) = \top$; and it satisfies $\nu_+ g \leq x^+$ iff $\nu_+ g \leq \sigma(x^-)^\partial$, iff $\sigma(x^-) \leq \nu_- g$. The remaining entries are the same computation. Both entries of every row and both bounds of every pair belong to $\chi^\pm \cup B^\pm$, so ι preserves its solutions. Finally, $(\iota\rho^\pm)(v^\pm) = (\nu_\mp\rho(v))^\partial = \nu_\pm\rho(v)$ by Lemma 2.2(1).

(3) If $\iota\sigma = \sigma$, then $\sigma(v^-) = \sigma(v^+)^\partial$ for every v, and Lemma 2.5 gives $\sigma = \rho^\pm$ for $\rho(v) = \nu_+\sigma(v^+)$, and the preservation of finiteness. □

Proof of Lemma 5.9. Write $\mathrm{maj}(L_1, L_2, L_3)$ for the set of words that lie in at least two of the sets L_j. It is symmetric and monotone for inclusion, and $\mathrm{maj}(L, L, L') = L$.

(1) Symmetry in the first two arguments is clear. If $t_j \leq t'_j$ for $j = 1, 2, 3$, then $L^0_{t_j} \subseteq L^0_{t'_j}$ and $L^1_{t'_j} \subseteq L^1_{t_j}$ by Theorem 3.2(2), so the same inclusions hold between the cones of the medians, and $\mathrm{med}(t_1, t_2, t_3) \leq \mathrm{med}(t'_1, t'_2, t'_3)$. The identity $\mathrm{med}(t, t, u) = t$ follows from $\mathrm{maj}(L, L, L') = L$.

(2) By Theorem 3.2(3), with f covariant, a word iv lies in $L^s_{f(a_i^j)_i}$ iff $v \in L^s_{a_i^j}$, and ε lies in no cone of a constructor tree. So the majority of the three cones $L^s_{f(a_i^j)_i}$ is the union over i of the sets $i\,\mathrm{maj}(L^s_{a_i^1}, L^s_{a_i^2}, L^s_{a_i^3})$, which is $L^s_{f(m_i)_i}$ for $m_i = \mathrm{med}(a_i^1, a_i^2, a_i^3)$; by Theorem 3.2(1) the two trees are equal. For duality, $L^s_{t^\partial} = L^{1-s}_t$, and maj is applied to each sign separately. For assignments of signed variables, $(\iota\,\mathrm{med}(\alpha,\beta,\gamma))(v^\pm) = \mathrm{med}(\alpha(v^\mp), \beta(v^\mp), \gamma(v^\mp))^\partial = \mathrm{med}((\iota\alpha)(v^\pm), (\iota\beta)(v^\pm), (\iota\gamma)(v^\pm))$.

(3) For $n = 0$ every tree is finite. Let $n \geq 1$, let t_1, t_2, t_3 be finite, and let h bound the lengths of their positions. A word w of length $h+1$ is a position of no t_j, so its longest prefix in the domain of t_j is a leaf, labeled $\bot$ or $\top$; thus $w \in L^0_{t_j} \cup L^1_{t_j}$ for each j. Two of the three trees put w in the same cone L^s, so w lies in the cone L^s of the median, and some prefix of w is a leaf of the median. Every proper prefix of a position is labeled f, so no word of length $h+2$ is a position of the median, which is therefore finite.

(4) Let α, β, γ solve a covariant system together with covariant ground bounds. A literal $u \leq v$ holds in their median $\mathrm{med}(\alpha,\beta,\gamma)$ by monotonicity. For $f(a_i)_i \leq u$, monotonicity gives $\mathrm{med}(f(\alpha(a_i))_i, f(\beta(a_i))_i, f(\gamma(a_i))_i) \leq \mathrm{med}(\alpha,\beta,\gamma)(u)$, and the left side is $f(\mathrm{med}(\alpha,\beta,\gamma)(a_i))_i$ by (2); the upper constructor literals are symmetric. The extreme literals hold since $\mathrm{med}(\bot,\bot,\bot) = \bot$ and $\mathrm{med}(\top,\top,\top) = \top$. For a lower bound $G \leq p$, $G = \mathrm{med}(G,G,G) \leq \mathrm{med}(\alpha,\beta,\gamma)(p)$, and upper bounds are symmetric. □

C.3 Covariant Finite-Bound Pinning

In this subsection f is covariant and $n \geq 1$; for $n = 0$, Lemma 5.10 is trivial because every tree is finite. We use the closure D^{cov}_χ, the path bounds Lo and Up, label clashes and cycle clashes of a covariant system without copy literals, as defined in Appendix A.1, and Theorem A.1(1): such a system has a solution iff it has no label clash, and a finite solution iff it has neither a label clash nor a cycle clash.

Path pins. For a covariant tree t, L^0_t is the set of words that meet a leaf $\top$ of t, and L^1_t the set of words that meet a leaf $\bot$. A *path pin* on a variable x is a covariant observation on x: one of the four conditions $w \in L^0_x$, $w \notin L^0_x$, $w \in L^1_x$ and $w \notin L^1_x$ on the value of x, for a word w. By Lemma 5.4, each is a bound by one of the four trees $\mathrm{emb}_\bot(w, \top)$, $\mathrm{emb}_\top(w, f(\top^n))$, $\mathrm{emb}_\top(w, \bot)$ and $\mathrm{emb}_\bot(w, f(\bot^n))$.

Lemma C.1 (Ground bounds as path pins). *For a finite ground tree g and a covariant tree t, $g \leq t$ iff $\pi \notin L^1_t$ for every position π of g labeled f and $\pi \in L^0_t$ for every position π of g labeled $\top$; and $t \leq g$ iff $\pi \notin L^0_t$ for every position π of g labeled f and $\pi \in L^1_t$ for every position π of g labeled $\bot$.*

Proof. For a tree t and a letter i, let t_i be the subtree of t at i if t is a constructor tree, and t itself if t is a leaf. Then $iv \in L^s_t$ iff $v \in L^s_{t_i}$, for every word v and sign s: for a constructor tree by Theorem 3.2(3), and for a leaf because its cones are Σ^* and $\varnothing$. We prove the first claim by induction on g; the second is symmetric. For $g = \bot$ both sides hold, and for $g = \top$ both sides say $t = \top$, that is, $\varepsilon \in L^0_t$. Let $g = f(g_i)_i$. Then $g \leq t$ iff $t \neq \bot$ and $g_i \leq t_i$ for every i: for $t = \top$ both hold, and for a constructor tree t this is the covariant rule. Now $t \neq \bot$ says $\varepsilon \notin L^1_t$, the condition at the root of g, and by induction and the description of t_i, $g_i \leq t_i$ is the condition at the positions $i\pi$ of g. □

So a finite set of finite ground bounds is equivalent to a finite set of path pins, and each path pin is a bound by one of the four pin trees.

Chains. Let x be a variable and w a word. A bound $\mathrm{emb}_b(w, g) \leq x$ by a lower pin tree, with $b = \bot$ and $g \in \{\top, f(\bot^n)\}$, is expressed by fresh variables $x_1, \ldots, x_{|w|}$ and a fresh filler o, with

$x_0 = x$, and the *chain*

$$f(a_i^{(j)})_i \le x_j \quad (j < |w|), \qquad o = \bot, \qquad \top \le x_{|w|} \text{ or } f(o,\dots,o) \le x_{|w|},$$

where $a_i^{(j)} = x_{j+1}$ for $i = w[j]$ and $a_i^{(j)} = o$ otherwise. An upper pin tree is expressed dually, with upper constructor literals, the filler literal $\top \le o$, and $x_{|w|} = \bot$ or $x_{|w|} \le f(o,\dots,o)$. By induction on $|w|$, using monotonicity of the covariant constructor in one direction and the trees t_i of the proof of Lemma C.1 in the other, an assignment satisfies the bound iff it extends to a solution of the chain: put $o = \bot$ or $\top$, and let x_j be the tree reached from the value of x along $w[:j]$, a leaf standing for itself below it. Give x_j the rank j and the filler the rank $|w| + 1$, and call the fresh variables of a lower chain *sources* and those of an upper chain *sinks*. Then every lower constructor literal of the chain has source children of a larger rank than its root, which is not a sink, and every upper constructor literal has sink children of a larger rank than its root, which is not a source.

Lemma C.2 (Ranked extensions [Zhao 2026, proof of Theorem "Finite countermodel transfer"]). *Let F_0 be a covariant system without copy literals over V_0, and let $F \supseteq F_0$ be a covariant system without copy literals over $V \supseteq V_0$, whose variables outside V_0 are sources or sinks, not both, with ranks in $\mathbb{N}$, the old variables in V_0 having rank 0. Suppose that every constructor literal of F that is not a literal of F_0 is a lower literal whose children are sources of a larger rank than its root, which is not a sink, or an upper literal whose children are sinks of a larger rank than its root, which is not a source. Then F has a cycle clash iff F_0 has one.*

Proof. Write $D, \mathrm{Lo}, \mathrm{Up}$ for the closure and the path bounds of F, and $D_0, \mathrm{Lo}_0, \mathrm{Up}_0$ for those of F_0. Upper constructor literals have no source children, and lower ones no sink children. By induction on the derivation of $D(a,b)$: (a) if b is a source, then $a = b$; (b) if a is a sink, then $b = a$; (c) if a and b are old, then $D_0(a,b)$. For transitivity, a middle variable between old endpoints is neither a source nor a sink, by (a) and (b). For the decomposition rule Eq. (10), the conclusion $D(a_i, b_i)$ has a child of an upper literal on the right and a child of a lower literal on the left, so (a) and (b) hold vacuously; if a_i and b_i are old, both literals have an old child, so they are literals of F_0, with old roots u, v; the induction hypothesis gives $D_0(u,v)$, hence $D_0(a_i, b_i)$. By induction on η, using (a) and (b): (d) if $\mathrm{Lo}(\eta, a, s)$ and s is a source, then a is a source of rank at least $\mathrm{rank}(s) + |\eta|$; (e) if $\mathrm{Up}(\eta, t, b)$ and t is a sink, then b is a sink of rank at least $\mathrm{rank}(t) + |\eta|$; and (f) between old variables, Lo and Up are Lo_0 and Up_0, since a new literal would lead into a source or out of a sink. In a cycle clash $\mathrm{Lo}(\eta, a, a)$, $D(a,b)$, $\mathrm{Up}(\eta, b, b)$ with $\eta \neq \varepsilon$, a is not a source by (d), b is not a sink by (e), a is not a sink since (b) would make $b = a$ a sink, and b is not a source since (a) would make $a = b$ a source. So both are old, and by (c) and (f) the clash is one of F_0. Conversely $D_0 \subseteq D$, $\mathrm{Lo}_0 \subseteq \mathrm{Lo}$ and $\mathrm{Up}_0 \subseteq \mathrm{Up}$. □

Proof of Lemma 5.10. Let F be a covariant system over V with a finite solution δ, let B be a finite set of finite ground bounds on V, and let ρ be a solution of $F \cup B$.

Copy literals. Let $F_0 = F^\flat$ be obtained by Lemma 2.7, applied to every copy literal of F. Then δ extends to a finite solution of F_0, ρ extends to a solution of F_0 that satisfies B, and every solution of F_0 restricts to a solution of F (Section 2.4).

Pins. By Lemma C.1, B is equivalent to a finite list $p_1, \dots, p_m$ of path pins on variables of V. Let F_k be F_0 together with the chains of $p_1, \dots, p_k$, with distinct fresh variables. We show by induction on k that F_k has a finite solution and that some solution of F_k satisfies $p_{k+1}, \dots, p_m$; for $k = 0$ these are δ and the extension of ρ. For the step, a solution of F_k that satisfies $p_{k+1}, \dots, p_m$ extends to a solution of F_{k+1} by the chain of p_{k+1}, and the extension still satisfies $p_{k+2}, \dots, p_m$, which concern old variables. So F_{k+1} has a solution, and no label clash. The system F_{k+1} is an extension of F_k as in Lemma C.2, with the variables of F_k as the old ones, so it has a cycle clash iff F_k has one; and F_k has none, because it has a finite solution. By Theorem A.1(1), F_{k+1} has a finite solution.

Conclusion. A finite solution of F_m satisfies every pin, by its chain, so its restriction is a finite solution of F_0 that satisfies B, and its further restriction is a finite solution of $F \cup B$. □

The proof extends the argument of [Zhao 2026, Theorem "Finite countermodel transfer"], where the bounds are the pin paths of a single word, to arbitrary finite ground bounds. The finite solution of F is used only to exclude cycle clashes among old variables; label clashes are excluded by the solution of $F \cup B$.

C.4 Closure under Intersection

Proof of Proposition 5.5. (1) Let $\mathcal{M}$ be a nonempty family of models of χ and $M = \bigcap \mathcal{M}$. If the premise of a rule lies in M, it lies in every member of $\mathcal{M}$, hence so does its conclusion; so M is closed under the rules. It contains Fact_χ, since every member does, and it is consistent, since it is contained in a member. For the projected family, let $S_j = U_j|_X$ with $U_j \in \mathrm{Mod}(\psi)$ for j in a nonempty index set; then $\bigcap_j U_j$ is a model of ψ with visible part $\bigcap_j S_j$. For assignments, $S_{\rho\sqcap\tau} = S_\rho \cap S_\tau$ by definition, so Lemma 3.5 turns the first two claims into the closure of solutions and of extensible visible assignments under $\sqcap$: if ρ and τ extend to solutions ρ' and τ' of ψ, then $\rho' \sqcap \tau'$ is a solution of ψ that extends $\rho \sqcap \tau$.

(2) The cones of $\bot \sqcap \top$ are $\varnothing \cap \Sigma^*$ and $\Sigma^* \cap \varnothing$, so it is Ω, all of whose words are positions labeled f; for $n \geq 1$ there are infinitely many. □

D Proofs for Summaries, Condition (U), and Extracted Instances

D.1 Summaries and the Lowest Point

Throughout, χ is a system over V, pure derivations are those of χ, that is, of the copy and constructor literals of $\chi^\pm$ (Section 6.1), and the automata are those of Definition 6.1.

We prove Lemmas 6.2 and 6.3 together: first the derivations given by the summaries, then the lowest point by induction along a derivation, and finally the converse directions of Lemma 6.2 as the special cases of the lowest point with an empty word. Recall that every set $P^\chi_p(w)$ is upward closed and every set $Q^\chi_q(w)$ downward closed under D_χ.

Step 1: summaries give derivations. We show the forward directions of Lemma 6.2(a)–(c), lifted to an arbitrary suffix σ. (a) Induct on the derivation of $D_\chi(a, b)$ by the rules of Definition 6.1, constructing a pure derivation from (a, σ) to (b, σ) for every σ at once. Reflexivity gives the empty derivation, a copy literal $a \leq b$ of $\chi^\pm$ one copy, and transitivity a concatenation. For the decomposition rule, which concludes $D_\chi(a_i, b_i)$ from $f(a_j)_j \leq u$, $D_\chi(u, v)$, and $v \leq f(b_j)_j$: push i from (a_i, σ) to $(u, i\sigma)$, follow the derivation from $(u, i\sigma)$ to $(v, i\sigma)$ given by the induction hypothesis, and pop i to (b_i, σ). (b) Induct on γ. For $\gamma = \varepsilon$, $P^\chi_p(\varepsilon) = \uparrow_\chi p$, and (a) applies. If $b \in P^\chi_p(\gamma i)$, there are $r \in P^\chi_p(\gamma)$ and a literal $r \leq f(b_j)_j$ of $\chi^\pm$ with $D_\chi(b_i, b)$; the induction hypothesis with $i\sigma$ in place of σ, a pop of i, and (a) compose to a derivation from $(p, \gamma i\sigma)$ to (b, σ). (c) Symmetrically, if $a \in Q^\chi_q(\beta i)$, there are $r \in Q^\chi_q(\beta)$ and a literal $f(a_j)_j \leq r$ with $D_\chi(a, a_i)$; (a), a push of i from (a_i, σ) to $(r, i\sigma)$, and the induction hypothesis compose.

Step 2: three facts on the lower automata. For signed variables q, q', words β and letters i:

(L1) if $D_\chi(q', q)$, then $Q^\chi_{q'}(\beta) \subseteq Q^\chi_q(\beta)$;

(L2) if $f(a_j)_j \leq q$ is a literal of $\chi^\pm$ with $a_i = q'$, then $Q^\chi_{q'}(\beta) \subseteq Q^\chi_q(i\beta)$;

(L3) if $q' \leq f(b_j)_j$ is a literal of $\chi^\pm$, then $Q^\chi_{q'}(i\beta) \subseteq Q^\chi_{b_i}(\beta)$.

All three follow by induction on β, since $\delta^\downarrow_j$ is monotone. For $\beta = \varepsilon$: (L1) is transitivity of D_χ; in (L2), an element a of $\downarrow_\chi q'$ lies in $\delta^\downarrow_i(\downarrow_\chi q)$, witnessed by q itself and the literal $f(a_j)_j \leq q$, since $D_\chi(a, a_i)$; in (L3), an element a of $\delta^\downarrow_i(\downarrow_\chi q')$ comes with r, $D_\chi(r, q')$, a literal $f(a'_j)_j \leq r$ and $D_\chi(a, a'_i)$, and

the decomposition rule applied to $f(a'_j)_j \le r$, $D_\chi(r, q')$ and $q' \le f(b_j)_j$ gives $D_\chi(a'_i, b_i)$, hence $a \in {\downarrow_\chi} b_i$.

Proof of Lemma 6.3. If $w[i:] = v[e:] = \sigma$ and $r \in P^\chi_{p,i}(w) \cap Q^\chi_{q,e}(v)$, then Step 1 gives pure derivations $(p, w) = (p, w[:i]\sigma) \to^*_\chi (r, \sigma) \to^*_\chi (q, v[:e]\sigma) = (q, v)$.

Conversely, consider a pure derivation from (p, w). We show by induction on its length that every atom (q', v') on it admits cuts $i \le |w|$ and $e \le |v'|$ with $w[i:] = v'[e:]$ and $P^\chi_{p,i}(w) \cap Q^\chi_{q',e}(v') \neq \varnothing$; for the last atom this is the claim. For the first atom take $i = e = 0$: $p \in {\uparrow_\chi} p \cap {\downarrow_\chi} p$. Let the invariant hold at (q', v') with cuts i, e and a signed variable r in the intersection, and let the next move lead to (q'', v'').

- A copy $q' \le q''$: then $v'' = v'$, and $r \in Q^\chi_{q'',e}(v'')$ by (L1); keep i and e.
- A push of j by a literal $f(a_k)_k \le q''$ with $a_j = q'$: then $v'' = jv'$, and $r \in Q^\chi_{q''}(j\, v'[:e]) = Q^\chi_{q'',e+1}(v'')$ by (L2); keep i and take $e + 1$.
- A pop of j by a literal $q' \le f(b_k)_k$: then $v' = jv''$ and $q'' = b_j$. If $e \ge 1$, then $v'[:e] = j\, v''[:e-1]$, and $r \in Q^\chi_{b_j,e-1}(v'')$ by (L3); keep i and take $e - 1$. If $e = 0$, then $r \in {\downarrow_\chi} q'$, so $q' \in P^\chi_{p,i}(w)$ by upward closure, and $w[i:] = v' = jv''$; hence $i < |w|$, $w[i] = j$, and $b_j \in \delta_j^{\uparrow}(P^\chi_{p,i}(w)) = P^\chi_{p,i+1}(w)$, witnessed by q' and the literal. Take $i + 1$ and $e = 0$, with the signed variable $b_j \in {\downarrow_\chi} b_j$.

In every case $w[i:] = v''[e:]$ for the new cuts. □

Proof of Lemma 6.2. The forward directions are Step 1 with $\sigma = \varepsilon$. For the converse of (b), apply Lemma 6.3 to a pure derivation from (p, γ) to (b, ε): the only cut of ε is 0, so $\gamma[i:] = \varepsilon$ forces $i = |\gamma|$, and some $r \in P^\chi_p(\gamma) \cap {\downarrow_\chi} b$ gives $b \in P^\chi_p(\gamma)$ by upward closure. The converse of (c) is symmetric, by downward closure, and that of (a) is the case $\gamma = \varepsilon$ of (b), since $P^\chi_a(\varepsilon) = {\uparrow_\chi} a$. □

Proof of Lemma 6.4. By Lemma 3.10, reversing a pure derivation and flipping every signed variable gives a pure derivation, and the dual of an atom has the same word. (1) By Lemma 6.2(a), $D_\chi(a, b)$ iff $(a, \varepsilon) \to^*_\chi (b, \varepsilon)$, iff $(\bar b, \varepsilon) \to^*_\chi (\bar a, \varepsilon)$, iff $D_\chi(\bar b, \bar a)$. Hence $r \in {\downarrow_\chi} \bar q$ iff $D_\chi(r, \bar q)$ iff $D_\chi(q, \bar r)$ iff $r \in \overline{{\uparrow_\chi} q}$. (2) By Lemma 6.2(c,b), $r \in Q^\chi_{\bar q}(w)$ iff $(r, \varepsilon) \to^*_\chi (\bar q, w)$, iff $(q, w) \to^*_\chi (\bar r, \varepsilon)$, iff $\bar r \in P^\chi_q(w)$. (3) By Lemma 6.2, $D_\chi(a, b)$ gives $(a, \varepsilon) \to^*_\chi (b, \varepsilon)$, $r \in P^\chi_p(w)$ gives $(p, w) \to^*_\chi (r, \varepsilon)$, and $r \in Q^\chi_q(w)$ gives $(r, \varepsilon) \to^*_\chi (q, w)$. The sign invariant of Lemma 3.6(2), extended to pure derivations, gives the three claims, since $\mathrm{par}(\varepsilon) = 0$. □

D.2 Least Closures and Deadness

Proof of Lemma 6.6. By definition, $(q, v) \in \mu_\chi$ iff $(q, v) \in \mathrm{Reach}_\chi(p, \varepsilon)$ for some fact (p, ε), iff, by Lemma 3.11, $(p, \alpha) \to^*_\chi (q, v)$ for some α. By Lemma 6.3, this holds iff there are cuts $i \le |\alpha|$ and $e \le |v|$ with $\alpha[:i] = v[e:]$ and $P^\chi_p(\alpha[:i]) \cap Q^\chi_{q,e}(v) \neq \varnothing$. If so, then $Q^\chi_{q,e}(v)$ meets $P^\chi_p(\gamma) \subseteq \mathrm{Fact}^{\uparrow}_\chi$ for $\gamma = \alpha[:i]$. Conversely, if $r \in P^\chi_p(\gamma) \cap Q^\chi_{q,e}(v)$ for a fact (p, ε), a word γ, and a cut e, take $\alpha = \gamma\, v[e:]$ and $i = |\gamma|$.

For the bound, if $r \in P^\chi_e(\gamma)$, then the state $P^\chi_e(\gamma)$ of the upper automaton of e, which has $4^{|V|}$ states, is reached by a word that visits no state twice, of length less than $4^{|V|}$. □

Proof of Lemma 6.10. We first show that

$$P^\chi_p(wz) = \bigcup_{r \in P^\chi_p(w)} P^\chi_r(z) \qquad \text{for all words } w, z. \tag{16}$$

For $z = \varepsilon$ this holds because $P^\chi_p(w)$ is upward closed under D_χ and $r \in {\uparrow_\chi} r = P^\chi_r(\varepsilon)$. The induction step from z to zi applies $\delta_i^{\uparrow}$, which distributes over unions.

(1) By Lemma 6.8(3) and Eq. (16), $\mathsf{DeadU}_\chi(p,w)$ iff some $r \in P^\chi_p(w)$ and some z have $P^\chi_r(z) \cap \mathsf{Neg}^\downarrow_\chi \neq \varnothing$, that is, by Lemma 6.8(3) for the atom (r,ε), iff some $r \in P^\chi_p(w)$ has $\mathsf{DeadU}_\chi(r,\varepsilon)$.

(2) This is Lemma 6.6 for the atom $(\bar p, w)$, together with $Q^\chi_{\bar p} = \overline{P^\chi_p}$ (Lemma 6.4(2)).

(3) Suppose $\mathsf{DeadT}_\chi(p,w)$. If $\mathsf{DeadU}_\chi(p,w)$, then (1) gives some $r \in P^\chi_p(w)$ with $\mathsf{DeadU}_\chi(r,\varepsilon)$, hence $r \in \mathsf{DeadT}^\varepsilon_\chi$. Otherwise $(\bar p, w\delta) \in \mu_\chi$ for some δ, and by (2) applied to $w\delta$ there is a cut $e \le |w\delta|$ with $\overline{P^\chi_p((w\delta)[:e])} \cap \mathsf{Fact}^\uparrow_\chi \neq \varnothing$. If $e \le |w|$, then $(\bar p, w) \in \mu_\chi$ by (2). If $e > |w|$, write $(w\delta)[:e] = wz$; by Eq. (16), some $r \in P^\chi_p(w)$ has $\overline{P^\chi_r(z)} \cap \mathsf{Fact}^\uparrow_\chi \neq \varnothing$, and (2) for the atom $(\bar r, z)$ gives $(\bar r, z) \in \mu_\chi$, so $r \in \mathsf{DeadT}^\varepsilon_\chi$.

Conversely, $(\bar p, w) \in \mu_\chi$ gives $\mathsf{DeadT}_\chi(p,w)$ with $\delta = \varepsilon$. Let $r \in P^\chi_p(w) \cap \mathsf{DeadT}^\varepsilon_\chi$; then $(p,w) \to^*_\chi (r,\varepsilon)$ by Lemma 6.2(b). If $\mathsf{DeadU}_\chi(r,\varepsilon)$, then $\mathsf{Reach}_\chi(r,\varepsilon) \subseteq \mathsf{Reach}_\chi(p,w)$ contains a forbidden root. If $(\bar r, \delta) \in \mu_\chi$, then duality reverses the derivation to $(\bar r, \varepsilon) \to^*_\chi (\bar p, w)$, Lemma 3.6(1) lifts it to $(\bar r, \delta) \to^*_\chi (\bar p, w\delta)$, and $(\bar p, w\delta) \in \mu_\chi$ because μ_χ is closed under the rules (Lemma 3.8(1)).

(4) The first part of DeadT_χ is inherited by prefixes by Lemma 6.8(4), and $(\bar p, wv\delta) = (\bar p, w(v\delta))$.

(5) By Lemma 6.8(3), $r \in \mathsf{DeadU}^\varepsilon_\chi$ iff some word z has $P^\chi_r(z) \cap \mathsf{Neg}^\downarrow_\chi \neq \varnothing$. This is the reachability of a set of states of the upper automaton of r, which has $4^{|V|}$ states, so words z of length less than $4^{|V|}$ suffice. Similarly, by (2) and Eq. (16), the set M of signed variables r with $(\bar r, \delta) \in \mu_\chi$ for some δ consists of the r with $P^\chi_r(z) \cap \overline{\mathsf{Fact}^\uparrow_\chi} \neq \varnothing$ for some word z, and the same bound applies. Finally $\mathsf{DeadT}^\varepsilon_\chi = \mathsf{DeadU}^\varepsilon_\chi \cup M$, and $\mathsf{Fact}^\uparrow_\chi$ is computable by Lemma 6.6. □

Proof of Theorem 6.11. By Corollary 3.13, (p,w) is dead iff (a) $\mathsf{DeadU}_\chi(p,w)$, or (b) $(\bar p, w\delta) \in \mu_\chi$ for some δ, or (c) (p,w) has a self-collision, that is, $(p, w\alpha) \to^*_\chi (\bar p, w\delta)$ for some words α and δ. Causes (a) and (b) together are $\mathsf{DeadT}_\chi(p,w)$, and the causes may overlap. It remains to show that a self-collision implies $\mathsf{DeadT}_\chi(p,w)$ or a self event, and that a self event implies a self-collision.

Assume a self-collision. By Lemmas 6.3 and 6.4(2), there are cuts $i \le |w\alpha|$ and $j \le |w\delta|$ with $(w\alpha)[:i] = (w\delta)[:j]$ and

$$P^\chi_p((w\alpha)[:i]) \cap \overline{P^\chi_p((w\delta)[:j])} \neq \varnothing.$$

In particular both sets $P^\chi_p((w\alpha)[:i])$ and $P^\chi_p((w\delta)[:j])$ are nonempty. If $i > |w|$, then $(w\alpha)[:i]$ is a proper extension of w with a nonempty upper set, and the pop bound (Lemma 6.8(2) with $k = |w|$) gives $\mathsf{DeadU}_\chi(p,w)$; the case $j > |w|$ is the same. Otherwise $i, j \le |w|$, and the equation reads $w[:i]\,\alpha = w[:j]\,\delta$, so $w[:i]$ and $w[:j]$ are prefixes of one word and hence comparable in the prefix order, and the displayed condition reads $P^\chi_{p,i}(w) \cap \overline{P^\chi_{p,j}(w)} \neq \varnothing$. By sign coherence (Lemma 6.4(3)), the elements of $P^\chi_{p,i}(w)$ have the sign $\mathrm{sgn}(p) \oplus \mathrm{par}(w[:i])$ and those of $\overline{P^\chi_{p,i}(w)}$ the opposite sign, so $i \neq j$. If $i < j$, then $w[:j]$ is the shorter suffix, so $w[:j] \preceq w[:i]$, and (i,j) is a self event. If $j < i$, flipping both sets gives $P^\chi_{p,j}(w) \cap \overline{P^\chi_{p,i}(w)} \neq \varnothing$ and $w[:i] \preceq w[:j]$, so (j,i) is a self event.

Conversely, let (i,j) be a self event of (p,w). Then $w[:i] = w[:j]\,\delta$ for some word δ, and the guard says, by Lemma 6.4(2), that $P^\chi_{p,i}(w) \cap Q^\chi_{\bar p}(w[:j]) \neq \varnothing$. Since $(w\delta)[:j] = w[:j]$ and $(w\delta)[:j] = w[:j]\,\delta = w[:i]$, Lemma 6.3 gives $(p,w) \to^*_\chi (\bar p, w\delta)$, a self-collision with $\alpha = \varepsilon$. □

D.3 Extracted Instances

Proof of Theorem 8.7. Write ζ_e for the state of $B = \mathcal{B}_p$ at the cut e of t. By Definition 8.5 and Lemma 7.3, $A_s(u) = Q^\varphi_{q,s}(u)$, $A'_s(u) = Q^\psi_{q,s}(u)$, $B'_e(t) = P^\psi_{p,e}(t)$, and $\zeta_e = (P^\varphi_{p,e}(t), [(\bar p, t[:e]) \in \mu_\varphi], P^\psi_{p,e}(t))$. We translate the conditions of Definition 8.1.

- Condition (a) says that $Q^\varphi_{q,s}(u) \cap \mathsf{Fact}^\uparrow_\varphi = \varnothing$ for every cut s, that is, $(q,u) \notin \mu_\varphi$ (Lemma 6.6).

- By Lemma 7.3(2), a cut that is rejected stays rejected at every later cut, so condition (b) says that the last cut of t is rejected: neither $\mathsf{DeadT}_\varphi(p,t)$ nor $\mathsf{DeadU}_\psi(p,t)$ holds.
- Condition (d) says that some cuts $s \le |u|$ and $e \le |t|$ have $t[e{:}] = u[s{:}]$ and $P^\psi_{p,e}(t) \cap Q^\psi_{q,s}(u) \neq \varnothing$; by Lemma 6.3 for ψ, this says $(p,t) \to^*_\psi (q,u)$.
- Assume (b). At the cut $J(t)$, dt fails, so $\mathsf{DeadU}_\varphi(p, t[{:}J(t)])$ fails, and the pop bound (Lemma 6.8(2)) gives $P^\varphi_{p,e}(t) = \varnothing$ for every $e > J(t)$. A guard $Q^\varphi_{q,s}(u) \cap P^\varphi_{p,e}(t) \neq \varnothing$ or a self guard $P^\varphi_{p,i}(t) \cap \overline{P^\varphi_{p,j}(t)} \neq \varnothing$ therefore forces $e \le J(t)$, respectively $j \le J(t)$. Hence, under (b), condition (c) says that there are no cuts $e \le |t|$ and $s \le |u|$ with $t[e{:}] \preceq u[s{:}]$ and $P^\varphi_{p,e}(t) \cap Q^\varphi_{q,s}(u) \neq \varnothing$, and condition (e) says that (p,t) has no self event in φ (Lemma 7.3(3)).

Let (u,t) be a hole. By (b) and (e), (p,t) is not dead by support and has no self event, so it is live by Theorem 6.11, since φ is satisfiable. By (d), $(p,t) \to^*_\psi (q,u)$. By (a) and (c), and by Lemma 6.14 for the live seed (p,t) and the atom (q,u), $(q,u) \notin \mathrm{Cl}_\varphi(p,t)$.

Conversely, let (p,t) be live in φ, $(p,t) \to^*_\psi (q,u)$, and $(q,u) \notin \mathrm{Cl}_\varphi(p,t)$. By Theorem 6.11, $\mathsf{DeadT}_\varphi(p,t)$ fails and (p,t) has no self event in φ; by (U) and Lemma 7.1, $\mathsf{DeadU}_\psi(p,t)$ fails. So (b) and (e) hold. Condition (d) holds by the derivation. Since $(q,u) \notin \mathrm{Cl}_\varphi(p,t)$, Lemma 6.14 gives $(q,u) \notin \mu_\varphi$, which is (a), and the absence of the cuts described above, which is (c).

(1) By Lemma 4.3(2), (II) fails iff some visible atom (p,t) that is live in φ and some visible atom (q,u) satisfy $(p,t) \to^*_\psi (q,u)$ and $(q,u) \notin \mathrm{Cl}_\varphi(p,t)$. By the equivalence just proved, this happens iff some I_{pq} has a hole.

(2) Let (u,t) be a hole of I_{pq}. Then (p,t) is live, so $S = \mathrm{Cl}_\varphi(p,t)$ is a model of φ (Lemma 3.8(2)). If $S = T|_X$ for a model T of ψ, then T contains (p,t) and is closed under the rules of ψ, so it contains (q,u), which is visible; hence $(q,u) \in S$, a contradiction. By Lemma 3.5, the solution of φ whose set of atoms is S has no extension to a solution of ψ. □

E Proofs for Non-Mortal Holes

This appendix contains the proofs omitted from Section 9: facts on infinite periodic words (Appendix E.1), the continuation identity and the decomposition (Appendix E.2), the breaker and thin states (Appendix E.3), survival (Appendix E.4), the pre-hole graph (Appendix E.5), the pump (Appendix E.6), and the three-case theorem (Appendix E.7). We keep the conventions of Section 9.

E.1 Infinite Periodic Words

Periods of finite words are as in Section 2.1, and the equivalence $\theta \preceq o\theta \iff \theta \preceq o^\omega$ for a nonempty word o is Lemma 2.1. We use the theorem of Fine and Wilf [Fine and Wilf 1965; Lothaire 2002] in its word form: a finite word with periods ϖ and ϖ' and length at least $\varpi + \varpi' - \gcd(\varpi,\varpi')$ has period $\gcd(\varpi,\varpi')$. It concerns equality of letters only and holds over every alphabet. The following facts on infinite words are used throughout.

Lemma E.1 (Infinite periodic words).

(a) Let ξ' be an infinite word with a period ϖ', and let $m \ge 0$. If $\xi'[m{:}]$ has a period ϖ, then ξ' has the period ϖ.

(b) Let ξ be purely periodic with least period ϖ. Every period of a suffix of ξ is a period of ξ, and therefore at least ϖ. For $i, j \ge 0$, $\xi[i{:}] = \xi[j{:}]$ if and only if $i \equiv j \pmod{\varpi}$. In particular, if $\xi = \alpha\lambda^\omega$ with $\lambda \neq \varepsilon$, then $\varpi \le |\lambda|$.

(c) Let o be a nonempty word. An infinite word whose finite prefixes are all prefixes of o^ω equals o^ω. If ξ' is an infinite word with $o\xi' = \xi'$, then $\xi' = o^\omega$; in particular ξ' has the period $|o|$.

(d) Let $\xi_1 = \alpha_1\lambda_1^\omega$ and $\xi_2 = \alpha_2\lambda_2^\omega$ with $\lambda_1, \lambda_2 \neq \varepsilon$. If ξ_1 and ξ_2 agree on their first $\max(|\alpha_1|, |\alpha_2|) + |\lambda_1| + |\lambda_2| - 1$ letters, then $\xi_1 = \xi_2$.

Proof. (a) Let $k \geq 0$, and choose $N \geq 0$ with $N\varpi' \geq m$. Using the period ϖ' of ξ' twice and the period ϖ of $\xi'[m:]$ once,

$$\begin{aligned}\xi'[k] &= \xi'[k + N\varpi'] = (\xi'[m:])[k + N\varpi' - m] \\ &= (\xi'[m:])[k + N\varpi' - m + \varpi] = \xi'[k + N\varpi' + \varpi] = \xi'[k + \varpi].\end{aligned}$$

(b) If a suffix $\xi[m:]$ has a period ϖ', then (a), applied to $\xi' = \xi$ with its period ϖ, shows that ξ has the period ϖ'; so $\varpi' \geq \varpi$. If $i \equiv j \pmod{\varpi}$, then $\xi[i:] = \xi[j:]$, since ϖ is a period of ξ. Conversely, let $\xi[i:] = \xi[j:]$, and let i' and j' be the remainders of i and j modulo ϖ. Then $\xi[i':] = \xi[i:]$ and $\xi[j':] = \xi[j:]$, so $\xi[i':] = \xi[j':]$. If $i' < j'$, this equation says that $\xi[i':]$ has the period $j' - i'$, which is then a period of ξ smaller than ϖ; this is impossible, and so is $j' < i'$. Hence $i' = j'$. Finally, $\xi[|\alpha|:] = \lambda^\omega$ has the period $|\lambda|$, so $\varpi \leq |\lambda|$.

(c) The first claim holds letter by letter. If $o\xi' = \xi'$, then $\xi' = o^k\xi'$ for every k, so ξ' begins with o^k for every k, and $\xi' = o^\omega$.

(d) Let $k = \max(|\alpha_1|, |\alpha_2|)$, $\varpi_1 = |\lambda_1|$, $\varpi_2 = |\lambda_2|$ and $g = \gcd(\varpi_1, \varpi_2)$. The suffixes $\xi_1[k:]$ and $\xi_2[k:]$ have the periods ϖ_1 and ϖ_2, respectively, and they share the prefix v of length $\varpi_1+\varpi_2-1 \geq \varpi_1+\varpi_2-g$, which therefore has both periods and, by the theorem of Fine and Wilf, the period g. As g divides ϖ_1 and $|v| \geq \varpi_1$, the prefix of length ϖ_1 of v is $(v[:g])^{\varpi_1/g}$, and so $\xi_1[k:] = (v[:g])^\omega$. Likewise $\xi_2[k:] = (v[:g])^\omega$. The first k letters agree by assumption, so $\xi_1 = \xi_2$. □

E.2 The Continuation Identity and the Decomposition

Proof of Lemma 9.1. (a) The automata read the same letters up to these cuts. Since $J(t) \leq |t|$, the cuts of $t\theta$ up to $J(t)$ carry the states of the same cuts of t, so $J(t\theta) = J(t)$.

(b) Let (s, e) be a cross comparison of $(u\theta, t\theta)$ with $e \leq J(t)$. Then $|u\theta| - s \geq |t\theta| - e \geq |\theta|$, because $e \leq |t|$; hence $s \leq |u|$. For $s \leq |u|$ and $e \leq |t|$ we have $(u\theta)[s:] = u[s:]\,\theta$ and $(t\theta)[e:] = t[e:]\,\theta$, so (s, e) is a cross comparison of $(u\theta, t\theta)$ if and only if $t[e:]\,\theta \preceq u[s:]\,\theta$. This relation implies $t[e:] \preceq u[s:]$: if $|t[e:]| \leq |u[s:]|$, then $t[e:]$ is a prefix of $u[s:]\,\theta$ of length at most $|u[s:]|$, and if $|t[e:]| > |u[s:]|$, then $t[e:]\,\theta$ is longer than $u[s:]\,\theta$. So it holds if and only if (s, e) is a cross comparison of (u, t) that θ keeps. For a self comparison, apply the same argument to the pair (t, t) and cuts $i < j \leq J(t)$.

(c) For a cross comparison (s, e) with overhang o we have $u[s:] = t[e:]\,o$, so $t[e:]\,\theta \preceq u[s:]\,\theta$ if and only if $\theta \preceq o\theta$, by canceling the common prefix $t[e:]$; the same holds for self comparisons. For $o \neq \varepsilon$ this is equivalent to $\theta \preceq o^\omega$ by Lemma 2.1. For $o = \varepsilon$ it always holds. □

Proof of the continuation facts. (K1) The events of $(u\theta, t\theta)$ are the cross comparisons (s, e) of $(u\theta, t\theta)$ with $e \leq J(t\theta) = J(t)$ whose states lie in G. By Lemma 9.1(b) they are the cross comparisons (s, e) of (u, t) with $e \leq J(t)$ that θ keeps, and by Lemma 9.1(a) their states are those in (u, t). The same argument applies to self events. (K2) is part of Lemma 9.1(c). (K3) An aligned comparison is never broken. If a word $\theta\theta'$ keeps a comparison with nonempty overhang o, then $\theta\theta' \preceq o^\omega$, hence $\theta \preceq o^\omega$, and θ keeps it. (K4) If $u[s:] = t[e:]$, then $(u\theta)[s:] = u[s:]\,\theta = t[e:]\,\theta = (t\theta)[e:]$, and the states of A' and B' at these cuts are unchanged. (K5) The states of A at the cuts $s \leq |u|$ of $u\theta$ are those of u, and its state at the cut $|u| + k$ is $A(A(u), \theta[:k])$. □

Proof of Lemma 9.3. Let (u_0, t_0) be a pre-hole completed by θ. By (K5), $u_0\theta$ is safe. By Lemma 9.1(a), $J(t_0\theta) = J(t_0)$ is defined. By (K4), $(u_0\theta, t_0\theta)$ has a head event. By (K1), its events are the events of (u_0, t_0) that θ keeps, and the self events of $t_0\theta$ are the self events of t_0 that θ keeps; there are none. So $(u_0\theta, t_0\theta)$ is a hole.

Conversely, let (u, t) be a hole and $J = J(t)$. It has a head event (s, e). Put $\gamma = t[:e]$ and $\sigma = u[s:] = t[e:]$. By (FH) and Lemma 8.4, $J \geq e = |\gamma|$. Hence $\theta = t[J:]$ has length $|t| - J \leq |t| - |\gamma| = |\sigma|$, so θ is a suffix of σ and therefore of u. Write $u = u_0\theta$ and $t = t_0\theta$. Then u_0 is safe, as a prefix of u,

and $J(t_0) = J = |t_0|$, since $B_i(t_0) = B_i(t)$ for $i \le |t_0|$. The head event satisfies $e = |\gamma| \le J = |t_0|$ and $s = |u| - |\sigma| \le |u| - |\theta| = |u_0|$, and $u_0[s{:}] = t_0[e{:}]$ because θ is a common suffix of $u[s{:}] = t[e{:}]$; by Lemma 9.1(a) the states at these cuts are unchanged, so (s, e) is a head event of (u_0, t_0). By (K1), the events of (u, t) are the events of (u_0, t_0) that θ keeps, and the self events of t are the self events of t_0 that θ keeps. Since (u, t) is a hole, θ breaks every obligation of (u_0, t_0), and by (K2) (u_0, t_0) has no aligned event. Thus (u_0, t_0) is a pre-hole, and θ, which is safe from $A(u_0)$ by (K5), completes it. □

E.3 The Breaker and Thin States

Proof of Lemma 9.6. Since q is branching, $\mathcal{G}_A$ has a path from q to a cyclic state r that has two distinct safe letters. Let α be the label of a shortest such path; it visits pairwise distinct states, so $|\alpha| < N_A$. Let λ be the label of a shortest cycle through r, so $1 \le |\lambda| \le N_A$. At least one of the two safe letters at r differs from $\lambda[0]$; call it ℓ. Let k be the least integer with $k|\lambda| \ge W + |\lambda|$, and put $\theta = \alpha\lambda^k\ell$. The word θ is safe from q: α leads to r, λ^k returns to r, and $A(r, \ell)$ is safe. By the choice of k, $k|\lambda| \le W + 2|\lambda| - 1$, so $|\theta| \le (N_A - 1) + (W + 2N_A - 1) + 1 < W + 3N_A$.

Let o be a word with $1 \le \Delta = |o| \le W$, and suppose that $\theta \preceq o^\omega$. Then θ has the period Δ, and so has its suffix $\theta' = \lambda^k\ell$. The prefix λ^k of θ' has the periods Δ and $|\lambda|$ and length $k|\lambda| \ge \Delta + |\lambda|$, so by the theorem of Fine and Wilf it has the period $g = \gcd(\Delta, |\lambda|)$. Since $k|\lambda| \ge \Delta$, the period Δ of θ' gives $\ell = \theta'[k|\lambda|] = \theta'[k|\lambda| - \Delta]$. The position $k|\lambda| - \Delta$ lies in λ^k and is divisible by g, so the period g of λ^k gives $\theta'[k|\lambda| - \Delta] = \theta'[0] = \lambda[0]$. Thus $\ell = \lambda[0]$, a contradiction.

For the second claim, let (u_0, t_0) be a pre-hole with $A(u_0)$ branching, and apply the first claim with $q = A(u_0)$ and $W = \max(|u_0|, |t_0|)$. Every obligation of (u_0, t_0) has a nonempty overhang o with $|o| \le W$ (Section 9.2). Since θ is not a prefix of o^ω, it breaks the obligation by Lemma 9.1(c). So θ completes (u_0, t_0). □

Proof of Lemma 9.8. (a) No state on a path from q in $\mathcal{G}_A$ is cyclic. Hence the $|\theta| + 1$ states visited by a word θ safe from q are pairwise distinct, since a repetition would close a cycle, and $|\theta| + 1 \le N_A$. The endpoint is mortal, since every state reachable from a mortal state is mortal.

(b) A cyclic state r has at least one safe letter, namely the first letter of a cycle through r. If it had two, q would be branching. So the unique safe letter at r is the first letter of every cycle through r, and it leads to the next state of such a cycle, which is cyclic.

(c) Let ξ be an infinite word safe from q, and let $r_0 = q, r_1, \ldots$ be the states it visits. Some state repeats, so some r_i is cyclic; let i be the least such index. By (b) and induction, every r_j with $j \ge i$ is cyclic, and $\xi[j]$ is the unique safe letter at r_j; so from i on the run is determined by its current state. By (b), each state of a cycle through r_i has the edge of that cycle as its only safe edge, so the run follows the cycle and returns to r_i; let $l \ge 1$ be least with $r_{i+l} = r_i$. Then $r_{j+l} = r_j$ for all $j \ge i$, and $\xi[i{:}] = \lambda^\omega$ with $\lambda = \xi[i{:}i + l]$. Put $\alpha = \xi[{:}i]$. The states $r_0, \ldots, r_{i-1}$ are not cyclic, so they are pairwise distinct and differ from the cyclic states $r_i, \ldots, r_{i+l-1}$. These are pairwise distinct as well: if $r_{i+m} = r_{i+m'}$ with $0 \le m < m' < l$, then by determinism the run has the period $m' - m$ from time $i + m$ on, so $r_{i+l-(m'-m)} = r_{i+l} = r_i$ with $1 \le l - (m' - m) < l$, contradicting the minimality of l. Hence $|\alpha| + |\lambda| = i + l \le N_A$, and $\alpha\lambda$ is safe from q with $A(q, \alpha\lambda) = A(q, \alpha)$, so $\xi = \alpha\lambda^\omega$ is a lasso of q. Since q is not mortal, some infinite word is safe from q, so q has a lasso.

(d) Let $r_0 = q, \ldots, r_{|\theta|}$ be the states visited by θ. If some r_i were cyclic, then by (b) and induction every r_m with $m \ge i$ would be cyclic, since $\theta[m]$ is a safe letter at r_m and hence the unique one; then $A(q, \theta) = r_{|\theta|}$ would be cyclic and not mortal. So no r_i is cyclic. Then the r_i are pairwise distinct, since a repetition would make a state cyclic, and $|\theta| < N_A$. □

E.4 Survival

Proof of Lemma 9.10. (a) By Lemma 9.1(c), a finite word keeps a comparison with nonempty overhang o if and only if it is a prefix of o^ω. So the comparison survives ξ if and only if every finite prefix of ξ is a prefix of o^ω, that is, by Lemma E.1(c), if and only if $\xi = o^\omega$. An aligned comparison is kept by every word.

(b) Suppose that $u[s{:}]\,\xi = t[e{:}]\,\xi$. The finite words $t[e{:}]$ and $u[s{:}]$ are prefixes of this infinite word, and $|t[e{:}]| \le |u[s{:}]|$, so $t[e{:}] \preceq u[s{:}]$ and (s, e) is a cross comparison. For every prefix θ of ξ, the words $t[e{:}]\,\theta$ and $u[s{:}]\,\theta$ are prefixes of the same infinite word, and the first is not longer, so $t[e{:}]\,\theta \preceq u[s{:}]\,\theta$: the comparison survives ξ. Conversely, let (s, e) be a cross comparison that survives ξ. For every n, $t[e{:}]\,\xi[{:}n] \preceq u[s{:}]\,\xi[{:}n]$, so the first $|t[e{:}]| + n$ letters of $u[s{:}]\,\xi$ are those of $t[e{:}]\,\xi$. As n is arbitrary, $u[s{:}]\,\xi = t[e{:}]\,\xi$. The claim on self comparisons is the case $u = t$, $s = i$, $e = j$.

(c) By (a), $\xi \ne o^\omega$. By Lemma E.1(d), applied to $\alpha\lambda^\omega$ and $\varepsilon\, o^\omega$, the two words differ at some position $i < |\alpha| + |\lambda| + |o| - 1$. The prefix of ξ of length $i + 1$ is not a prefix of o^ω, so it breaks the comparison by Lemma 9.1(c), and by (K3) so does every longer prefix, in particular the prefix of length $|\alpha| + |\lambda| + |o| - 1$. □

E.5 The Pre-Hole Graph

For words v and z, the *aligned pairs* of (v, z) are the pairs of cuts (s, e), $s \le |v|$ and $e \le |z|$, with $v[s{:}] = z[e{:}]$. Let $\mathrm{h}(v, z) \in \{0, 1\}$ be 1 if and only if some aligned pair (s, e) of (v, z) has $(A'_s(v), B'_e(z)) \in H$, and let $\mathrm{g}(v, z) \in \{0, 1\}$ be 1 if and only if some aligned pair has $(A_s(v), B_e(z)) \in G$.

Lemma E.2 (Aligned pairs under extension). *For words v, z and letters ℓ, ℓ':*

(a) the only aligned pair of (v, ε) is $(|v|, 0)$, and the only aligned pair of (ε, z) is $(0, |z|)$;
(b) the aligned pairs of $(v\ell, z\ell')$ are $(|v\ell|, |z\ell'|)$ and, if $\ell = \ell'$, the aligned pairs of (v, z).

Consequently $\mathrm{h}(v\ell, z\ell') = 1$ if and only if $(A'(v\ell), B'(z\ell')) \in H$, or $\ell = \ell'$ and $\mathrm{h}(v, z) = 1$; and likewise for g*, with A, B and G.*

Proof. (a) The suffixes of ε are empty, and the only empty suffix of v is $v[|v|{:}]$. (b) The pair of last cuts is aligned. Let (s, e) be another aligned pair. If $s = |v\ell|$, then the common suffix is empty and $e = |z\ell'|$; so $s \le |v|$ and, likewise, $e \le |z|$. Then $(v\ell)[s{:}] = v[s{:}]\,\ell$ and $(z\ell')[e{:}] = z[e{:}]\,\ell'$, which are equal if and only if $\ell = \ell'$ and $v[s{:}] = z[e{:}]$. The states at the cuts $s \le |v|$ and $e \le |z|$ are those of v and z. □

The graph $\mathcal{P}_E$. Let E be a set of safe states of A. The nodes of $\mathcal{P}_E$ are the tuples $\kappa = (r_A, r_B, r_{A'}, r_{B'}, h, g, \phi)$ with $r_M \in S_M$ for each automaton M, bits $h, g \in \{0, 1\}$, and a phase $\phi \in \{0, 1, 2\}$; there are $12aba'b'$ of them. Every edge is labeled by a pair of words of length at most one. From a node κ there are the following edges.

- A *spine edge*, labeled (ℓ, ε), if $\phi = 0$ and $A(r_A, \ell)$ is safe. It leads to the node with $A(r_A, \ell)$ and $A'(r_{A'}, \ell)$ in place of r_A and $r_{A'}$, with $h = 1$ if and only if $(A'(r_{A'}, \ell), r_{B'}) \in H$, with $g = 1$ if and only if $(A(r_A, \ell), r_B) \in G$, and with phase 0.
- A *target edge*, labeled (ε, ℓ'), if $\phi = 1$ and $r_B \notin \mathrm{Rej}$. It leads to the node with $B(r_B, \ell')$ and $B'(r_{B'}, \ell')$ in place of r_B and $r_{B'}$, with $h = 1$ if and only if $(r_{A'}, B'(r_{B'}, \ell')) \in H$, with $g = 1$ if and only if $(r_A, B(r_B, \ell')) \in G$, and with phase 1.
- A *joint edge*, labeled (ℓ, ℓ'), if $A(r_A, \ell)$ is safe and $r_B \notin \mathrm{Rej}$. It leads to the node with all four states advanced, with $h = 1$ if and only if $(A'(r_{A'}, \ell), B'(r_{B'}, \ell')) \in H$ or $\ell = \ell'$ and the old h is 1, with g defined in the same way from A, B, G and the old g, and with phase 2.

A path *spells* the pair of words obtained by concatenating the first and the second components of its labels. For words v and z and a phase ϕ, put

$$\mathrm{node}(v, z, \phi) = (A(v), B(z), A'(v), B'(z), \mathrm{h}(v, z), \mathrm{g}(v, z), \phi).$$

The *initial* nodes are $\mathrm{node}(\varepsilon, \varepsilon, \phi)$ for $\phi \in \{0, 1, 2\}$, provided that 0 is safe; the *final* nodes are those with $r_A \in E$, $r_B \in \mathrm{Rej}$, $h = 1$ and $g = 0$.

Lemma E.3 (Invariant). *If a path from an initial node spells (v, z) and ends at the node κ, then $\kappa = \mathrm{node}(v, z, \phi)$ for some phase ϕ, the word v is safe, $B_e(z) \notin \mathrm{Rej}$ for all $e < |z|$, and $z = \varepsilon$ if $\phi = 0$, and $v = \varepsilon$ if $\phi = 1$.*

Proof. By induction on the length of the path. An initial node $\mathrm{node}(\varepsilon, \varepsilon, \phi)$ satisfies the claim, since 0 is safe. Let the claim hold for a path that spells (v, z) and ends at $\kappa = \mathrm{node}(v, z, \phi)$. A spine edge requires $\phi = 0$, so $z = \varepsilon$, and by Lemma E.2(a) it leads to $\mathrm{node}(v\ell, \varepsilon, 0)$; the word $v\ell$ is safe, since the new state of A is safe. A target edge requires $\phi = 1$, so $v = \varepsilon$; it leads to $\mathrm{node}(\varepsilon, z\ell', 1)$ by Lemma E.2(a), and $B_{|z|}(z\ell') = B(z) = r_B$ is not in Rej. A joint edge leads to $\mathrm{node}(v\ell, z\ell', 2)$ by Lemma E.2(b), and safety and non-rejection follow as before. The phase 2 imposes no condition. □

Lemma E.4 (Correctness of the pre-hole graph). *The graph $\mathcal{P}_E$ has a path from an initial node to a final node that spells (v, z) if and only if (v, z) is a pre-hole with $A(v) \in E$.*

Proof. Let such a path end at the final node κ. By Lemma E.3, $\kappa = \mathrm{node}(v, z, \phi)$, the word v is safe, and $B_e(z) \notin \mathrm{Rej}$ for $e < |z|$; as κ is final, $A(v) \in E$, $B(z) \in \mathrm{Rej}$, so that $J(z) = |z|$, $\mathrm{h}(v, z) = 1$, so that (v, z) has a head event, and $\mathrm{g}(v, z) = 0$. Since every cut $e \le |z| = J(z)$ is admissible for an event, $\mathrm{g}(v, z) = 0$ says that (v, z) has no aligned event. So (v, z) is a pre-hole with $A(v) \in E$.

Conversely, let (v, z) be a pre-hole with $A(v) \in E$. Suppose first that $|v| \ge |z|$, and put $m = |v| - |z|$. Start at $\mathrm{node}(\varepsilon, \varepsilon, 0)$, which is initial since v is safe; take spine edges labeled $(v[0], \varepsilon), \ldots, (v[m-1], \varepsilon)$, and then joint edges labeled $(v[m+i], z[i])$ for $i < |z|$. By Lemma E.2, the path is at $\mathrm{node}(v[:m'], z[:e'], \phi)$ after it has spelled $(v[:m'], z[:e'])$. Every edge is allowed: the new states of A are states of the safe word v, and the joint edges start at states $B_i(z)$ with $i < |z| = J(z)$, which are not in Rej. The path ends at $\mathrm{node}(v, z, \phi)$, which is final because (v, z) is a pre-hole with $A(v) \in E$. If $|z| > |v|$, start at $\mathrm{node}(\varepsilon, \varepsilon, 1)$, take target edges for the first $|z| - |v|$ letters of z, which start at states $B_i(z)$ with $i < |z|$, and continue with joint edges as before. □

Proof of Proposition 9.12. Let (u_0, t_0) be a pre-hole with $A(u_0) \in E$. By Lemma E.4, $\mathcal{P}_E$ has a path from an initial node to a final node. A shortest such path visits no node twice, since a repeated node would allow us to cut out a cycle; so it has fewer than $12aba'b'$ edges. Each edge contributes at most one letter to each word, so the path spells a pair (u_1, t_1) with $|u_1|, |t_1| < 12aba'b'$, which is a pre-hole with $A(u_1) \in E$ by Lemma E.4. For decidability, test every pair of words over the finite alphabet Σ of these lengths. □

E.6 The Pump

Throughout this subsection, ξ is a purely periodic infinite word with least period ϖ. We prove Theorem 9.14 in five steps: a reformulation of survivor-freeness, the structure of rests and periodic ends, with a bound on the periodic end of the target, non-self-similar words, a deletion lemma, and four deletions together with a count.

E.6.1 Matched Pairs and Good Pairs.

Definition E.5 (Matched pairs, good pairs). Let u and t be words. A *matched pair* of (u, t) is a pair of cuts (s, e), $s \le |u|$ and $e \le |t|$, with $|t| - e \le |u| - s$ and $u[s:] \, \xi = t[e:] \, \xi$. A *matched self pair* of t is a pair of cuts $i < j \le |t|$ with $t[i:] \, \xi = t[j:] \, \xi$. The pair (u, t) is *good* if

(G1) u is safe;
(G2) $B_e(t) \notin \mathrm{Rej}$ for every $e < |t|$, and $B(t) \in \mathrm{Rej}$;
(G3) (u, t) has a head event;
(G4) $(A_s(u), B_e(t)) \notin G$ for every matched pair (s, e) of (u, t);
(G5) $(B_i(t), B_j(t)) \notin G_\mathrm{s}$ for every matched self pair (i, j) of t.

Lemma E.6 (Good pairs). *A pair (u, t) is good if and only if it is a pre-hole for which ξ is survivor-free.*

Proof. Condition (G2) says that $J(t) = |t|$; so every cut of t is admissible for events and self events. Let (u, t) be good. Every aligned pair of cuts is a matched pair, so by (G4) (u, t) has no aligned event; with (G1)–(G3), it is a pre-hole. If an event (s, e) survived ξ, then, being a cross comparison, it would satisfy $|t| - e \le |u| - s$, and by Lemma 9.10(b) $u[s{:}]\,\xi = t[e{:}]\,\xi$; so it would be a matched pair carrying G, contrary to (G4). Likewise a surviving self event would be a matched self pair carrying G_s, contrary to (G5). So ξ is survivor-free for (u, t).

Conversely, let (u, t) be a pre-hole for which ξ is survivor-free. Conditions (G1)–(G3) hold. By Lemma 9.10(b), a matched pair is a cross comparison that survives ξ; if it carried G, it would be an event, since $e \le |t| = J(t)$, that survives ξ. Likewise a matched self pair carrying G_s would be a surviving self event. So (G4) and (G5) hold. □

E.6.2 Rests and Periodic Ends. Let w be a word. A cut s of w is *periodic* if $w[s{:}]\,\xi$ has the period ϖ. For $s \le s'$ the word $w[s'{:}]\xi$ is a suffix of $w[s{:}]\,\xi$, so every cut after a periodic cut is periodic; the cut $|w|$ is periodic, because ξ has the period ϖ. Let R_w be the least periodic cut. The cuts $s \ge R_w$ are periodic, and the cuts $s < R_w$, the *rest cuts*, are not. Put $T_w = |w| - R_w$, and call $|w| - s$ the *end distance* of the cut s; the periodic cuts are the cuts of end distance at most T_w.

Lemma E.7 (Rests and periodic ends). *Let u and t be words.*

(a) For periodic cuts s of u and e of t, $u[s{:}]\,\xi = t[e{:}]\,\xi$ if and only if $|u| - s \equiv |t| - e \pmod{\varpi}$.
(b) If $s \le |u|$, $e \le |t|$ and $u[s{:}]\,\xi = t[e{:}]\,\xi$, then $\max(R_u - s, 0) = \max(R_t - e, 0)$. In particular, s is periodic if and only if e is, and if s is a rest cut, then e is a rest cut with $R_t - e = R_u - s$.
(c) If $T_u \ne T_t$, then the end distance $|u| - s = |t| - e$ of every aligned pair (s, e) of (u, t) is at most $\min(T_u, T_t)$.

Proof. (a) Let $k = |u| - s$ and $k' = |t| - e$, and choose N with $N\varpi \ge \max(k, k')$. The infinite word $\xi' = u[s{:}]\,\xi$ has the period ϖ, so $\xi' = \xi'[N\varpi{:}] = \xi[N\varpi - k{:}]$. Likewise $t[e{:}]\,\xi = \xi[N\varpi - k'{:}]$. By Lemma E.1(b), these are equal if and only if $N\varpi - k \equiv N\varpi - k'$, that is, $k \equiv k' \pmod{\varpi}$.

(b) Put $m_u = \max(R_u - s, 0)$ and $m_t = \max(R_t - e, 0)$, and suppose that $m_u > m_t$; the other case is symmetric. Then $s + m_t < R_u \le |u|$, and $e \le e + m_t \le |t|$ with $e + m_t \ge R_t$. Removing the first m_t letters of both infinite words gives $u[s + m_t{:}]\xi = t[e + m_t{:}]\xi$. The right side has the period ϖ, because $e + m_t$ is periodic; so $s + m_t$ is a periodic cut, which contradicts $s + m_t < R_u$. The remaining claims follow, because s is a rest cut exactly when $R_u - s > 0$.

(c) Let (s, e) be an aligned pair and $k = |u| - s = |t| - e$. Then $u[s{:}]\,\xi = t[e{:}]\,\xi$. If s were a rest cut, then by (b) so would be e, with $R_u - s = R_t - e$, and $T_u = k - (R_u - s) = k - (R_t - e) = T_t$, which is excluded. So s is periodic, and by (b) so is e; that is, $k \le T_u$ and $k \le T_t$. □

For a rest cut j of u, a cut e of t with $u[j{:}]\,\xi = t[e{:}]\,\xi$ is a *partner* of j; by Lemma E.7(b), it is the rest cut $e = R_t - (R_u - j)$, so j has at most one partner. Partners of rest cuts of t are defined in the same way. A rest cut is *partnered* if it has a partner and *unpartnered* otherwise. Let C_u and U_u be the sets of partnered and unpartnered rest cuts of u, and C_t and U_t those of t.

Lemma E.8 (Periodic target end). *Let t be a word with $B_e(t) \notin \mathrm{Rej}$ for every $e < |t|$ and $B(t) \in \mathrm{Rej}$, as in condition* (G2). *Then $T_t < N_B\,\varpi$.*

Proof. Put $L = |t|$, and suppose that two periodic cuts $i < j$ of t have the same state of B and end distances that agree modulo ϖ. By Lemma E.7(a), $t[i{:}]\,\xi = t[j{:}]\,\xi$, and comparing the first $L - j$ letters gives $t[i{:}i + L - j] = t[j{:}]$. By determinism,

$$B_{L-(j-i)}(t) = B(B_i(t), t[j{:}]) = B(B_j(t), t[j{:}]) = B_L(t) \in \mathsf{Rej},$$

although $L - (j - i) < L$. This contradicts the hypothesis. Hence the $T_t + 1$ periodic cuts e of t have pairwise distinct signatures $(B_e(t), (L - e) \bmod \varpi)$, of which there are at most $N_B\varpi$; so $T_t + 1 \le N_B\varpi$. □

E.6.3 Non-Self-Similar Words. An infinite word ξ' is *non-self-similar* if $z\xi' \neq \xi'$ for every nonempty finite word z.

Lemma E.9 (Non-self-similar words).

(a) If j is a rest cut of a word w, then $w[j{:}]\,\xi$ is non-self-similar.
(b) Let ξ' be non-self-similar. If $z_1\xi' = z_2\xi'$ for finite words z_1 and z_2, then $z_1 = z_2$. If z is a nonempty finite word and $k \ge 0$, then $z\xi' \neq \xi'[k{:}]$.
(c) Let ξ' be non-self-similar and t a word. At most one cut e of t satisfies $t[e{:}]\,\xi = \xi'$.
(d) Let v and v' be words, and let j be a rest cut of v such that $v[j{:}]\,\xi \neq v'[e{:}]\,\xi$ for every cut e of v'. Then $z\,v[j{:}]\,\xi \neq v'[k{:}]\,\xi$ for every finite word z and every cut k of v'.

Proof. (a) Let $\xi' = w[j{:}]\,\xi$, and suppose that $z\xi' = \xi'$ with $z \neq \varepsilon$. Then ξ' has the period $|z|$ by Lemma E.1(c). Its suffix $\xi'[|w| - j{:}] = \xi$ has the period ϖ, so ξ' has the period ϖ by Lemma E.1(a). Then j is a periodic cut, a contradiction.

(b) Let $|z_1| \ge |z_2|$. Both words are prefixes of $z_1\xi' = z_2\xi'$, so $z_1 = z_2 z$ for a word z; canceling z_2 gives $z\xi' = \xi'$, so $z = \varepsilon$. If $z\xi' = \xi'[k{:}]$, then $(\xi'[{:}k]\,z)\xi' = \xi'[{:}k]\,\xi'[k{:}] = \xi'$ with $\xi'[{:}k]\,z$ nonempty, a contradiction.

(c) If $e < e'$ are cuts with $t[e{:}]\,\xi = t[e'{:}]\xi = \xi'$, then $t[e{:}e']\xi' = t[e{:}]\,\xi = \xi'$ with $t[e{:}e']$ nonempty, a contradiction.

(d) Let $\xi' = v[j{:}]\,\xi$, and suppose that $z\xi' = v'[k{:}]\,\xi$. If $|z| \le |v'| - k$, then $z = v'[k{:}k + |z|]$, and canceling z gives $\xi' = v'[k + |z|{:}]\xi$, contrary to the hypothesis. If $|z| > |v'| - k$, then $z = v'[k{:}]\,z'$ with z' nonempty, and canceling $v'[k{:}]$ gives $z'\xi' = \xi$. Then $\xi' = \xi[|z'|{:}]$ has the period ϖ, and j is a periodic cut, a contradiction. □

E.6.4 The Deletion Lemma. Let $u = xwy$ and $t = x'w'y'$ with $w \neq \varepsilon$ or $w' \neq \varepsilon$, and put $\bar{u} = xy$ and $\bar{t} = x'y'$. A cut s of $\bar{u}$ is *deep* if $s < |x|$ and *shallow* otherwise; its *image* is $\hat{s} = s$ if s is deep and $\hat{s} = s + |w|$ if s is shallow. Cuts e of $\bar{t}$ are deep if $e < |x'|$, shallow otherwise, with images $\hat{e}$ defined in the same way with w'. The image map is strictly increasing. For a deep cut s, $\bar{u}[s{:}] = x[s{:}]\,y$ and $u[\hat{s}{:}] = x[s{:}]\,wy$; for a shallow cut $s = |x| + k$, $\bar{u}[s{:}] = u[\hat{s}{:}] = y[k{:}]$, and the end distances agree: $|\bar{u}| - s = |u| - \hat{s}$. The same holds for $\bar{t}$. Put $Y = y\xi$ and $Y' = y'\xi$.

Lemma E.10 (Deletion). *Let (u, t) be good, and suppose:*

(D1) each of A and A' has the same state at the cuts $|x|$ and $|xw|$ of u, and each of B and B' has the same state at the cuts $|x'|$ and $|x'w'|$ of t;
(D2) the image $(\hat{s}, \hat{e})$ of every matched pair (s, e) of $(\bar{u}, \bar{t})$ is a matched pair of (u, t), and the image $(\hat{i}, \hat{j})$ of every matched self pair (i, j) of $\bar{t}$ is a matched self pair of t;
(D3) some head event (s, e) of (u, t) is shallow in both words, that is, $s \ge |xw|$ and $e \ge |x'w'|$, or deep in both words at equal distance from factors of equal length, that is, $s \le |x|$, $e \le |x'|$, $|x| - s = |x'| - e$ and $|w| = |w'|$.

Then $(\bar{u}, \bar{t})$ is good, $A(\bar{u}) = A(u)$, and $|\bar{u}| + |\bar{t}| < |u| + |t|$.

Proof. First, for $M \in \{A, A'\}$ and every cut s of $\bar{u}$, $M_s(\bar{u}) = M_{\hat{s}}(u)$. For $s \le |x|$ the two words have the common prefix $x[:s]$. For a shallow cut $s = |x| + k$, the automaton M reads x and then $y[:k]$ in $\bar{u}$, and x, w and then $y[:k]$ in u; by (D1), $M(0, xw) = M(0, x)$, so both runs end in the same state. Likewise $M_e(\bar{t}) = M_{\hat{e}}(t)$ for $M \in \{B, B'\}$ and every cut e of $\bar{t}$. Moreover, the states of A' and B' at cuts $s \le |x|$ of $\bar{u}$ and $e \le |x'|$ of $\bar{t}$ are those at the same cuts of u and t, because the prefixes coincide.

(G1) Every $A_s(\bar{u}) = A_{\hat{s}}(u)$ is safe. (G2) For $e < |\bar{t}|$ we have $\hat{e} < |t|$, so $B_e(\bar{t}) = B_{\hat{e}}(t) \notin \mathrm{Rej}$. The last cut $|\bar{t}|$ is shallow with image $|t|$, so $B(\bar{t}) = B(t) \in \mathrm{Rej}$; in the same way $A(\bar{u}) = A(u)$.

(G3) Let (s, e) be the head event of (D3). If it is shallow in both words, then $\bar{s} = s - |w|$ and $\bar{e} = e - |w'|$ are shallow cuts with images s and e, so $\bar{u}[\bar{s}:] = u[s:] = t[e:] = \bar{t}[\bar{e}:]$, and the states of A' and B' there are those at s and e; so $(\bar{s}, \bar{e})$ is a head event of $(\bar{u}, \bar{t})$. Otherwise $u[s:] = x[s:]\, wy$ equals $t[e:] = x'[e:]\, w'y'$, where $|x[s:]| = |x'[e:]|$ and $|w| = |w'|$; comparing factors of equal lengths gives $x[s:] = x'[e:]$, $w = w'$ and $y = y'$. Hence $\bar{u}[s:] = x[s:]\, y = x'[e:]\, y' = \bar{t}[e:]$, and since $s \le |x|$ and $e \le |x'|$, the states of A' and B' at these cuts are those in u and t. So (s, e) is a head event of $(\bar{u}, \bar{t})$.

(G4) Let (s, e) be a matched pair of $(\bar{u}, \bar{t})$. By (D2), $(\hat{s}, \hat{e})$ is a matched pair of (u, t), so $(A_{\hat{s}}(\bar{u}), B_e(\bar{t})) = (A_{\hat{s}}(u), B_{\hat{e}}(t)) \notin G$. (G5) Let (i, j) be a matched self pair of $\bar{t}$. By (D2), $(\hat{i}, \hat{j})$ is a matched self pair of t, so $(B_i(\bar{t}), B_j(\bar{t})) = (B_{\hat{i}}(t), B_{\hat{j}}(t)) \notin G_s$.

Finally, $|\bar{u}| + |\bar{t}| = |u| + |t| - |w| - |w'|$, and one of the deleted factors is nonempty. □

A pair whose cuts are both shallow satisfies (D2) automatically: its suffixes and end distances are those of its image. Since the image map is increasing, a self pair $i < j$ with i shallow has j shallow. So (D2) follows from the following five implications, which are sufficient conditions for it, for deep cuts $s < |x|$ of $\bar{u}$ and $e < |x'|$ of $\bar{t}$, shallow cuts $|x| + k$ of $\bar{u}$ and $|x'| + k$ of $\bar{t}$, and deep cuts $i < j < |x'|$ of $\bar{t}$.

(i) Both cuts deep: if $|x'[e:]| + |y'| \le |x[s:]| + |y|$ and $x[s:]\, Y = x'[e:]\, Y'$, then $|x'[e:]| + |w'| + |y'| \le |x[s:]| + |w| + |y|$ and $x[s:]\, wY = x'[e:]\, w'Y'$.

(ii) Spine cut deep, target cut shallow: if $|y'| - k \le |x[s:]| + |y|$ and $x[s:]\, Y = y'[k:]\, \xi$, then $x[s:]\, wY = y'[k:]\, \xi$. The length condition of the image, $|y'| - k \le |x[s:]| + |w| + |y|$, then holds, since only the spine side grows.

(iii) Spine cut shallow, target cut deep: if $|x'[e:]| + |y'| \le |y| - k$ and $y[k:]\, \xi = x'[e:]\, Y'$, then $|x'[e:]| + |w'| + |y'| \le |y| - k$ and $y[k:]\, \xi = x'[e:]\, w'Y'$.

(iv) Self pair, both cuts deep: if $x'[i:]\, Y' = x'[j:]\, Y'$, then $x'[i:]\, w'Y' = x'[j:]\, w'Y'$.

(v) Self pair, first cut deep, second cut $|x'| + k$ shallow: if $x'[i:]\, Y' = y'[k:]\, \xi$, then $x'[i:]\, w'Y' = y'[k:]\, \xi$.

Lemma E.11 (Two ways to lift matched pairs).

(a) Absorption. *If $wY = Y$, then* (ii) *holds.*

(b) Rigidity. *If Y' is non-self-similar, then the premises of* (iv) *and* (v) *never hold, so* (iv) *and* (v) *hold.*

Proof. (a) If $wY = Y$, then $x[s:]\, wY = x[s:]\, Y$, which gives (ii).

(b) In (iv), $x'[i:]\, Y' = x'[j:]\, Y'$ would give $x'[i:] = x'[j:]$ by Lemma E.9(b), although the two words have different lengths. In (v), $y'[k:]\, \xi = Y'[k:]$ because $k \le |y'|$, and $x'[i:]\, Y' = Y'[k:]$ with $x'[i:]$ nonempty contradicts Lemma E.9(b). □

E.6.5 The Four Deletions. Put

$$N_1 = aa'\varnothing, \qquad N_2 = 2aba'b', \qquad N_3 = aa', \qquad N_4 = bb'.$$

For a pair (u, t) and a cut s of u, let the *head bit* $\eta(s)$ be 1 if some head event (s', e') of (u, t) has $s' \ge s$, and 0 otherwise; it does not increase with s.

Lemma E.12 (Four deletions). *Let (u, t) be good, and suppose that one of the following conditions holds:*

(1) $T_u \geq T_t + N_1$;
(2) $|C_u| > N_2$;
(3) $|U_u| > N_3$;
(4) $|U_t| > N_4$.

Then there is a good pair $(\bar{u}, \bar{t})$ with $A(\bar{u}) = A(u)$ and $|\bar{u}| + |\bar{t}| < |u| + |t|$.

Proof. In each case we choose $u = xwy$ and $t = x'w'y'$, check (D1)–(D3), and apply Lemma E.10; for (D2) we check the implications (i)–(v), using Lemma E.11 where possible. When $x' = w' = \varepsilon$ and $y' = t$, the word $\bar{t} = t$ has no deep cut, so only (ii) has to be checked, and (D1) for B and B' is trivial. When $x = w = \varepsilon$ and $y = u$, only (iii), (iv) and (v) have to be checked, and (D1) for A and A' is trivial.

(1) Spine-end deletion. For $0 \leq \delta \leq N_1$ let $s_\delta = |u| - T_t - \delta$; it is a periodic cut of u, since $T_t + \delta \leq T_t + N_1 \leq T_u$. The signature $(A_{s_\delta}(u), A'_{s_\delta}(u), \delta \bmod \varpi)$ takes at most N_1 values, so some $\delta_1 < \delta_2$ have the same signature. Put $x = u[{:}s_{\delta_2}]$, $w = u[s_{\delta_2}{:}s_{\delta_1}]$, $y = u[s_{\delta_1}{:}]$, $x' = w' = \varepsilon$ and $y' = t$. (D1) holds by the signature, and $wY = Y$ by Lemma E.7(a), since the end distances $T_t + \delta_2$ and $T_t + \delta_1$ agree modulo ϖ; so (ii) holds by absorption. For (D3), note that $T_u \neq T_t$, as $N_1 \geq 1$. By Lemma E.7(c), every head event (s, e) has end distance at most T_t, so $s \geq |u| - T_t \geq s_{\delta_1} = |xw|$, and $e \geq 0 = |x'w'|$.

(2) Synchronous rest deletion. For $j \in C_u$ let $e(j)$ be its partner, $e(j) = R_t - (R_u - j)$. The signature $(A_j(u), A'_j(u), B_{e(j)}(t), B'_{e(j)}(t), \eta(j))$ takes at most N_2 values, so some $j_1 < j_2$ in C_u have the same signature. Put $e_1 = e(j_1)$ and $e_2 = e(j_2)$, so that $e_2 - e_1 = j_2 - j_1 > 0$, and let $x = u[{:}j_1]$, $w = u[j_1{:}j_2]$, $y = u[j_2{:}]$, $x' = t[{:}e_1]$, $w' = t[e_1{:}e_2]$ and $y' = t[e_2{:}]$. Then $Y = u[j_2{:}]\xi = t[e_2{:}]\xi = Y'$; call this word ξ'. It is non-self-similar by Lemma E.9(a). Moreover $w\xi' = u[j_1{:}]\xi = t[e_1{:}]\xi = w'\xi'$ and $|w| = |w'|$, so $w = w'$. (D1) holds by the signature. For (D2), (iv) and (v) hold by rigidity. In (i), the premise $x[s{:}]\,\xi' = x'[e{:}]\,\xi'$ gives $x[s{:}] = x'[e{:}]$ by Lemma E.9(b), hence $x[s{:}]\,w\xi' = x'[e{:}]\,w'\xi'$, and the length condition gains $|w| = |w'|$ on both sides. The premise of (ii) never holds: $x[s{:}]\,\xi' = y'[k{:}]\,\xi = \xi'[k{:}]$ with $x[s{:}]$ nonempty contradicts Lemma E.9(b). Neither does that of (iii), for the same reason. For (D3), suppose first that $\eta(j_2) = 1$, so some head event (s, e) has $s \geq j_2$. If $e < e_2$, then $u[s{:}]\,\xi = t[e{:}]\,\xi = t[e{:}e_2]\xi'$ with $t[e{:}e_2]$ nonempty, while $u[s{:}]\,\xi = \xi'[s - j_2{:}]$, which contradicts Lemma E.9(b). So $e \geq e_2 = |x'w'|$ and $s \geq j_2 = |xw|$. Otherwise $\eta(j_1) = \eta(j_2) = 0$, so every head event (s, e) has $s < j_1$. Removing the first $j_1 - s$ letters from $u[s{:}] = t[e{:}]$ gives $u[j_1{:}] = t[e + j_1 - s{:}]$, so $t[e + j_1 - s{:}]\xi = u[j_1{:}]\xi = t[e_1{:}]\xi$. This word is non-self-similar by Lemma E.9(a), so $e + j_1 - s = e_1$ by Lemma E.9(c). Hence $s \leq |x|$, $e \leq |x'|$, $|x| - s = |x'| - e$, and $|w| = |w'|$.

(3) Spine rest deletion. The pair $(A_j(u), A'_j(u))$ takes at most N_3 values on U_u, so some $i < j$ in U_u have the same pair. Put $x = u[{:}i]$, $w = u[i{:}j]$, $y = u[j{:}]$, $x' = w' = \varepsilon$ and $y' = t$. (D1) holds. The premise of (ii), $x[s{:}]\,(u[j{:}]\,\xi) = t[k{:}]\,\xi$, never holds, by Lemma E.9(d) with $v = u$, $v' = t$ and the unpartnered rest cut j. For (D3), every head event (s, e) has $s \geq j$: otherwise removing the first $j - s$ letters from $u[s{:}] = t[e{:}]$ gives $u[j{:}] = t[e + j - s{:}]$, and j would be partnered. So $s \geq |xw|$, and $e \geq 0 = |x'w'|$.

(4) Target rest deletion. The pair $(B_e(t), B'_e(t))$ takes at most N_4 values on U_t, so some $i < j$ in U_t have the same pair. Put $x = w = \varepsilon$, $y = u$, $x' = t[{:}i]$, $w' = t[i{:}j]$ and $y' = t[j{:}]$. (D1) holds. The word $Y' = t[j{:}]\,\xi$ is non-self-similar by Lemma E.9(a), so (iv) and (v) hold by rigidity. The premise of (iii), $u[k{:}]\,\xi = x'[e{:}]\,(t[j{:}]\,\xi)$, never holds, by Lemma E.9(d) with $v = t$, $v' = u$ and the unpartnered rest cut j. For (D3), every head event (s, e) has $e \geq j = |x'w'|$, since otherwise $t[j{:}] = u[s + j - e{:}]$ and j would be partnered; and $s \geq 0 = |xw|$. □

No deletion inside the periodic end of t is needed: by Lemma E.8, that periodic end is already short in every good pair.

E.6.6 Counting and the Proof of the Pump.

Lemma E.13 (Counting). *Let (u,t) be a good pair for which none of the conditions (1)–(4) of Lemma E.12 holds. Then $|u|, |t| < N_1 + N_2 + N_3 + N_4 + b\varpi$.*

Proof. Map each $e \in C_t$ to its partner. By Lemma E.7(b) the partner is the rest cut $R_u - (R_t - e)$ of u, which is partnered; so the map is an injection of C_t into C_u, and $|C_t| \le |C_u|$. Since (2), (3) and (4) fail, $R_u = |C_u| + |U_u| \le N_2 + N_3$ and $R_t = |C_t| + |U_t| \le N_2 + N_4$. By Lemma E.8, $T_t \le b\varpi - 1$, and since (1) fails, $T_u \le T_t + N_1 - 1 \le b\varpi + N_1 - 2$. Hence $|u| = R_u + T_u \le N_1 + N_2 + N_3 + b\varpi - 2$ and $|t| = R_t + T_t \le N_2 + N_4 + b\varpi - 1$, and both are less than $N_1 + N_2 + N_3 + N_4 + b\varpi$. □

Proof of Theorem 9.14. Put $K = N_1 + N_2 + N_3 + N_4 + b\varpi = 2aba'b' + aa'\varpi + b\varpi + aa' + bb'$. By Lemma E.6, the pair (u,t) is good. Among the good pairs (u',t') with $A(u') = A(u)$ choose one with $|u'| + |t'|$ least. If $|u'| \ge K$ or $|t'| \ge K$, then by Lemma E.13 one of the conditions (1)–(4) of Lemma E.12 holds for (u',t'), and Lemma E.12 gives a good pair with the same endpoint and a smaller total length, a contradiction. Hence $|u'|, |t'| < K$, and (u',t') is a pre-hole for which ξ is survivor-free, by Lemma E.6. If $\varpi \le N_A = a$, then $aa'\varpi \le a^2a'$ and $b\varpi \le ab$, so $K \le \mathsf{P}(a,b,a',b')$. □

E.7 The Three-Case Theorem

The bound L_{nm}. Since $a,b,a',b' \ge 1$, we have $ab, aa', bb' \le aba'b'$, so $\mathsf{P} \le 5aba'b' + a^2a' \le 6a^2ba'b'$. Hence $\max(12aba'b', \mathsf{P}) \le 12aba'b' + a^2a'$ and $\max(12aba'b', \mathsf{P}) \le 12a^2ba'b'$, which give

$$\mathsf{L}_{\mathrm{nm}}(a,b,a',b') \le 48aba'b' + 4a^2a' + 6a \qquad \text{and} \qquad \mathsf{L}_{\mathrm{nm}}(a,b,a',b') \le 48a^2ba'b' + 6a \le 54a^2ba'b'.$$

Proof of Theorem 9.17. We first show that (1) or (2) yields a short hole; this part does not use (FH). Put $K_0 = 12aba'b'$ and $K = \max(K_0, \mathsf{P}(a,b,a',b'))$.

Suppose that (1) holds for a pre-hole (u_0,t_0), and let $q = A(u_0)$. By Proposition 9.12 with $E = \{q\}$, there is a pre-hole (u_1,t_1) with $A(u_1) = q$ and $|u_1|, |t_1| \le K_0 - 1$. By Lemma 9.6 it is completed by a word θ with $|\theta| \le K_0 - 1 + 3a$, and by Lemma 9.3 $(u_1\theta, t_1\theta)$ is a hole with $|u_1\theta| + |t_1\theta| = |u_1| + |t_1| + 2|\theta| \le 4(K_0 - 1) + 6a < 4K + 6a$.

Suppose that (2) holds for a pre-hole (u_0,t_0) and an infinite word ξ safe from the thin state $q = A(u_0)$. By Lemma 9.8(c), ξ is a lasso $\alpha\lambda^\omega$ of q. If ξ is not purely periodic, then Proposition 9.12 with $E = \{q\}$ gives a pre-hole (u_1,t_1) with $A(u_1) = q$ and $|u_1|, |t_1| \le K_0 - 1$, and ξ is survivor-free for it by Lemma 9.13. If ξ is purely periodic, its least period is at most $|\lambda| \le a$ by Lemma E.1(b), and Theorem 9.14 gives a pre-hole (u_1,t_1) for which ξ is survivor-free, with $A(u_1) = q$ and $|u_1|, |t_1| \le \mathsf{P}(a,b,a',b') - 1$. In both cases $|u_1|, |t_1| \le K - 1$, and ξ is a lasso of $A(u_1)$ that is survivor-free for (u_1,t_1). By Corollary 9.11, the prefix θ of ξ of length $\max(|u_1|, |t_1|) + a - 1 \le K + a - 2$ completes (u_1,t_1), and by Lemma 9.3 $(u_1\theta, t_1\theta)$ is a hole with $|u_1\theta| + |t_1\theta| \le 2(K-1) + 2(K + a - 2) < 4K + 6a$.

In particular, (1) or (2) implies that I has a hole, and (3) does so trivially. Conversely, let I satisfy (FH), and let (u,t) be a hole. By Lemma 9.3, $(u,t) = (u_0\theta, t_0\theta)$, where $\theta = t[J(t):]$ completes the pre-hole (u_0,t_0). Let $q = A(u_0)$, which is safe. Then $A(u) = A(q,\theta)$, and $J(t) = J(t_0) = |t_0|$, so $|t| - J(t) = |\theta|$. If q is branching, (1) holds. If q is mortal, then by Lemma 9.8(a) $|\theta| < N_A$ and $A(u)$ is mortal, so (u,t) is a mortal hole. Let q be thin. If $A(u)$ is not mortal, some infinite word ξ' is safe from $A(u)$, so $\xi = \theta\xi'$ is safe from q. Every obligation of (u_0,t_0) is broken by θ, which is a prefix of ξ, and so does not survive ξ; hence (2) holds with ξ. If $A(u)$ is mortal, then $|\theta| < N_A$ by Lemma 9.8(d), and (u,t) is a mortal hole. □

F Proofs for Mortal Holes

This appendix defines the constants of Theorem 10.14 and proves the results of Section 10 whose proofs were omitted or outlined there. Appendix F.1 proves Lemmas 10.4 and 10.5, and Appendix F.2 proves Lemma 10.10. Appendices F.3 to F.8 prove Theorem 10.14, following the structure of the proof of Theorem 2.20 by Zhao [2026] as recalled in Appendix A.4; each subsection states what it reuses and what it adds. Appendix F.9 proves the closed-form estimates and Corollary 10.16.

What is reused unchanged: the alignment lemma (Lemma 2.17); the label facts Eq. (4) and the desubstitution lemma (Lemma 2.24); the facts on ordinary and symbolic states, start prefixes and entry states (Appendices A.3 and A.4); the comparison-preserving map lemma, the shape of level maps and the isolated exchange for holes (Lemmas A.4 to A.6); the propagation of boundary pairs (Lemma A.7(2)); the shortening of unary holes (Lemma A.8); and the column summaries, column shortening and agreement under derivation (Lemmas A.9 and A.10). What is new is that every step that replaces the witness, namely isolated exchange, the top shortening, column shortening and supported compression, also preserves the positive condition. Nothing below uses finiteness of an alphabet: block alphabets are infinite, and only state sets, supports, type spaces and summary values must be finite.

The constants. The bound $\mathsf{Len}^\sharp(N,d)$ of Theorem 10.14 and the constants behind it are as follows. For integers $N \geq 1$, $d \geq 0$, $M \geq 1$ and $k \geq 0$ put

$$\begin{aligned} \mathsf{C}^\sharp(N,d) &= N^N\, 2^{5N^2+10N(N+d)(d+1)+7N+1}, & \mathsf{A}^\sharp(N,d) &= d+1+\mathsf{C}^\sharp(N,d),\\ \Xi^\sharp(N,d) &= 6N^2+10N(N+d)(d+1)+7N+3d+4, & \mathsf{G}^\sharp(N,d) &= 2+\mathsf{A}^\sharp(N,d)\,3^d,\\ \mathrm{pcol}(M,k) &= \mathrm{col}(M,k)^2\, 2^{2M(M+1)k}\, 3^k, & & \\ \mathsf{Len}^\sharp(N,d) &= 3(N+d+1)\,\mathrm{pcol}\big(N+d+1, \mathsf{G}^\sharp(N,d)\big)^d, & & \end{aligned} \tag{17}$$

with col from Eq. (14). They extend the constants Eq. (14) of the plain-entailment proof by the new tables and fields: $\mathsf{C}^\sharp(N,d)$ bounds the anchored exchange types at one level, $\mathsf{A}^\sharp(N,d)$ the protected and isolated letters of a normal hole at one level, $\mathsf{G}^\sharp(N,d)$ its dictionaries, $\mathrm{pcol}(M,k)$ the positive column summaries of k blocks over a monitor with M states, and $\mathsf{Len}^\sharp(N,d)$ is the resulting length bound; $\Xi^\sharp(N,d)$ is the exponent of its closed form. The functions are nondecreasing in each argument, and the alphabet enters none of them.

F.1 The Separator Word

Proof of Lemma 10.4. (i) Let $y \leq m$. The suffix $w[y{:}]$ contains $\sharp$ at offset $m-y$. The suffix $w[x{:}]$ is longer, and its letter at offset $m-y$ is $w[x+m-y]$ with $x+m-y < m$, a letter of Σ. Hence $w[y{:}] \not\preceq w[x{:}]$.

(ii) Here $w[m+1+e{:}] = t[e{:}]$ is a word over Σ and $w[x{:}] = u[x{:}]\,\sharp\, t$. If $|t[e{:}]| \leq |u[x{:}]|$, the prefix test reads only letters of $u[x{:}]$, so it is the test $t[e{:}] \preceq u[x{:}]$. Otherwise $t[e{:}] \not\preceq u[x{:}]$ by length, and also $t[e{:}] \not\preceq w[x{:}]$, because the letter of $w[x{:}]$ at offset $m-x$ is $\sharp$, while the letter of $t[e{:}]$ at that offset lies in Σ.

(iii) Here $w[m+1+i{:}] = t[i{:}]$ and $w[m+1+j{:}] = t[j{:}]$.

(iv) The letter $\sharp$ occurs in w only at m. The continuation letter of a comparison $(x, m+1+e)$ is $w[x+|w|-m-1-e] = w[x+|t|-e]$, so the comparison is anchored iff $x+|t|-e = m$, that is, iff $|t[e{:}]| = |u[x{:}]|$. By (ii), the pair is a comparison iff $t[e{:}] \preceq u[x{:}]$, which together with equal lengths says $t[e{:}] = u[x{:}]$. Conversely, $t[e{:}] = u[x{:}]$ gives equal lengths and, by (ii), a comparison. □

We prove Lemma 10.5 for the reader $\mathcal{R}_I^{\Pi}$ and the relation Π constructed in Section 10.2. The cap N_A of the counter is the mortal bound of Definition 9.16, not the size of $\mathcal{R}_I^{\Pi}$; for $N_A = 1$ no state $\mathrm{done}(r', k)$ is rejected, as required by $|t| - J(t) < 1$.

Proof of Lemma 10.5. The reader and Π are computable, since mortality is decided by reachability in the safe graph of A (Definition 9.5).

Runs. The states after the separator have no transition on $\sharp$ other than into sink, which is absorbing. So a word without $\sharp$ ends in a state $\mathrm{pre}(\cdot)$ or in sink, and a word with two occurrences of $\sharp$ ends in sink. Let $w = u\sharp t$ with $u, t \in \Sigma^*$ and $m = |u|$. If u is not safe, the run enters sink within u and stays there. If u is safe, then by induction the state at a cut $x \le m$ is $\mathrm{pre}(A_x(u), A'_x(u))$, and the separator leads to act of the start states of B and B' iff $A(u)$ is mortal, and to sink otherwise. In the first case, by induction on e, the state at the cut $m + 1 + e$ is $\mathrm{act}(B_e(t), B'_e(t))$ while no earlier cut of t is rejected by B, that is, for $e \le J(t)$ if $J(t)$ is defined and for all e otherwise; and it is $\mathrm{done}(B'_e(t), \min(e - J(t), N_A))$ for $e > J(t)$.

(i) There are $N_A N_{A'}$ states $\mathrm{pre}(q, q')$, $N_B N_{B'}$ states $\mathrm{act}(r, r')$, $N_{B'} N_A$ states $\mathrm{done}(r', k)$, and one sink.

(iii) and rejection. By the description of runs, the last state of a word is rejected iff the word is $u\sharp t$ with $u, t \in \Sigma^*$, u is safe, $A(u)$ is mortal, $J(t)$ is defined, and $|t| - J(t) \le N_A - 1$: if $|t| = J(t)$, the run ends in $\mathrm{act}(B_{J(t)}(t), \cdot)$ with $B_{J(t)}(t) \in \mathrm{Rej}$; if $|t| > J(t)$, it ends in a state done whose counter $\min(|t| - J(t), N_A)$ is at most $N_A - 1$ iff $|t| - J(t) \le N_A - 1$. No state pre is rejected, and the states $\mathrm{act}(B_e(t), \cdot)$ with $e < J(t)$ are not, so the first rejected cut is $m + 1 + J(t)$.

(ii) Let $w = u\sharp t$ be rejected, as just described. By Lemma 10.4, every comparison of w has the form $(x, m+1+e)$ with $x \le m$, a cross comparison (x, e) of (u, t), or $(m+1+i, m+1+j)$ with $i < j$, a comparison (i, j) of t. Admitted pairs join a state pre to a state act, or two states act; the states act occur exactly at the cuts $m+1+e$ with $e \le J(t)$. So a comparison $(x, m+1+e)$ is admitted iff $e \le J(t)$ and $(A_x(u), B_e(t)) \in G$, that is, iff (x, e) is an event of (u, t); and a comparison $(m+1+i, m+1+j)$ is admitted iff $j \le J(t)$ and $(B_i(t), B_j(t)) \in G_{\mathrm{s}}$, that is, iff (i, j) is a self event of t. Hence w is a hole iff (u, t) has no event and t has no self event. By Lemma 10.4(iv), the anchored comparisons of w are the pairs $(x, m + 1 + e)$ with $u[x:] = t[e:]$. Their first state is $\mathrm{pre}(A_x(u), A'_x(u))$, and their second state is $\mathrm{act}(B_e(t), B'_e(t))$ or $\mathrm{done}(B'_e(t), k)$, with B'-component $B'_e(t)$ in both cases, also when $e > J(t)$. So Π holds on such a pair iff $(A'_x(u), B'_e(t)) \in H$, and an anchored comparison in Π is exactly a head event. Altogether, w is a Π-hole iff $w = u\sharp t$ where (u, t) satisfies conditions (a)–(e) of Definition 8.1, $A(u)$ is mortal and $|t| - J(t) < N_A$, that is, iff (u, t) is a mortal hole. A word that is not rejected is not a hole. □

F.2 Positive Desubstitution

Throughout this subsection, $\mathcal{M}$ is a lettered monitor with a positive channel $\Theta = (F, R, S, S', \Phi)$, ℓ is a marker, and $v = X_0 \ell X_1 \ell \cdots X_{k-1} \ell$ is the expansion of a nonempty word $v' = [X_0] \cdots [X_{k-1}]$ of blocks avoiding ℓ. We use the notation of the label facts Eq. (4): the run $\zeta'_0 = \star, \ldots, \zeta'_k$ of v' in $\mathcal{M}_\ell$, the cuts $a_i = |X_0 \ell \cdots X_{i-1} \ell|$ at which the blocks begin, and $b_i = a_i + |X_i|$; in addition $a_k = |v|$. The letter of v at the position $a_i + g$, for $0 \le g \le |X_i|$, is $(X_i \ell)[g]$. We write L and L' for the labels of v and v'.

Lemma F.1 (Transport of continuation letters). *Let (x, y) be a comparison of v with $y < |v|$, and let $i < i' < k$, $X_i = \hat{x}\hat{y}$ and $X_{i'} = \hat{x}'\hat{y}$ be as in Lemma 2.17, so that $x = a_i + |\hat{x}|$, $y = a_{i'} + |\hat{x}'|$, and $(i + 1, i' + 1)$ is a comparison of v'. Put $r = (i + 1) + k - (i' + 1)$, the position of the continuation letter of $(i + 1, i' + 1)$ in v'. Then $i < r < k$, $x + |v| - y = a_r$, and the continuation letter of (x, y) is $v[a_r] = (X_r \ell)[0]$.*

Proof. Since $\hat{y}$ ends both blocks, $b_i - x = |\hat{y}| = b_{i'} - y$. Hence

$$|v| - y = |\hat{y}| + 1 + \sum_{h=i'+1}^{k-1} (|X_h| + 1) = |\hat{y}| + 1 + (a_k - a_{i'+1}).$$

Since $(i+1, i'+1)$ is a comparison of v', $X_{i+1+m} = X_{i'+1+m}$ for $0 \le m < k - i' - 1$, so the sum equals $a_r - a_{i+1}$, where $r = i + k - i'$. With $x + |\hat{y}| + 1 = b_i + 1 = a_{i+1}$ this gives $x + |v| - y = a_r$. Finally $i < r$ because $i' < k$, and $r < k$ because $i < i'$. □

For $i' + 1 = k$ the comparison $(i+1, k)$ of v' is terminal and $r = i + 1$: the continuation letter is $v'[i+1]$, as in clause (S′).

Proof of Lemma 10.10. The second claim follows from the first and Lemma 2.24. For the first, recall from Eq. (4) that $L_{a_i+g} = lab_g(\zeta'_i, X_i)$ for $g \le |X_i|$, $L_{|v|} = (\zeta'_k, \ell)$, and $L'_{i+1} = (\zeta'_i, X_i)$, with $\zeta'_k = \delta_\ell(\zeta'_{k-1}, X_{k-1})$. Every cut $x < |v|$ lies in exactly one interval $[a_i, b_i]$, and then $x = a_i + g$ with $g \le |X_i|$ and $v[x] = (X_i\ell)[g]$.

($\Rightarrow$) Suppose $\mathrm{Pos}_\Theta(v)$, and consider the clause that holds.

- (F): $F(L_{|v|}) = F(\delta_\ell(\zeta'_{k-1}, X_{k-1}), \ell)$ is the first disjunct of $F_\ell(\zeta'_{k-1}, X_{k-1}) = F_\ell(L'_k)$, which is clause (F) of v'.
- (R): a comparison (x, y) with $y < |v|$, $R(L_x, L_y)$ and $v[x + |v| - y] \in \Phi$. Take $i < i'$, $\hat{x}$, $\hat{x}'$ and $\hat{y}$ as in Lemma 2.17. Then $L_x = lab_{|\hat{x}|}(\zeta'_i, X_i)$ and $L_y = lab_{|\hat{x}'|}(\zeta'_{i'}, X_{i'})$, so $R(L_x, L_y)$ witnesses $R_\ell(L'_{i+1}, L'_{i'+1})$. By Lemma F.1, the continuation letter of (x, y) is $(X_r\ell)[0]$, so $X_r \in \Phi_\ell$, where $v'[r]$ is the continuation letter of the comparison $(i+1, i'+1)$ of v'. If $i' + 1 < k$, this is clause (R) of v'. If $i' + 1 = k$, then $r = i + 1$, and since $S'_\ell = R_\ell$, this is clause (S′) of v' at the cut $i + 1 < k$.
- (S) or (S′): a cut $x < |v|$ with $S(L_x, L_{|v|})$, or with $S'(L_x, L_{|v|})$ and $v[x] \in \Phi$. Write $x = a_i + g$. Then $L_x = lab_g(\zeta'_i, X_i)$, $L_{|v|} = (\zeta'_k, \ell)$ and $v[x] = (X_i\ell)[g]$, so $pt(\zeta'_i, X_i; \zeta'_k)$ holds. If $i < k - 1$, this is $S_\ell((\zeta'_i, X_i), (\zeta'_{k-1}, X_{k-1})) = S_\ell(L'_{i+1}, L'_k)$, clause (S) of v' at the cut $i + 1 < k$. If $i = k - 1$, it is the second disjunct of $F_\ell(L'_k)$, clause (F) of v'.

($\Leftarrow$) Suppose $\mathrm{Pos}_{\Theta_\ell}(v')$. All relations of Θ_ℓ are false at $\triangleright = L'_0$, so a witnessing cut x' of v' satisfies $x' \ge 1$; write $x' = i + 1$.

- (F): $F_\ell(L'_k)$. Either $F(\zeta'_k, \ell) = F(L_{|v|})$, clause (F) of v; or $pt(\zeta'_{k-1}, X_{k-1}; \zeta'_k)$, which gives some $g \le |X_{k-1}|$ with $S(L_x, L_{|v|})$, or with $S'(L_x, L_{|v|})$ and $v[x] \in \Phi$, at the cut $x = a_{k-1} + g \le b_{k-1} < |v|$: clause (S) or (S′) of v.
- (S): $S_\ell(L'_{i+1}, L'_k)$ with $i + 1 < k$, that is, $pt(\zeta'_i, X_i; \zeta'_k)$. As in the previous case this gives clause (S) or (S′) of v at a cut $x = a_i + g \le b_i < |v|$.
- (R) or (S′): a comparison $(i+1, i'+1)$ of v' with $R_\ell(L'_{i+1}, L'_{i'+1})$ and continuation letter $v'[r] \in \Phi_\ell$, where either $i' + 1 < k$ and $r = i + k - i'$ (clause (R)), or $i' + 1 = k$, $r = i + 1$, and the comparison is the terminal pair $(i+1, k)$, which is a comparison because $i + 1 < k$ (clause (S′), since $S'_\ell = R_\ell$). In both cases $i < i' < k$. The relation R_ℓ gives decompositions $X_i = \hat{x}\hat{y}$ and $X_{i'} = \hat{x}'\hat{y}$ with $R(lab_{|\hat{x}|}(\zeta'_i, X_i), lab_{|\hat{x}'|}(\zeta'_{i'}, X_{i'}))$. Put $x = a_i + |\hat{x}|$ and $y = a_{i'} + |\hat{x}'|$. Then $y \le b_{i'} < |v|$, and by Lemma 2.17 the pair (x, y) is a comparison of v; by the label facts $R(L_x, L_y)$ holds; and by Lemma F.1 its continuation letter is $(X_r\ell)[0]$, which lies in Φ because $X_r \in \Phi_\ell$. This is clause (R) of v.

This proves the equivalence. □

F.3 Anchored Exchange and Bounded Dictionaries

Fix a lettered monitor $\mathcal{M}_0$ with $N \ge 1$ states and a positive channel Θ_0 on it, and a positive hole w of depth $D \le d$ with hierarchy (w_i, ℓ_i), monitors $\mathcal{M}_i$ and channels $\Theta_i = (F_i, R_i, S_i, S'_i, \Phi_i)$ as in Corollary 10.11. We use the notions of step 2 of Appendix A.4: the dictionaries $\mathcal{A}_i$ and $g_i = |\mathcal{A}_i|$,

the start counts s_i and start prefixes, the entry states U_i with $|U_i| \le N + s_i \le N + d$, the protected letters P_i with $|P_i| \le s_i + 1 \le d + 1$, the boundary pairs BP_i with $b_i = |\mathrm{BP}_i|$, and isolated letters. Recall that every occurrence of an unprotected letter has an ordinary entry state.

Definition F.2 (Anchored exchange type). Let $1 \le i < D$. The *anchored exchange type* of an unprotected letter a of w_i consists of its exchange type (Appendix A.4) together with the following tables of $(\mathcal{M}_i, \Theta_i)$:

- the row $F_i(\zeta, a)$ for every ordinary state ζ;
- for each $Y \in \{R_i, S_i, S_i'\}$, the self table $Y((\zeta, a), (\zeta', a))$ for ordinary ζ, ζ';
- for each such Y, the protected rows $Y((\zeta, a), (\vartheta, z))$ and $Y((\vartheta, z), (\zeta, a))$ for ordinary ζ, $\vartheta \in U_i$ and $z \in P_i$;
- for each such Y, the initial row $Y(\triangleright, (\zeta, a))$ for ordinary ζ, which is false at every level $i \ge 1$;
- the bit $[a \in \Phi_i]$.

Lemma F.3 (Anchored exchange types). *At every level $1 \le i < D \le d$ there are at most $\mathrm{C}^\sharp(N, d)$ anchored exchange types.*

Proof. Ordinary states transition to ordinary states, so the transition table has at most N^N values. The rows of T_i^{fin} and F_i take $2N$ bits; the self tables of the five relations $\Lambda_i, \Lambda_i^{\mathrm{fin}}, R_i, S_i, S_i'$ take $5N^2$ bits, their protected rows $10N|U_i||P_i| \le 10N(N+d)(d+1)$ bits, and their initial rows $5N$ bits; and the bit of Φ_i is one more. Hence there are at most $N^N 2^{5N^2 + 10N(N+d)(d+1) + 7N + 1} = \mathrm{C}^\sharp(N, d)$ types. □

Lemma F.4 (Anchored isolated exchange). *Let $1 \le i < D$, let $a \ne b$ be isolated letters of w_i with the same anchored exchange type, where the level-zero expansion of b is not longer than that of a, let h replace a by b and fix every other letter, and let w' be the expansion of $h(w_i)$ with the original lower markers. Then w' is a positive hole of $(\mathcal{M}_0, \Theta_0)$ of depth D, no longer than w, with every start count unchanged, no dictionary larger than the corresponding one of w, and level-i dictionary $\mathcal{A}_i \setminus \{a\}$.*

Proof. The anchored type contains the exchange type, so a and b have the same exchange type, and by Lemma A.6 the map h satisfies the hypotheses of Lemma A.5: it identifies no unequal predecessor pair of $v = w_i$, so v and $h(v)$ have exactly the same comparisons (Lemma A.4); it fixes the start prefix; it preserves the state run of v in $\mathcal{M}_i$; and $h(v)$ is a hole of $\mathcal{M}_i$. Hence w' is a hole of $\mathcal{M}_0$ of depth D with the stated start counts, dictionaries and length, and its markers below i are $\ell_0, \dots, \ell_{i-1}$, with $w_i' = h(v)$. So the monitors and channels of w' through level i are $(\mathcal{M}_j, \Theta_j)_{j \le i}$, and by Lemma 10.10, applied at the levels $i - 1, \dots, 0$, it remains to show $\mathrm{Pos}_{\Theta_i}(h(v))$.

Write L and L^h for the labels of v and of $h(v)$ in $\mathcal{M}_i$. The runs coincide, so L_c^h is L_c with its letter mapped by h.

Claim: for every comparison (x, y) of v and each $Y \in \{R_i, S_i, S_i'\}$, $Y(L_x^h, L_y^h) \iff Y(L_x, L_y)$. If $x = 0$, then $L_x = L_x^h = \triangleright$, and L_y is unchanged unless $v[y-1] = a$, when the initial row applies. Let $x \ge 1$. If neither $v[x-1]$ nor $v[y-1]$ is a, both labels are unchanged. If both are a, the self table applies, both entry states being ordinary. If exactly one of them is a, the other letter c differs from a, so (x, y) is left maximal. If $y < |v|$, the ordered pair of predecessor letters lies in BP_i, and isolation of a makes c protected. If $y = |v|$, then $v[y-1] = \ell_i$ is protected and differs from a, so $v[x-1] = a$ and $c = \ell_i$. In both cases $c \in P_i$ is read from a state in U_i, $c \ne b$ because b is unprotected, and the protected row of Y in the appropriate orientation applies. This proves the claim.

Now fix a clause witnessing $\mathrm{Pos}_{\Theta_i}(v)$.

- (F): the final label $L_{|v|}$ has the protected letter ℓ_i, so $L_{|v|}^h = L_{|v|}$.

- (R): a comparison (x, y) with $y < |v|$. It is a comparison of $h(v)$, the claim keeps $R_i(L_x, L_y)$, and the continuation letter is at the same position $x + |v| - y$, since h preserves lengths. That letter is either unchanged or changed from a to b, and $[a \in \Phi_i] = [b \in \Phi_i]$.
- (S) or (S′): a cut $x < |v|$. The pair $(x, |v|)$ is a terminal comparison of both words, the claim keeps the value of S_i or S'_i, and for (S′) the letter $h(v)[x]$ lies in Φ_i iff $v[x]$ does.

Hence $\mathrm{Pos}_{\Theta_i}(h(v))$. □

The merge is not injective, so it gives no agreement of the derived data above level i. After every merge we recompute the hierarchy, the towers, the protected sets and the anchored types of the new word; its positivity in the new towers follows from Corollary 10.11.

A positive hole is *anchored-normal* if at every level $1 \le i < D$ distinct isolated letters have distinct anchored exchange types, computed in its own hierarchy and towers.

Lemma F.5 (Normal form and bounded dictionaries).

(1) Every positive hole of depth D has an anchored-normal positive hole of depth D, no greater length, the same start counts, and no larger dictionary at any level.

(2) Let w be an anchored-normal positive hole of depth $D \le d$, and put $\mathrm{A}^\sharp = \mathrm{A}^\sharp(N, d)$. Then $g_D = 1$, and for $1 \le i < D$,

$$g_i \le \mathrm{A}^\sharp + 2b_i, \qquad 2b_i \le (\mathrm{A}^\sharp - 1)(3^{D-1-i} - 1), \qquad g_i \le 1 + (\mathrm{A}^\sharp - 1)3^{D-1-i} \le \mathrm{G}^\sharp(N, d).$$

Consequently, every positive hole of depth $D \le d$ has a positive hole of depth D and no greater length with $g_i \le \mathrm{G}^\sharp(N, d)$ for every $1 \le i \le D$.

Proof. (1) Use the potential $\sum_{j=0}^{D} g_j$. While anchored normality fails at some level $1 \le i < D$, merge an offending pair by Lemma F.4, in the direction of the shorter level-zero expansion. The result is a positive hole of depth D, no longer, with the same start counts; no dictionary grows and the level-i dictionary shrinks, so the potential decreases strictly. The process terminates in an anchored-normal positive hole.

(2) The top level w_D is unary, so $g_D = 1$, and $b_D = 0$ because a unary word has no left-maximal comparison. Let $1 \le i < D$. Partition $\mathcal{A}_i$ into the protected letters, at most $d + 1$; the isolated letters, at most one per anchored type by normality, hence at most $\mathrm{C}^\sharp(N, d)$ by Lemma F.3; and the remaining letters, each of which is unprotected and not isolated, hence a component of a pair in BP_i. The two projections of BP_i have at most $2b_i$ elements, so $g_i \le \mathrm{A}^\sharp + 2b_i$. By the propagation in Lemma A.7(2), which holds for every nonempty word, $b_j \le b_{j+1} + g_{j+1} - 1$ for every j. Hence $b_{D-1} \le b_D + g_D - 1 = 0$, and for $1 \le i < D - 1$, $b_i \le b_{i+1} + \mathrm{A}^\sharp + 2b_{i+1} - 1 = 3b_{i+1} + \mathrm{A}^\sharp - 1$. By downward induction on i, with $r = D - 1 - i$,

$$2b_i \le 3(\mathrm{A}^\sharp - 1)(3^{r-1} - 1) + 2(\mathrm{A}^\sharp - 1) = (\mathrm{A}^\sharp - 1)(3^r - 1),$$

the base case $r = 0$ having both sides zero. Then $g_i \le \mathrm{A}^\sharp + 2b_i \le 1 + (\mathrm{A}^\sharp - 1)3^r \le 2 + \mathrm{A}^\sharp 3^d = \mathrm{G}^\sharp(N, d)$.

The last assertion follows from (1) and (2), since $g_D = 1 \le \mathrm{G}^\sharp(N, d)$. □

F.4 The Unary Top Level

Lemma F.6 (Positive unary words). *Let $\mathcal{M}$ be a lettered monitor with state set Q, Θ a positive channel on it, and z^m a positive hole of $(\mathcal{M}, \Theta)$. Then $z^{m'}$ is a positive hole of $(\mathcal{M}, \Theta)$ for some $1 \le m' \le 3|Q|$.*

Proof. Every pair of cuts $x < y$ of a unary word is a comparison, and its continuation letter is z. Fix a clause witnessing $\mathrm{Pos}_\Theta(z^m)$, and let W be the set consisting of m and the cuts of the witness other than 0: the cuts x and y for (R), the cut x for (S) and (S′), and none for (F); so $W \cup \{0\}$ has at most four elements.

Let $i < i' \le m-1$ be cuts with equal states such that no cut of W lies in $\{i+1,\dots,i'\}$, and delete the letters at the positions $i,\dots,i'-1$. The cuts of the resulting word $z^{m-(i'-i)}$ correspond to the cuts $c \le i$ and $c > i'$ of z^m, by $c \mapsto c$ and $c \mapsto c-(i'-i)$, an order-preserving map that keeps the labels: the state at the new cut i is the state at the old cut i'. The last cut maps to the last cut. Every pair of cuts of a unary word is a comparison, so every comparison of the new word has the labels of a comparison of z^m, terminal iff that one is, and the final label is unchanged; hence the new word is a hole. The cuts of W are kept, with their labels and their order, and since every pair of cuts is a comparison with continuation letter z, the witnessing clause holds for the images of its cuts. So the shorter word is again a positive hole, and W maps onto the corresponding set of the new witness.

Repeat such deletions while possible; the length decreases, so this terminates, in a positive hole $z^{m'}$ with $m' \ge 1$, because the last letter is never deleted. Let $c < c'$ be consecutive elements of $W \cup \{0\}$. The states at the cuts $c,\dots,c'-1$ are pairwise distinct: two equal ones at $i < i'$ would have $i' \le c'-1 \le m'-1$ and no cut of W in $\{i+1,\dots,i'\}$, so a further deletion would be possible. Hence $c'-c \le |Q|$, and, as there are at most three such gaps and they cover $[0,m']$, $m' \le 3|Q|$. □

F.5 Agreement with Channels

We extend agreement Eq. (13) to channels. Let $\mathcal{U}$ and $\mathcal{V}$ be lettered monitors with positive channels $\Theta^{\mathcal{U}} = (F^{\mathcal{U}}, R^{\mathcal{U}}, S^{\mathcal{U}}, S'^{\mathcal{U}}, \Phi^{\mathcal{U}})$ and $\Theta^{\mathcal{V}}$, let E be a finite set of letters of $\mathcal{U}$, and ρ a map from E into the alphabet of $\mathcal{V}$. We write $(\mathcal{U},\Theta^{\mathcal{U}}) \equiv_{E,\rho} (\mathcal{V},\Theta^{\mathcal{V}})$ if $\mathcal{U} \equiv_{E,\rho} \mathcal{V}$ and, for every $a \in E$, every state ζ, and all labels L, L' whose letters lie in E, the start label included,

$$F^{\mathcal{V}}(\zeta,\rho(a)) \iff F^{\mathcal{U}}(\zeta,a), \qquad \rho(a) \in \Phi^{\mathcal{V}} \iff a \in \Phi^{\mathcal{U}},$$

and $Y^{\mathcal{V}}(\widehat{\rho}L, \widehat{\rho}L') \iff Y^{\mathcal{U}}(L,L')$ for $Y \in \{R,S,S'\}$.

Agreement preserves positive holes. Let v be a nonempty word over E. Agreement of the monitors gives the same states along v and $\rho(v)$, labels that correspond under $\widehat{\rho}$, and holes that correspond. Since ρ is injective on E, the words v and $\rho(v)$ have the same letter equalities, hence the same comparisons, and the continuation letter of a comparison of $\rho(v)$ is the image of that of v. Each clause of the positive condition therefore holds for v iff it holds for $\rho(v)$, and v is a positive hole of $(\mathcal{U},\Theta^{\mathcal{U}})$ iff $\rho(v)$ is a positive hole of $(\mathcal{V},\Theta^{\mathcal{V}})$. *Agreements compose*, by composing the maps.

Lemma F.7 (Agreement under derivation). *Suppose $(\mathcal{U},\Theta^{\mathcal{U}}) \equiv_{E,\rho} (\mathcal{V},\Theta^{\mathcal{V}})$ and $\ell \in E$. For every finite family $\mathcal{F}$ of blocks over E avoiding ℓ, the letterwise map ρ^* gives*

$$(\mathcal{U}_\ell,\Theta^{\mathcal{U}}_\ell) \equiv_{\mathcal{F},\rho^*} (\mathcal{V}_{\rho(\ell)},\Theta^{\mathcal{V}}_{\rho(\ell)}),$$

and the image blocks avoid $\rho(\ell)$.

Proof. The agreement of the derived monitors and the avoidance of $\rho(\ell)$ are Lemma A.10. Its proof shows, by induction along a block, that for $X \in \mathcal{F}$ and $\zeta \in Q \sqcup \{\star\}$ the shifted blocks of X and $\rho^*(X)$ have the same states, so that $lab_g(\zeta,\rho^*X) = \widehat{\rho}\, lab_g(\zeta,X)$ for every g and the exits coincide. The letters $(X\ell)[g]$ lie in E, and $(\rho^*(X)\rho(\ell))[g] = \rho((X\ell)[g])$. Hence:

- $\rho^*(X) \in \Phi^{\mathcal{V}}_{\rho(\ell)}$ iff $\rho((X\ell)[0]) \in \Phi^{\mathcal{V}}$ iff $(X\ell)[0] \in \Phi^{\mathcal{U}}$ iff $X \in \Phi^{\mathcal{U}}_\ell$;
- by injectivity, the decompositions $X = xy$, $X' = x'y$ correspond exactly to the decompositions of ρ^*X and ρ^*X' with a common suffix, and R agrees at the corresponding labels, whose letters lie in E or which are $\triangleright$; so R_ℓ, and hence S'_ℓ, agree;
- the positive terminal test $pt(\zeta,X;\zeta'')$ reads S and S' at the labels $lab_g(\zeta,X)$ and (ζ'',ℓ), which correspond to $lab_g(\zeta,\rho^*X)$ and $(\zeta'',\rho(\ell))$, and the letters $(X\ell)[g]$; so it agrees, and with equal exits so do S_ℓ and, using agreement of F at the letter $\ell \in E$, F_ℓ.

All relations at the new start label are false on both sides. □

F.6 Positive Column Summaries

Fix a lettered monitor $\mathcal{M}$ with state set Q, $M = |Q|$, start ζ_0, a positive channel $\Theta = (F, R, S, S', \Phi)$, a marker ℓ, and blocks $X_1, \dots, X_k$ avoiding ℓ. We use the summary, the column word and the notation $lab_0(\zeta)$ of step 4 of Appendix A.4; block indices are a, b, entry states ζ, ζ' range over $Q \sqcup \{\star\}$, and final states ζ'' over Q.

Fields. Let $\mathcal{M}^\Theta$ be the lettered monitor with the states, start and transitions of $\mathcal{M}$, the final target F, and the relations R and S in place of Λ and Λ^{fin}. The *positive summary* of $(X_1, \dots, X_k)$ consists of the following fields:

Field	Meaning
sum	the summary of Appendix A.4 of $(X_1, \dots, X_k)$ for $\mathcal{M}$
sum^Θ	the summary of Appendix A.4 of $(X_1, \dots, X_k)$ for $\mathcal{M}^\Theta$
$tSL_a(\zeta, \zeta'')$	some $g < \lvert X_a\rvert$ has $S'(lab_g(\zeta, X_a), (\zeta'', \ell))$ and $X_a[g] \in \Phi$
$pend_a(\zeta, \zeta'')$	$S'(lab_{\lvert X_a\rvert}(\zeta, X_a), (\zeta'', \ell))$
fst_a	*none* if $X_a = \varepsilon$, and otherwise the bit $[X_a[0] \in \Phi]$

In sum^Θ, the field $term_a(\zeta, \zeta'')$ says that some $g \le |X_a|$ has $S(lab_g(\zeta, X_a), (\zeta'', \ell))$, and the field $adm(a, b, \zeta, \zeta')$ says that some $X_a = xy$, $X_b = x'y$ has $R(lab_{|x|}(\zeta, X_a), lab_{|x'|}(\zeta', X_b))$; its exits and equality bits repeat those of *sum*. The field *pend* records an S'-test at the last position of the shifted block, whose letter is not known yet: it is the next letter of the block, or the marker if the block ends there. Counting assignments, the positive summary takes at most $\mathrm{pcol}(M, k)$ values: each of the two summaries takes at most $\mathrm{col}(M, k)$ values (Appendix A.5); the fields *tSL* and *pend* give $2^{2M(M+1)k}$, since a ranges over k blocks, ζ over $M + 1$ entry states and ζ'' over M states; and *fst* gives 3^k.

The automaton. The tuple of empty blocks has the initial summaries of Appendix A.4 for $\mathcal{M}$ and for $\mathcal{M}^\Theta$, tSL_a false, $pend_a(\zeta, \zeta'') = S'(lab_0(\zeta), (\zeta'', \ell))$, and $fst_a = none$. Reading a column γ updates both summaries as in Appendix A.4, and the other fields as follows, simultaneously and from the old values. If γ_a is absent, the fields of block a are unchanged. If $\gamma_a = z$, then, with $e = exit_a(\zeta)$ the old exit,

$$tSL'_a(\zeta, \zeta'') = tSL_a(\zeta, \zeta'') \vee \big(pend_a(\zeta, \zeta'') \wedge z \in \Phi\big), \qquad pend'_a(\zeta, \zeta'') = S'((e, z), (\zeta'', \ell)),$$
$$fst'_a = fst_a \text{ if } fst_a \neq none, \text{ and } [z \in \Phi] \text{ otherwise.}$$

Lemma F.8 (Positive summaries).

(1) After every prefix of a valid column word, every field equals its meaning for the current tuple.

(2) The final positive summary of $(X_1, \dots, X_k)$ determines the derived channel Θ_ℓ on the letters $X_1, \dots, X_k$, at every state of $\mathcal{M}_\ell$. With $term^\Theta$ and adm^Θ the fields of sum^Θ and

$$T_a(\zeta, \zeta'') = term^\Theta_a(\zeta, \zeta'') \vee tSL_a(\zeta, \zeta'') \vee \big(pend_a(\zeta, \zeta'') \wedge \ell \in \Phi\big),$$

it is given by

$$\begin{aligned} X_a \in \Phi_\ell &\iff fst_a = 1 \ \vee \ (fst_a = none \wedge \ell \in \Phi), \\ R_\ell((\zeta, X_a), (\zeta', X_b)) \iff S'_\ell((\zeta, X_a), (\zeta', X_b)) &\iff adm^\Theta(a, b, \zeta, \zeta'), \\ S_\ell((\zeta, X_a), (\zeta', X_b)) &\iff T_a(\zeta, exit_b(\zeta')), \\ F_\ell(\zeta, X_a) &\iff F(exit_a(\zeta), \ell) \vee T_a(\zeta, exit_a(\zeta)). \end{aligned}$$

In particular S_ℓ has no F-term. All relations at the start label of $\mathcal{M}_\ell$ are false by definition.

Proof. (1) Induction on the number of columns read. For the two summaries this is the correctness of the column automaton of Appendix A.4, applied to $\mathcal{M}$ and to $\mathcal{M}^{\Theta}$. The initial values of the other fields are their meanings for empty blocks: the shifted block of ε read from ζ has the single label $lab_0(\zeta)$, and no position $g < 0$ exists. An absent entry leaves the prefix of its track unchanged. Let track a read z, so that its prefix X becomes Xz. The labels of the shifted block of Xz are those of X at the positions $g \le |X|$, and the new last label $lab_{|X|+1}(\zeta, Xz) = (exit_a(\zeta), z)$. The letter at position $|X|$ of Xz is z, and the letters at the positions $g < |X|$ are unchanged. This proves the updates of *tSL* and *pend*: the new position $|X|$ of *tSL* carries the old pending test *pend* and the letter z. By validity, letters are appended to a track in order, so the first letter of a nonempty track is the letter read when *fst* leaves *none*.

(2) By (1), the final fields are their meanings for $(X_1, \dots, X_k)$. In Definition 10.9, $(X_a \ell)[0]$ is $X_a[0]$ if $X_a \neq \varepsilon$ and ℓ otherwise, which gives Φ_ℓ. The field adm^{Θ} is the defining formula of $R_\ell = S'_\ell$. The positive terminal test is $pt(\zeta, X_a; \zeta'') = T_a(\zeta, \zeta'')$, because $(X_a \ell)[g]$ is $X_a[g]$ for $g < |X_a|$ and ℓ for $g = |X_a|$. Finally $\delta_\ell(\zeta', X_b) = exit_b(\zeta')$. □

The summary of $\mathcal{M}$ alone does not suffice, as Example 10.13 shows. We now give the field *pend* in that example.

Example F.9 (Example 10.13 continued: the pending test). In Example 10.13 the top letter $[ab]$ is shortened as the single track ab over the level-1 monitor $\mathcal{M}_1$ and channel Θ_1, with the marker $c = [\sharp 1]$. For every state ζ'' of $\mathcal{M}_1$, the shifted block of c read from ζ'' reads $0\sharp 1$ from an ordinary state, or $\sharp 1$ from the start, and in both cases the state before the final 1 is 4. After the prefix a, the last label of the shifted track is $(2, a)$ from an ordinary entry, or $(\star_1, a)$ from the fresh entry, and the only common suffix of the blocks ε and $\sharp 1$ is empty. So *pend*, the value of $S'_1 = R_1$ at that label and the final label (ζ'', c), compares the states $\delta(2, 0) = 0$, or the start state 0, with $\delta(4, 1) = 2$; as $(0, 2) \notin \Pi$, *pend* is false everywhere. After the prefix ab the last label is $(0, b)$, and the common suffix 1 of the blocks 1 and $\sharp 1$ compares the states $\delta(0, 0) = 3$ and 4; as $(3, 4) \in \Pi$, *pend* is true everywhere. This is the original witness $(1, 4)$, and it enters F_2 through the disjunct $pend \wedge c \in \Phi_1$ of Lemma F.8(2), since the expansion of c starts with $\sharp$. Hence the positive summaries after a and after ab differ, and the column holding b is kept. At level 0 the tracks ε, 1 and $\sharp 1$ admit no deletion either: their column word has two columns, each of which is the first column of a nonempty track, where its field *fst* changes. The compression returns w itself.

Lemma F.10 (Positive column shortening). *Let $\mathcal{F} = \{X_1, \dots, X_k\}$ be a set of k distinct blocks avoiding ℓ. There are subsequences $\tau(X_a)$ of the blocks X_a with $|\tau(X_a)| < \mathrm{pcol}(M, k)$ for every a, such that τ is injective on $\mathcal{F}$, the blocks $\tau(X_a)$ avoid ℓ, and*

$$(\mathcal{M}_\ell, \Theta_\ell) \equiv_{\mathcal{F},\tau} (\mathcal{M}_\ell, \Theta_\ell).$$

Proof. The new positive summary is a function of the old one and the column, and there are at most $\mathrm{pcol}(M, k)$ positive summaries. If the column word has at least $\mathrm{pcol}(M, k)$ columns, two of its cuts carry the same positive summary. Deleting the columns between them leaves a valid column word, and the same final summary, because the remaining columns are read from the same summary. Repeating, we reach fewer than $\mathrm{pcol}(M, k)$ columns; the tracks are subsequences of the blocks and avoid ℓ. Equal final summaries give equal equality bits, so τ preserves equality and inequality of blocks and is injective on $\mathcal{F}$; by Lemma A.9 and Appendix A.4, the summary of $\mathcal{M}$ determines $\mathcal{M}_\ell$ on the letters, and by Lemma F.8(2) the positive summary determines Θ_ℓ on them, at every state and at labels with letters in $\mathcal{F}$ or the start label. Both sides are the same monitor and channel, with the same states and start. □

F.7 Supported Compression

For markers $\boldsymbol{\ell} = (\ell_0, \ldots, \ell_{j-1})$ write $(\mathcal{M}_0^{\boldsymbol{\ell}}, \Theta_0^{\boldsymbol{\ell}}) = (\mathcal{M}_0, \Theta_0)$ and $(\mathcal{M}_{h+1}^{\boldsymbol{\ell}}, \Theta_{h+1}^{\boldsymbol{\ell}}) = ((\mathcal{M}_h^{\boldsymbol{\ell}})_{\ell_h}, (\Theta_h^{\boldsymbol{\ell}})_{\ell_h})$, and use valid families and the expansions $\mathrm{Exp}_{\boldsymbol{\ell},j}$ as in Appendix A.4.

Lemma F.11 (Supported compression with channels). *Fix a nonempty word w with hierarchy markers $\boldsymbol{\ell}$ and dictionaries $\mathcal{A}_h$, and $j \le d$. Suppose $|\mathcal{A}_h| \le g$ for $1 \le h \le j$, and put $\kappa_\sharp = \mathrm{pcol}(N + d + 1, g)$. For every set $E_j \subseteq \mathcal{A}_j$ there are new markers $\widetilde{\boldsymbol{\ell}} = (\widetilde{\ell}_0, \ldots, \widetilde{\ell}_{j-1})$ and a map ρ_j, injective on E_j, such that*

$$(\mathcal{M}_j^{\boldsymbol{\ell}}, \Theta_j^{\boldsymbol{\ell}}) \equiv_{E_j,\rho_j} (\mathcal{M}_j^{\widetilde{\boldsymbol{\ell}}}, \Theta_j^{\widetilde{\boldsymbol{\ell}}}), \qquad |\mathrm{Exp}_{\widetilde{\boldsymbol{\ell}},j}(\rho_j(a))| \le \kappa_\sharp^j \quad (a \in E_j),$$

and the image family is valid for the new markers.

Proof. This is the induction of Lemma A.11, with positive summaries and agreement with channels. For $j = 0$ take the identity: a letter expands to itself, of length $1 = \kappa_\sharp^0$, whatever the size of E_0.

For the step from j to $j+1 \le d$, let $E_{j+1} \subseteq \mathcal{A}_{j+1}$, so $|E_{j+1}| \le g$. Put $(\mathcal{U}, \Theta^{\mathcal{U}}) = (\mathcal{M}_j^{\boldsymbol{\ell}}, \Theta_j^{\boldsymbol{\ell}})$, which has $N + j$ states. The blocks in E_{j+1} are distinct words over $\mathcal{A}_j$ avoiding ℓ_j. Lemma F.10 gives subsequences $\tau(X)$ with $|\tau(X)| < \mathrm{pcol}(N + j, |E_{j+1}|) \le \kappa_\sharp$, by monotonicity, and $(\mathcal{U}_{\ell_j}, \Theta_{\ell_j}^{\mathcal{U}}) \equiv_{E_{j+1},\tau} (\mathcal{U}_{\ell_j}, \Theta_{\ell_j}^{\mathcal{U}})$. Put

$$E_j = \{\ell_j\} \cup \bigcup_{X \in E_{j+1}} \mathrm{alph}(\tau(X)) \;\subseteq\; \mathcal{A}_j.$$

The induction hypothesis gives markers $\widetilde{\ell}_0, \ldots, \widetilde{\ell}_{j-1}$, the monitor and channel $(\mathcal{V}, \Theta^{\mathcal{V}}) = (\mathcal{M}_j^{\widetilde{\boldsymbol{\ell}}}, \Theta_j^{\widetilde{\boldsymbol{\ell}}})$, and ρ_j with $(\mathcal{U}, \Theta^{\mathcal{U}}) \equiv_{E_j,\rho_j} (\mathcal{V}, \Theta^{\mathcal{V}})$ and expansions of length at most $\kappa_\sharp^j$. Set $\widetilde{\ell}_j = \rho_j(\ell_j)$ and $\rho_{j+1}(X) = \rho_j^*(\tau(X))$ for $X \in E_{j+1}$. By Lemma F.7, applied to the family $\tau(E_{j+1})$ of blocks over E_j avoiding ℓ_j, $(\mathcal{U}_{\ell_j}, \Theta_{\ell_j}^{\mathcal{U}}) \equiv_{\tau(E_{j+1}),\rho_j^*} (\mathcal{V}_{\widetilde{\ell}_j}, \Theta_{\widetilde{\ell}_j}^{\mathcal{V}})$, and composition with the column agreement gives $(\mathcal{M}_{j+1}^{\boldsymbol{\ell}}, \Theta_{j+1}^{\boldsymbol{\ell}}) \equiv_{E_{j+1},\rho_{j+1}} (\mathcal{M}_{j+1}^{\widetilde{\boldsymbol{\ell}}}, \Theta_{j+1}^{\widetilde{\boldsymbol{\ell}}})$. The image blocks avoid $\widetilde{\ell}_j$ and their letters lie in the valid family $\rho_j(E_j)$, so the new family is valid. Each new block followed by its marker has at most $\kappa_\sharp$ letters, each expanding to at most $\kappa_\sharp^j$ letters, so every new letter expands to at most $\kappa_\sharp^{j+1}$ letters. No dictionary or normality property of an intermediate compressed word is used. □

F.8 Proof of the Anchored Compression Theorem

Let $\mathcal{M}_0$ be a lettered monitor with $N \ge 1$ states and Θ_0 a positive channel on it, and suppose that $(\mathcal{M}_0, \Theta_0)$ has a positive hole of depth at most d. By Lemma F.5, it has a positive hole w of some depth $D \le d$ with $g_i \le \mathrm{G}^\sharp(N, d)$ for $1 \le i \le D$. Keep w, its hierarchy (w_i, ℓ_i) and its dictionaries fixed, and put $\kappa_\sharp = \mathrm{pcol}(N + d + 1, \mathrm{G}^\sharp(N, d)) \ge 1$.

The top level. By Corollary 10.11, the unary word $w_D = z^m$ is a positive hole of $(\mathcal{M}_D, \Theta_D)$, and $\mathcal{M}_D$ has $N + D$ states. By Lemma F.6, $z^{m'}$ is a positive hole of $(\mathcal{M}_D, \Theta_D)$ for some $1 \le m' \le 3(N+D)$.

Compression. Apply Lemma F.11 with $j = D$, $g = \mathrm{G}^\sharp(N, d)$ and $E_D = \{z\}$. It gives markers $\widetilde{\boldsymbol{\ell}}$ and ρ_D with $(\mathcal{M}_D, \Theta_D) \equiv_{\{z\},\rho_D} (\mathcal{M}_D^{\widetilde{\boldsymbol{\ell}}}, \Theta_D^{\widetilde{\boldsymbol{\ell}}})$. Agreement preserves positive holes, so $\widetilde{w}_D = \rho_D(z)^{m'}$ is a positive hole of $(\mathcal{M}_D^{\widetilde{\boldsymbol{\ell}}}, \Theta_D^{\widetilde{\boldsymbol{\ell}}})$.

Expansion. For $i < D$ let $\widetilde{w}_i$ be the expansion of $\widetilde{w}_{i+1}$ with the marker $\widetilde{\ell}_i$. By validity, $\widetilde{w}_{i+1}$ is a nonempty word of blocks avoiding $\widetilde{\ell}_i$, so Lemma 10.10 shows, by downward induction, that $\widetilde{w}_i$ is a positive hole of $(\mathcal{M}_i^{\widetilde{\boldsymbol{\ell}}}, \Theta_i^{\widetilde{\boldsymbol{\ell}}})$. At $i = 0$, $\widetilde{w} = \widetilde{w}_0$ is a positive hole of $(\mathcal{M}_0, \Theta_0)$. Its length is

$$m' \, |\mathrm{Exp}_{\widetilde{\boldsymbol{\ell}},D}(\rho_D(z))| \le 3(N + D)\, \kappa_\sharp^D \le 3(N + d + 1)\, \kappa_\sharp^d = \mathrm{Len}^\sharp(N, d).$$

Depth. Each $\widetilde{w}_i$ with $i < D$ ends with its marker $\widetilde{\ell}_i$, and cutting it at that marker returns $\widetilde{w}_{i+1}$. So the canonical hierarchy of $\widetilde{w}$ is $\widetilde{w}_0, \ldots, \widetilde{w}_D$ through level D, and $\widetilde{w}_D$ is unary. Hence $\mathrm{depth}(\widetilde{w}) \le D \le d$.

If $D = 0$, there is no compression: w is unary and Lemma F.6 gives a positive hole of length at most $3N \le \mathsf{Len}^\sharp(N, 0)$ and depth 0.

This proves the general statement of Theorem 10.14. For a reader with a relation Π, apply it to the lettered monitor of the reader and the base channel Θ_Π, whose positive holes are the Π-holes (Lemma 10.8); the monitor has the N states of the reader, and a nonempty word has the same depth whichever monitor reads it. The estimates are Lemma F.12 below.

F.9 Estimates and Decision Procedures

Lemma F.12 (Closed form). *For $N \ge 1$ and $d \ge 0$, with $\Xi^\sharp = \Xi^\sharp(N, d)$,*

$$\mathsf{G}^\sharp(N, d) \le 2^{\Xi^\sharp} \qquad \text{and} \qquad \mathsf{Len}^\sharp(N, d) \le 2^{2^{5\Xi^\sharp+6}}.$$

Proof. Put $\Xi_0 = \Xi^\sharp - 3d - 3 = 6N^2 + 10N(N + d)(d + 1) + 7N + 1$. Since $N \le 2^N$, we have $N^N \le 2^{N^2}$, so $\mathsf{C}^\sharp(N, d) \le 2^{\Xi_0}$. For nonnegative x and y, $2^x + 2^y \le 2^{x+y+1}$; with $d + 1 \le 2^{d+1}$ this gives $\mathsf{A}^\sharp(N, d) \le 2^{\Xi_0+d+2}$. Using $3^d \le 2^{2d}$,

$$\mathsf{G}^\sharp(N, d) = 2 + \mathsf{A}^\sharp(N, d)\, 3^d \le 2 + 2^{\Xi_0+3d+2} \le 2^{\Xi_0+3d+3} = 2^{\Xi^\sharp}.$$

Put $H = \Xi^\sharp$, $M = N + d + 1$, $g = \mathsf{G}^\sharp(N, d)$ and $P = 2^H$. Then $M + 1 \le H$, so M, $M + 1$ and d are at most P, and $g \le P$. Replacing M by 2^M in the first factor of Eq. (14) gives $\mathrm{col}(M, g) \le 2^{2M(M+1)g+g^2+(M+1)^2g^2} \le 2^{4P^4} = 2^{2^{4H+2}}$. Moreover $2^{M(M+1)g} \le 2^{P^3} \le 2^{2^{4H+2}}$ and $3^g \le 2^{2g} \le 2^{2^{4H+2}}$. Hence the five factors of $\mathrm{pcol}(M, g) = \mathrm{col}(M, g)^2\, (2^{M(M+1)g})^2\, 3^g$ give $\mathrm{pcol}(M, g) \le 2^{5 \cdot 2^{4H+2}} \le 2^{2^{4H+5}}$. Finally $3M \le 2^{H+2}$ and $d \le 2^H$, so

$$\mathsf{Len}^\sharp(N, d) = 3M\, \mathrm{pcol}(M, g)^d \le 2^{H+2+d\, 2^{4H+5}} \le 2^{2^{5H+5}+2^{5H+5}} = 2^{2^{5H+6}}. \qquad \square$$

Proof of Lemma 10.18. Put $N_0 = 2^\alpha$ and $D_0 = 2^{\alpha+\beta+1}$. Then $\bar{N}_I \le N_0$ and $d_I = 2\bar{N}_I N_A - 1 < D_0$, so, by monotonicity and Lemma F.12, $\mathsf{Len}_I^\sharp \le \mathsf{Len}^\sharp(N_0, D_0) \le 2^{2^{5\Xi^\sharp(N_0,D_0)+6}}$. Since $\alpha \ge 3$ and $\beta \ge 0$, we have $N_0 \ge 8$, $D_0 \ge 16$ and $N_0 \le D_0/2$. Hence $10N_0(N_0 + D_0)(D_0 + 1) \le 10N_0 \cdot \frac{3}{2}D_0 \cdot \frac{17}{16}D_0 < 16N_0D_0^2$, $6N_0^2 \le 3N_0D_0 \le \frac{3}{16}N_0D_0^2$, and $7N_0 + 3D_0 + 4 \le N_0D_0 \le \frac{1}{16}N_0D_0^2$, so that

$$\Xi^\sharp(N_0, D_0) \le 17N_0D_0^2 = 17 \cdot 2^{3\alpha+2\beta+2} \qquad \text{and} \qquad 5\Xi^\sharp(N_0, D_0) + 6 \le 340 \cdot 2^{3\alpha+2\beta} + 6 < 2^{3\alpha+2\beta+9}.$$

Therefore $\mathsf{Len}_I^\sharp \le 2^{2^{2^{3\alpha+2\beta+9}}} = E_3(3\alpha + 2\beta + 9)$. $\square$

Proof of Corollary 10.16. Compute $\mathsf{Len}_I^\sharp$, and test every pair (u, t) of words over Σ with $|u| + 1 + |t| \le \mathsf{Len}_I^\sharp$: check conditions (a)–(e) of Definition 8.1 over the finitely many cuts, decide whether $A(u)$ is mortal by reachability in the safe graph, and check $|t| - J(t) < N_A$. A pair that passes all tests is a mortal hole. If a mortal hole exists, Theorem 10.15 supplies one among the tested pairs. $\square$

G Proofs for Constrained Types

Proof of Proposition 12.1. A valuation of the visible variables that satisfies C_2 is the same as a valuation η of the free variables together with an instance of S_2 under η. Since $\bar{a}_1$ is disjoint from the visible variables, an extension of this valuation to $\bar{a}_1$ that satisfies C_1 is the same as an instance of S_1 under η, and R_{12} together with the domain condition says that this instance is below the given instance of S_2. This proves (1). If the domain condition fails, no instance of S_1 is below an instance of S_2, so $S_1 \le S_2$ holds iff S_2 has no instance. $\square$

H Proofs for Complexity and Sharper Bounds (Not Mechanized)

This appendix proves the results of Section 13 whose proofs were omitted there. Like that section, it is not mechanized, and nothing outside Section 13 uses it.

H.1 The Smaller Unit Reader and the Rejection Clock

Proof of Lemma 13.2. (1) The bit b never returns from 1 to 0, and if du fails at a cut, it fails at every later cut (Lemma 7.3(2)). So the replaced states are closed under the transitions of $\mathcal{R}_{\mathrm{U}}(p)$, and a run that meets one of them stays among them. None of them is rejected, since a rejected state has $b = 0$ and satisfies du. Hence a word rejected by $\mathcal{R}_{\mathrm{U}}(p)$ meets no replaced state, and a word rejected by $\widehat{\mathcal{R}}_p$ never meets the absorbing state $\bot$. In both cases the run consists of kept states, between which the transitions are unchanged, so the word has the same run in both readers and is rejected by both. Each comparison of the word is tested on states of this run, which carry the same self guard in both readers. So the two readers reject the same words, with the same states at every cut and the same admitted comparisons, and they have the same holes.

(2) If two prefixes of a word are rejected by $\widehat{\mathcal{R}}_p$, they are rejected by $\mathcal{R}_{\mathrm{U}}(p)$ by (1); by betweenness for $\mathcal{R}_{\mathrm{U}}(p)$ (Lemma 7.6), so is every intermediate prefix, and by (1) again it is rejected by $\widehat{\mathcal{R}}_p$. A word rejected by $\widehat{\mathcal{R}}_p$ has, by (1), the same first rejected cut and the same admitted comparisons in both readers, so the admission cone of $\mathcal{R}_{\mathrm{U}}(p)$ transfers.

(3) A kept state after a word u is $\zeta = (P_p^{\varphi}(u), 0, P_p^{\psi}(u))$ by Lemma 7.3(1), and $\mathrm{du}(\zeta)$ gives $P_p^{\psi}(u) \cap \mathsf{DeadU}_{\psi}^{\varepsilon} \neq \varnothing$, so $P_p^{\psi}(u) \neq \varnothing$. By Lemma 6.4(3), applied in φ and in ψ to the same seed p and the same word u, all elements of $P_p^{\varphi}(u)$ and of $P_p^{\psi}(u)$ have the sign $\mathrm{sgn}(p) \oplus \mathrm{par}(u)$, which the nonempty set $P_p^{\psi}(u)$ determines. For each sign there are at most 2^v sets $P \subseteq X^{\pm}$ and $2^s - 1$ nonempty sets $P' \subseteq (X \cup Z)^{\pm}$ of that sign. Adding $\bot$ gives at most $1 + 2^{v+1}(2^s - 1) = 2^{v+s+1} - 2^{v+1} + 1 < 2^{v+s+1}$ states. □

Proof of Lemma 13.3. Let w be rejected by $\mathcal{R}_{\mathrm{U}}(p)$, and let $\zeta_j = (P_j, b_j, P'_j)$ be the state of $\mathcal{B}_p$ at the cut j, so that $P_j = P_{p,j}^{\varphi}(w)$ is also the state of $\mathcal{A}_p$ there. At the last cut, $b = 0$ and du holds, so by Lemma 7.3(2) and the monotonicity of the bit, $b_j = 0$ and $\mathrm{du}(\zeta_j)$ hold at every cut j. Hence $\zeta_j \in \mathsf{Rej}_{\mathrm{U}}$ iff $\mathrm{dt}(\zeta_j)$ fails, iff $P_j \cap \mathsf{DeadT}_{\varphi}^{\varepsilon} = \varnothing$, which is rejection by $\mathcal{A}_p$ at the cut j. So w is rejected by $\mathcal{A}_p$, with the same first rejected cut, and $r(w)$ is the same for both readers. By sign coherence (Lemma 6.4(3)), every reachable state of $\mathcal{A}_p$ is empty or a nonempty set of elements of $X^{\pm}$ of one sign, so $\mathcal{A}_p$ has at most $2(2^v - 1) + 1 = 2^{v+1} - 1$ states. Theorem 2.19 holds for every reader and every nonempty rejected word, without any condition on the admission relation, so for $w \neq \varepsilon$ it gives $\mathrm{depth}(w) \leq 2(2^{v+1} - 1)(r(w) + 1) - 1$. □

H.2 Tree Classes and Signed Systems

For the first proof we use the terminology of Appendix B: for a tree t, a word w *meets* a leaf v of t if $v \preceq w$; by the chain property (Section 3), a word meets at most one leaf, a word that meets no leaf is a position labeled f, and a proper prefix of a position is never a leaf; and $w \in L_t^s$ iff w meets a leaf of t of sign s (Eq. (5)).

Proof of Lemma 13.6. A word w is a position labeled f iff it meets no leaf, that is, iff $w \notin L_t^0 \cup L_t^1$. The positions of t are ε and the children of its positions labeled f, so t has at most $1 + nk$ positions if it has k positions labeled f; hence t is finite iff $\Sigma^* \setminus (L_t^0 \cup L_t^1)$ is finite.

For regularity, let w be a position of t and $w^{-1}t$ the subtree of t at w. The proper prefixes of w are not leaves, so wv meets a leaf of t iff it meets a leaf wv' with v' a leaf of $w^{-1}t$, and the sign of

wv' in t is the sign of v' in $w^{-1}t$ plus $\mathrm{par}(w)$. Hence

$$L^{s}_{w^{-1}t} = w^{-1}L_t^{s\oplus \mathrm{par}(w)} \qquad \text{for every position } w \text{ of } t \text{ and every sign } s. \tag{18}$$

If w is not a position of t, then a proper prefix of w is a leaf, of some sign s, and $w^{-1}L_t^s = \Sigma^*$ and $w^{-1}L_t^{1-s} = \varnothing$. Suppose L_t^0 and L_t^1 are regular. By the Myhill–Nerode theorem they have finitely many left quotients, so by Eq. (18) the subtrees of t have finitely many leaf pairs, and by Theorem 3.2(1) t has finitely many distinct subtrees. Conversely, if t has finitely many distinct subtrees, then by Eq. (18) and the case of non-positions, the pair $(w^{-1}L_t^0, w^{-1}L_t^1)$ takes finitely many values as w ranges over all words, so both languages have finitely many left quotients and are regular. □

Proof of Lemma 13.7. If ρ is a $\mathcal{D}$-solution of χ, then $\rho^{\pm}$ is a $\mathcal{D}$-solution of $\chi^{\pm}$ by Lemma 2.5. For the converse let σ be a $\mathcal{D}$-solution of $\chi^{\pm}$.

For $n = 0$ there are no children, so the literals of $\chi^{\pm}$ of sign + in Table 2 are the literals of χ with every variable v replaced by v^+, and the order is the same chain for every variance. Hence $v \mapsto \sigma(v^+)$ solves χ, and the three classes coincide.

Let $n \geq 1$, and let $\chi^\flat$ be obtained by Lemma 2.7, with a fresh variable $y = y_{uv}$ for each copy literal $u \leq v$ and the literals $f(a_i)_i \leq y$ and $y \leq f(b_i)_i$. We extend σ to a $\mathcal{D}$-solution of $(\chi^\flat)^{\pm}$. The four signed literals of the two new literals are

$$f(a_i^{c(i)})_i \leq y^+, \quad y^- \leq f(a_i^{\bar{c}(i)})_i, \quad y^+ \leq f(b_i^{c(i)})_i, \quad f(b_i^{\bar{c}(i)})_i \leq y^-.$$

Put $\sigma(y^+) = f(\sigma(a_i^{c(i)}))_i$ and $\sigma(y^-) = f(\sigma(b_i^{\bar{c}(i)}))_i$. The first and fourth literals hold with equality. The other two hold iff $\sigma(a_i^{c(i)}) \leq \sigma(b_i^{c(i)})$ and $\sigma(b_i^{\bar{c}(i)}) \leq \sigma(a_i^{\bar{c}(i)})$ for every i. For $i \geq 1$ we have $a_i = b_i = u$, so both sides are equal. For $i = 0$ these are the two inequalities $\sigma(u^+) \leq \sigma(v^+)$ and $\sigma(v^-) \leq \sigma(u^-)$, in either case of $c(0)$, and they hold because $u^+ \leq v^+$ and $v^- \leq u^-$ are the signed literals of $u \leq v$. The new values lie in $\mathcal{D}$.

So $(\chi^\flat)^{\pm}$ is $\mathcal{D}$-satisfiable, and in particular satisfiable. By Theorem A.1(2), $\chi^\flat$ has a solution, and a finite one if $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$; by Theorem A.1(3) it has a regular one. In each case $\chi^\flat$ has a $\mathcal{D}$-solution for the variance c, whose restriction to the variables of χ is a $\mathcal{D}$-solution of χ by Lemma 2.7.

Finally, if χ has a solution, then so has $\chi^\flat$ by Lemma 2.7, hence a regular one by Theorem A.1(3), and its restriction is a regular solution of χ. □

I The Mechanized Proof

This appendix records the representation conventions, module layout, and generated correspondence with the paper. The scope, trusted statements, and commands for checking the development are given in Section 14.

Conventions. Variable carriers are finite: the public interface uses finite initial segments, and the core separates visible and hidden variables with the sum type $X \oplus Z$. Systems are lists of literals, interpreted conjunctively; ordering and repeated literals do not affect satisfaction. Regular trees are unfoldings of finite graphs, and finite trees are interpretations of finite terms. Subtrees are optional: a path outside the domain returns `none`, so subtree identities can quantify over all paths. A default letter supplies an inhabited alphabet where a total canonical hierarchy or a periodic stream requires one; the periodic stream of a nonempty word is independent of that choice. Bounded searches are the executable form of the decision procedures. A decidability claim is a computable definition that returns a `Decidable` instance built from these searches; a conditional claim is a computable function from decision procedures to decision procedures. The checker rejects entries that are not computable. These conventions recur in the `Form:` notes of `DeciENSSE/Paper/*.lean`.

I.1 Module Map

Paths in Table 9 are relative to `DeciENSSE/`, except for the shared library `DeciNSSE/`. A module name denotes its `.lean` file.

Table 9. Module map of the library.

Folder or module	Content
`Semantics`, `Main`, `NSSE`, `Plain`	Public syntax, entailment relations, existential deciders, and plain entailment.
`Terms`, `Flattening`, `Subsumption`, `TopFree`	Type expressions, flattening, constrained types and subsumption, and the gadget for the signature without top.
`Core/Main`, `Core/Nullary`, `Core/Basic`	Core deciders with the arity dispatch, root-label enumeration at arity zero, prefix sets, and covariant pushdown rules.
`Core/Full/`	Full-signature systems, signed translation, sandwich elimination, observations, ground bounds, and graph equations.
`Core/Mixed/`	Leaf cones, Horn closure, the three conditions, summaries, unit readers, extraction, tree-class transfers, and decision procedures.
`Core/Automata/`	Automaton moves, saturation, and powerset constructions.
`Core/Words/`, `Core/Words/Self/`	Two-word instances, streams, periodicity, pre-holes, lassos, pumps, self events, and the combined hole bound.
`Core/Words/Mortal/`	Separator words, the Π-reader, depth, and the mortal bound.
`Anchored/`	Positive channels, exchange types, dictionaries, column summaries, anchored compression, and its closed form.
`Bounds`, `Bounds/`, `Certificate`, `Counterexamples`	State counts, closed forms, counterexample words, and countermodels.
`Support/`, `Examples/`	Auxiliary tree and graph constructions and interfaces to the shared library; proved and evaluated examples.
`Paper`, `Paper/`	Contract declarations for the numbered results.
`DeciNSSE/`	Shared tree semantics, satisfiability, readers, hierarchies, and bounded hole search.

I.2 Correspondence

The following table is generated from the contract declarations. Regenerate its rows from the repository root with:

```
python3 scripts/contract.py --table paper/sections/lean-table.tex
```

The first column identifies the result and, where applicable, its numbered item; the second gives the fully qualified Lean declaration. The last column reproduces its `Form:` note, including local statement packaging beyond the global conventions above. For example, consistency separates avoidance of forbidden roots from the absence of exclusion pairs; self comparisons specialize cross comparisons to equal words; and conjunctions or structures collect claims made together in the paper. These are statement-form conventions, not alternative proof routes. Only existing contract entries produce rows; coverage is checked by the strict command in Section 14.

Table 10. Generated paper–Lean correspondence. Statement-form notes are reproduced from the contract.

Result	Lean declaration	Form note
Lemma C.1	`DeciENSSE.Paper.lem_tc_ground_pins`	The recursive list groundPins enumerates constructor and extreme-leaf positions; a Boolean selects lower or upper bounds, and PathPin.Holds interprets their observations. The equivalence also holds at arity zero.
Lemma C.2	`DeciENSSE.Paper.lem_tc_ranked`	Finite variable sets are initial segments; Ranked.Extension packages the embedding, source/sink partition, and rank assumptions. Old ranks need not be zero; the result also holds at arity zero.
Lemma F.1	`DeciENSSE.Paper.lem_mo_transport`	The expansion is expand l v' with block cuts cutP l v' i; only the data of the alignment lemma are assumed, from which (x, y) is a comparison; letters are optional list entries. Marker avoidance is not required.
Lemma F.3	`DeciENSSE.Paper.lem_mo_types`	The number of types is the cardinality of the type space, the codomain of anchoredType at the protected letters; the bound holds at every level below the depth of every nonempty word. Level zero is included; Inhabited supplies the default letter of the canonical hierarchy, which is unused because the word and its hierarchy levels are nonempty.
Lemma F.4	`DeciENSSE.Paper.lem_mo_exchange`	The letter map h is identify a b; dictionaries are compared through alphCard and start counts through sCount. The canonical hierarchy uses an Inhabited alphabet; a letter is available from the nonempty hole, and the default is never used.
Lemma F.5(1)	`DeciENSSE.Paper.lem_mo_dictionary_1`	Anchored normality is PosNormal; dictionaries are compared through alphCard and start counts through sCount. The canonical hierarchy uses an Inhabited alphabet; a letter is available from the nonempty hole, and the default is never used.
Lemma F.5(2)	`DeciENSSE.Paper.lem_mo_dictionary_2`	g_i is alphCard w i and b_i is the cardinality of boundaryPairs; A-sharp and G-sharp are anchoredAllowance and anchoredDictionaryBound. The unnumbered consequence is included. Inhabited supplies the default letter of the canonical hierarchy, which is unused on nonempty holes.
Lemma F.6	`DeciENSSE.Paper.lem_mo_unary`	The unary word z^m is List.replicate m z.
Lemma F.7	`DeciENSSE.Paper.lem_mo_agreement`	Agreement with channels is PosAgree, between monitors with a shared core type and possibly different alphabets; the support and the family of blocks are lists.
Lemma F.8(1)	`DeciENSSE.Paper.lem_mo_summary_1`	A column word is a list of columns, maps from Fin k to Option Gamma with none as padding, and its current tuple is dec; the fields form a PosColumnState and their meanings are psumm; the prefixes of a valid word are its takes. State finiteness and marker avoidance are not required for this invariant.

Result	Lean declaration	Form note
Lemma F.8(2)	`DeciENSSE.Paper.lem_mo_summary_2`	The final positive summary is psumm of the tuple, by item (1); determination is stated as equal summaries giving equal rows of the derived channel, followed by the explicit formulas; the first-letter field is an optional Prop. State finiteness and marker avoidance are not required for these formulas.
Lemma F.10	`DeciENSSE.Paper.lem_mo_columns`	The family is a list of at most k blocks; subsequences are sublists, and injectivity is stated on the list.
Lemma F.11	`DeciENSSE.Paper.lem_mo_compression`	Markers are sequences over all levels, of which only those below j are used; Exp is expandTo, validity of the image family is Admissible, and E_j is a duplicate-free list. Inhabited supplies the default letter of the canonical hierarchy, which is unused because w and its hierarchy levels are nonempty.
Lemma F.12	`DeciENSSE.Paper.lem_mo_closed_form`	G-sharp, L-sharp and Xi-sharp are anchoredDictionaryBound, anchoredLengthBound and anchoredExponent; N at least 1 is not needed.
Lemma E.1(1)	`DeciENSSE.Paper.lem_nm_infinite_1`	StreamPeriod also permits the trivial zero shift; the original period p remains positive. The alphabet need not be finite.
Lemma E.1(2)	`DeciENSSE.Paper.lem_nm_infinite_2`	Under the least-period hypothesis on xi, periods of suffixes are quantified directly; choosing k = |alpha| and q = |lambda| gives the lasso consequence. The suffix also retains the least period. The alphabet need not be finite.
Lemma E.1(3)	`DeciENSSE.Paper.lem_nm_infinite_3`	A default alphabet letter presents periodic streams; its value is irrelevant for nonempty periods. The alphabet need not be finite.
Lemma E.1(4)	`DeciENSSE.Paper.lem_nm_infinite_4`	Lassos are represented by their eventual periods and prefix lengths. The alphabet need not be finite.
Lemma E.2(1)	`DeciENSSE.Paper.lem_nm_hit_1`	The alphabet need not be finite.
Lemma E.2(2)	`DeciENSSE.Paper.lem_nm_hit_2`	The raw cut characterization is paired with the AlignedHit identity for an arbitrary relation, which includes both guard and head hit bits. The alphabet need not be finite. The automaton state carriers need not be finite.
Lemma E.3	`DeciENSSE.Paper.lem_nm_graph_invariant`	PreValid packages the node identity, spine safety, target openness and the two phase conditions. Neither a finite alphabet nor (FH) is needed. E may be any finite set of states.
Lemma E.4	`DeciENSSE.Paper.lem_nm_graph_correct`	Neither a finite alphabet nor (FH) is needed. E may be any finite set of states.
Lemma E.6	`DeciENSSE.Paper.lem_nm_good`	Survivor-freeness is the conjunction of its cross and self parts. Neither a finite alphabet nor (FH) is needed. The equivalence does not require periodicity of the stream.
Lemma E.7(1)	`DeciENSSE.Paper.lem_nm_ends_1`	The words are the suffixes at the two periodic cuts; their lengths are the end distances. The alphabet need not be finite.

Result	Lean declaration	Form note
Lemma E.7(2)	`DeciENSSE.Paper.lem_nm_ends_2`	Rest distances are natural subtractions; equality also identifies periodic and rest cuts. Only the two periodic-split certificates and the suffix equality are assumed; p need not be positive or least. The alphabet need not be finite.
Lemma E.7(3)	`DeciENSSE.Paper.lem_nm_ends_3`	The minimum bound is a conjunction of the two inequalities. The alphabet need not be finite.
Lemma E.8	`DeciENSSE.Paper.lem_nm_target_end`	TargetOpen packages all proper cuts; the periodic end has length t.length - rt. Neither a finite alphabet nor (FH) is needed.
Lemma E.9(1)	`DeciENSSE.Paper.lem_nm_rigid_1`	The alphabet need not be finite.
Lemma E.9(2)	`DeciENSSE.Paper.lem_nm_rigid_2`	The alphabet need not be finite.
Lemma E.9(3)	`DeciENSSE.Paper.lem_nm_rigid_3`	Uniqueness is equality of two cuts with the same non-self-similar suffix stream. The alphabet need not be finite. Cut uniqueness does not require periodicity of the appended stream.
Lemma E.9(4)	`DeciENSSE.Paper.lem_nm_rigid_4`	The alphabet need not be finite.
Lemma E.10(1)	`DeciENSSE.Paper.lem_nm_deletion_1`	The three numbered premises are jointly required in each entry. D1 uses loop equations, D2 is expanded into the five deep/shallow implications (wholly shallow pairs lift automatically), and D3 is the surviving-head disjunction. The output is the specified deleted pair. Neither a finite alphabet nor (FH) is needed. The deletion lemma holds for arbitrary streams.
Lemma E.10(2)	`DeciENSSE.Paper.lem_nm_deletion_2`	The three numbered premises are jointly required in each entry. D1 uses loop equations, D2 is expanded into the five deep/shallow implications (wholly shallow pairs lift automatically), and D3 is the surviving-head disjunction. The output is the specified deleted pair. Neither a finite alphabet nor (FH) is needed. The deletion lemma holds for arbitrary streams.
Lemma E.10(3)	`DeciENSSE.Paper.lem_nm_deletion_3`	The three numbered premises are jointly required in each entry. D1 uses loop equations, D2 is expanded into the five deep/shallow implications (wholly shallow pairs lift automatically), and D3 is the surviving-head disjunction. The output is the specified deleted pair. Neither a finite alphabet nor (FH) is needed. The deletion lemma holds for arbitrary streams.
Lemma E.11(1)	`DeciENSSE.Paper.lem_nm_dischargers_1`	Absorption is stated for an arbitrary prefix and target stream, encompassing implication (ii). Periodicity of the stream is not needed. The alphabet need not be finite.

Result	Lean declaration	Form note
Lemma E.11(2)	`DeciENSSE.Paper.lem_nm_dischargers_2`	The two impossible premises are stated directly for deep-deep and deep-shallow self pairs. The appended stream need not be periodic. The alphabet need not be finite.
Lemma E.12(1)	`DeciENSSE.Paper.lem_nm_four_1`	N1 is the product-state cardinality; Shrinks packages a good pair with the same endpoint and smaller total length. Neither a finite alphabet nor (FH) is needed.
Lemma E.12(2)	`DeciENSSE.Paper.lem_nm_four_2`	P may be any finite set of partnered rest cuts, including all of C_u; N2 is its signature-space cardinality. Shrinks packages the conclusion. Neither a finite alphabet nor (FH) is needed.
Lemma E.12(3)	`DeciENSSE.Paper.lem_nm_four_3`	P may be any finite set of unpartnered spine rest cuts, including U_u; N3 is its signature-space cardinality. Shrinks packages the conclusion. No periodic-split certificate for the target is needed. Neither a finite alphabet nor (FH) is needed.
Lemma E.12(4)	`DeciENSSE.Paper.lem_nm_four_4`	P may be any finite set of unpartnered target rest cuts, including U_t; N4 is its signature-space cardinality. Shrinks packages the conclusion. No periodic-split certificate for the spine is needed. Neither a finite alphabet nor (FH) is needed.
Lemma E.13	`DeciENSSE.Paper.lem_nm_count`	partnered and unpartnered enumerate C and U; signature cardinalities express N1 through N4, and pumpBound is their sum plus b times the period. Neither a finite alphabet nor (FH) is needed.
Theorem A.1(1)	`DeciENSSE.Paper.thm_bg_nsse_sat_1`	Variables are Fin k; finite solutions are finite terms. The cycle witness bound is k * k.
Theorem A.1(2)	`DeciENSSE.Paper.thm_bg_nsse_sat_2`	Variables are Fin k, with signed variables Fin (2 * k); finite solutions are finite terms.
Theorem A.1(3)	`DeciENSSE.Paper.thm_bg_nsse_sat_3`	Variables are Fin k and regular solutions are finite-graph assignments.
Lemma A.2(1)	`DeciENSSE.Paper.lem_bg_cores_1`	Ordinary cores are original-word positions; symbolic cores are the iterated-option start states of a monitor. A default letter defines the total hierarchy.
Lemma A.2(2)	`DeciENSSE.Paper.lem_bg_cores_2`	IsGreatest supplies the maximum ordinary core; the set identity includes the asserted comparisons. A default letter defines the total hierarchy.
Lemma A.3	`DeciENSSE.Paper.lem_bg_exchange_types`	ClosedType is the finite type of exchange tables; normalTypeBound is C(N,d). A default letter defines the total hierarchy. The bound applies to the entire exchange-table type for any nonempty word satisfying the depth inequalities; holehood is not needed.
Lemma A.4	`DeciENSSE.Paper.lem_bg_comparison_map`	NoCollapse states the predecessor-pair condition with zero-based letter indices.

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Lemma A.5	`DeciENSSE.Paper.lem_bg_level_maps`	LevelMapResult packages the hole, marker, dictionary, higher-map, start-count and unary-level claims; depth and dictionary cardinalities are also explicit. A default letter defines the total hierarchy. This structural result does not require finitely many monitor states.
Lemma A.6	`DeciENSSE.Paper.lem_bg_isolated_exchange`	The first conjunct lists the hypotheses of lem:bg-level-maps for the map identify x y: no collapse of predecessor pairs, the mapped-level hole, dictionary containment, the fixed start prefix and the unchanged run of cores. LevelMapResult includes the unchanged start counts and unary levels; letterWeight is the level-zero expansion length. A default letter defines the total hierarchy.
Lemma A.7(1)	`DeciENSSE.Paper.lem_bg_dictionary_1`	A default letter defines the total hierarchy; Normal packages distinct exchange types of distinct isolated letters.
Lemma A.7(2)	`DeciENSSE.Paper.lem_bg_dictionary_2`	Boundary propagation is the equivalent subtraction-free inequality b_i + 1 ≤ b_(i+1) + g_(i+1); normalAllowance is A(N,d). A default letter defines the total hierarchy.
Lemma A.7(3)	`DeciENSSE.Paper.lem_bg_dictionary_3`	normalAllowance and dictionaryBound are A and G; the final existential applies to every hole of depth at most d. A default letter defines the total hierarchy.
Lemma A.8	`DeciENSSE.Paper.lem_bg_unary`	
Lemma A.9	`DeciENSSE.Paper.lem_bg_columns`	Tuples are Fin k indexed; a supported letter map expresses the replacement, whose Agree field records injectivity and preservation of all derived-monitor tables; terminal-summary bits are retained explicitly. Deletion is stated for a factorization at the two cuts. Raw tuple shortening does not require marker avoidance; the supported replacement preserves avoidance when it is assumed.
Lemma A.10	`DeciENSSE.Paper.lem_bg_agreement`	Finite supports and families are lists; the conclusion explicitly includes avoidance of the mapped marker. Agreement under derivation also holds for infinite state carriers.
Lemma A.11	`DeciENSSE.Paper.lem_bg_compression`	E is a duplicate-free list of supported level-j letters; PositiveDictionaryBound, Agree and Admissible package dictionary bounds, all-state agreement with injectivity, and recursive marker validity. descentFactor N d g is col(N+d+1,g). The recursion is explicit: supports Es h with Es j = E, shortenings tau h from column shortening and level maps rho h, with rho (h+1) X = (tau h X).map (rho h) and the new marker of level h equal to the image of the old one under rho h, at every level h below j. A default letter defines the total hierarchy.

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Lemma A.12	`DeciENSSE.Paper.lem_bg_estimate`	dictionaryExponent is Xi, dictionaryBound is G, and holeLengthBound N is lengthBound N d_N; the estimates hold even at N = 0.
Proposition 12.1(1)	`DeciENSSE.Paper.prop_ap_subsumption_1`	The consequent, the conjunction of C_1 and R_12, is Scheme.subsumptionGoal, whose context part ranges over the common domain, which is Dom Gamma_1 here; the class C is an arbitrary predicate on trees. For the bottom-constructor signature, instantiate C with the top-free trees of the intended class; the variable carriers need not be finite.
Proposition 12.1(2)	`DeciENSSE.Paper.prop_ap_subsumption_2`	The class C is an arbitrary predicate on trees; a solution of C_2 assigns the free variables and the bound variables of S_2. For the bottom-constructor signature, instantiate C with the top-free trees of the intended class; the variable carriers need not be finite.
Corollary 12.2	`DeciENSSE.Paper.cor_ap_decidable`	Decidability is a computable pair of triples of the public deciders, over the full signature at every variance (arrow types are n = 2 with Examples.arrow) and over the bottom-constructor signature; the agreement of regular and arbitrary trees, for both signatures, is the propositional component.
Example 12.3	`DeciENSSE.Paper.ex_ap_identity`	Both schemes have no free variables and empty contexts; the claims hold for every predicate C on trees, without closure assumptions. The parity claim is about the flattened consequent of S_1 ≤ S_2.
Example 12.4	`DeciENSSE.Paper.ex_ap_pottier`	The class D is closed under the constants and the constructor. Proving k ≤ a through the lower bound means entailing that k is below the arrow from a to the bottom tree; the signed literals are SignedC at the signed variables a+ = nu_0 a and a- = nu_1 a.
Example 12.5	`DeciENSSE.Paper.ex_ap_selfapp`	The class C is closed under the constants and the constructor; the two schemes have no free variables and empty contexts.
Definition 11.1(1)	`DeciENSSE.Paper.def_as_certificate_1`	Words are the type CounterexampleWord with a kind index; the entry characterizes the (I0)-words of each length L. The type does not assume the systems satisfiable.
Definition 11.1(2)	`DeciENSSE.Paper.def_as_certificate_2`	Words are the type CounterexampleWord with a kind index; the entry characterizes the (U)-words of each length L. The type and characterization do not assume the systems satisfiable.
Definition 11.1(3)	`DeciENSSE.Paper.def_as_certificate_3`	Words are the type CounterexampleWord with a kind index; the entry characterizes the hole words of each length L and includes the length |u| + |t| + 1 of the separator word u#t. The type and characterization do not assume the systems satisfiable.

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Theorem 11.2(1)	`DeciENSSE.Paper.thm_as_main_1`	Decidability is a computable triple of the public deciders, for arbitrary, regular and finite trees, on flat systems over finite ordinals; expressions are decided by decideExistsEntailsTerm and its variants (thm_intro_main).
Theorem 11.2(2)	`DeciENSSE.Paper.thm_as_main_2`	Decidability is a computable triple of the public deciders for the bottom-constructor signature with f covariant, over top-free arbitrary, regular and finite trees, on flat systems over finite ordinals.
Theorem 11.2(3)	`DeciENSSE.Paper.thm_as_main_3`	D ranges over the three tree classes; the observations of a word are a list of signed leaf atoms. Independence of n and c is the form of the bound.
Theorem 11.2(4)	`DeciENSSE.Paper.thm_as_main_4`	Systems over the bottom-constructor signature are flat systems read at the covariant variance with top-free values in D; the gadget pair is gadgetAnte and gadgetCons over complete lists xs and zs, with the visible carrier Sum X X and the hidden carrier Sum Z Z.
Lemma 11.3(1)	`DeciENSSE.Paper.lem_as_countermodel_1`	A and O are CounterexampleWord.seeds and observations; mu_phi and Cl_phi are by definition the hulls of the empty set and of a singleton, and the decoding rho_A has the atom set H_phi(A). The satisfiability of psi is not needed.
Lemma 11.3(2)	`DeciENSSE.Paper.lem_as_countermodel_2`	The satisfiability of psi is not needed; rho_A is identified by its atom set H_phi(A).
Lemma 11.3(3)	`DeciENSSE.Paper.lem_as_countermodel_3`	B is the list CounterexampleWord.bounds of ground bounds with finite-term trees; the union of phi and B is SatT together with the bounds; no extension exists even with arbitrary values. The satisfiability of psi is not needed.
Lemma 11.3(4)	`DeciENSSE.Paper.lem_as_countermodel_4`	Systems over the bottom-constructor signature are read at the covariant variance with top-free values; the gadget pair is gadgetAnte and gadgetCons over complete lists xs and zs, and the restriction is the composition of rho with Sum.inl. No counterexample word or satisfiability assumption on the gadget consequent is required; the input literals may use the full signature, interpreted on top-free values.
Proposition 11.5(1)	`DeciENSSE.Paper.prop_as_bounds_1`	No satisfiability hypothesis is needed.
Proposition 11.5(2)	`DeciENSSE.Paper.prop_as_bounds_2`	holeBound(N_U, d_{N_U}) is holeLengthBound at the N_U states of the unit reader, the type TargetState X Z; the satisfiability of psi is not needed.
Proposition 11.5(3)	`DeciENSSE.Paper.prop_as_bounds_3`	(I0) and the satisfiability of psi are not needed; Lnm and Len-sharp are extractedBoundSelf at the state counts of extractSelf, and condI0Bound is the bound of (1). This item includes the closing claims; that the mortal bound is the largest is proved for the closed forms, E3(10v+10s+17) ≤ E3(10v+12s+15), and every exact bound is at most the mortal closed form.

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Corollary 11.6	`DeciENSSE.Paper.cor_as_plain`	Plain entailment is existential entailment with no hidden variables and the consequent query x y (bfQuery over the bottom-constructor signature). Decidability is a computable pair of triples of the public deciders; the two equivalences are the propositional component, with Sat_fin(phi) a solution by finite terms.
Lemma 2.1	`DeciENSSE.Paper.lem_bg_periodic`	The infinite repetition is a periodic stream; its arbitrary default letter is immaterial for a nonempty period.
Lemma 2.2(1)	`DeciENSSE.Paper.lem_bg_normalization_1`	
Lemma 2.2(2)	`DeciENSSE.Paper.lem_bg_normalization_2`	The chained equivalence is two equivalences with the same left-hand side.
Lemma 2.2(3)	`DeciENSSE.Paper.lem_bg_normalization_3`	
Lemma 2.2(4)	`DeciENSSE.Paper.lem_bg_normalization_4`	Subtrees are optional, so the identity also covers words outside the domain; regularity means finitely many subtrees.
Lemma 2.3	`DeciENSSE.Paper.lem_bg_bf_order`	TopFree expresses the smaller signature; constructor positions are tested by their labels.
Lemma 2.5	`DeciENSSE.Paper.lem_bg_signed`	Dual coherence is the equation signDual A = A; finite and regular trees use finite-term and finite-graph presentations. Existence and uniqueness express the bijection. The class-preservation claims for decode hold without dual coherence.
Lemma 2.6	`DeciENSSE.Paper.lem_bg_flattening`	C is an arbitrary tree predicate containing the values of the input subexpressions, expressed by EvalClosed; the three standard tree classes D are specializations, stated separately. Over the bottom-constructor signature at every variance, values are top-free trees of the class and the flattened systems have no top literal. Size is the number of variable occurrences (at most 6 per symbol) and of names (at most 1 per symbol).
Lemma 2.7	`DeciENSSE.Paper.lem_bg_sandwich`	Positive arity is a NeZero instance; sandwichArgs selects argument zero and repeats the filler elsewhere.
Theorem 2.9(1)	`DeciENSSE.Paper.thm_bg_nsse_1`	The variable set V is finite, as in the definition of a system; finite-graph presentations express regular satisfiability.
Theorem 2.9(2)	`DeciENSSE.Paper.thm_bg_nsse_2`	Decidability is a computable definition, the executable finite-satisfiability procedure Core.Mixed.decideSatFin on systems over a finite variable carrier.
Lemma 2.14	`DeciENSSE.Paper.lem_bg_extension`	The result holds without finiteness assumptions on the alphabet or the state carrier.
Theorem 2.15	`DeciENSSE.Paper.thm_bg_brt`	BoundedRejectedTail existentially selects a hole and its first rejected cut, with tail length at most the displayed bound. The alphabet need not be finite.

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Lemma 2.17	`DeciENSSE.Paper.lem_bg_alignment`	Strict cut inequalities are hypotheses; the two comparison predicates are expressed by their suffix-prefix conditions.
Lemma 2.18	`DeciENSSE.Paper.lem_bg_return_gap`	ReturnGap quantifies the rejected word, its first rejected cut, and the comparison with equal states. The return-gap statement needs neither finite states nor a finite alphabet.
Theorem 2.19	`DeciENSSE.Paper.thm_bg_depth`	J is the first rejected cut, so the tail length is w.length - J; a default letter defines the total canonical hierarchy. The alphabet need not be finite.
Theorem 2.20	`DeciENSSE.Paper.thm_bg_search`	InL d expresses depth at most d; lettered holes are nonempty by definition. Both monitor and reader forms are included; a default letter defines the total hierarchy. The reader version, like the monitor version, does not require a finite alphabet.
Corollary 2.21	`DeciENSSE.Paper.cor_bg_hole_decision`	holeLengthBound N is lengthBound N (2 * N * (4 * N - 1) - 1).
Lemma 2.24	`DeciENSSE.Paper.lem_bg_desub`	The one-step equivalence is oriented from derivative to expansion; the canonical hierarchy uses a default letter. Cores Q j is the iterated option state type.
Theorem 4.2(1)	`DeciENSSE.Paper.thm_ch_characterization_1`	Visible and hidden carriers need not be finite.
Theorem 4.2(2)	`DeciENSSE.Paper.thm_ch_characterization_2`	Includes the unnumbered satisfiability-dispatch equivalence. Visible and hidden carriers need not be finite.
Lemma 4.3(1)	`DeciENSSE.Paper.lem_ch_pure_1`	Visible and hidden carriers need not be finite.
Lemma 4.3(2)	`DeciENSSE.Paper.lem_ch_pure_2`	Visible and hidden carriers need not be finite.
Lemma 5.1	`DeciENSSE.Paper.lem_tc_extension`	Consistency separates forbidden roots from exclusion pairs; visible equality is inclusion because seeds are retained. The decoded extension is also least in the information order. Visible and hidden carriers may be infinite.
Lemma 5.3	`DeciENSSE.Paper.lem_tc_two_seeds`	The asserted and denied sets are a list of at most two signed observations. The result also holds for infinite visible and hidden carriers.
Lemma 5.4	`DeciENSSE.Paper.lem_tc_pins`	The four covariant cases are packaged by Observation.bound; signed observations package the mixed cases. Bounds contain finite terms.
Proposition 5.5(1)	`DeciENSSE.Paper.prop_ch_meet_1`	Models and projected models are sets of signed atoms; their intersection closure packages the assignment-level consequence. Intersection closure also holds over arbitrary variable carriers.
Proposition 5.5(2)	`DeciENSSE.Paper.prop_ch_meet_2`	Positive arity is witnessed by a child index; infinitude is absence of a finite-term presentation.
Lemma 5.6	`DeciENSSE.Paper.lem_tc_pinning_regular`	The variable set V is finite, as in the definition of a system; regular bounds are bindings to a finite indexed family of finite graphs.

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Theorem 5.7	`DeciENSSE.Paper.thm_tc_pinning`	Bounds are a finite list of finite-term comparisons; finiteness means having a finite-term presentation.
Lemma 5.8(1)	`DeciENSSE.Paper.lem_tc_signed_1`	System satisfaction and satisfaction of the bound list are paired as a conjunction.
Lemma 5.8(2)	`DeciENSSE.Paper.lem_tc_signed_2`	
Lemma 5.8(3)	`DeciENSSE.Paper.lem_tc_signed_3`	Finiteness is existence of finite-term presentations; the inverse is the library function decode. Finiteness of decode does not require dual coherence.
Lemma 5.9(1)	`DeciENSSE.Paper.lem_tc_median_1`	Monotonicity in the three arguments is packaged as simultaneous monotonicity.
Lemma 5.9(2)	`DeciENSSE.Paper.lem_tc_median_2`	
Lemma 5.9(3)	`DeciENSSE.Paper.lem_tc_median_3`	Finite trees are trees with a finite domain.
Lemma 5.9(4)	`DeciENSSE.Paper.lem_tc_median_4`	System closure and closure of arbitrary lower and upper ground bounds are packaged separately.
Lemma 5.10	`DeciENSSE.Paper.lem_tc_pinning_covariant`	Bounds are a finite list of finite-term comparisons.
Theorem 5.12(1)	`DeciENSSE.Paper.thm_tc_regular_1`	The variable set V is finite, as in the definition of a system; regularity is existence of a finite-graph presentation.
Theorem 5.12(2)	`DeciENSSE.Paper.thm_tc_regular_2`	ClassExtension quantifies the regular visible assignment and its extension.
Theorem 5.12(3)	`DeciENSSE.Paper.thm_tc_regular_3`	This item also packages the theorem's unnumbered entailment equivalence; regularity uses finite-graph presentations.
Theorem 5.13(1)	`DeciENSSE.Paper.thm_tc_finite_1`	Finiteness is existence of a finite-term presentation. Visible and hidden carriers may be infinite.
Theorem 5.13(2)	`DeciENSSE.Paper.thm_tc_finite_2`	ClassExtension packages extension of finite assignments; this item also includes the theorem's unnumbered finite dispatch equivalence. Only the visible carrier is required to be finite.
Corollary 5.14	`DeciENSSE.Paper.cor_tc_decidable`	The conditional is a computable transformer at fixed finite carriers. A decision procedure over arbitrary trees yields one over regular trees through the regular equivalence, and one over finite trees through the finite dispatch, whose two finite-satisfiability tests run Core.Mixed.decideSatFin. Both equivalences are included.
Lemma 5.16(1)	`DeciENSSE.Paper.lem_tc_gadget_1`	
Lemma 5.16(2)	`DeciENSSE.Paper.lem_tc_gadget_2`	Omega is the unfolding of the one-state graph. Finite trees use finite terms; height counts constructors (at arity zero the witness is still f()).

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Theorem 5.17	`DeciENSSE.Paper.thm_tc_gadget`	The three tree classes are a conjunction. Fresh variables are sum copies; lists enumerate all visible and hidden variables. Top-free values interpret the smaller signature. The equivalences also allow full-syntax systems interpreted with top-free values; the paper uses their specialization without top literals.
Corollary 5.18	`DeciENSSE.Paper.cor_tc_bf`	Fin-indexed public syntax, with k visible and h hidden variables. Each conditional is a computable transformer: a decision procedure for the covariant problem over the full signature, on the doubled carriers, yields one for the bottom-constructor signature in the same tree class, through the reduction toCoreTopFree and splitTopFree. The three reductions and the regular/arbitrary equivalence are included. The size claims are the doubled carriers Sum (Fin k) (Fin k) and Sum (Fin h) (Fin h) and the translated lengths, two more literals per visible or hidden variable.
Lemma 8.4	`DeciENSSE.Paper.lem_ex_head_before_J`	J is an option; its defined value is supplied as an equality. The alphabet need not be finite.
Lemma 8.6	`DeciENSSE.Paper.lem_ex_fh`	
Theorem 8.7(1)	`DeciENSSE.Paper.thm_ex_reduction_1`	This item includes the theorem's unnumbered pointwise equivalence; signed variables are pairs and visible variables use Sum.inl.
Theorem 8.7(2)	`DeciENSSE.Paper.thm_ex_reduction_2`	Models are sets of signed atoms; 'restrict' restricts a model to the visible variables and 'configSet' is the atom set of an assignment.
Corollary 8.8	`DeciENSSE.Paper.cor_ex_entailment`	
Example 1.1	`DeciENSSE.Paper.ex_intro_shift`	The running systems use top-free binary trees at the covariant order, with an unused visible b in phi_1 and phi_3. The subtree-position identity allows arbitrary trees; the subtree characterization requires only a top-free x; the explicit witness theorem for phi_2 also holds over the full signature. Kpos and Suf are sets of words, the shifted subtree is subtreeBot, and the nested bound of phi_3 is stated semantically.
Theorem 1.2	`DeciENSSE.Paper.thm_intro_main`	Decidability is a computable pair of triples of the public deciders for systems of type expressions, over the full signature at every variance and over the bottom-constructor signature with f covariant. The certificate claims, for flat systems in each tree class D, are the propositional component; E3(O(s)) is E3(22s + 15), and E3(44s + 15) over the bottom-constructor signature, where the word is one of the gadget pair gadgetAnte and gadgetCons over complete lists xs and zs.
Theorem 3.2(1)	`DeciENSSE.Paper.thm_lf_representation_1`	LeafPair packages disjointness and closure under arbitrary right extensions.

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Theorem 3.2(2)	`DeciENSSE.Paper.thm_lf_representation_2`	
Theorem 3.2(3)	`DeciENSSE.Paper.thm_lf_representation_3`	The four constant-cone equalities are expressed by membership for either sign.
Lemma 3.5	`DeciENSSE.Paper.lem_lf_clauses`	The singleton-system clause equivalence includes the automatic extension and exclusion conditions; uniqueness and existence express the bijection. The correspondence also holds for infinite variable carriers.
Lemma 3.6(1)	`DeciENSSE.Paper.lem_lf_moves_1`	Includes the stated extension to pure derivations. The variable carrier need not be finite.
Lemma 3.6(2)	`DeciENSSE.Paper.lem_lf_moves_2`	Includes the stated sign invariant for pure derivations. The variable carrier need not be finite.
Lemma 3.8(1)	`DeciENSSE.Paper.lem_lf_closure_1`	Facts, seeds, closure and leastness are separate conjuncts. The variable carrier need not be finite.
Lemma 3.8(2)	`DeciENSSE.Paper.lem_lf_closure_2`	Consistency is Avoids paired with Consistent; the general leastness clause also applies to the empty and singleton seed sets. The variable carrier need not be finite.
Lemma 3.8(3)	`DeciENSSE.Paper.lem_lf_closure_3`	Consistency separates forbidden roots from exclusion pairs; Exclusion fixes the positive-first orientation. The variable carrier need not be finite.
Lemma 3.8(4)	`DeciENSSE.Paper.lem_lf_closure_4`	The variable carrier need not be finite.
Lemma 3.8(5)	`DeciENSSE.Paper.lem_lf_closure_5`	Sum types separate visible and hidden variables; consistency is Avoids paired with Consistent. The equivalence holds for every seed set, without the paper's redundant closure assumptions. Visible and hidden carriers may be infinite.
Lemma 3.10	`DeciENSSE.Paper.lem_lf_duality`	The variable carrier need not be finite.
Lemma 3.11	`DeciENSSE.Paper.lem_lf_ext_first`	The variable carrier need not be finite.
Lemma 3.11	`DeciENSSE.Paper.lem_lf_extensions`	The variable carrier need not be finite.
Lemma 3.12	`DeciENSSE.Paper.lem_lf_collision`	The variable carrier need not be finite.
Corollary 3.13(1)	`DeciENSSE.Paper.cor_lf_dead_1`	This item states sufficiency of cause (a) and its pure form; the full disjunction equivalence is in item (3). The variable carrier need not be finite.
Corollary 3.13(2)	`DeciENSSE.Paper.cor_lf_dead_2`	This item states sufficiency of cause (b); the full disjunction equivalence is in item (3). The variable carrier need not be finite.
Corollary 3.13(3)	`DeciENSSE.Paper.cor_lf_dead_3`	Includes the full three-cause equivalence and the pure form of cause (c). The variable carrier need not be finite.
Lemma 10.4(1)	`DeciENSSE.Paper.lem_mo_sep_1`	The separator word is sepWord u t over Option alpha, with none as the separator. Alphabet finiteness is not required.

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Lemma 10.4(2)	`DeciENSSE.Paper.lem_mo_sep_2`	Cross comparisons are TwoWordInstance.Cross, which also records the cut bounds. Alphabet finiteness is not required.
Lemma 10.4(3)	`DeciENSSE.Paper.lem_mo_sep_3`	The comparison equivalence holds over any alphabet; finiteness is not required.
Lemma 10.4(4)	`DeciENSSE.Paper.lem_mo_sep_4`	An anchored comparison is a comparison whose continuation letter, at the cut x + length w - y, is the separator; aligned cross comparisons are TwoWordInstance.Aligned. Alphabet finiteness is not required.
Lemma 10.5(1)	`DeciENSSE.Paper.lem_mo_pi_reader_1`	The reader is Mortal.reader on all states of RState, itself a computable definition. This item also packages the computability of the relations as a computable definition, whose values are the decision procedures Self.instDecidableAdm, Mortal.instDecidableRej and Mortal.instDecidablePi for admission, rejection and Pi.
Lemma 10.5(2)	`DeciENSSE.Paper.lem_mo_pi_reader_2`	A Pi-hole is written out as a nonempty reader hole with an anchored comparison in Pi; the admissions are Self.Adm, the cross and self guards together.
Lemma 10.5(3)	`DeciENSSE.Paper.lem_mo_pi_reader_3`	Stated for the mortal hole (u, t) of item (2); the rejected tail is the number of letters after the first rejected cut.
Lemma 10.6	`DeciENSSE.Paper.lem_mo_depth`	The depth bound d_I is written out with the state count of the reader.
Lemma 10.8	`DeciENSSE.Paper.lem_mo_base`	The lettered monitor of a reader is ofReader and the base channel is baseChan, whose relation is also false into the start label, which no comparison reaches; the first claim holds for every word; a Pi-hole is written out as a nonempty reader hole with an anchored comparison in Pi. Neither alphabet nor state finiteness is required.
Lemma 10.10	`DeciENSSE.Paper.lem_mo_desub`	The expansion of v' is expand l v'; the derived monitor and channel are der and posDer. State finiteness is not required.
Corollary 10.11	`DeciENSSE.Paper.cor_mo_tower`	The canonical hierarchy is hierOf with markers markOf, which needs an inhabited alphabet; the towers are tower and posTower, and their recursion is included; the number of states is not used.
Theorem 10.14	`DeciENSSE.Paper.thm_mo_anchored`	L-sharp, G-sharp and Xi-sharp are anchoredLengthBound, anchoredDictionaryBound and anchoredExponent. Alphabets are inhabited for the canonical hierarchy; finiteness of the alphabet and N at least 1 are not needed. A Pi-hole is written out as in lem:mo-base.
Theorem 10.15	`DeciENSSE.Paper.thm_mo_short`	L-sharp of I is anchoredLengthBound at the state count of the reader and the depth bound of lem:mo-depth, written out.

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Corollary 10.16	`DeciENSSE.Paper.cor_mo_decidable`	Decidability is a computable definition, the executable procedure Self.decideHasMortalHole, together with the bounded search over words of length at most L-sharp of I that it performs.
Theorem 10.17	`DeciENSSE.Paper.thm_mo_decide`	Lnm is pumpHoleBound and L-sharp of I is written out as in thm:mo-short. Decidability is a computable definition for instances with (FH): the executable bounded search SelfTwoWordInstance.decideBoundedHole at the bound selfHoleBound, transported along hasHole_iff_bounded.
Lemma 10.18	`DeciENSSE.Paper.lem_mo_estimate`	N-bar is the cardinality of the reader-state type Mortal.State, and Len-sharp is mortalHoleBound at the four cardinalities; this numerical estimate requires no automaton instance and also permits empty state carriers.
Corollary 10.19	`DeciENSSE.Paper.cor_mo_extracted`	Len-sharp of I_pq is extractedBoundSelf mortalHoleBound X Z, the mortal bound at the state counts of extractSelf, the same for all p and q; the separator word u#t has length |u| + |t| + 1.
Lemma 9.1(1)	`DeciENSSE.Paper.lem_nm_continuation_1`	State preservation is quantified over an arbitrary automaton, and thus includes all four readers on either word. Neither a finite alphabet nor (FH) is needed.
Lemma 9.1(2)	`DeciENSSE.Paper.lem_nm_continuation_2`	Cross includes both cut bounds, including s ≤ |u|; self comparisons specialize the same identity to u = t and strict cuts. Neither a finite alphabet nor (FH) is needed.
Lemma 9.1(3)	`DeciENSSE.Paper.lem_nm_continuation_3`	Keeps is defined by the overhang prefix test; self comparisons specialize cross comparisons to equal words. The periodic stream of a nonempty overhang takes its default letter from the overhang itself, so the alphabet may be empty; it need not be finite.
Lemma 9.3	`DeciENSSE.Paper.lem_nm_decomposition`	Completion is split into cross and self parts; the rejected cut is the length of the pre-hole target, whose J equals that length by definition. The alphabet need not be finite.
Lemma 9.6	`DeciENSSE.Paper.lem_nm_breaker`	The periodic prefix test is written $\theta \le$prefix oθ, equivalent to $\theta \le$prefix o^ω for nonempty o; completion has cross and self parts. Neither a finite alphabet nor (FH) is needed.
Lemma 9.8(1)	`DeciENSSE.Paper.lem_nm_thin_1`	Neither a finite alphabet nor (FH) is needed.
Lemma 9.8(2)	`DeciENSSE.Paper.lem_nm_thin_2`	Neither a finite alphabet nor (FH) is needed.
Lemma 9.8(3)	`DeciENSSE.Paper.lem_nm_thin_3`	A default alphabet letter presents periodic streams; its value is irrelevant for nonempty periods. Neither a finite alphabet nor (FH) is needed.
Lemma 9.8(4)	`DeciENSSE.Paper.lem_nm_thin_4`	Neither a finite alphabet nor (FH) is needed.

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Lemma 9.10(1)	`DeciENSSE.Paper.lem_nm_survival_1`	Non-alignment is equivalent to a nonempty overhang for a comparison; self comparisons are the strict equal-word specialization. A default alphabet letter presents periodic streams; its value is irrelevant for nonempty periods. The alphabet need not be finite.
Lemma 9.10(2)	`DeciENSSE.Paper.lem_nm_survival_2`	Self comparisons are the specialization u = t and s < e. The alphabet need not be finite.
Lemma 9.10(3)	`DeciENSSE.Paper.lem_nm_survival_3`	The overhang length is offset; self comparisons are the specialization u = t and s < e. A default alphabet letter presents periodic streams; its value is irrelevant for nonempty periods. The alphabet need not be finite.
Corollary 9.11	`DeciENSSE.Paper.cor_nm_lasso_completion`	Completion is split into cross completion and self completion. A default alphabet letter presents periodic streams; its value is irrelevant for nonempty periods. Neither a finite alphabet nor (FH) is needed. The completion implication does not require a pre-hole hypothesis.
Proposition 9.12	`DeciENSSE.Paper.prop_nm_short_prehole`	E is a finite set of states; the result holds without requiring its members to be safe. Decidability is a computable definition, the executable bounded search TwoWordInstance.preHoleExistsDecidable over pairs of words of length at most 12aba'b'. (FH) is not needed.
Lemma 9.13	`DeciENSSE.Paper.lem_nm_aperiodic`	Self comparisons specialize cross comparisons to equal words; survivor-freeness has cross and self parts. Neither a finite alphabet nor (FH) is needed.
Theorem 9.14	`DeciENSSE.Paper.thm_nm_pump`	pumpBound is the displayed polynomial and pumpCapacity is Pcap; survivor-freeness has cross and self parts. Neither a finite alphabet nor (FH) is needed.
Theorem 9.17(1)	`DeciENSSE.Paper.thm_nm_three_cases_1`	State carriers are finite initial segments; pumpHoleBound is Lnm. This item gives branching completion with its bound, without (FH).
Theorem 9.17(2)	`DeciENSSE.Paper.thm_nm_three_cases_2`	State carriers are finite initial segments; survivor-freeness has cross and self parts. Includes the exact prefix claim for a completed hole and the Lnm bound without (FH).
Theorem 9.17(3)	`DeciENSSE.Paper.thm_nm_three_cases_3`	NonMortalCase is the disjunction of the branching and thin survivor-free pre-hole cases. Includes the theorem's full equivalence and its original-hole mortal alternative. The alphabet need not be finite.
Corollary 9.18	`DeciENSSE.Paper.cor_nm_short_holes`	State carriers are finite initial segments and pumpHoleBound is Lnm.
Lemma 6.2(1)	`DeciENSSE.Paper.lem_sm_summaries_1`	Pure moves are represented in the covariant signed system.
Lemma 6.2(2)	`DeciENSSE.Paper.lem_sm_summaries_2`	Pure moves are represented in the covariant signed system.
Lemma 6.2(3)	`DeciENSSE.Paper.lem_sm_summaries_3`	Pure moves are represented in the covariant signed system.

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Lemma 6.3	`DeciENSSE.Paper.lem_sm_lowest_point`	
Lemma 6.4(1)	`DeciENSSE.Paper.lem_sm_duality_1`	
Lemma 6.4(2)	`DeciENSSE.Paper.lem_sm_duality_2`	
Lemma 6.4(3)	`DeciENSSE.Paper.lem_sm_duality_3`	
Lemma 6.6	`DeciENSSE.Paper.lem_sm_mu`	The bound 4^|V| is the cardinality of the signed powerset; PF is the finite set defined by this bounded search, and it equals the set of the paper, defined with arbitrary words. The computability claim is a computable definition: membership in that set is decided by membership in PF.
Lemma 6.8(1)	`DeciENSSE.Paper.lem_sm_roots_1`	Root deadness is expanded as reachability of a forbidden root.
Lemma 6.8(2)	`DeciENSSE.Paper.lem_sm_roots_2`	Root deadness is expanded as reachability of a forbidden root.
Lemma 6.8(3)	`DeciENSSE.Paper.lem_sm_roots_3`	Root deadness is expanded as reachability of a forbidden root.
Lemma 6.8(4)	`DeciENSSE.Paper.lem_sm_roots_4`	Root deadness is expanded as reachability of a forbidden root.
Lemma 6.10(1)	`DeciENSSE.Paper.lem_sm_dead_states_1`	The root-dead region is represented by the computable state predicate negReachS rather than an intersection with a set of variables.
Lemma 6.10(2)	`DeciENSSE.Paper.lem_sm_dead_states_2`	flipHitS is intersection of the flipped summary with PF.
Lemma 6.10(3)	`DeciENSSE.Paper.lem_sm_dead_states_3`	The support-dead region is the disjunction of negReachS and muReachS on the current state; the last disjunct is opposite-atom membership in Mu.
Lemma 6.10(4)	`DeciENSSE.Paper.lem_sm_dead_states_4`	
Lemma 6.10(5)	`DeciENSSE.Paper.lem_sm_dead_states_5`	The computability claim for the two finite sets of signed variables is a computable definition: membership is decided by the executable procedures decideRootDeadEmpty and decideSupportDeadEmpty; root deadness is reachability of a forbidden root.
Theorem 6.11	`DeciENSSE.Paper.thm_sm_dead`	
Lemma 6.14	`DeciENSSE.Paper.lem_sm_membership`	
Lemma 6.15	`DeciENSSE.Paper.lem_sm_head_prefix`	Root deadness is expanded as reachability of a forbidden root.
Proposition 6.16(1)	`DeciENSSE.Paper.prop_sm_decide_1`	Forbidden roots are enumerated by negList; MuCollision is simultaneous membership of both signs. The bound is the cardinality of the product of the two flagged powerset readers, equal to 2^(4|V|+2). Decidability of satisfiability is a computable definition, the executable procedure decideSatMixed.

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Proposition 6.16(2)	`DeciENSSE.Paper.prop_sm_decide_2`	FailureI0 is consequent membership and antecedent nonmembership at a visible signed atom. The bound is the cardinality of the product of the two flagged powerset readers, equal to 2^(2|X|+2|X+Z|+2). Decidability of (I0) is a computable definition, the executable procedure decideCondI0.
Lemma 7.1	`DeciENSSE.Paper.lem_un_dead`	Absence of a live root-dead seed is written as a universally quantified implication.
Lemma 7.3(1)	`DeciENSSE.Paper.lem_un_source_1`	The Boolean component is specified by its truth predicate; the state count is the cardinality of the product carrier.
Lemma 7.3(2)	`DeciENSSE.Paper.lem_un_source_2`	
Lemma 7.3(3)	`DeciENSSE.Paper.lem_un_source_3`	The comparison includes its strict cut bounds explicitly.
Lemma 7.5	`DeciENSSE.Paper.lem_un_holes`	Includes the consequent universal characterization of (U); signed variables index the readers.
Lemma 7.6	`DeciENSSE.Paper.lem_un_rejected_path`	RejectedPath packages betweenness and the admission cone.
Theorem 7.9	`DeciENSSE.Paper.thm_un_decidable`	N_U is the cardinality of TargetState; holeLengthBound N is holeBound(N, d_N). Decidability is a computable definition for satisfiable antecedents, the executable procedure decideCondU. The closed form is stated for a nonempty visible carrier, with v = |X| and s = |X| + |Z|.